

Reporte: Core vs Divergent Metabolismo + Clínica + Supervivencia

Pacientes Core: 1111 | Divergentes: 55

Variables metabólicas analizadas: 248 | Significativas ($p_{adj} < 0.05$): 74

Soluciones FBA: FBA, pFBA, L1w, L2, L2w

Estructura: Clínica → Metabolismo → Integración → Supervivencia

Tamaño de efecto: Cliff's . | ≥ 0.147 pequeño · ≥ 0.330 mediano · ≥ 0.474 grande

SECCIÓN I

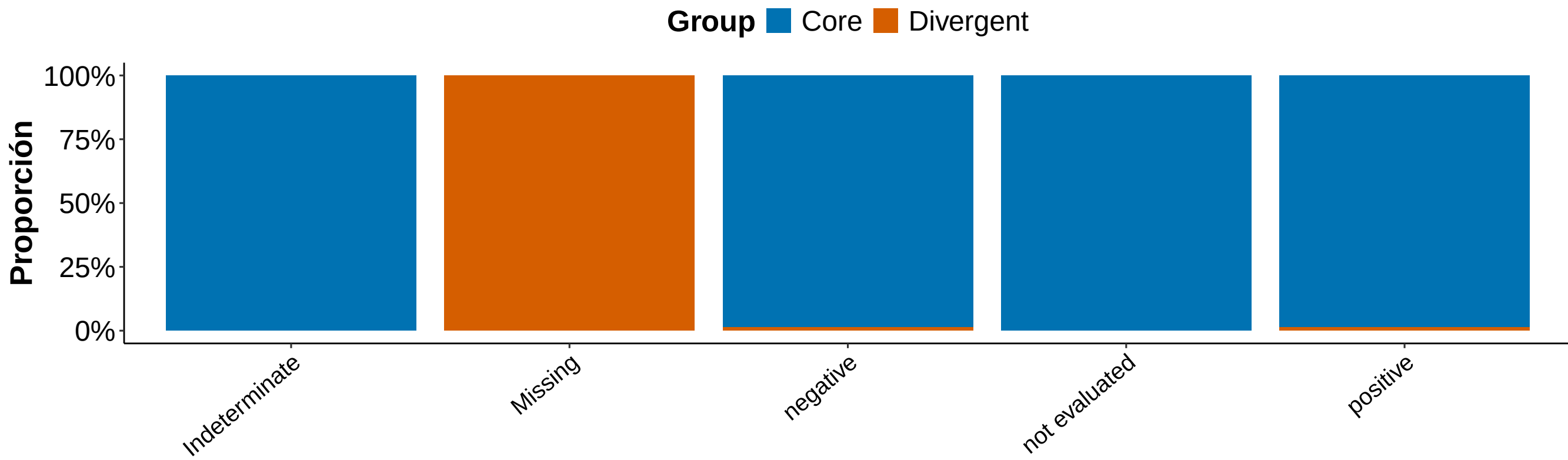
Diferencias Clínicas: Core vs Divergent

Macro-categorías: Biología Tumoral | Estadío | Historia Terapéutica | Factores del Paciente

Clínica Categórica – .A. Biología Tumoral

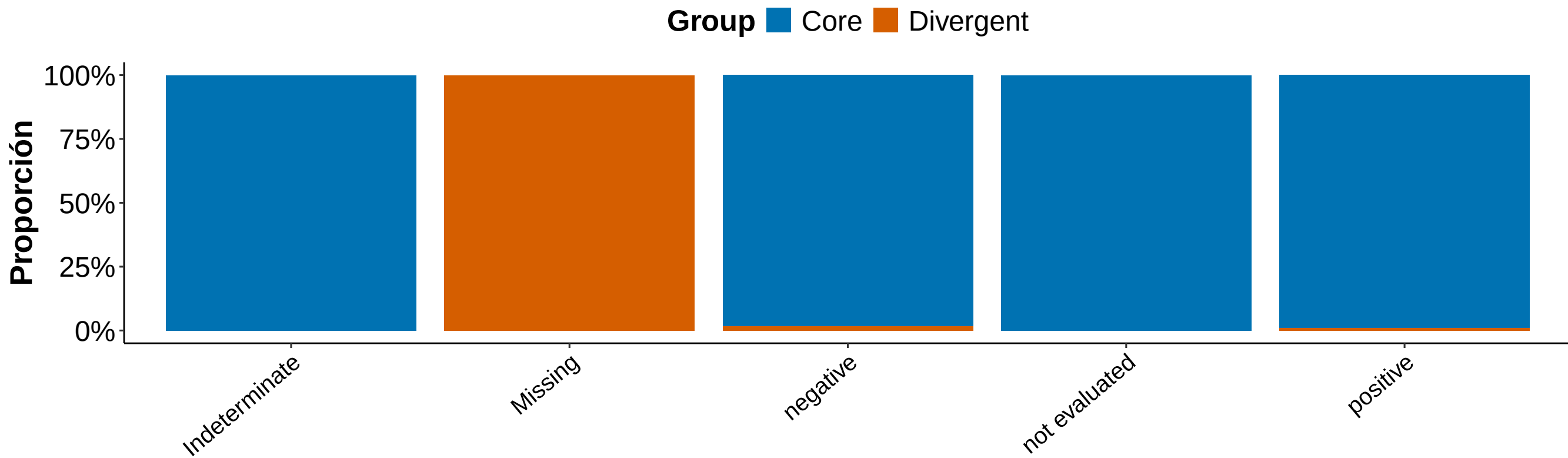
ER

Chi² p: 1.55e-184



PR

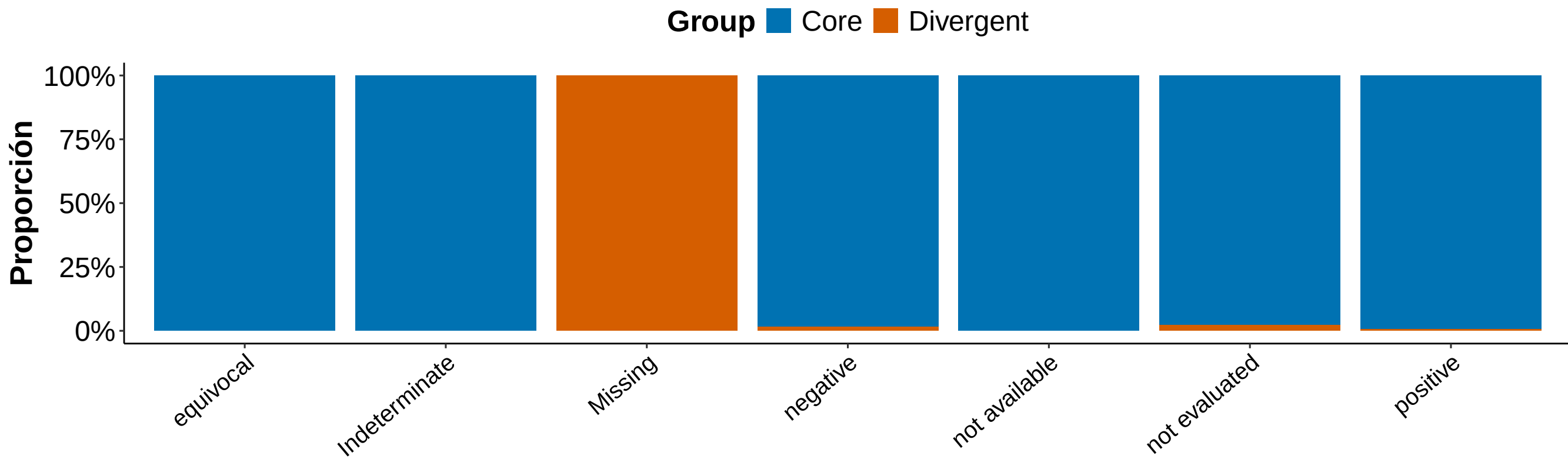
Chi² p: 1.4e-184



Clínica Categórica – .A. Biología Tumoral

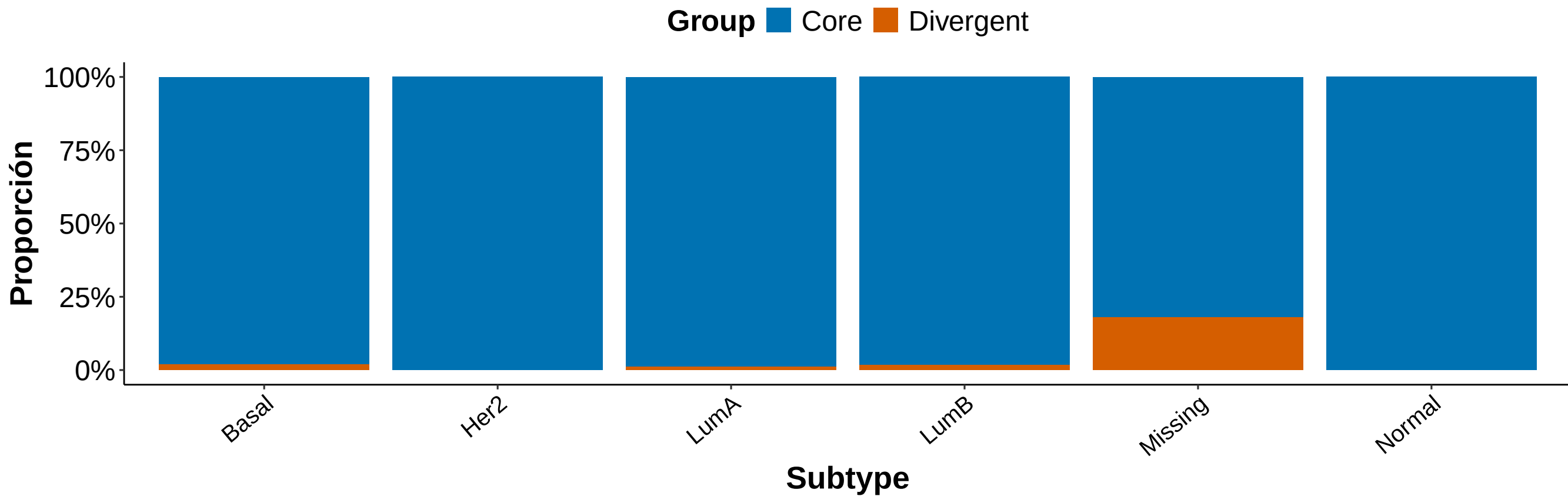
HER2

Chi² p: 1.89e-182



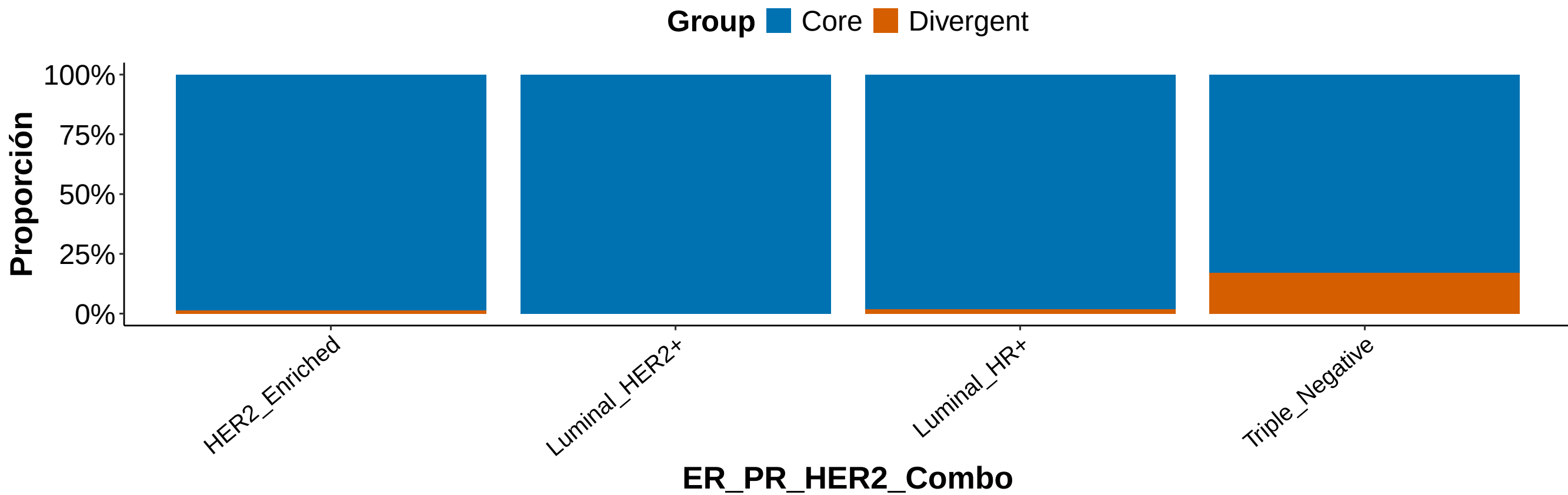
Subtype

Chi² p: 1.43e-24



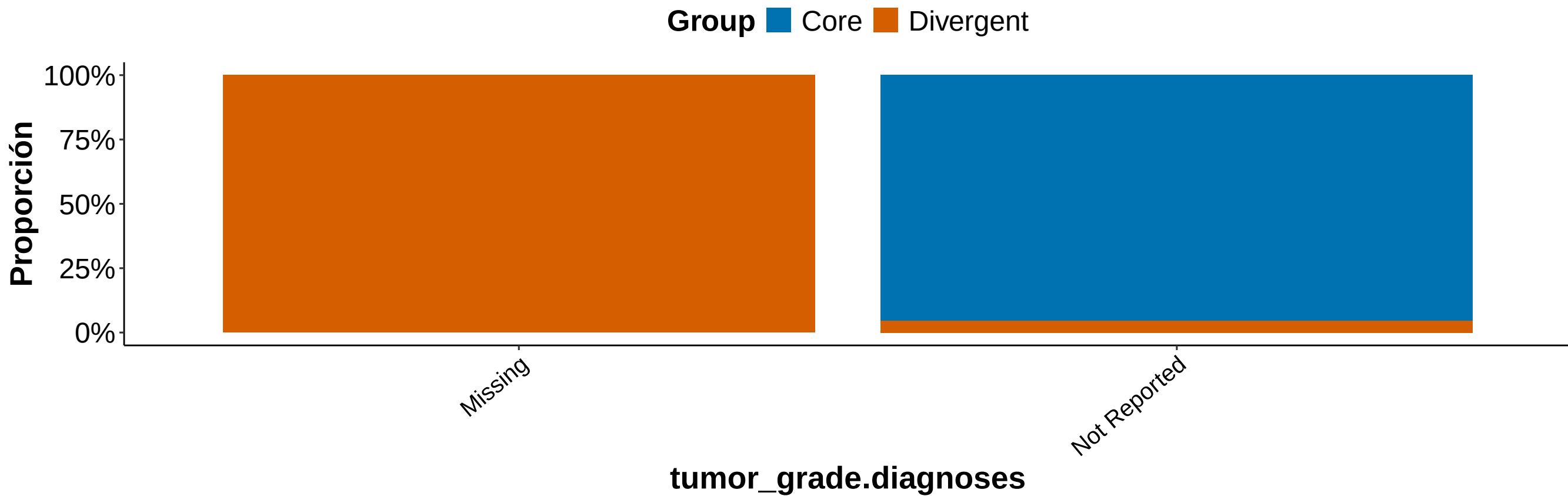
ER_PR_HER2_Combo

Chi² p: 8.16e-24



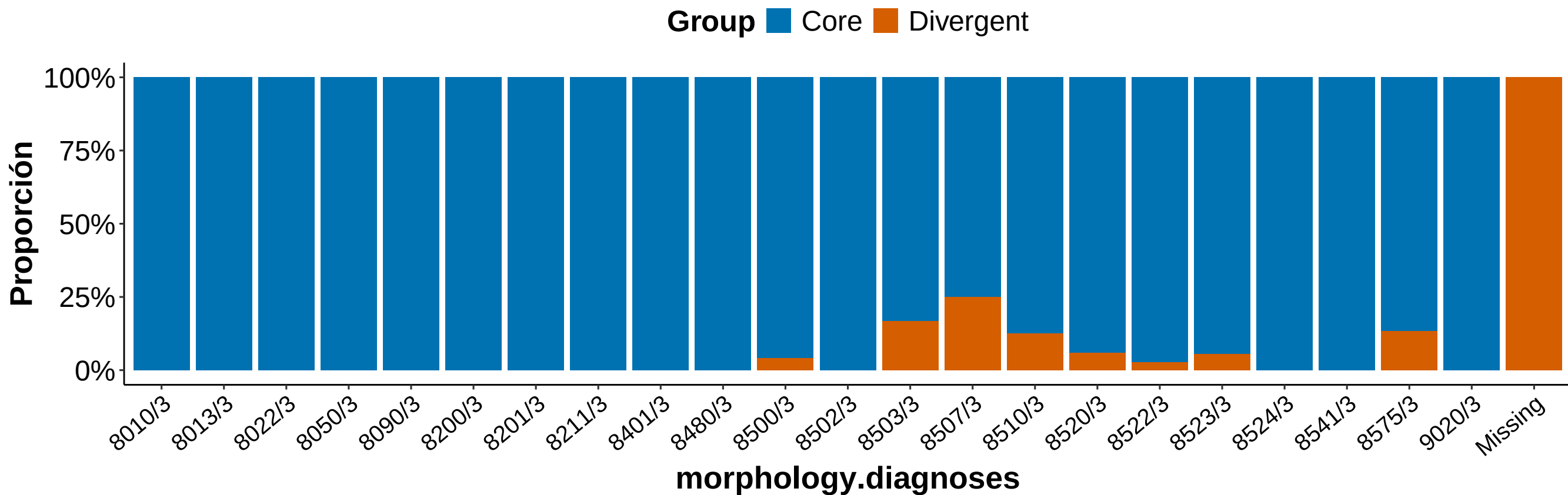
tumor_grade.diagnoses

Chi² p: 0.0326



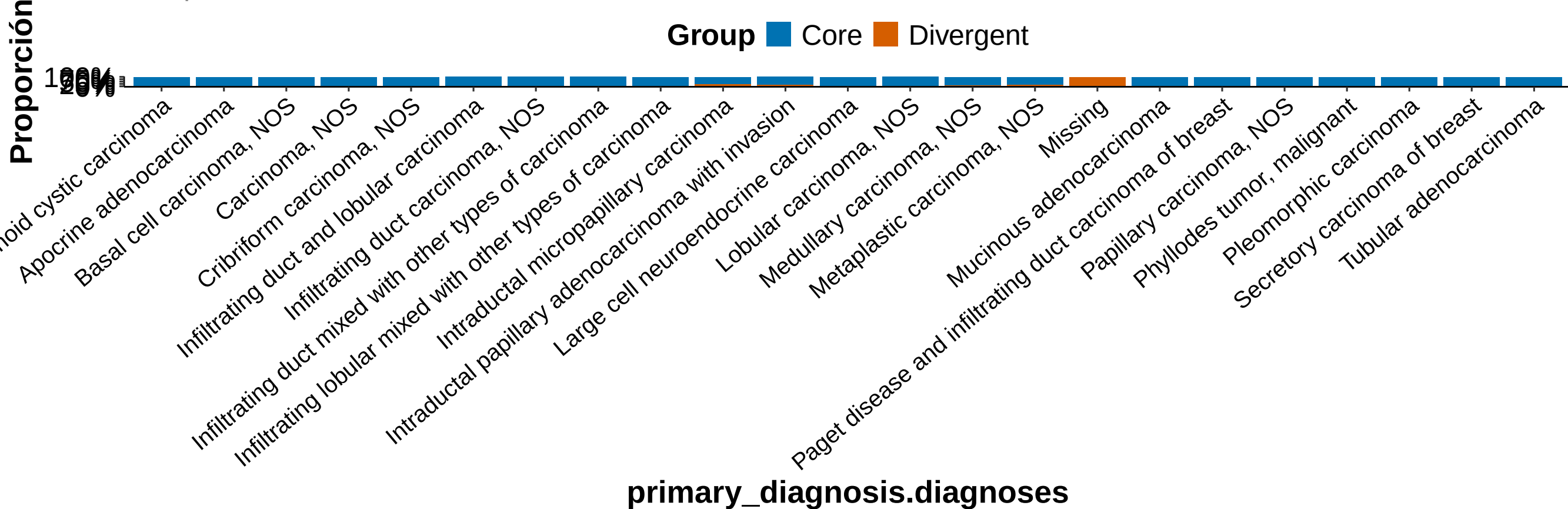
morphology.diagnoses

Chi² p: 0.0649



primary_diagnosis.diagnoses

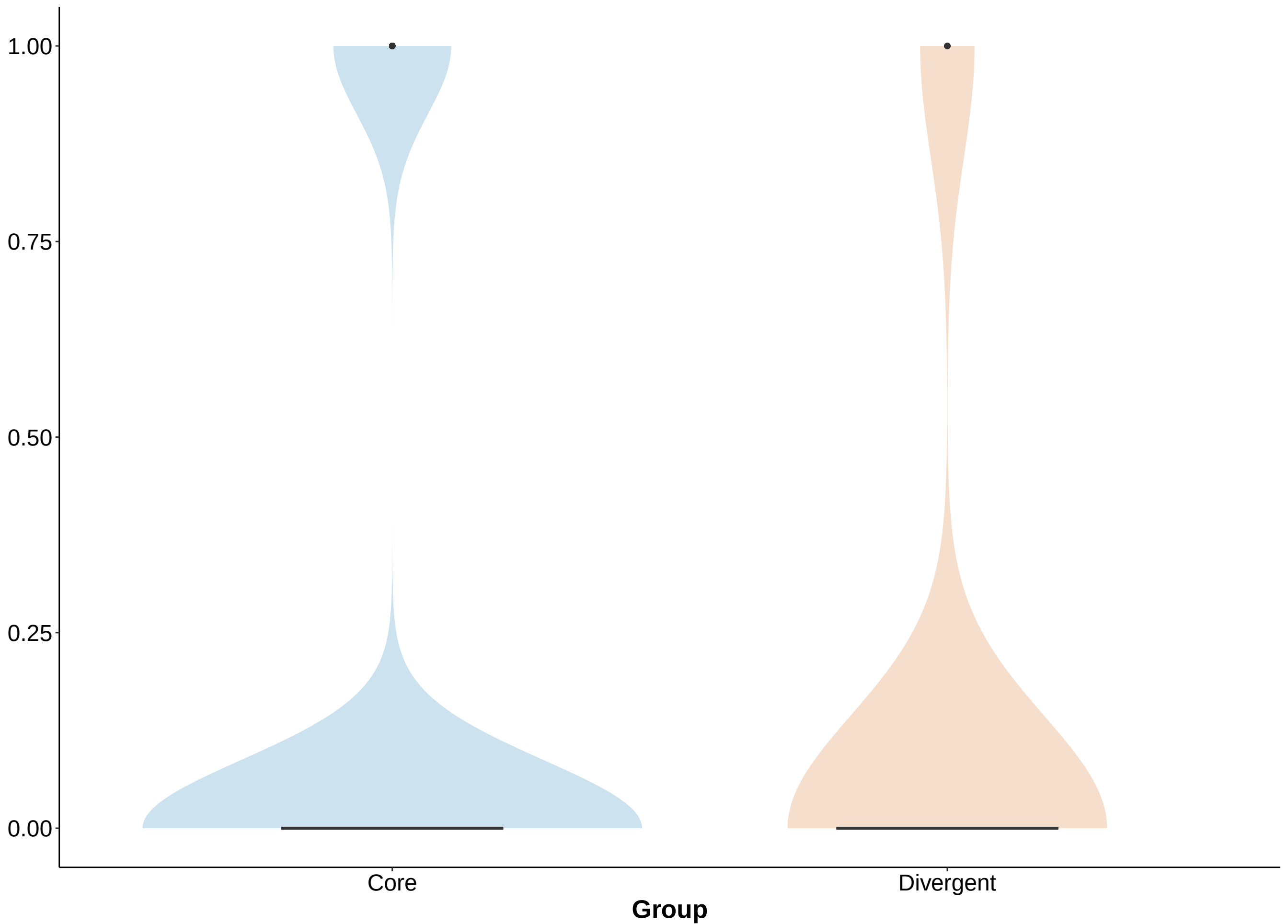
Chi² p: 0.0649



Clínica Numérica – . B. Estadío y Carga Tumoral

Metastasis_Flag

p-adj: 0.403 | .=-0.05 (Negligible)



Cliff's .: Clínica: . B. Estadío y Carga Tumoral

Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core1.00.5.00.51.0



Metastasis_Flag

ns .=-0.05

Cliff's Δ – Clínica: .B. Estadío y Carga Tumoral

Cliff's Δ + Δ% mediana | azul = mayor en Core · naranja = mayor en Divergent

Metastasis_Flag

ns Δ=-0.05 (NA%)

-0.6

Grande

Mediano

Pequeño

0

Pequeño

Mediano

Grande

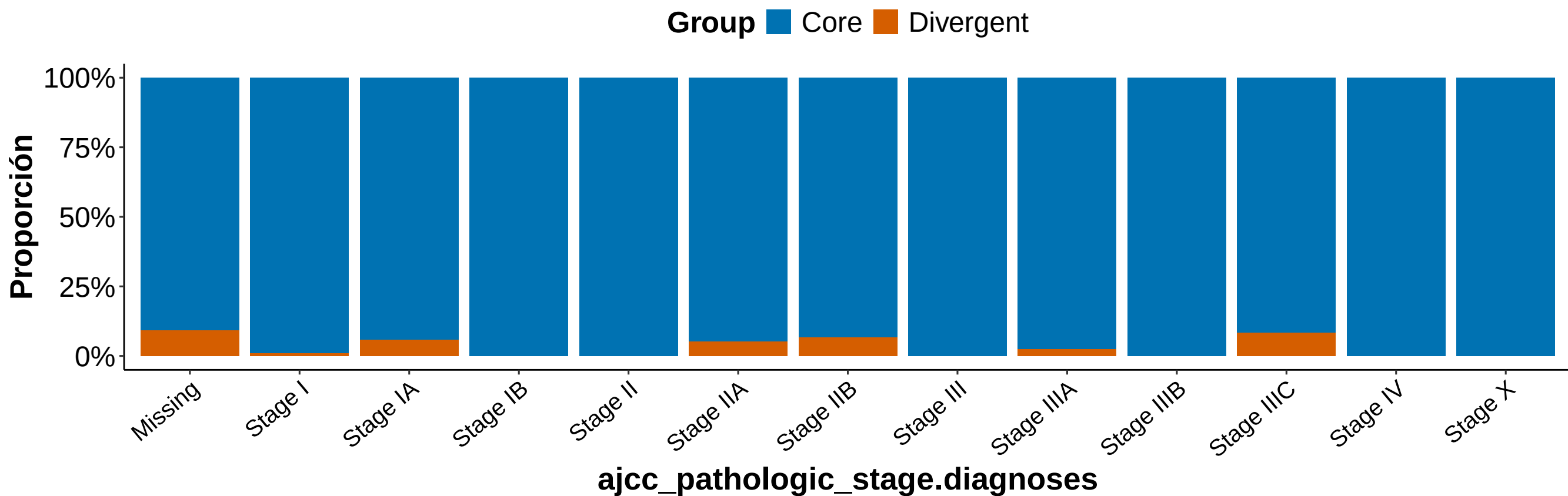
0.6

Cliff's delta [Core vs Divergent]

Clínica Categórica – . B. Estadío y Carga Tumoral

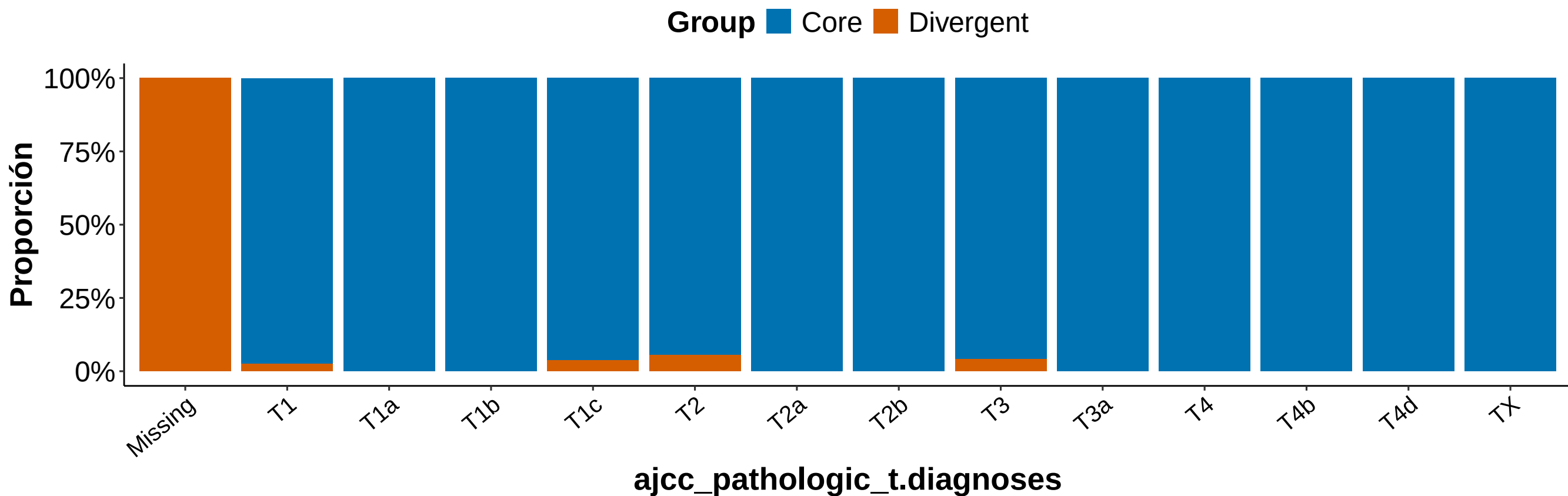
ajcc_pathologic_stage.diagnoses

Chi² p: 0.309



ajcc_pathologic_t.diagnoses

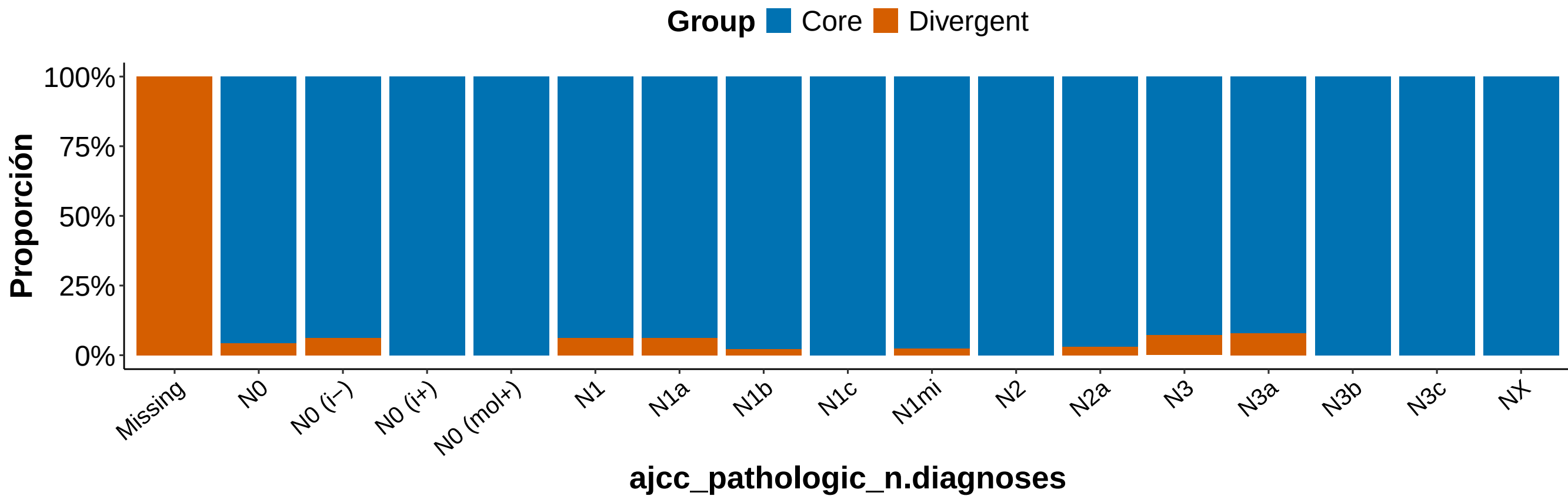
Chi² p: 0.018



Clínica Categórica – . B. Estadío y Carga Tumoral

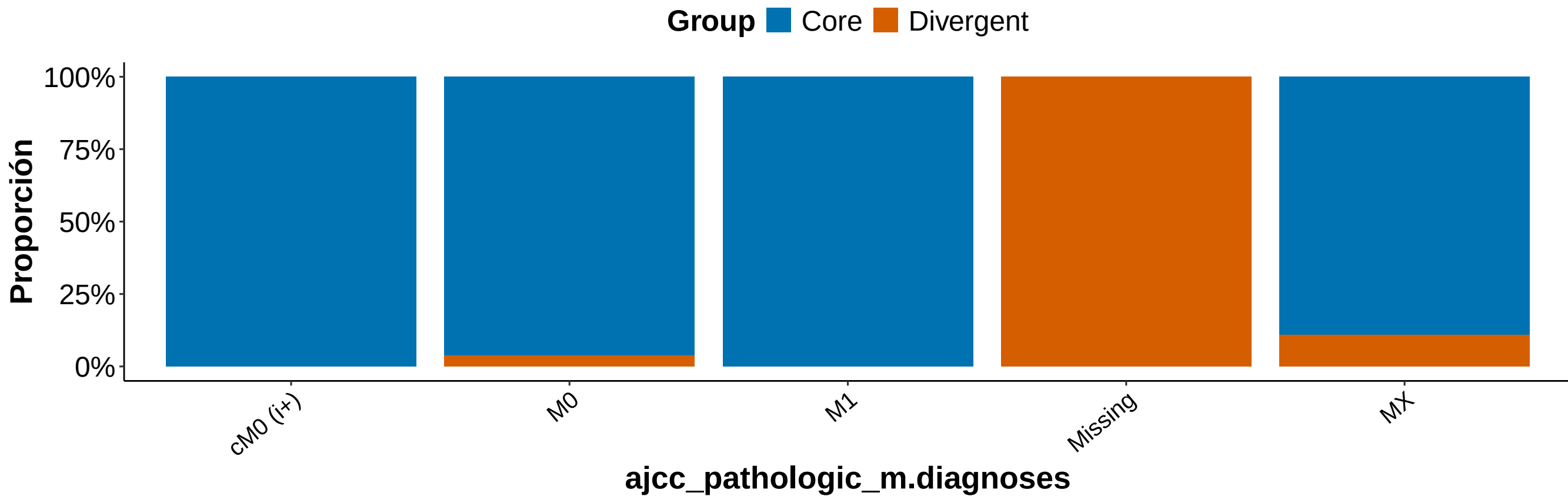
ajcc_pathologic_n.diagnoses

Chi² p: 0.0118

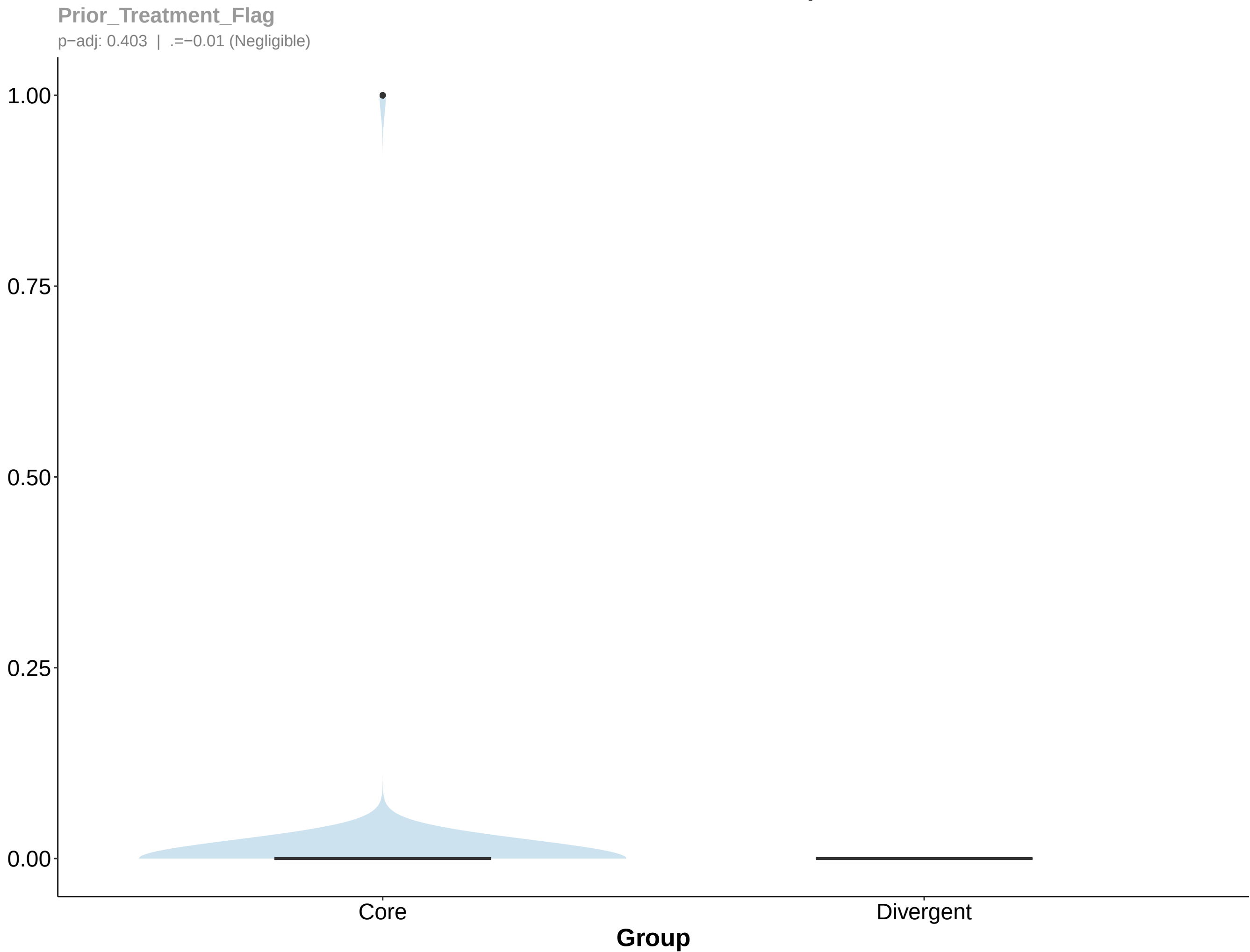


ajcc_pathologic_m.diagnoses

Chi² p: 1.29e-07



Clínica Numérica – . C. Historia Terapéutica



Cliff's .: Clínica: . C. Historia Terapéutica

Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05



Prior_Treatment_Flag

ns .=-0.01

Cliff's . – Clínica: . C. Historia Terapéutica

Cliff's . + .% mediana | azul = mayor en Core · naranja = mayor en Divergent

Prior_Treatment_Flag

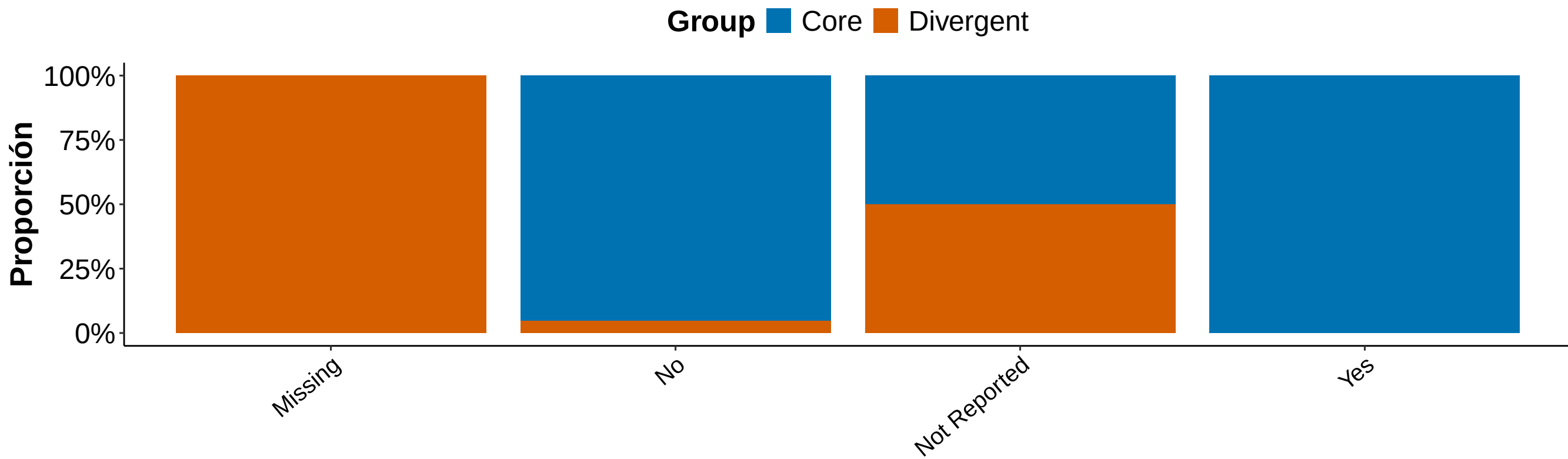
ns .=-0.01 (NA%)

-0.6 Grande Mediano Pequeño 0 Pequeño Mediano Grande 0.6

Cliff's delta [Core vs Divergent]

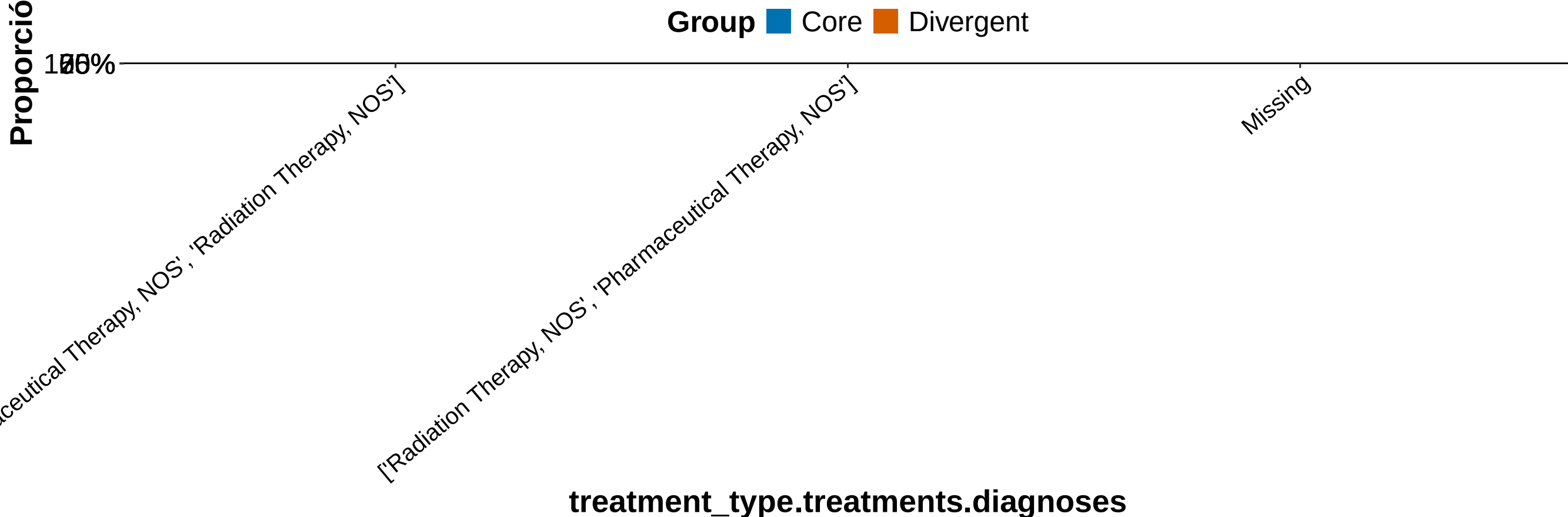
prior_treatment.diagnoses

Chi² p: 1.35e-06



treatment_type.treatments.diagnoses

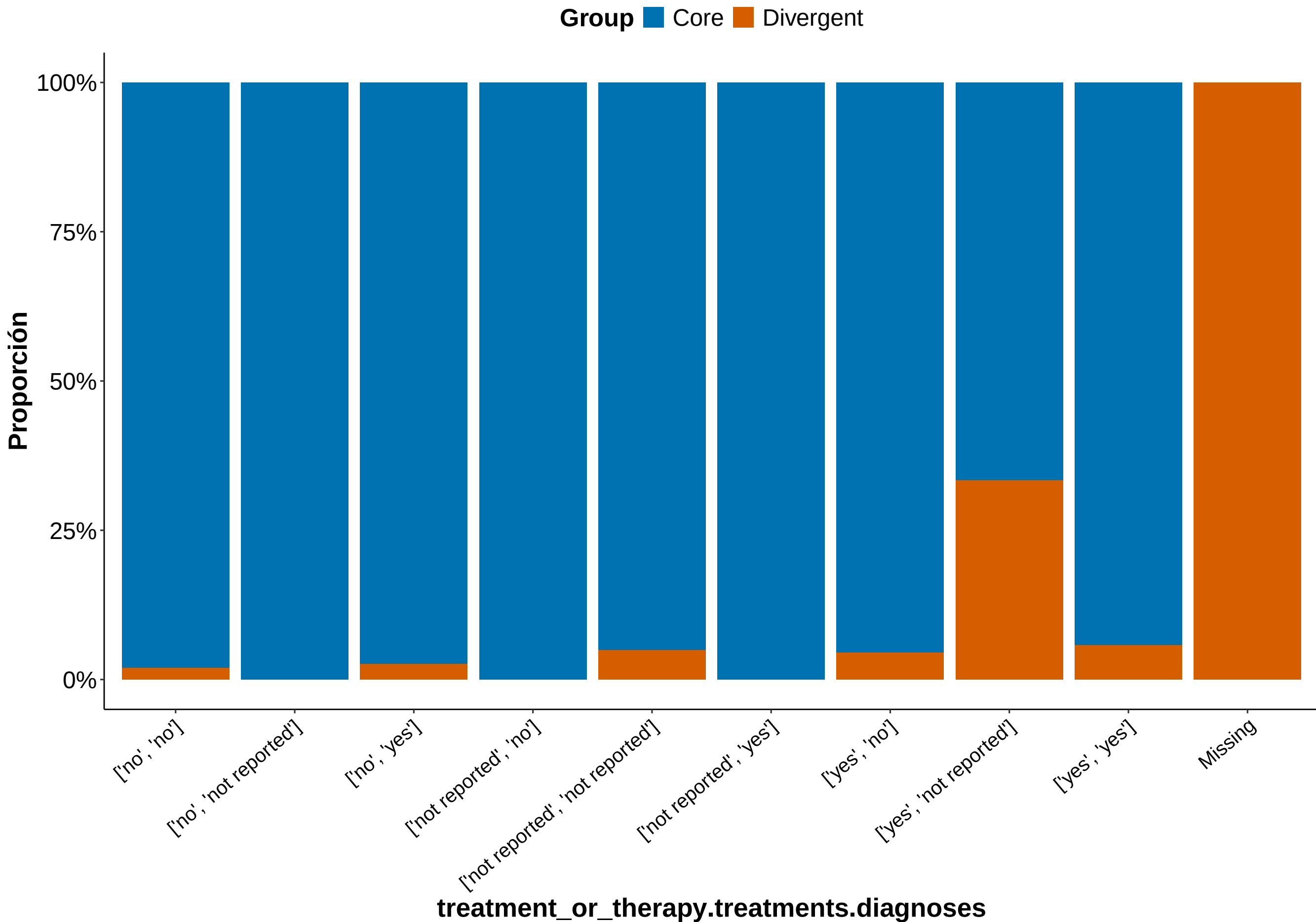
Chi² p: 3.78e-05



Clínica Categórica – . C. Historia Terapéutica

treatment_or_therapy.treatments.diagnoses

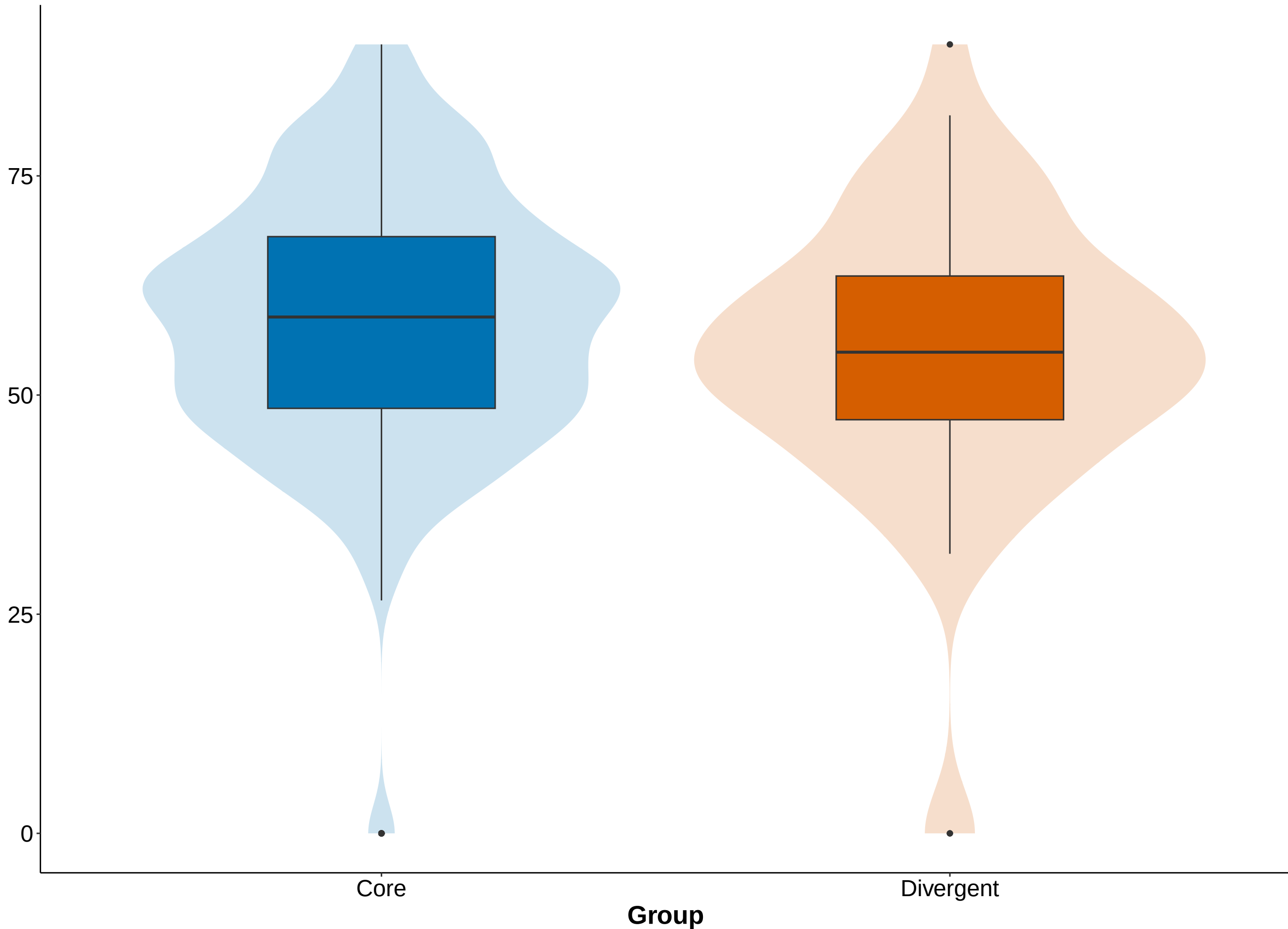
Chi² p: 0.000281



Clínica Numérica – . D. Factores del Paciente

age_at_diagnosis.diagnoses

p-adj: 0.136 | .=-0.14 (Negligible) | .%: -6.8



Cliff's .: Clínica: . D. Factores del Paciente

Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core1.00.0.00.51.0



age_at_diagnosis.diagnoses

ns .=-0.14

Cliff's . – Clínica: . D. Factores del Paciente

Cliff's . + .% mediana | azul = mayor en Core · naranja = mayor en Divergent

age_at_diagnosis.diagnoses

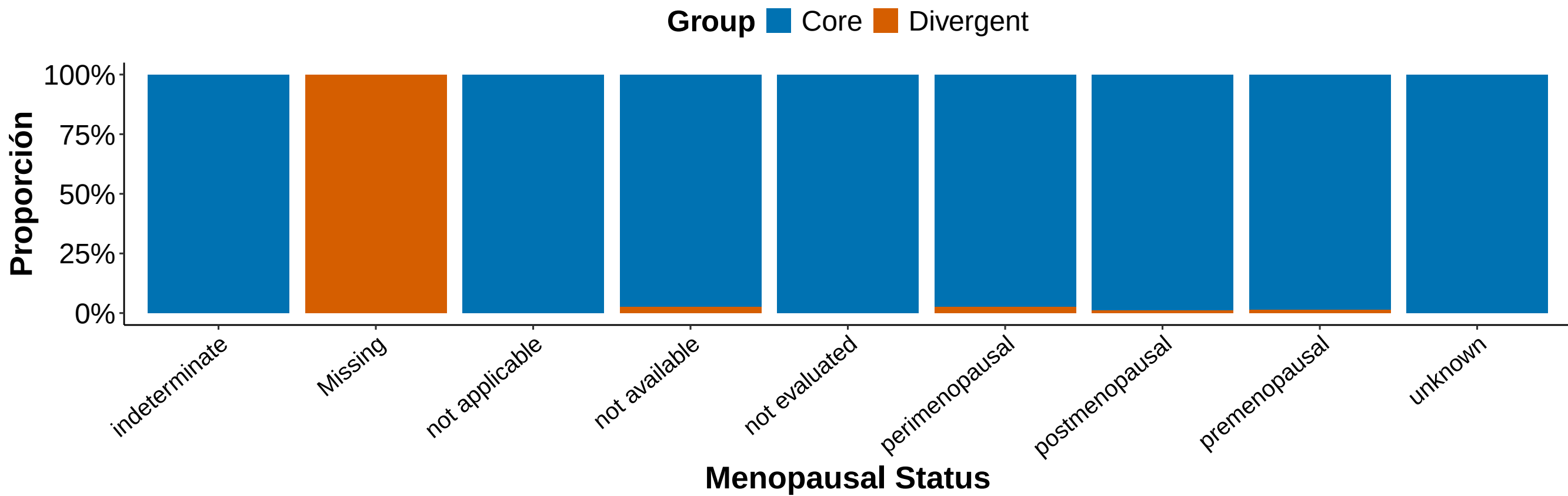
ns .=-0.14 (-6.8%)



Cliff's delta [Core vs Divergent]

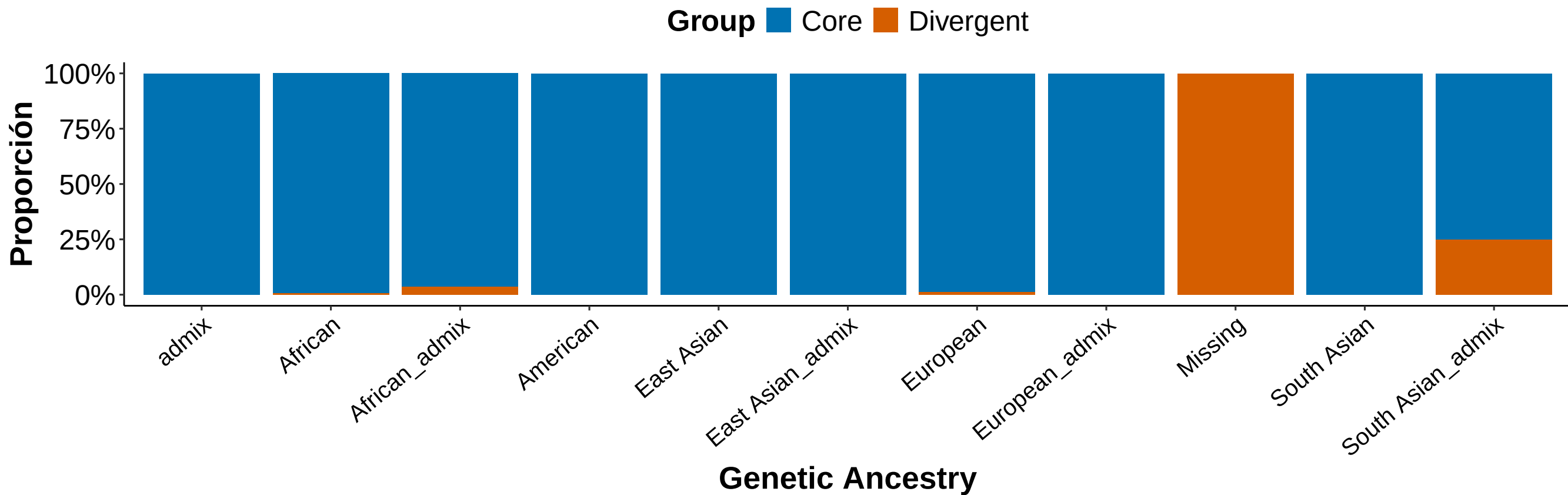
Menopausal Status

Chi² p: 3.67e-180



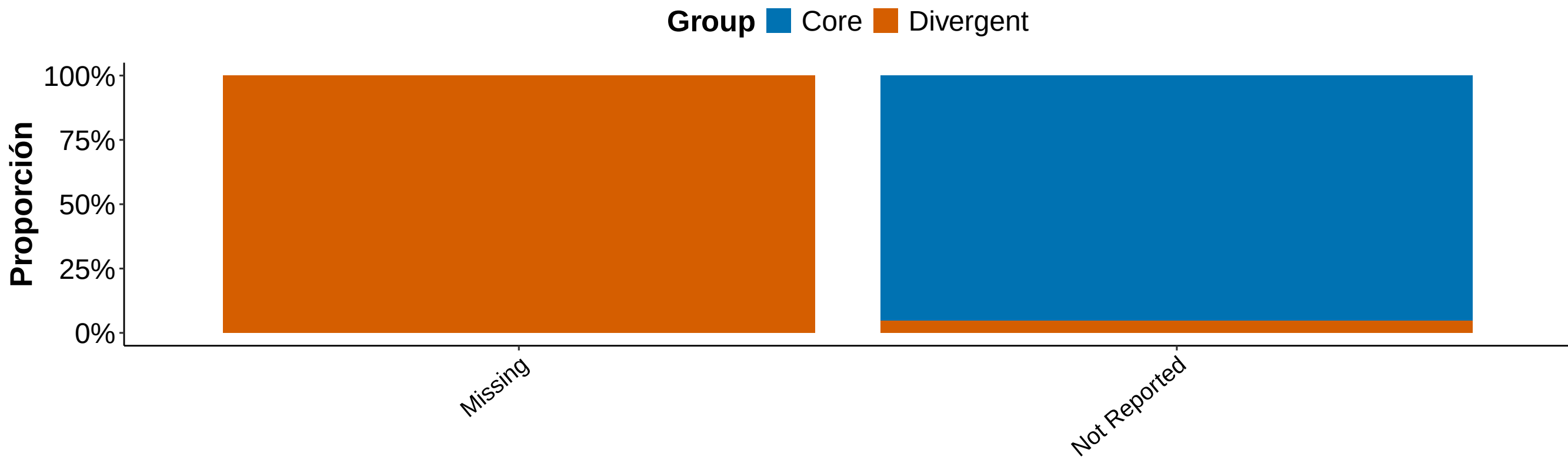
Genetic Ancestry

Chi² p: 2.86e-179



alcohol_history.exposures

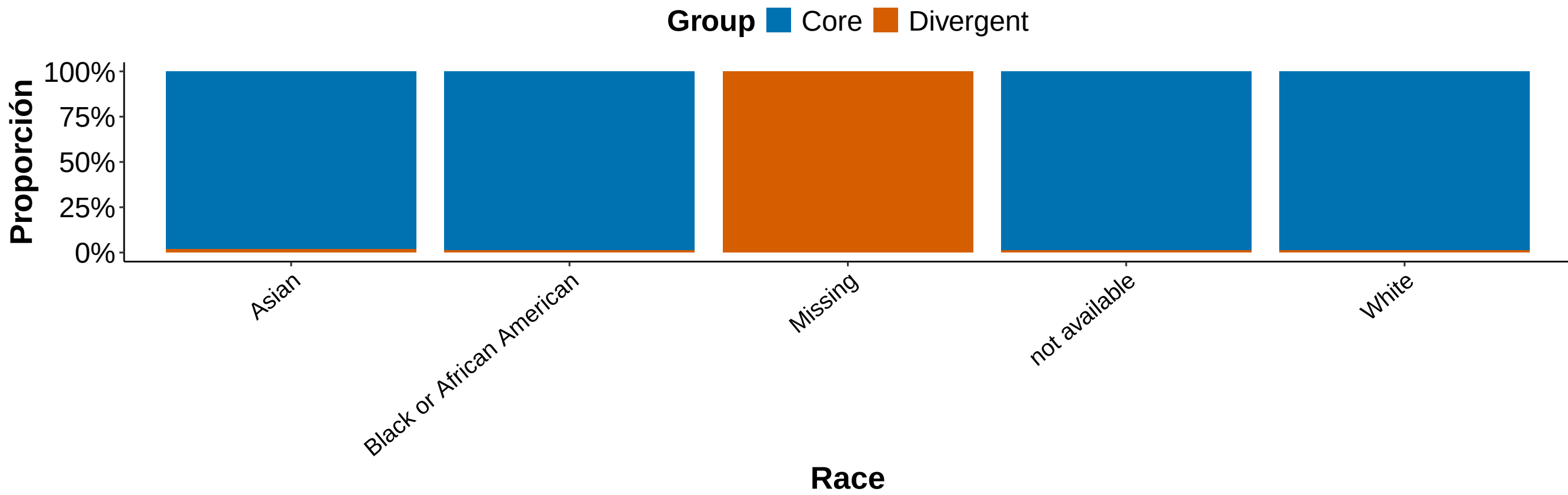
Chi² p: 0.0326



alcohol_history.exposures

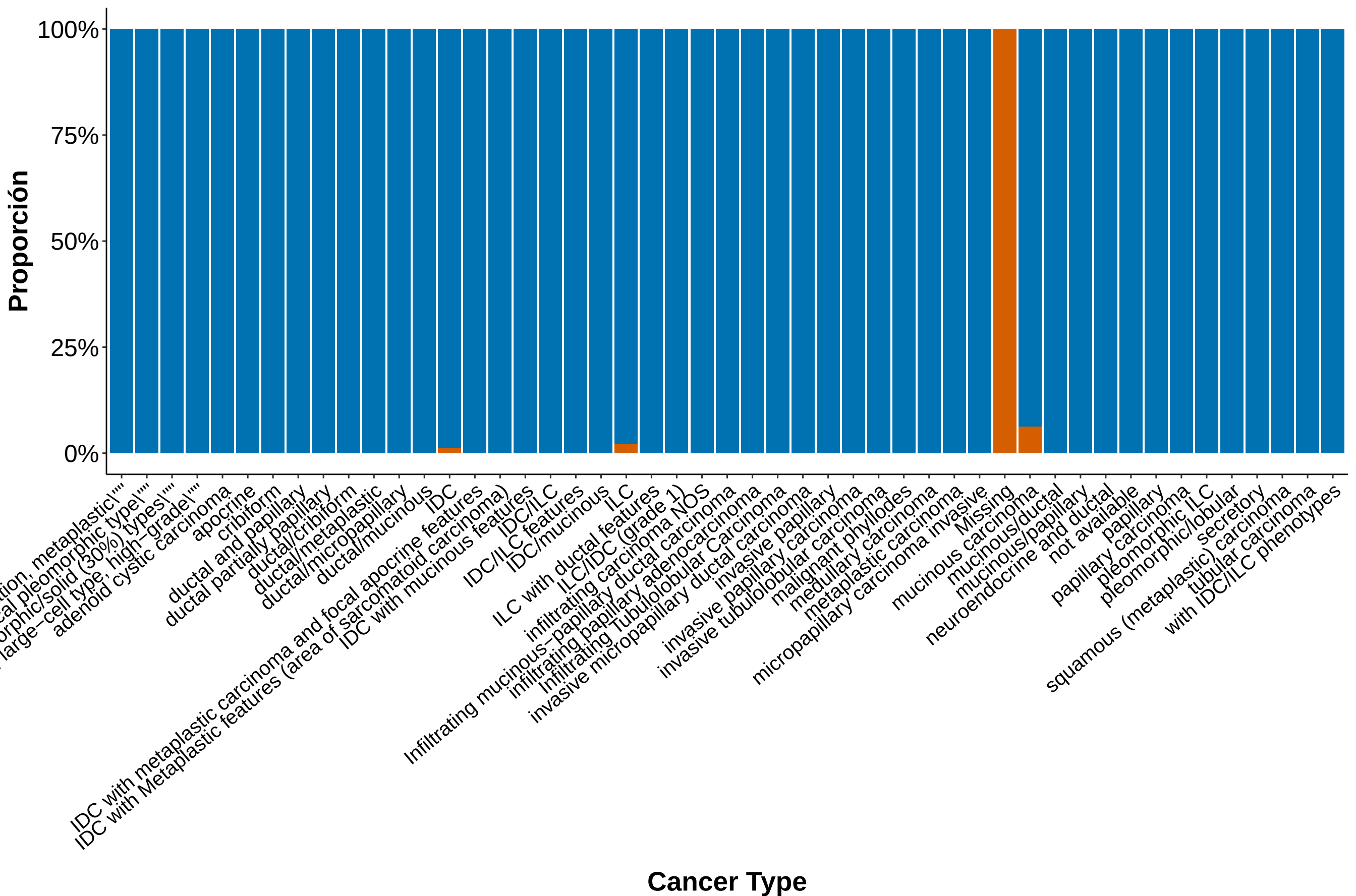
Race

Chi² p: 1.7e-184



Chi² p: 2.82e-149

Group ■ **Core** ■ **Divergent**



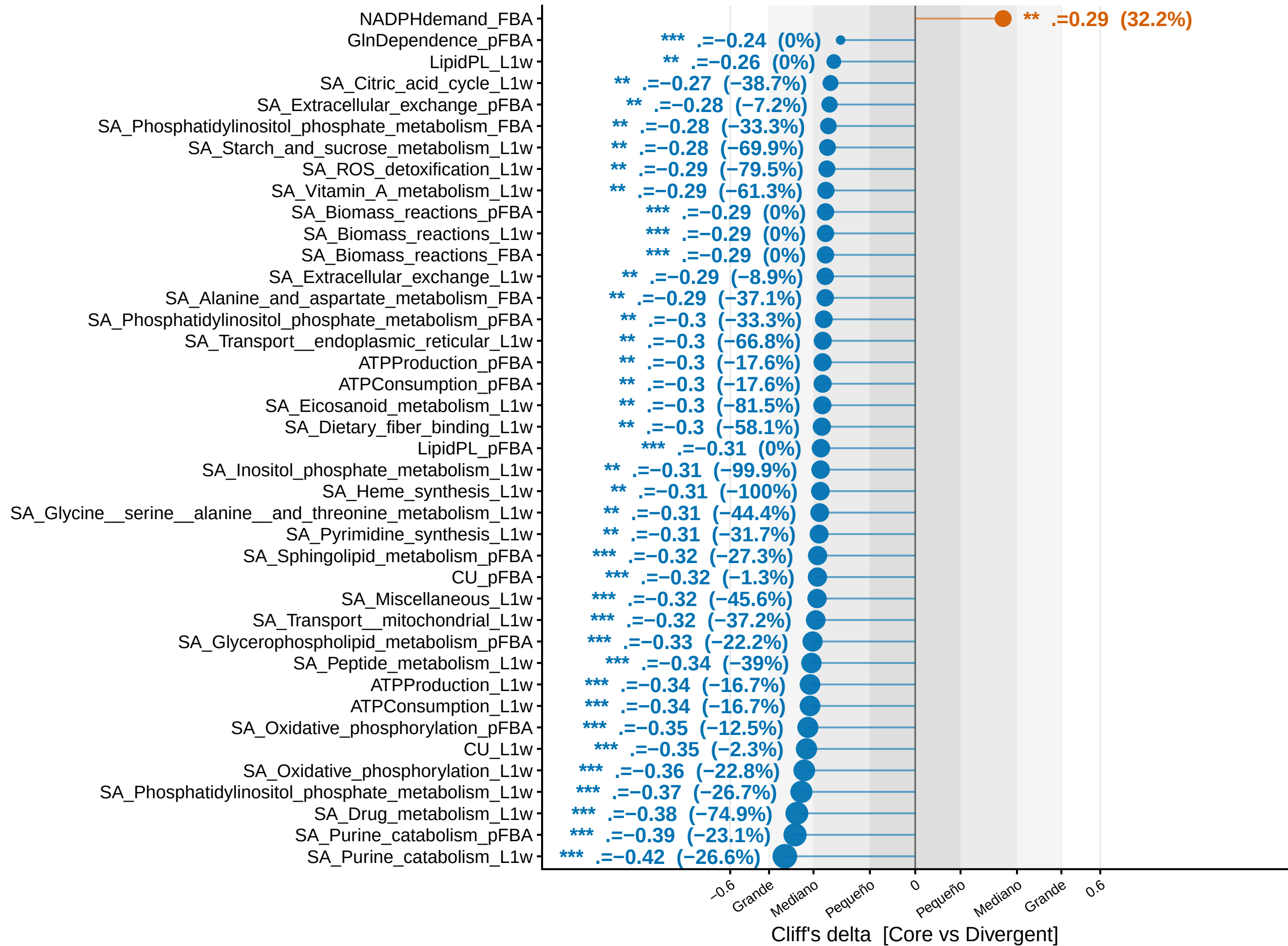
SECCIÓN II

Diferencias Metabólicas: Core vs Divergent

5 bloques funcionales: Energía | Redox | Lípidos | Dependencias | Reprogramación Sistémica

Top 40 variables metabólicas – Cliff's .

Cliff's . + .% mediana | azul = mayor en Core · naranja = mayor en Divergent



Cliff's .: . 1. Energía y Bioenergética

Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core

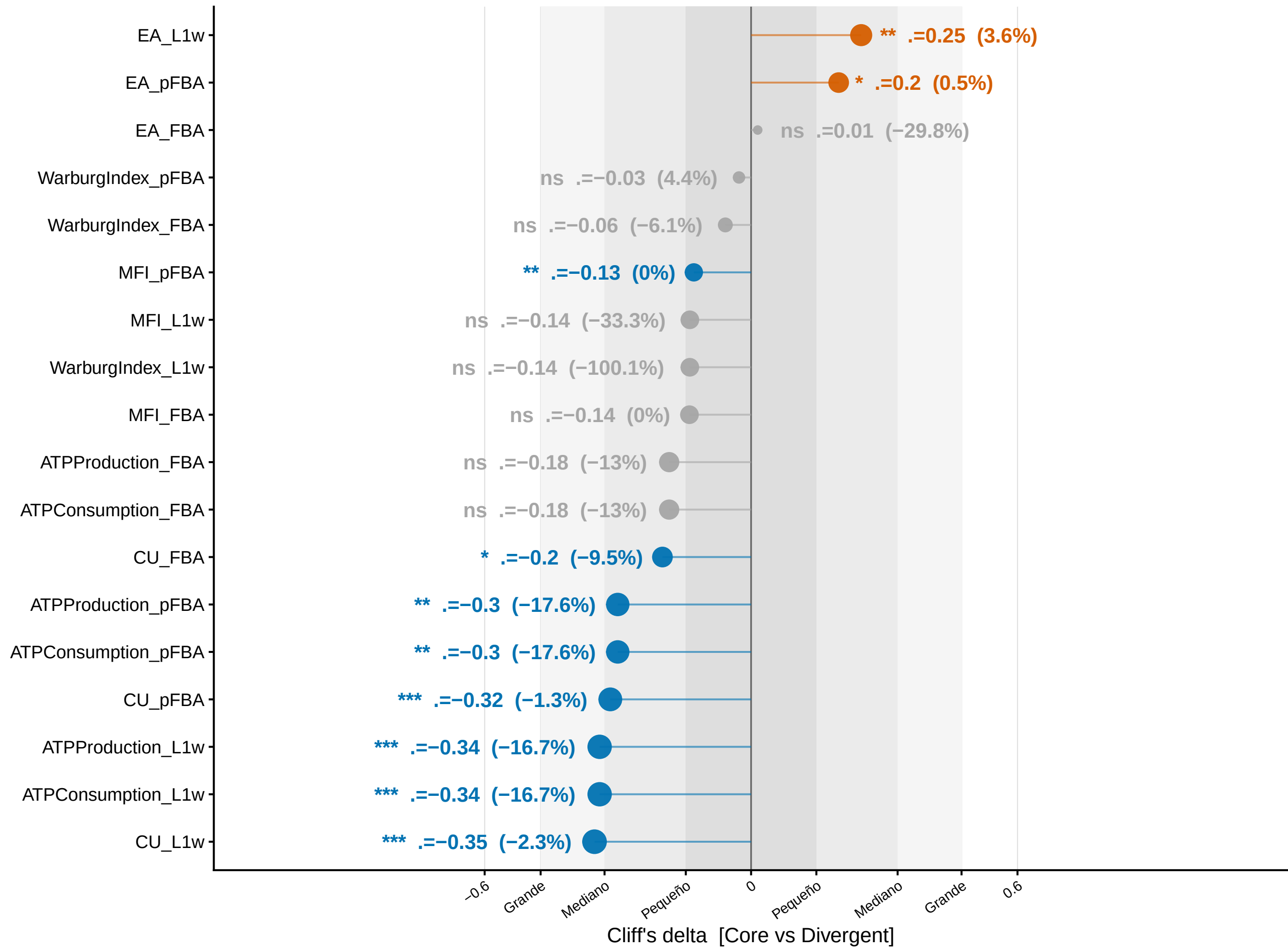


-1.0 -0.5 0.0 0.5 1.0

EA_L1w	** .=-0.25
EA_pFBA	* .=-0.2
EA_FBA	ns .=-0.01
WarburgIndex_pFBA	ns .=-0.03
WarburgIndex_FBA	ns .=-0.06
MFI_pFBA	** .=-0.13
MFI_L1w	ns .=-0.14
WarburgIndex_L1w	ns .=-0.14
MFI_FBA	ns .=-0.14
ATPProduction_FBA	ns .=-0.18
ATPConsumption_FBA	ns .=-0.18
CU_FBA	* .=-0.2
ATPProduction_pFBA	** .=-0.3
ATPConsumption_pFBA	** .=-0.3
CU_pFBA	*** .=-0.32
ATPProduction_L1w	*** .=-0.34
ATPConsumption_L1w	*** .=-0.34
CU_L1w	*** .=-0.35

Cliff's . – .1. Energía y Bioenergética

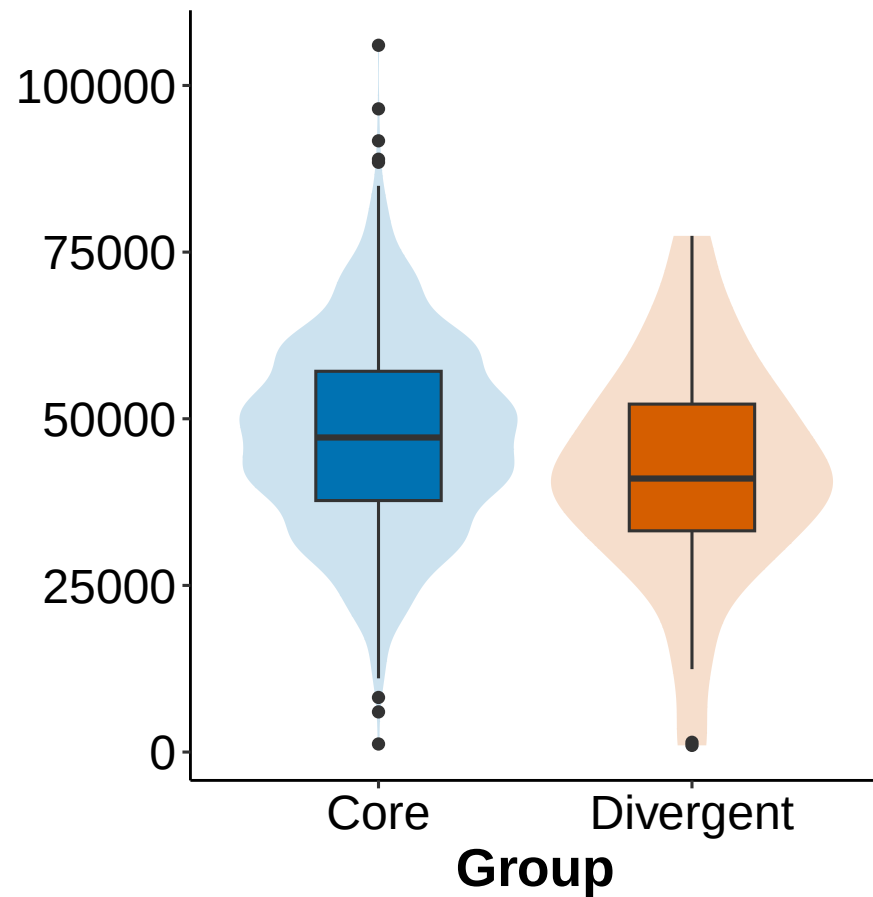
Cliff's . + .% mediana | azul = mayor en Core · naranja = mayor en Divergent



Metabolismo – . 1. Energía y Bioenergética

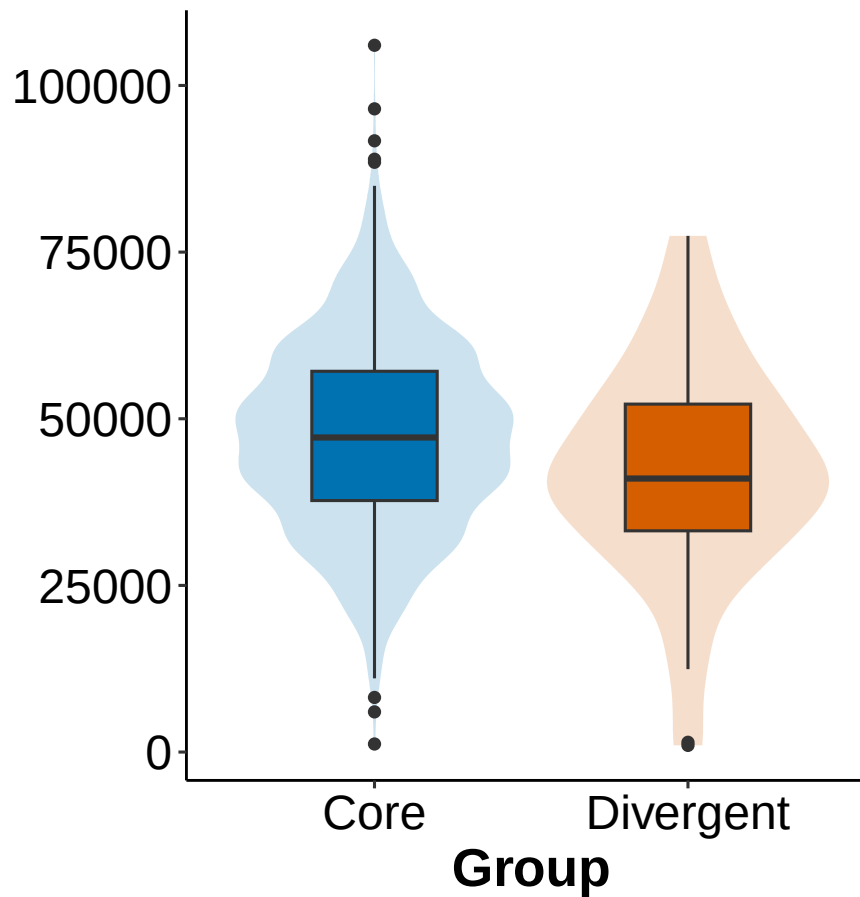
ATPConsumption_FBA

p-adj: 0.0636 | .=-0.18 (Pequeño) | .%: -13



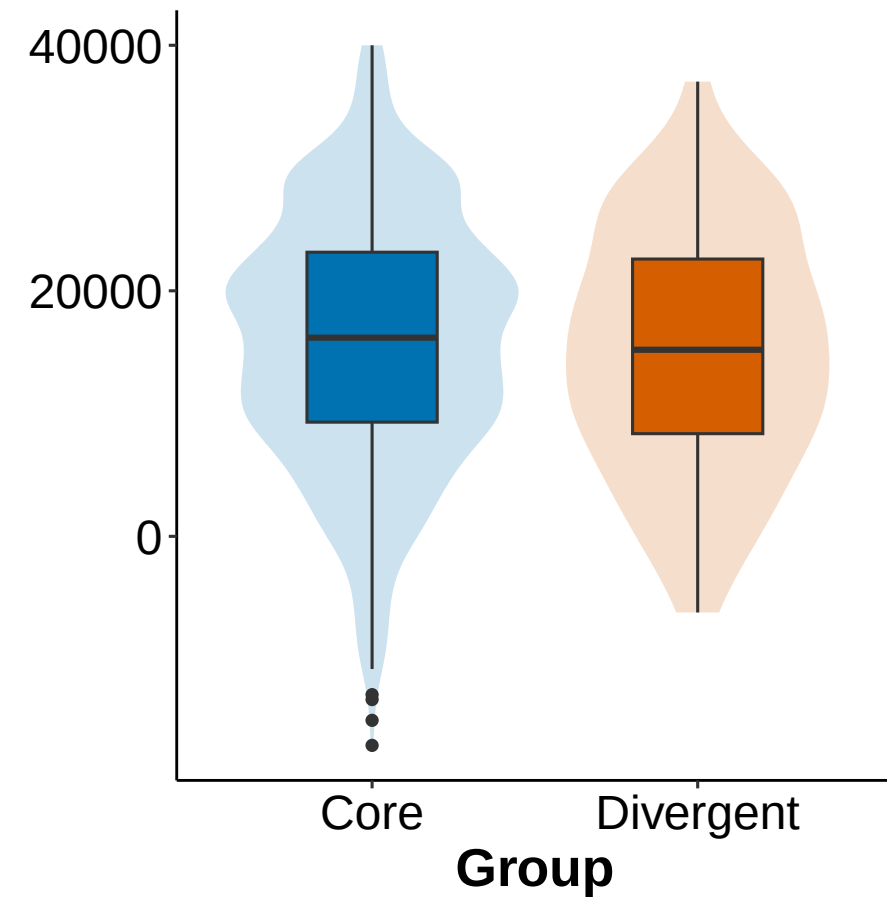
ATPProduction_FBA

p-adj: 0.0636 | .=-0.18 (Pequeño) | .%: -13



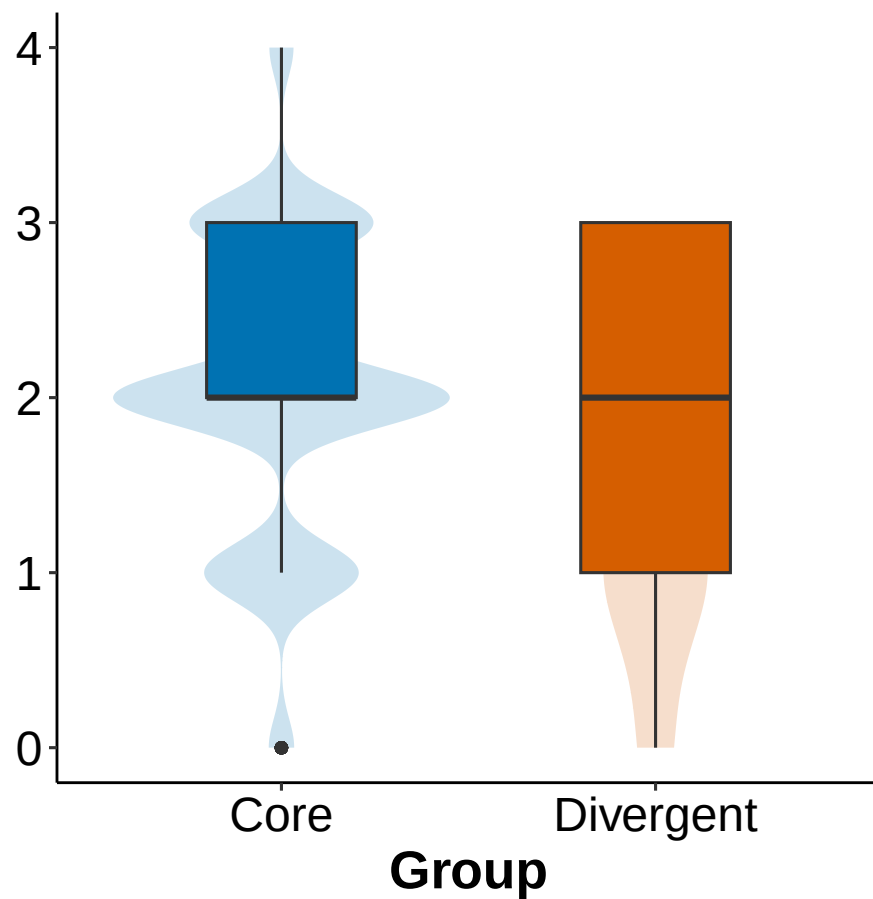
WarburgIndex_FBA

p-adj: 0.68 | .=-0.06 (Negligible) | .%: -6.1



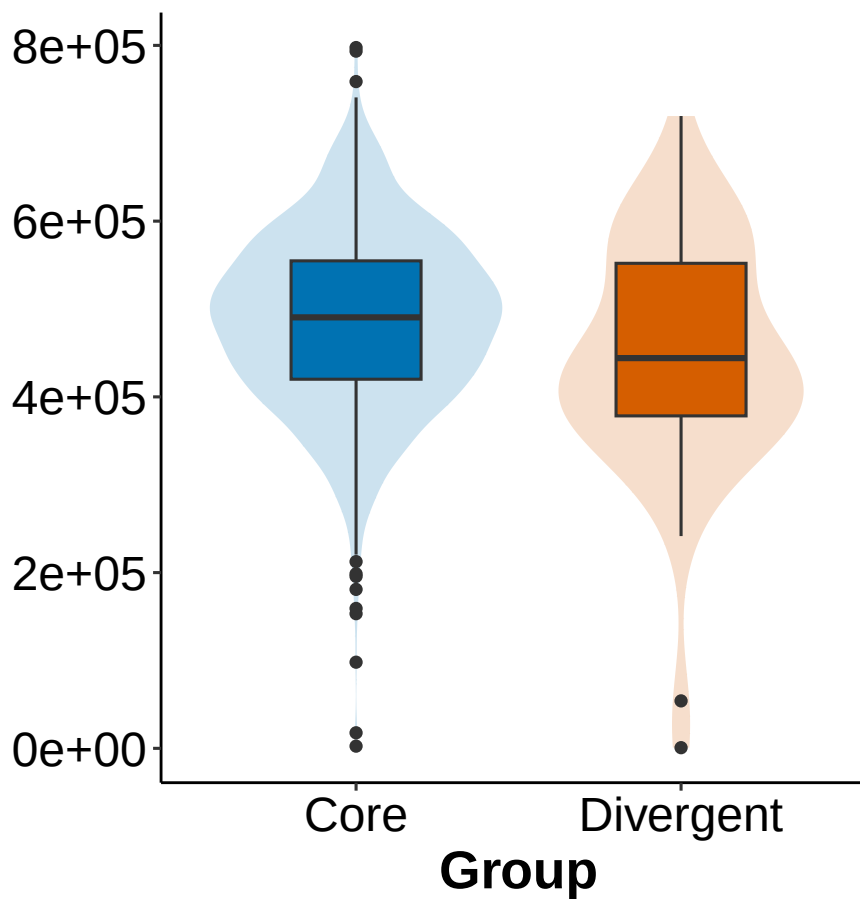
MFI_FBA

p-adj: 0.152 | .=-0.14 (Negligible) | .%: 0



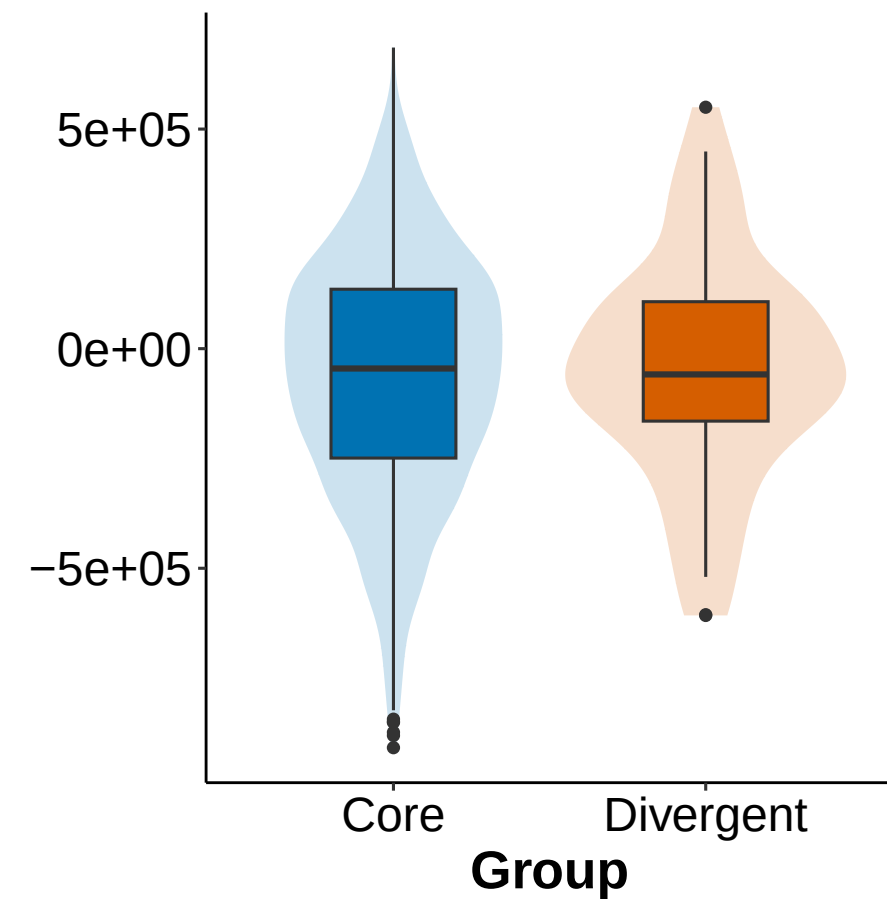
CU_FBA

p-adj: 0.042 | .=-0.2 (Pequeño) | .%: -9.5



EA_FBA

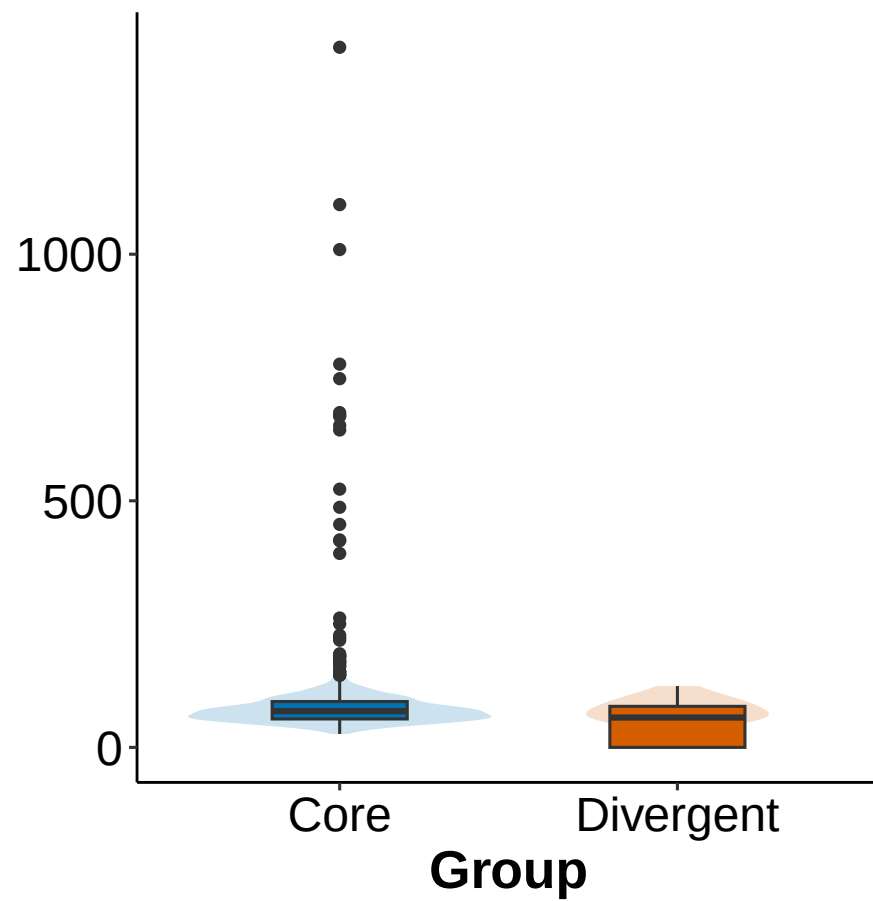
p-adj: 0.917 | .=0.01 (Negligible) | .%: -29.8



Metabolismo – . 1. Energía y Bioenergética

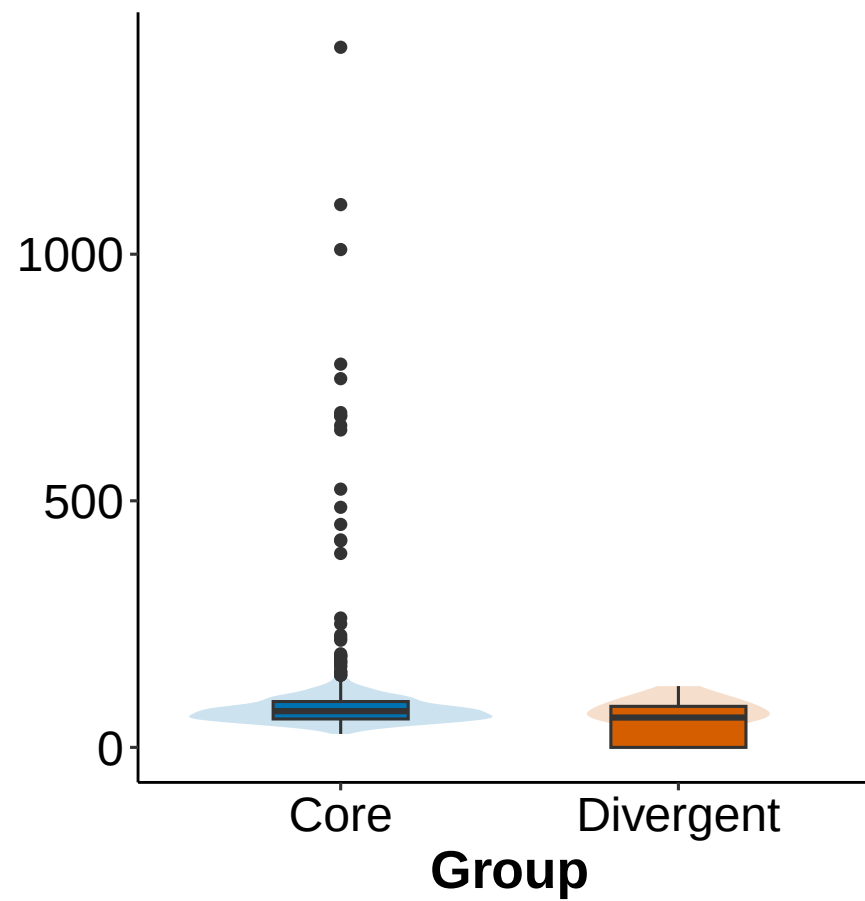
ATPConsumption_pFBA

p-adj: 0.00143 | .=-0.3 (Pequeño) | .%: -17.6



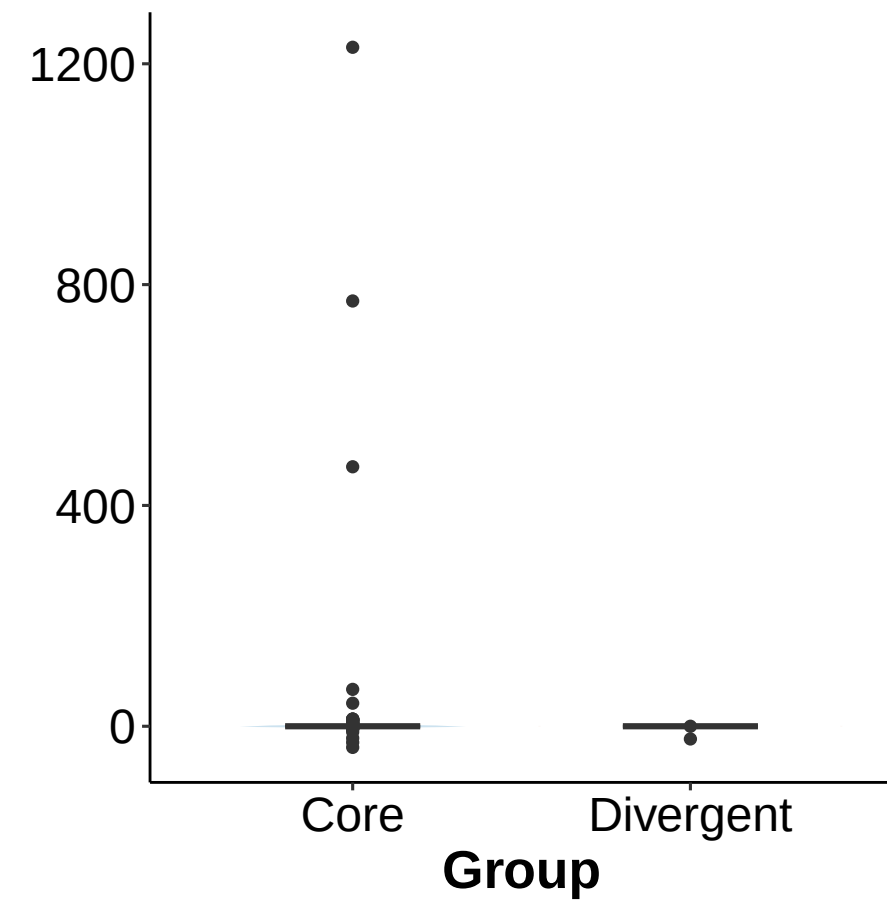
ATPProduction_pFBA

p-adj: 0.00143 | .=-0.3 (Pequeño) | .%: -17.6



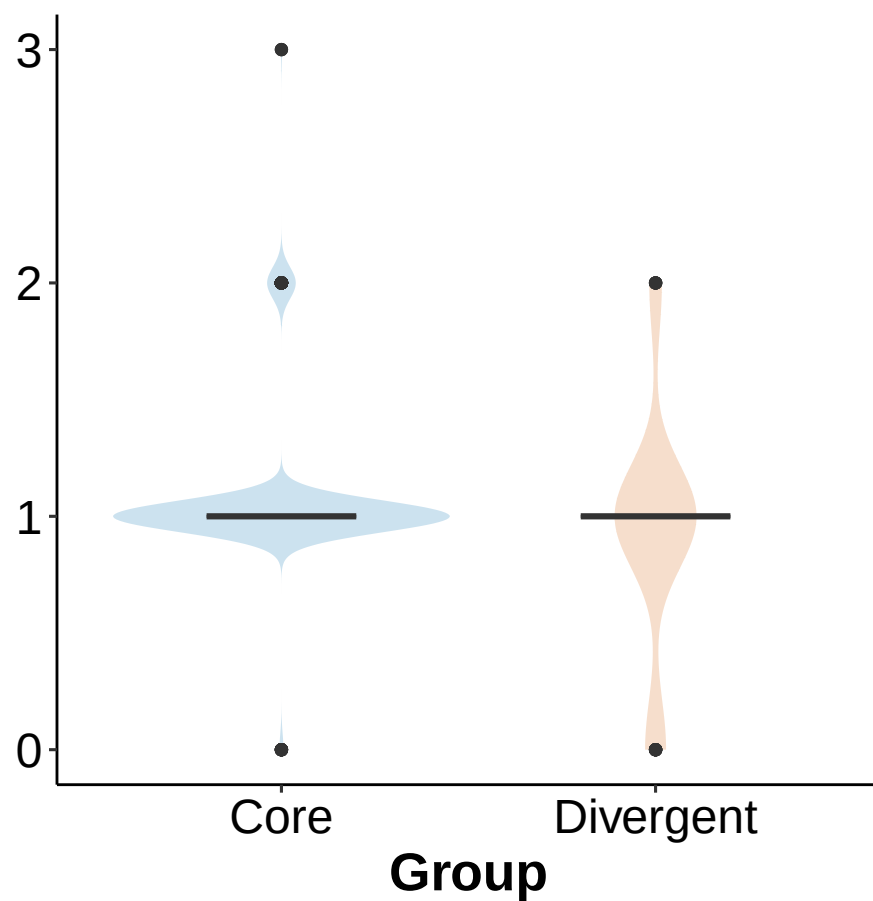
WarburgIndex_pFBA

p-adj: 0.839 | .=-0.03 (Negligible) | .%: 4.4



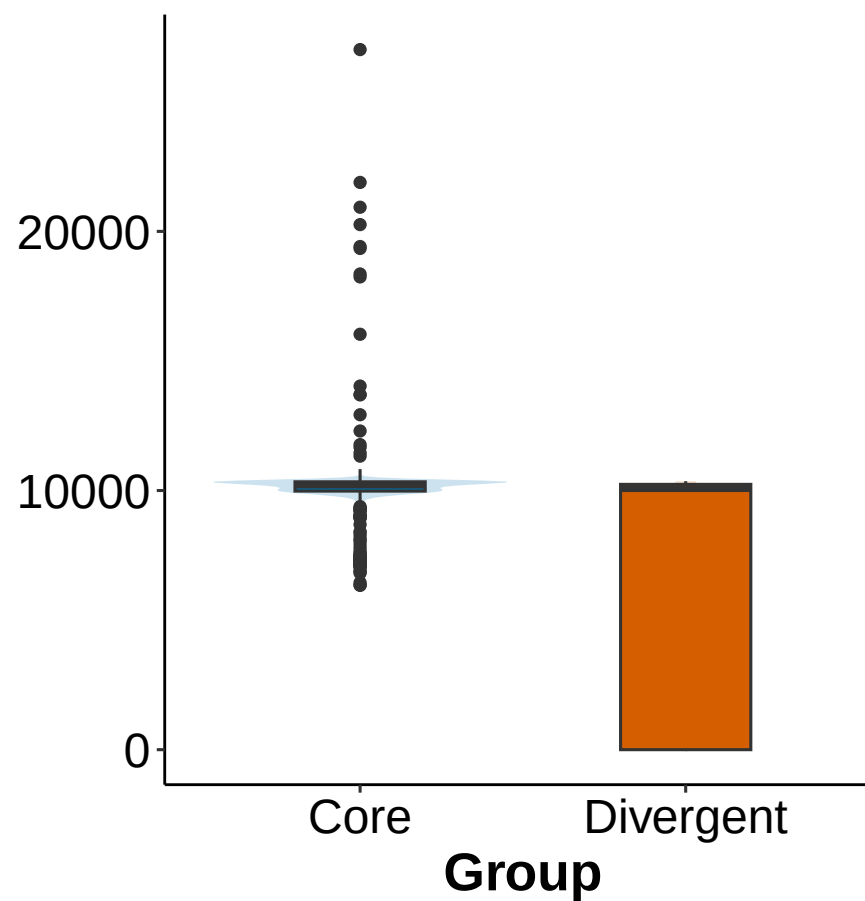
MFI_pFBA

p-adj: 0.00899 | .=-0.13 (Negligible) | .%: 0



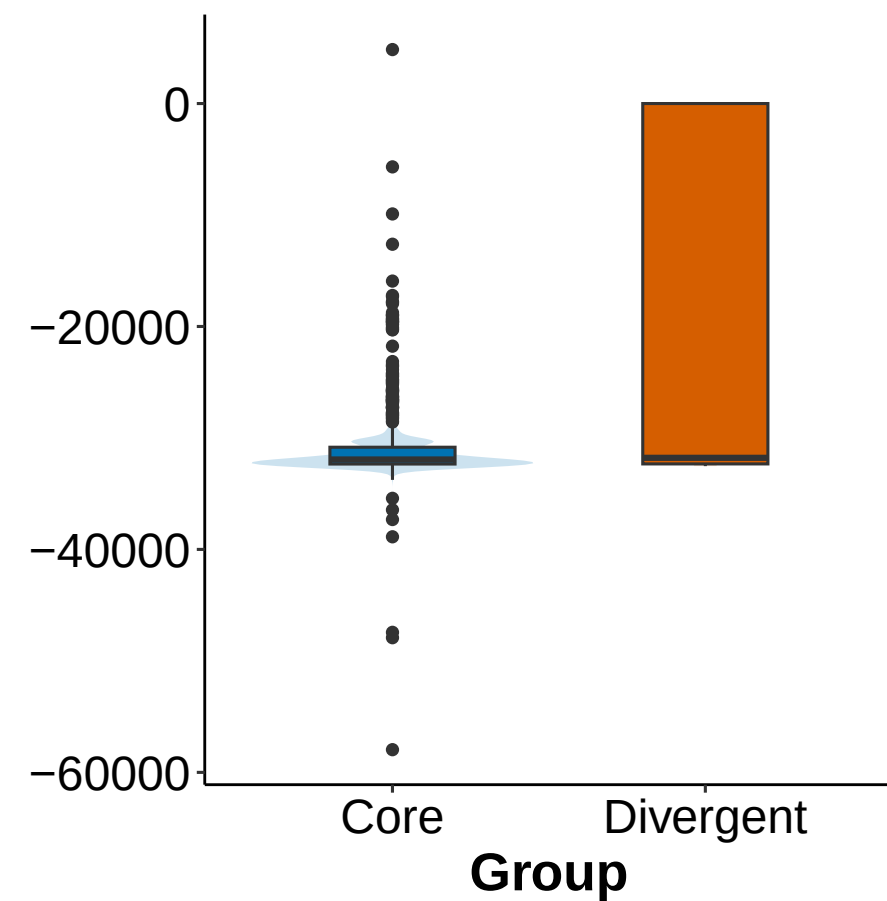
CU_pFBA

p-adj: 0.000878 | .=-0.32 (Pequeño) | .%: -1



EA_pFBA

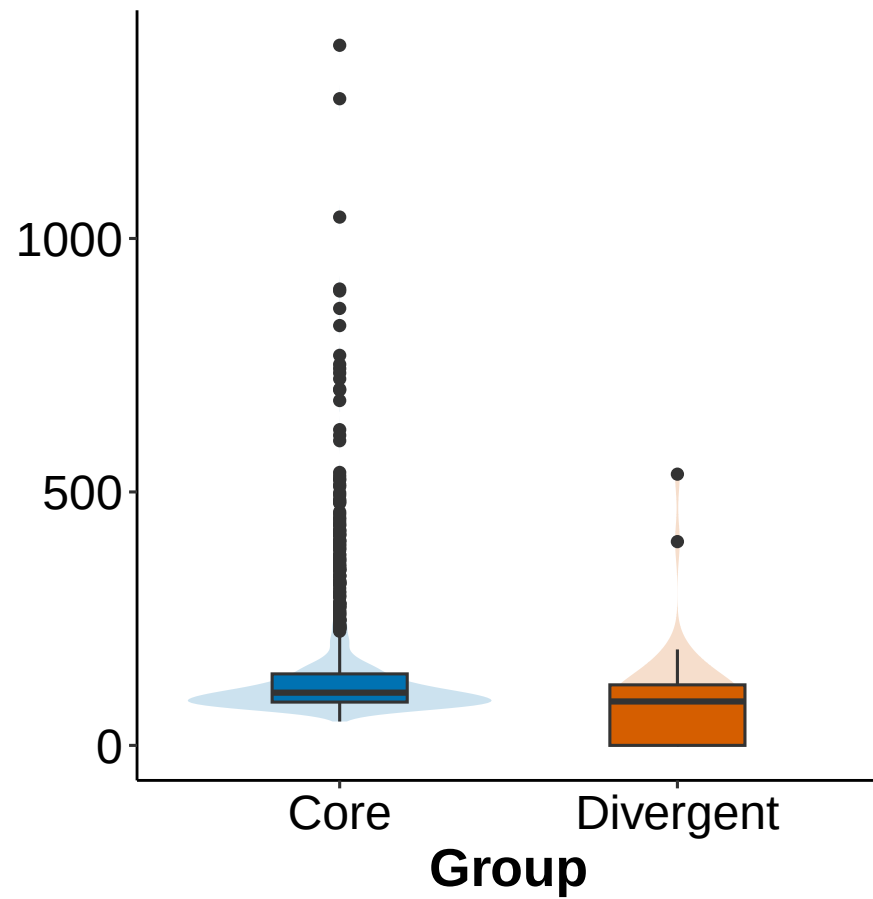
p-adj: 0.045 | .=-0.2 (Pequeño) | .%: 0.5



Metabolismo – . 1. Energía y Bioenergética

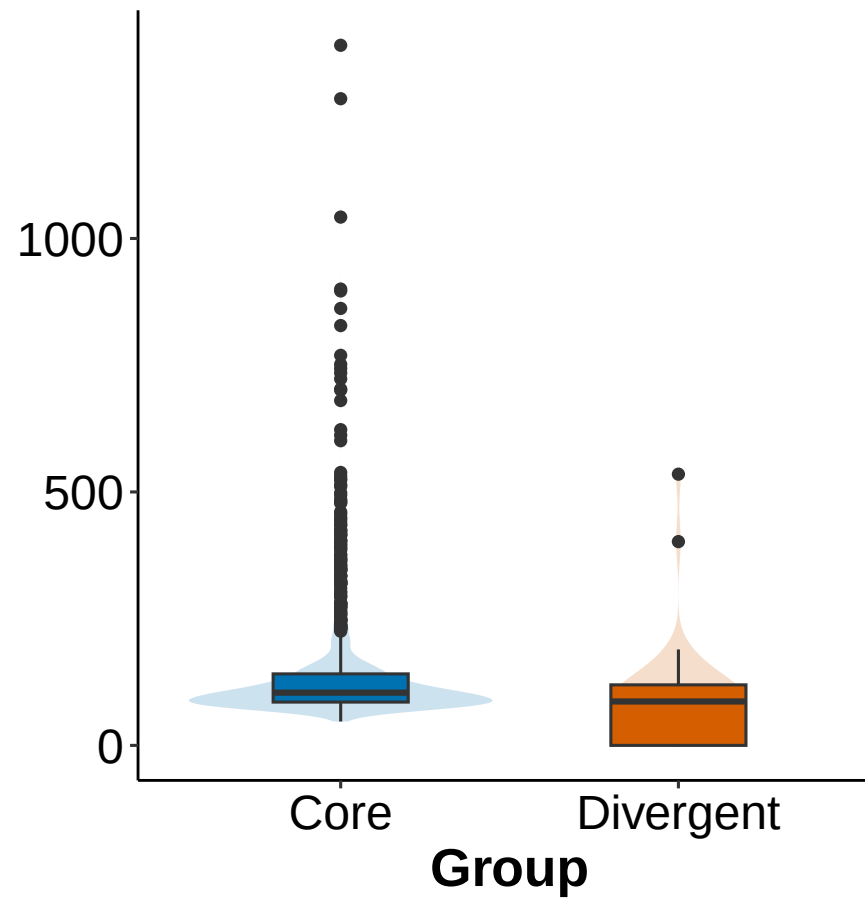
ATPConsumption_L1w

p-adj: 0.000339 | .=-0.34 (Mediano) | .%: -16.1



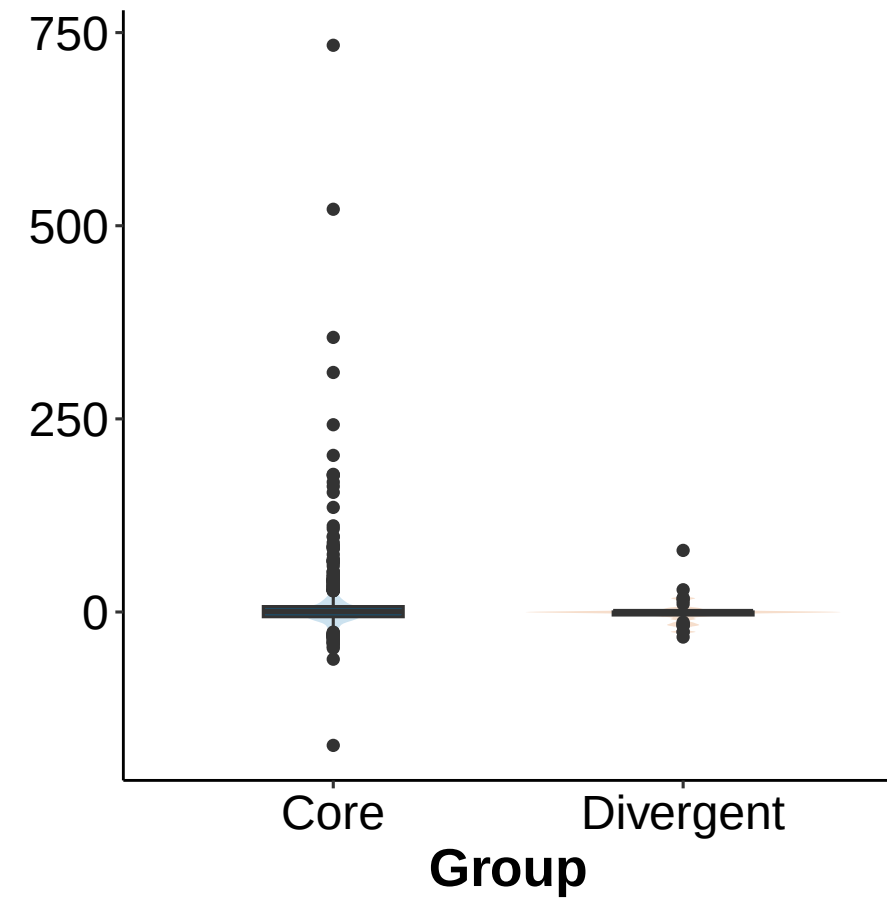
ATPProduction_L1w

p-adj: 0.000339 | .=-0.34 (Mediano) | .%: -16.1



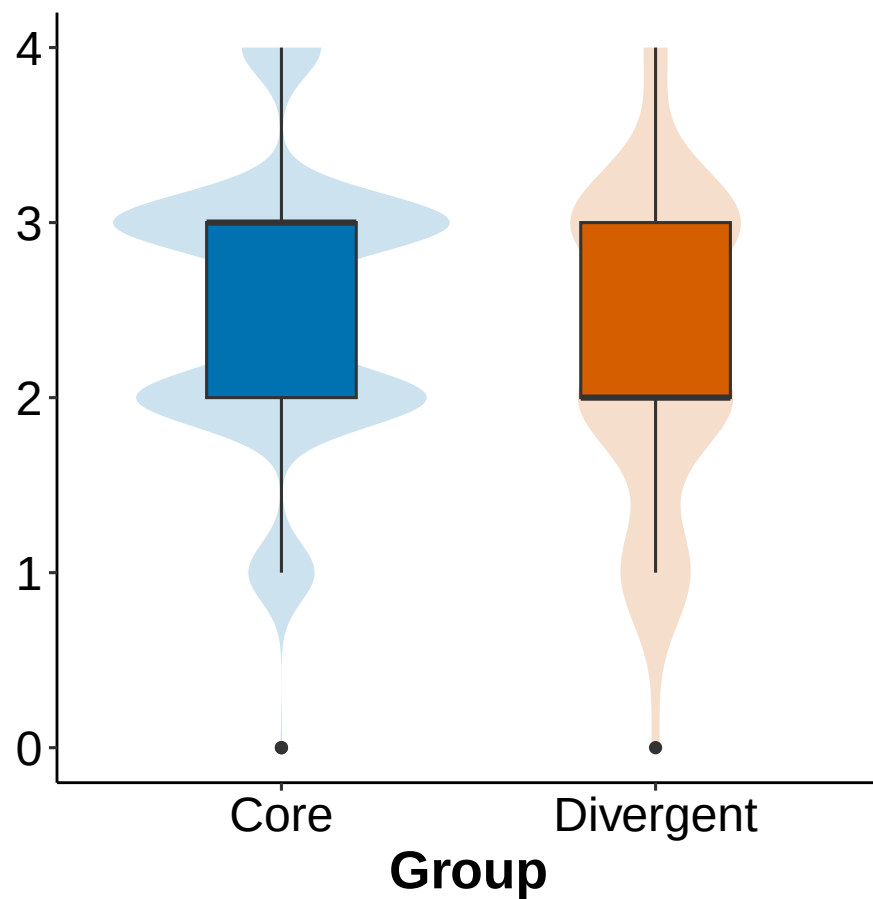
WarburgIndex_L1w

p-adj: 0.192 | .=-0.14 (Negligible) | .%: -100.1



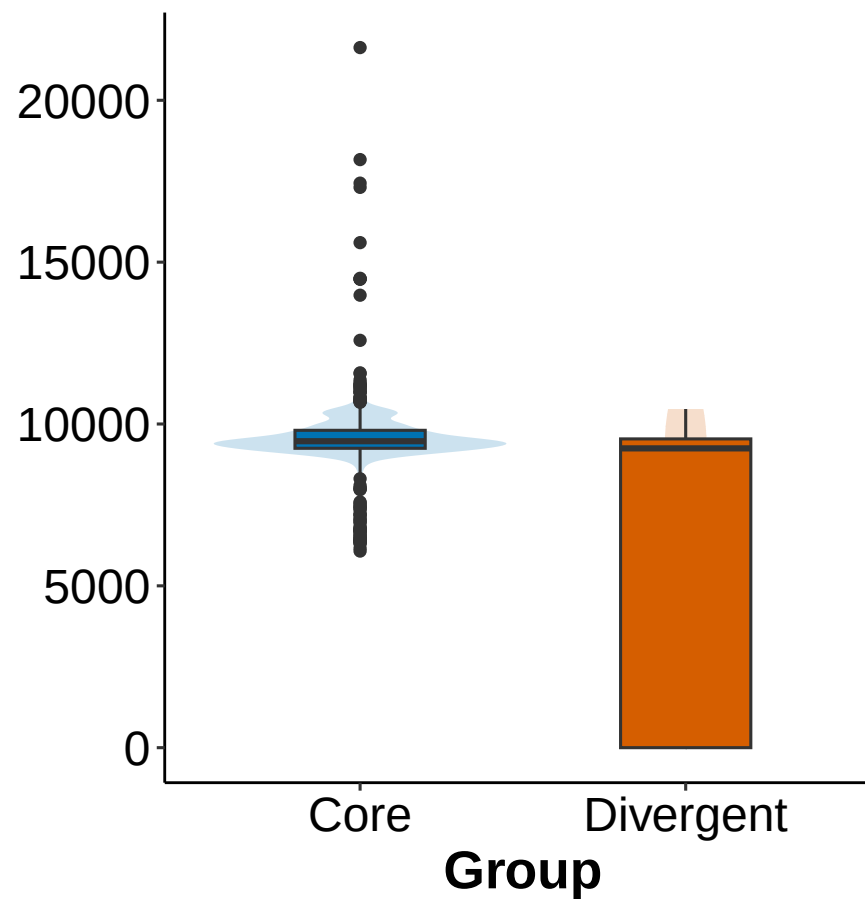
MFI_L1w

p-adj: 0.152 | .=-0.14 (Negligible) | .%: -33.3



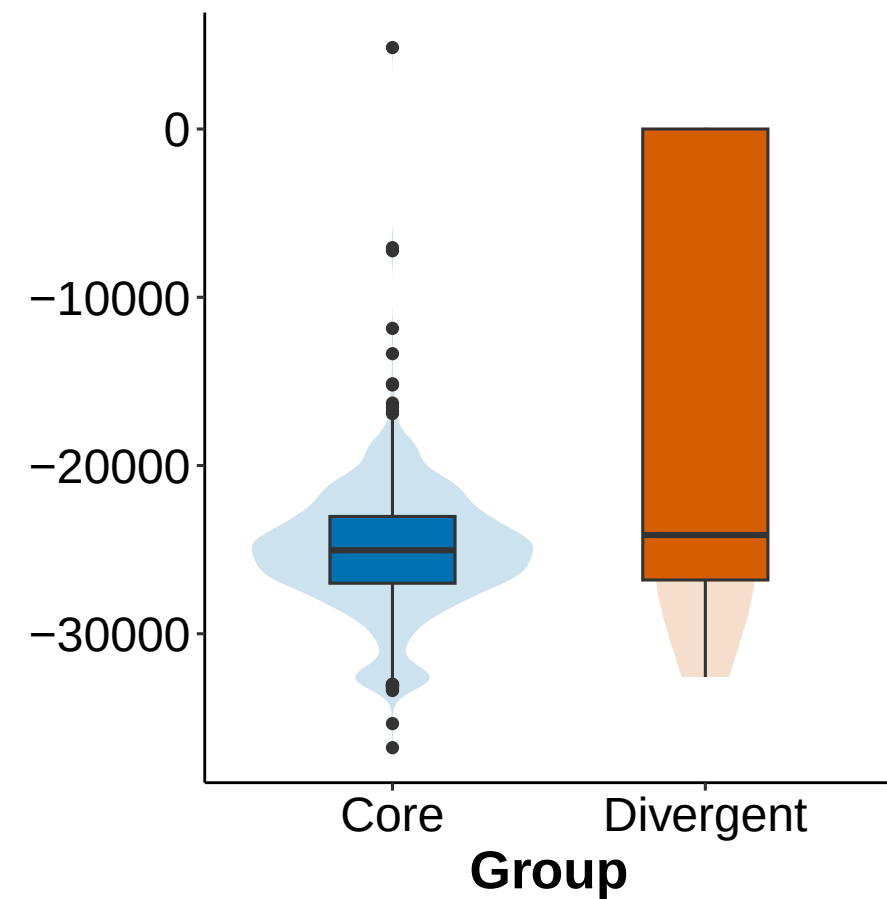
CU_L1w

p-adj: 0.000223 | .=-0.35 (Mediano) | .%: -2.1



EA_L1w

p-adj: 0.00899 | .=-0.25 (Pequeño) | .%: 3.6



Cliff's .: . 2. Estado Redox y Anabolismo

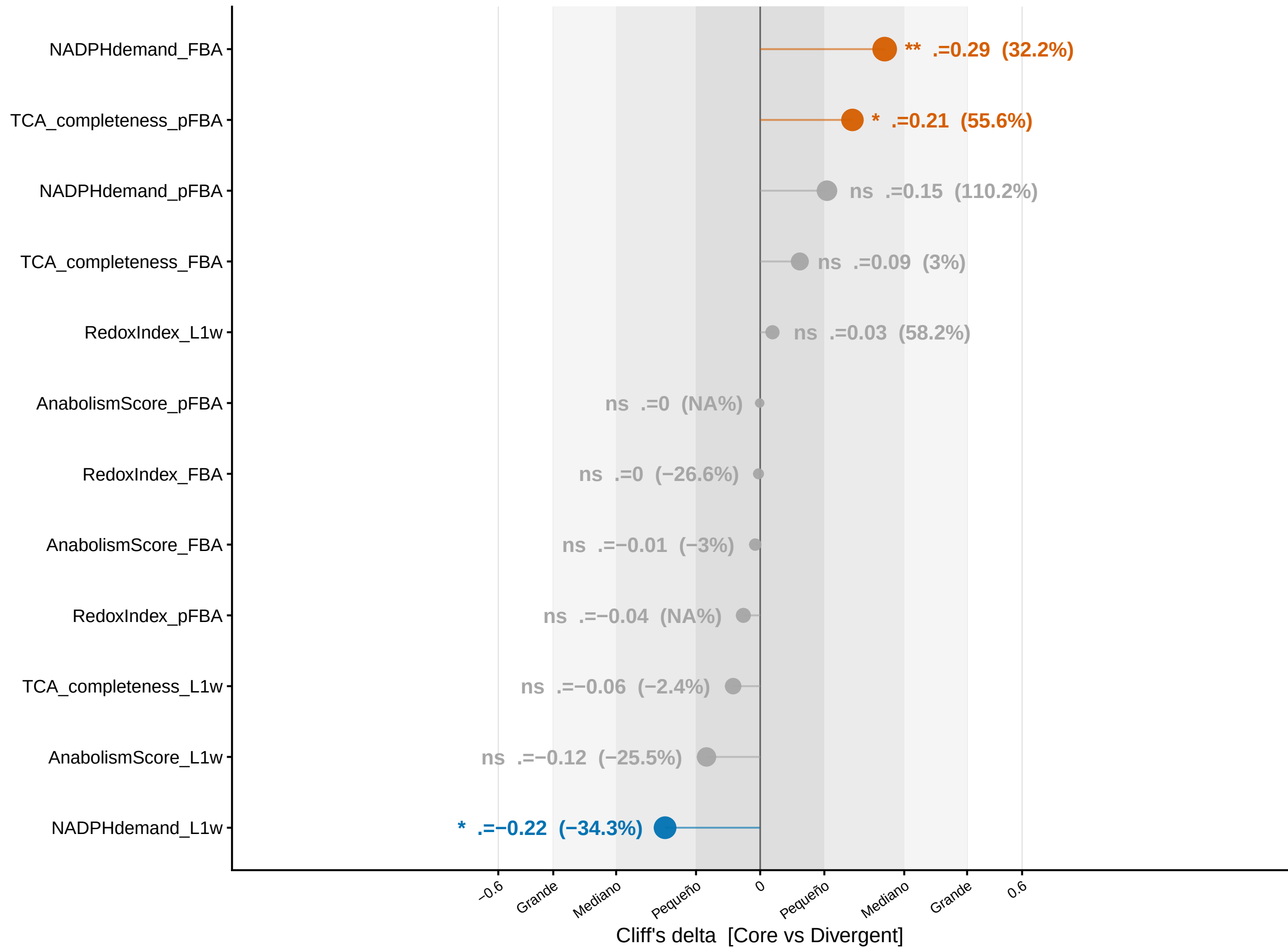
Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05



NADPHdemand_FBA	** .=-0.29
TCA_completeness_pFBA	* .=-0.21
NADPHdemand_pFBA	ns .=-0.15
TCA_completeness_FBA	ns .=-0.09
RedoxIndex_L1w	ns .=-0.03
AnabolismScore_pFBA	ns .=-0
RedoxIndex_FBA	ns .=-0
AnabolismScore_FBA	ns .=-0.01
RedoxIndex_pFBA	ns .=-0.04
TCA_completeness_L1w	ns .=-0.06
AnabolismScore_L1w	ns .=-0.12
NADPHdemand_L1w	* .=-0.22

Cliff's . – . 2. Estado Redox y Anabolismo

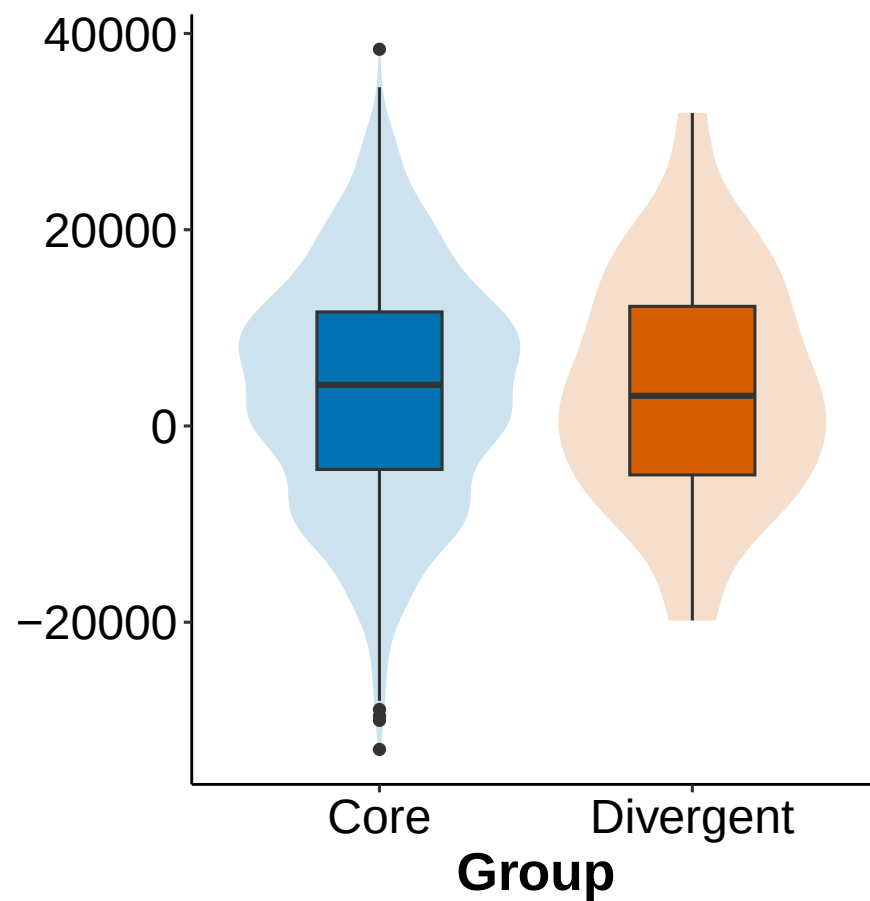
Cliff's . + .% mediana | azul = mayor en Core · naranja = mayor en Divergent



Metabolismo – . 2. Estado Redox y Anabolismo

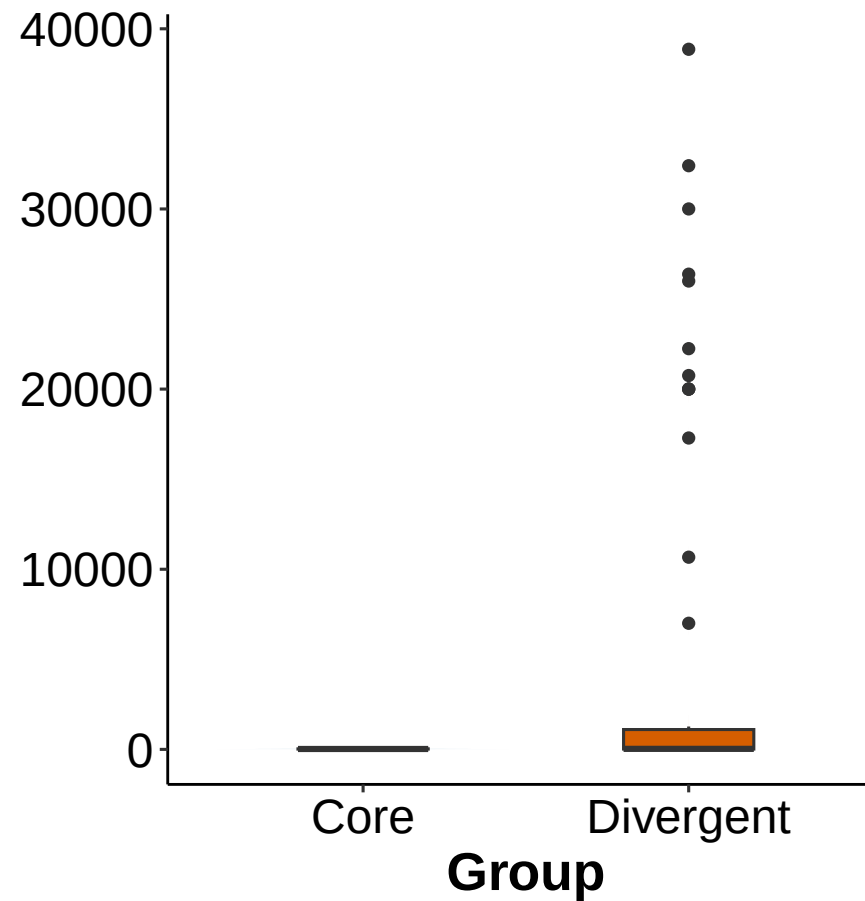
RedoxIndex_FBA

p-adj: 0.969 | .=0 (Negligible) | .%: -26.6



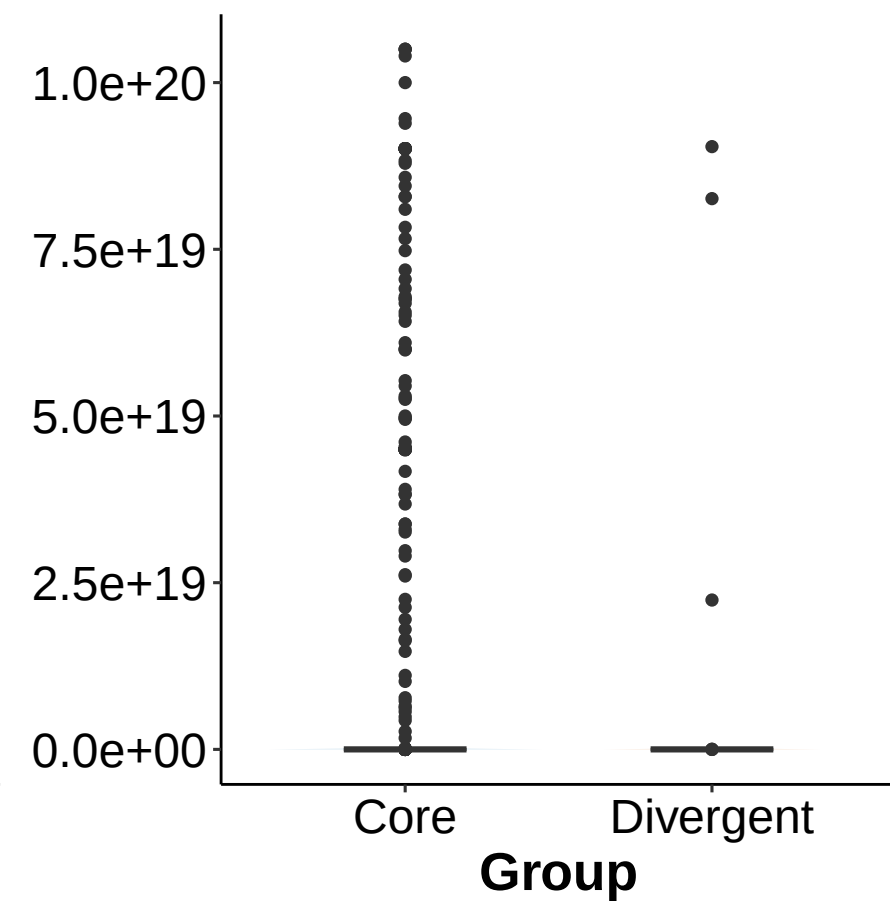
NADPHdemand_FBA

p-adj: 0.0024 | .=0.29 (Pequeño) | .%: 32.2



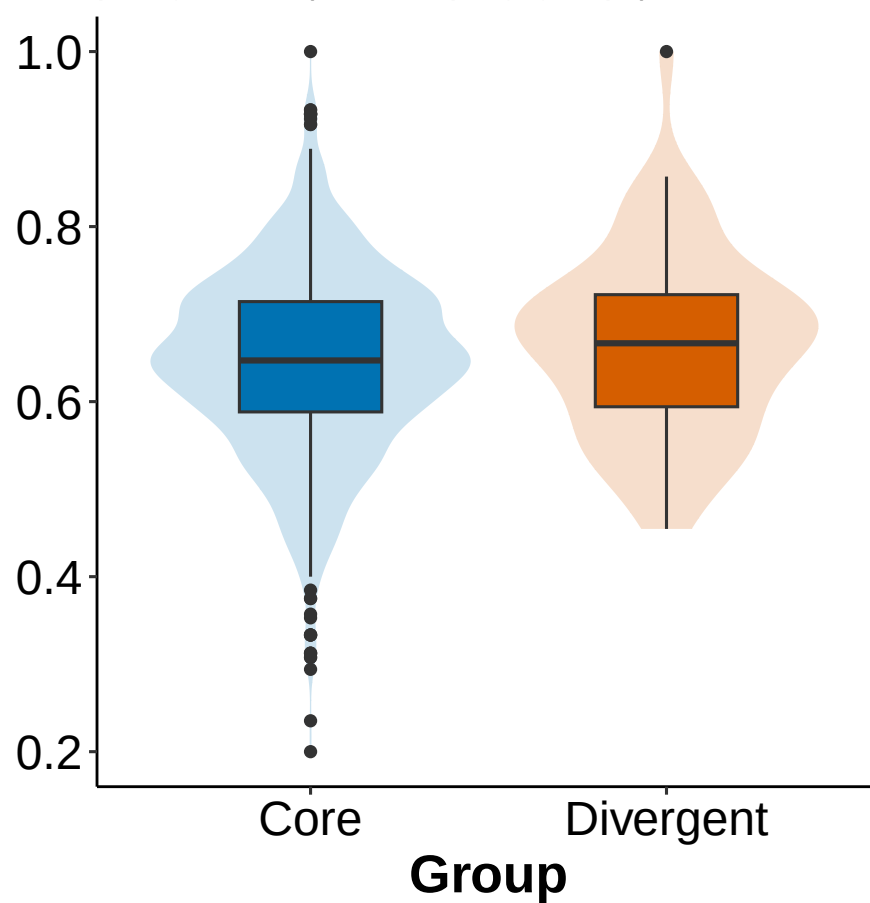
AnabolismScore_FBA

p-adj: 0.943 | .=-0.01 (Negligible) | .%: -3



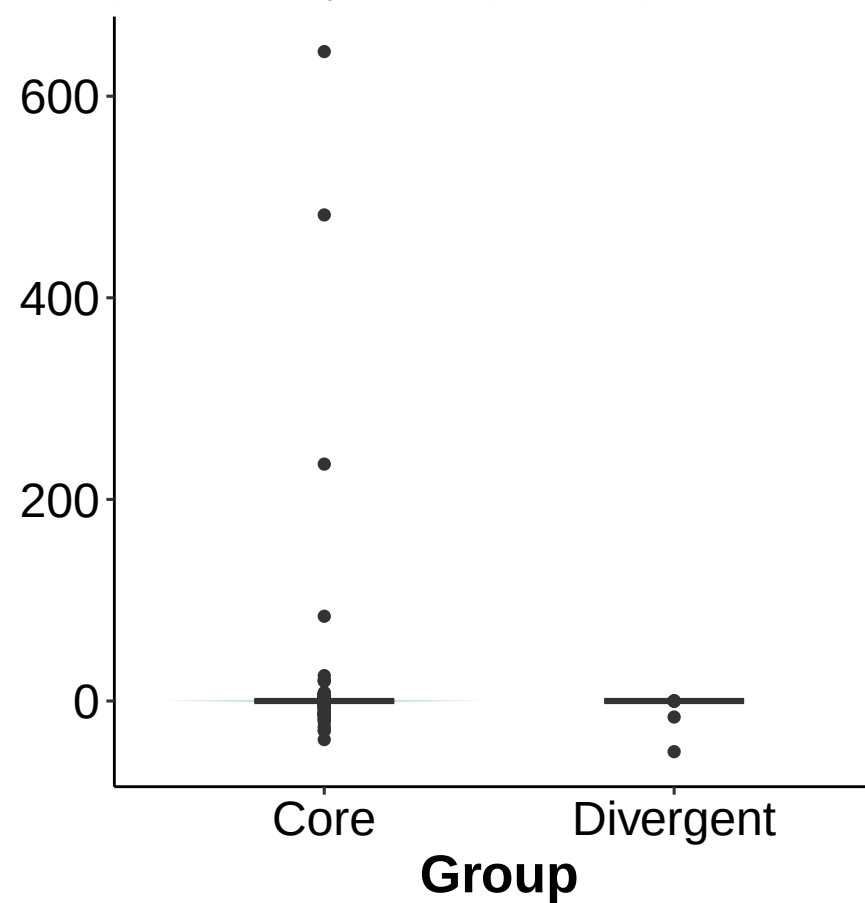
TCA_completeness_FBA

p-adj: 0.457 | .=0.09 (Negligible) | .%: 3



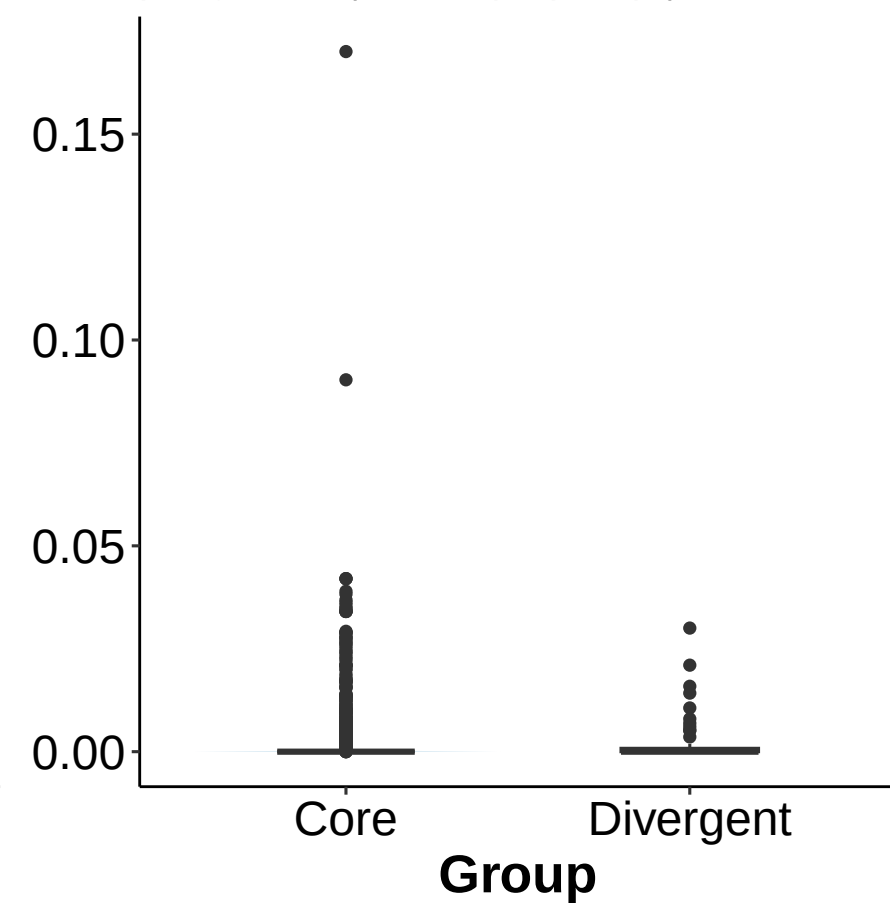
RedoxIndex_pFBA

p-adj: 0.789 | .=-0.04 (Negligible)

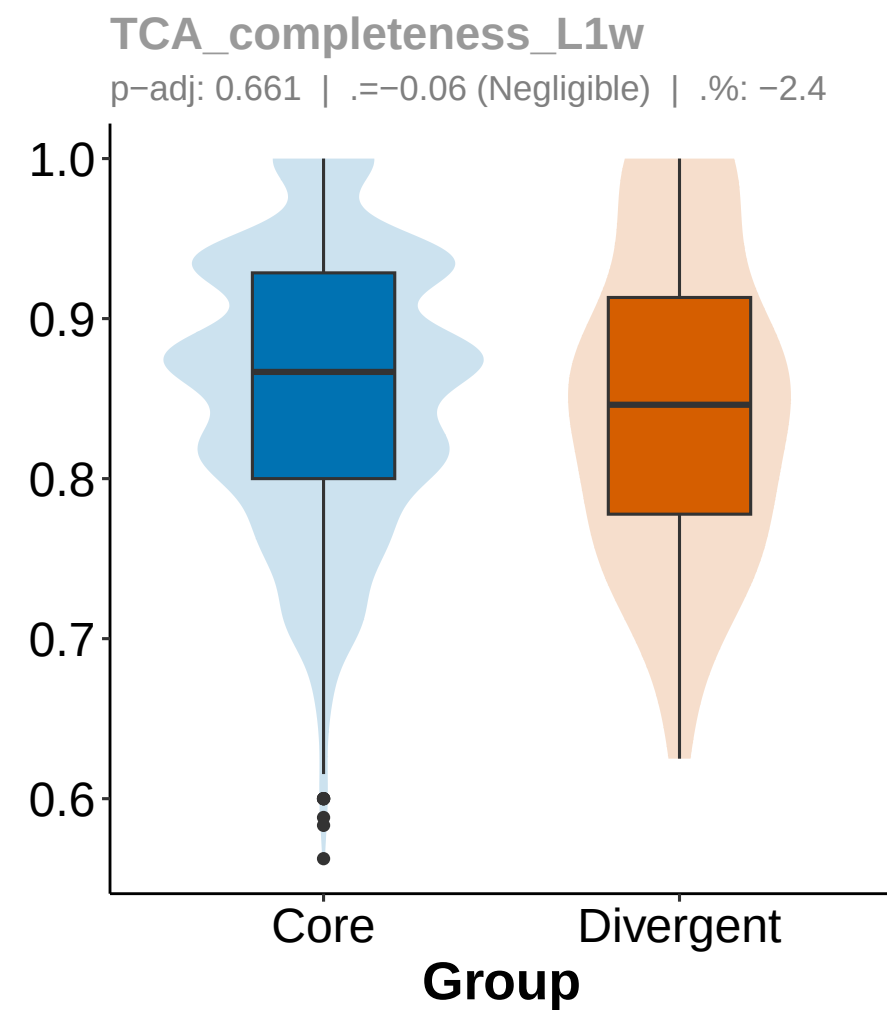
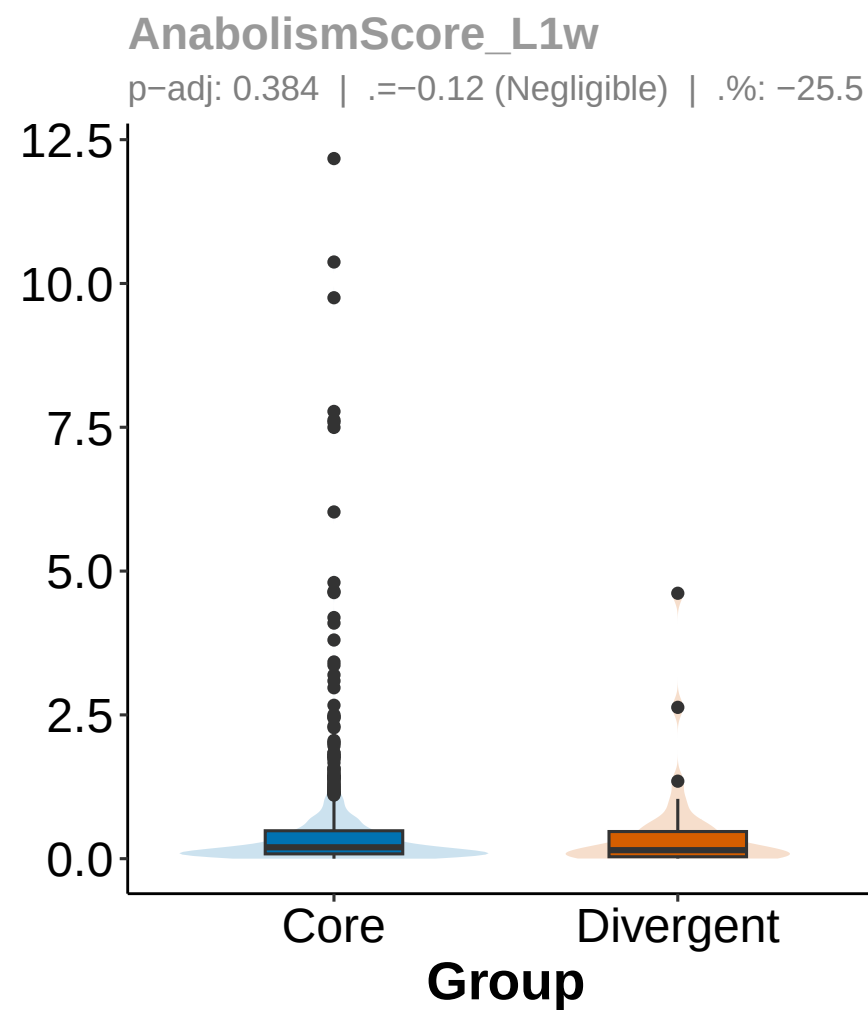
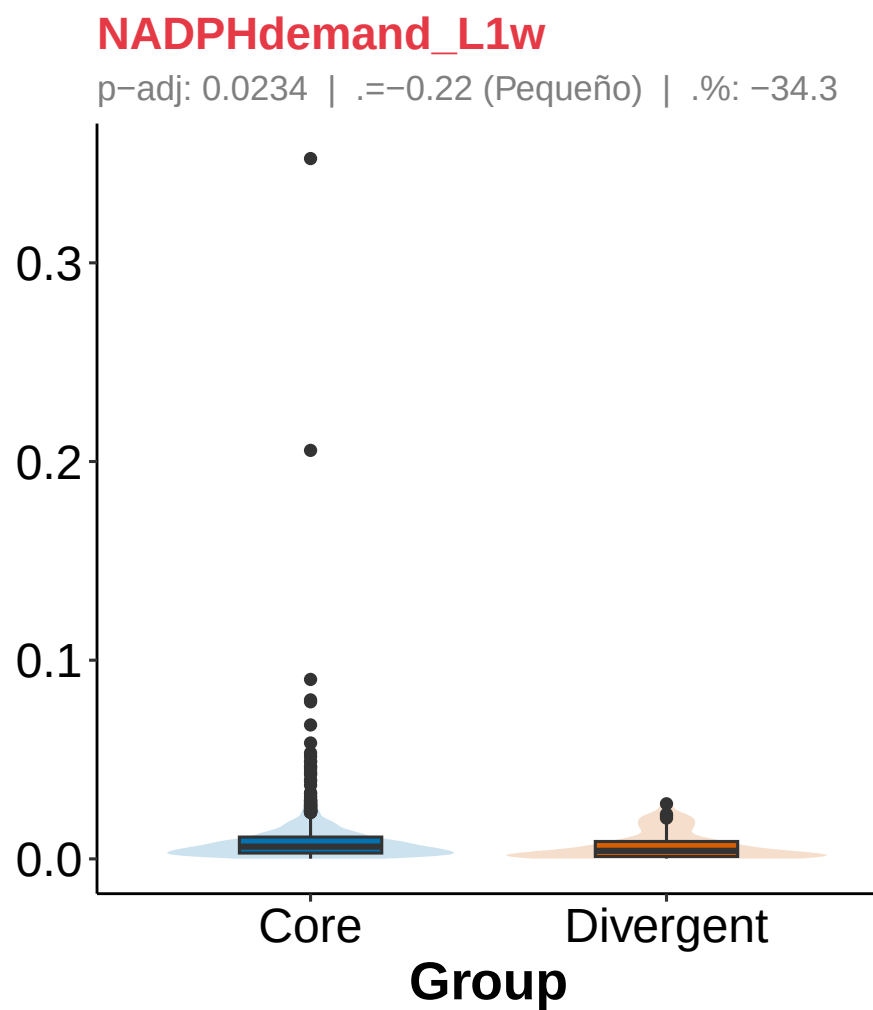
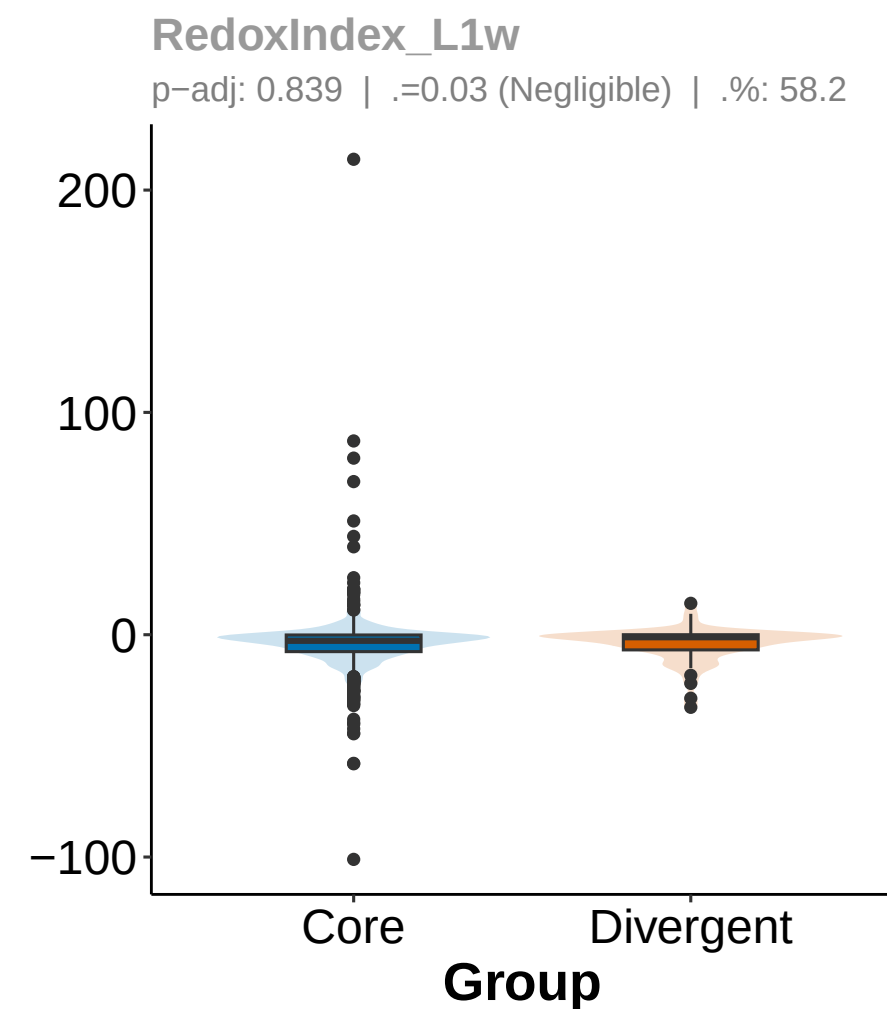
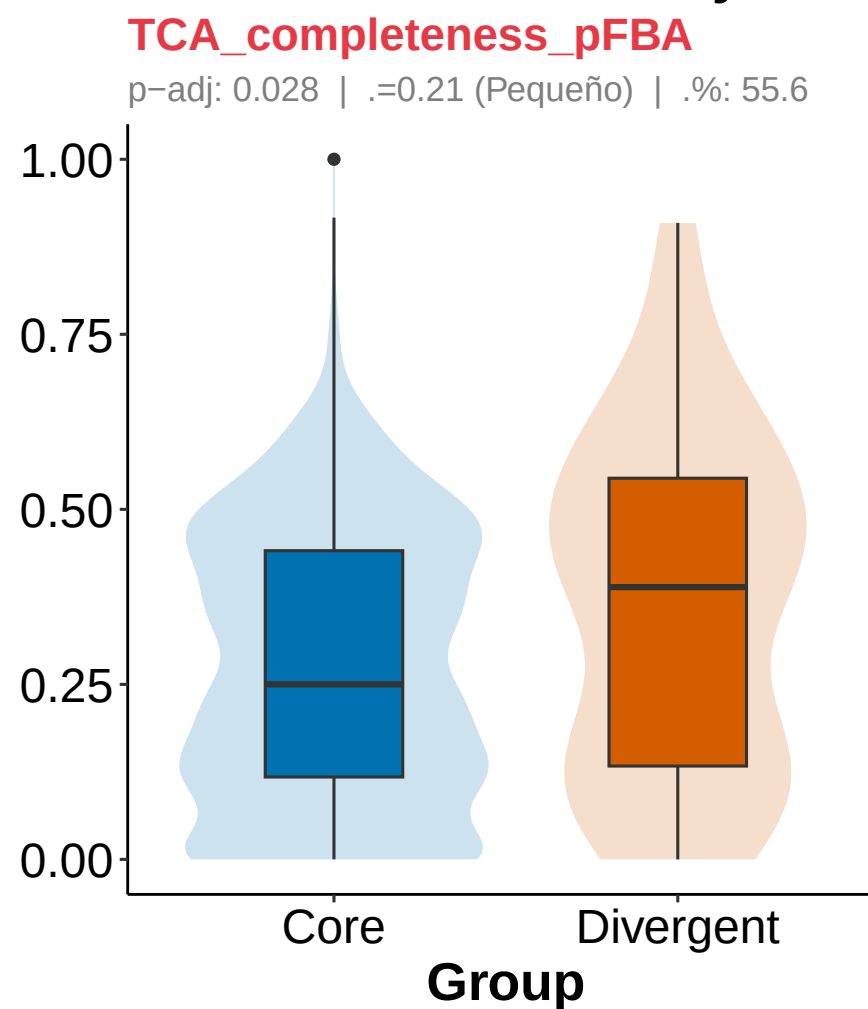
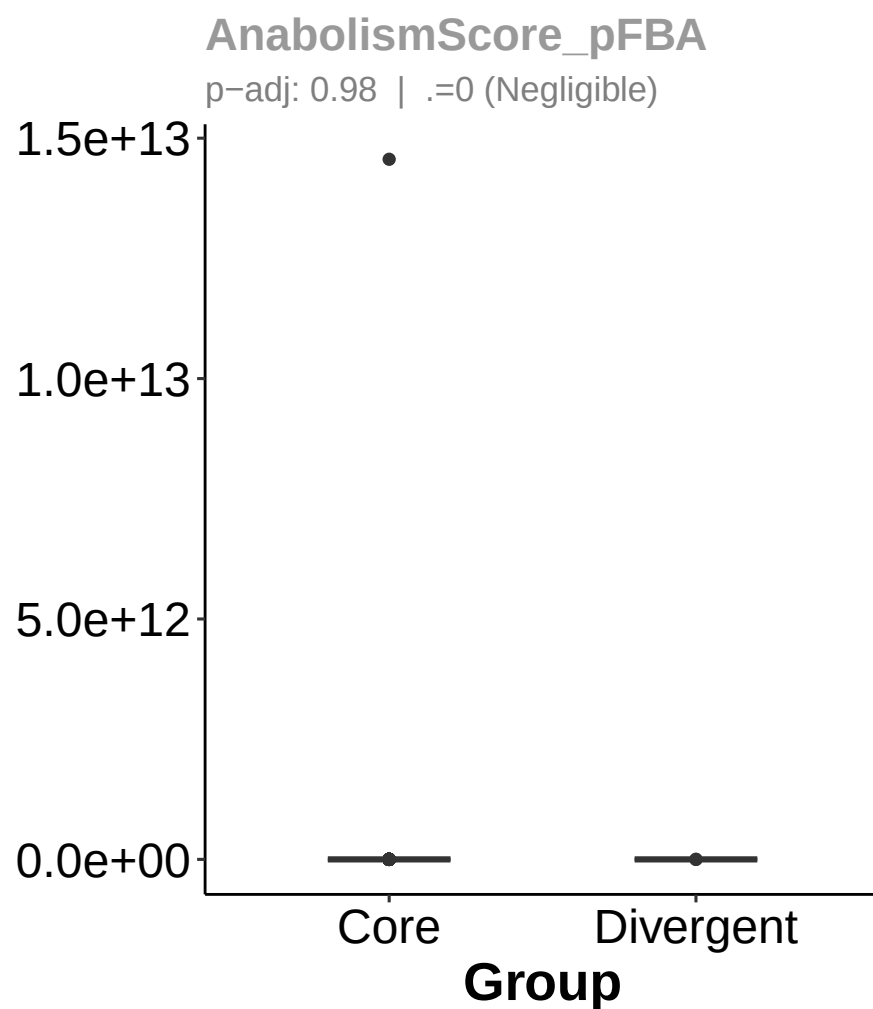


NADPHdemand_pFBA

p-adj: 0.119 | .=0.15 (Pequeño) | .%: 110.2

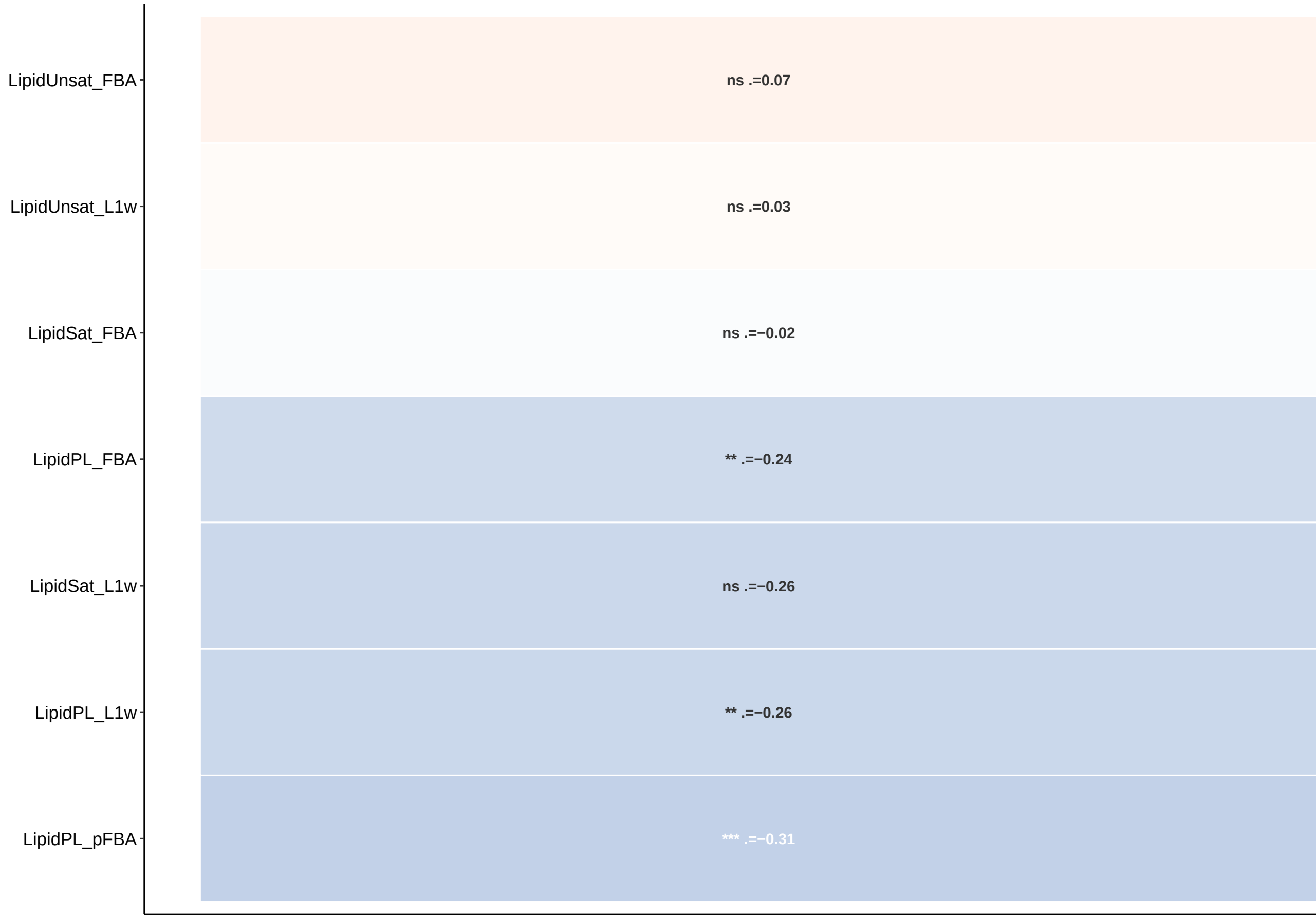


Metabolismo – . 2. Estado Redox y Anabolismo



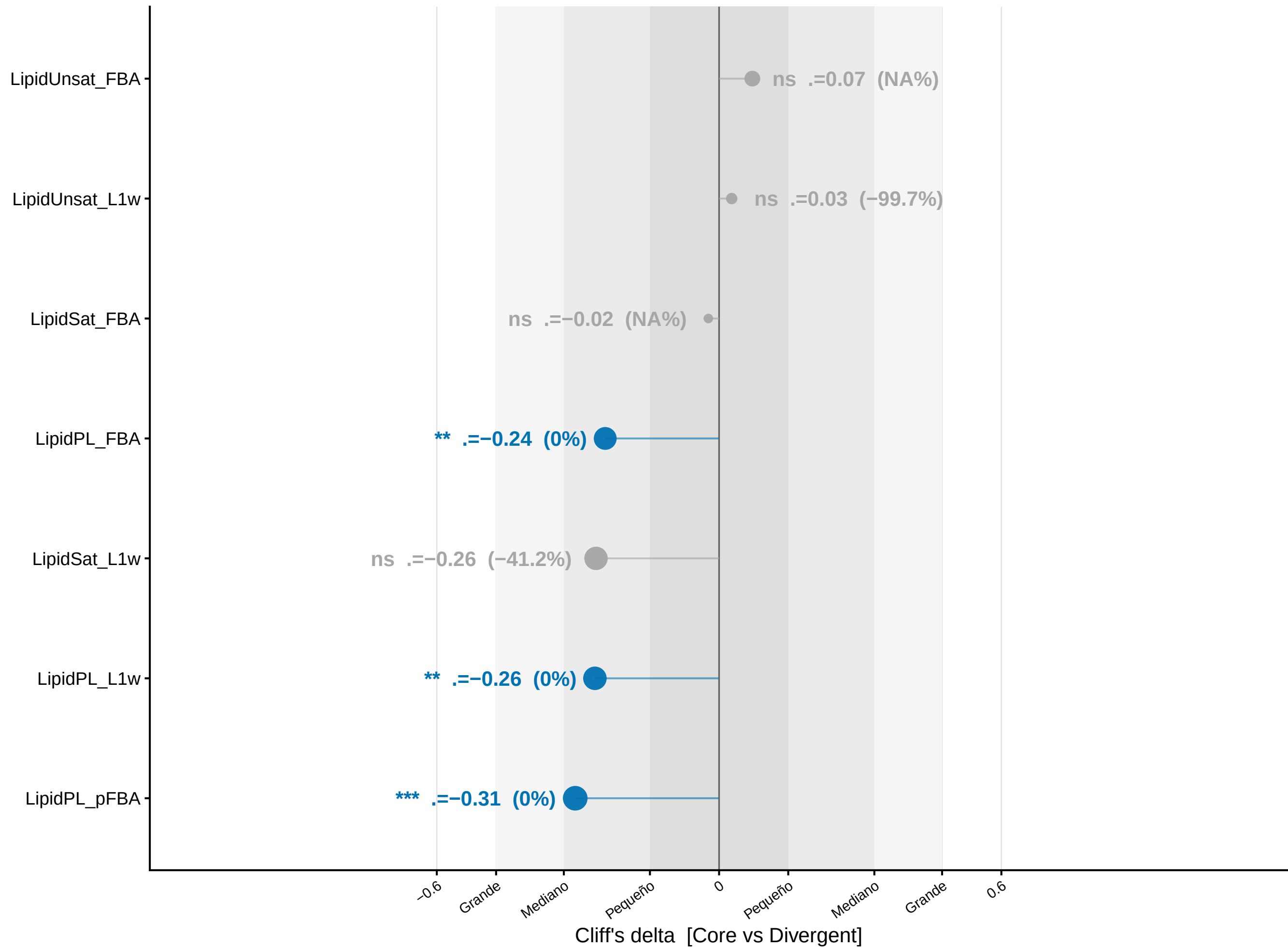
Cliff's .: . 3. Metabolismo Lipídico

Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

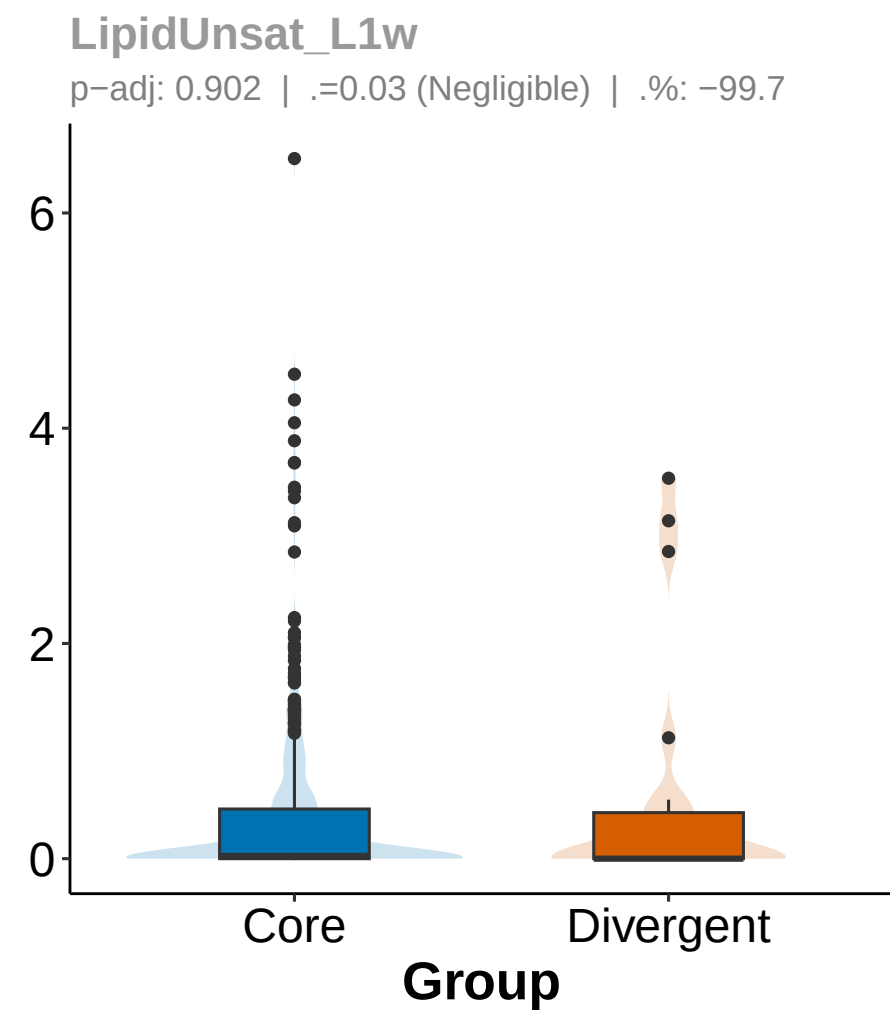
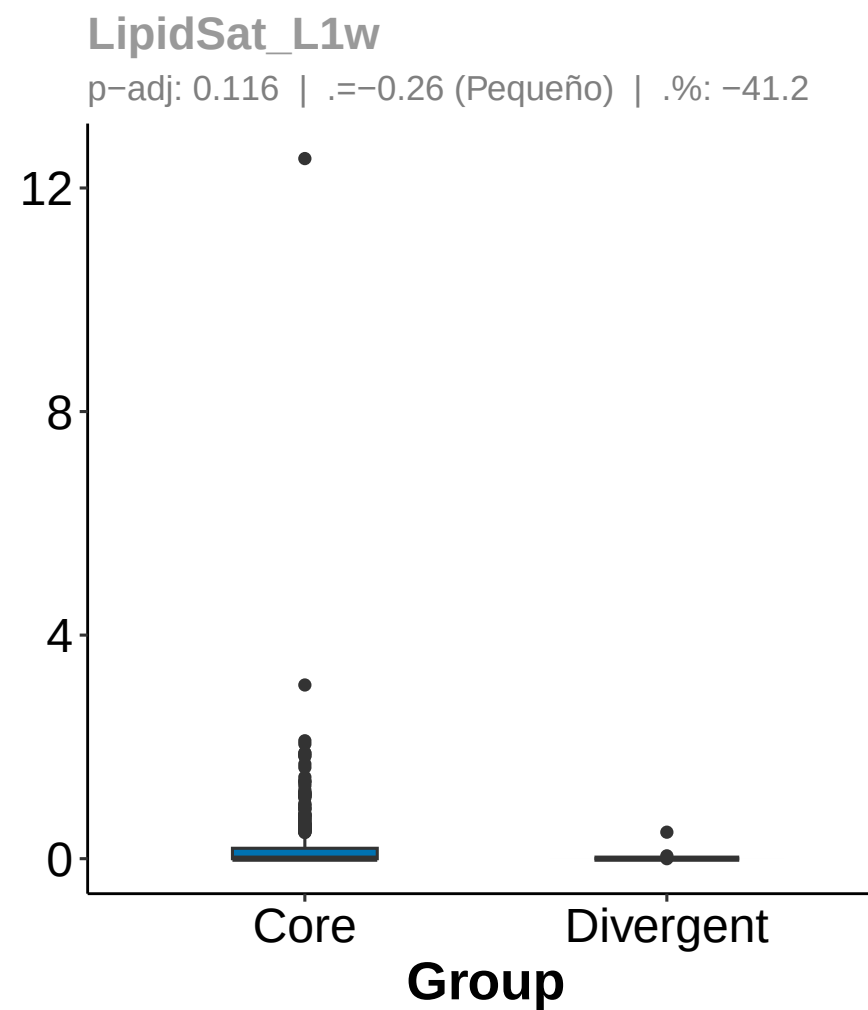
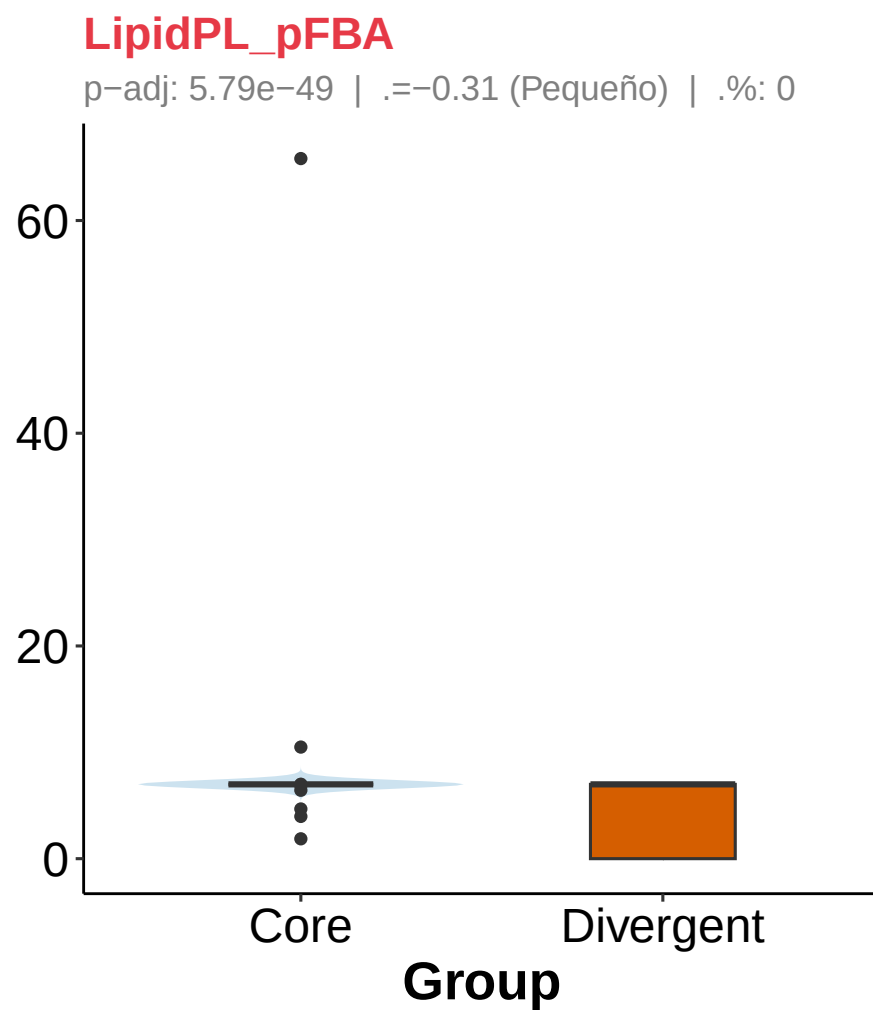
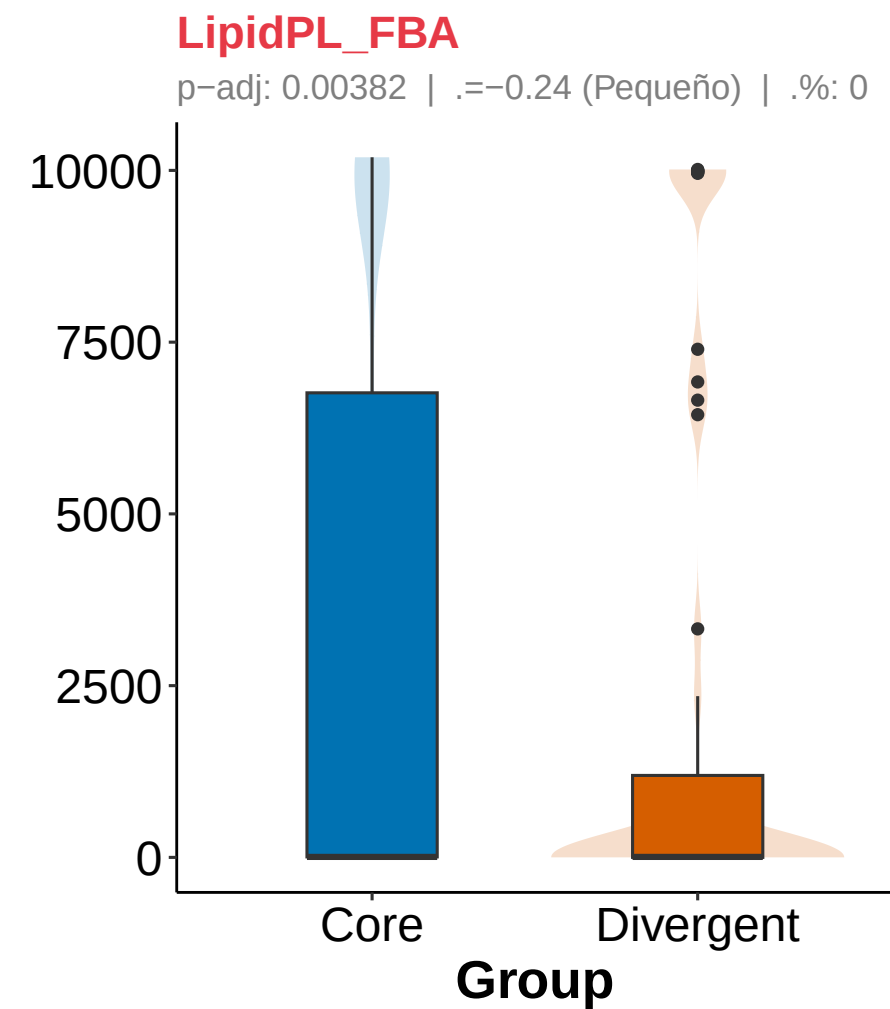
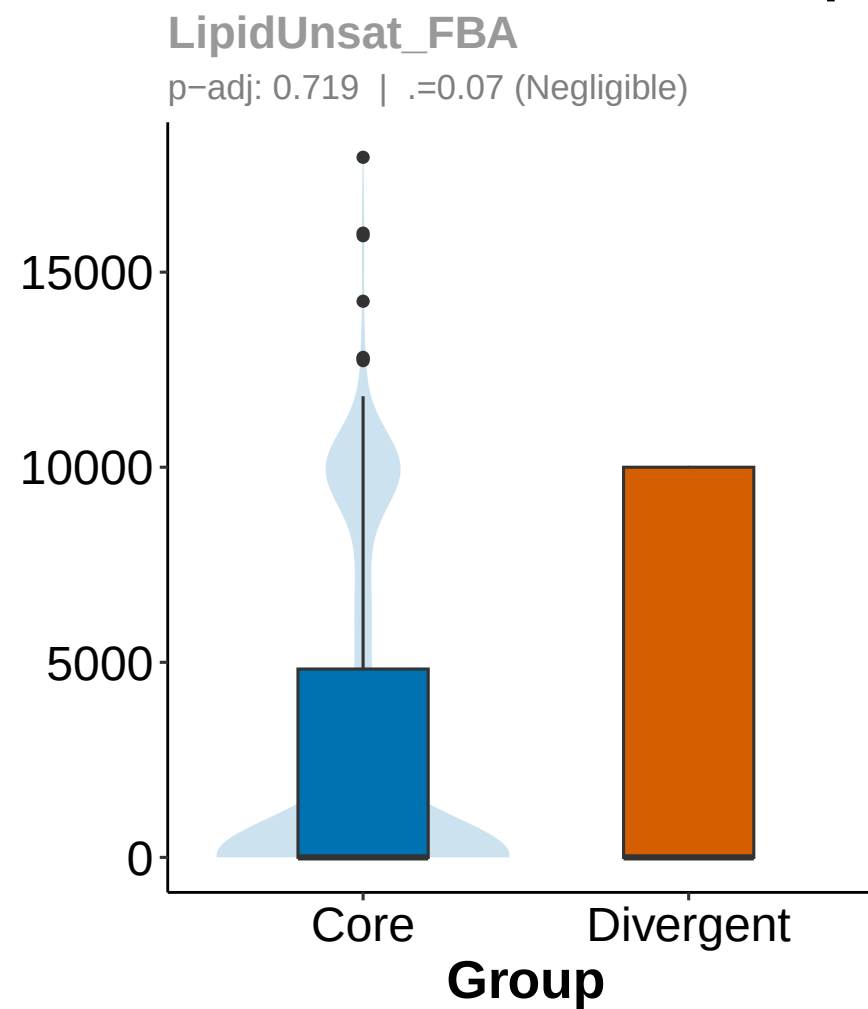
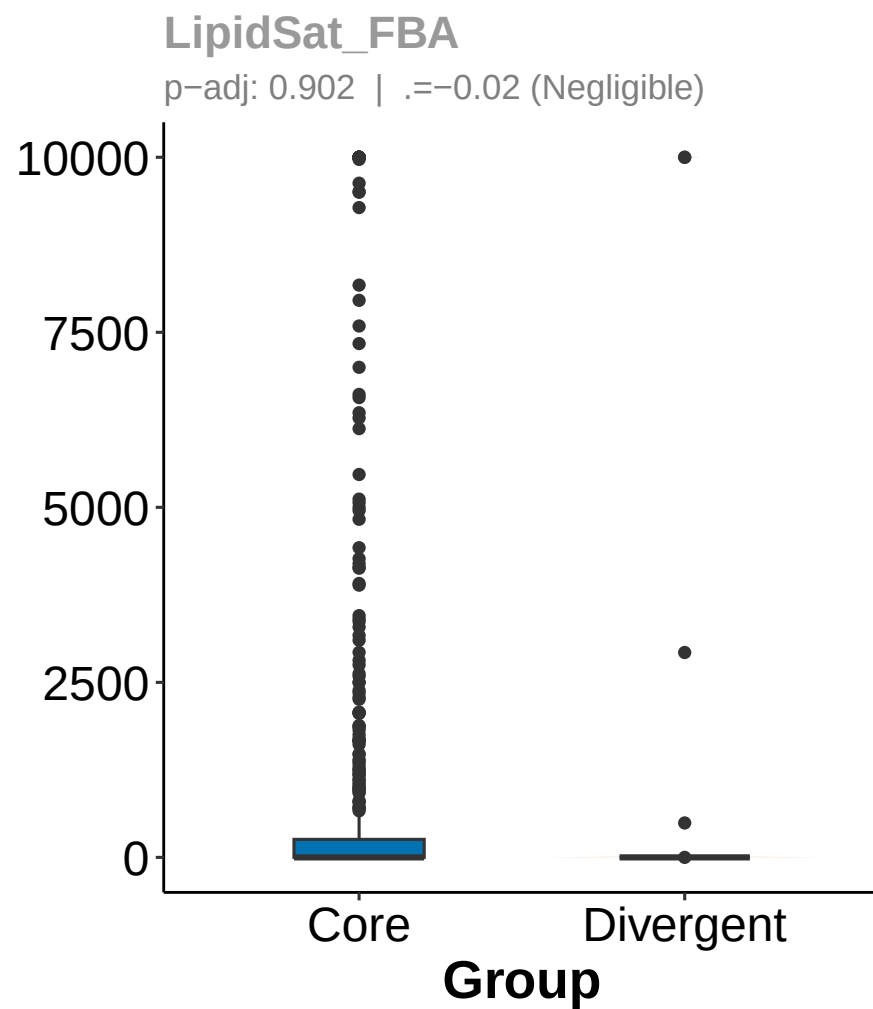


Cliff's . – . 3. Metabolismo Lipídico

Cliff's . + .% mediana | azul = mayor en Core · naranja = mayor en Divergent



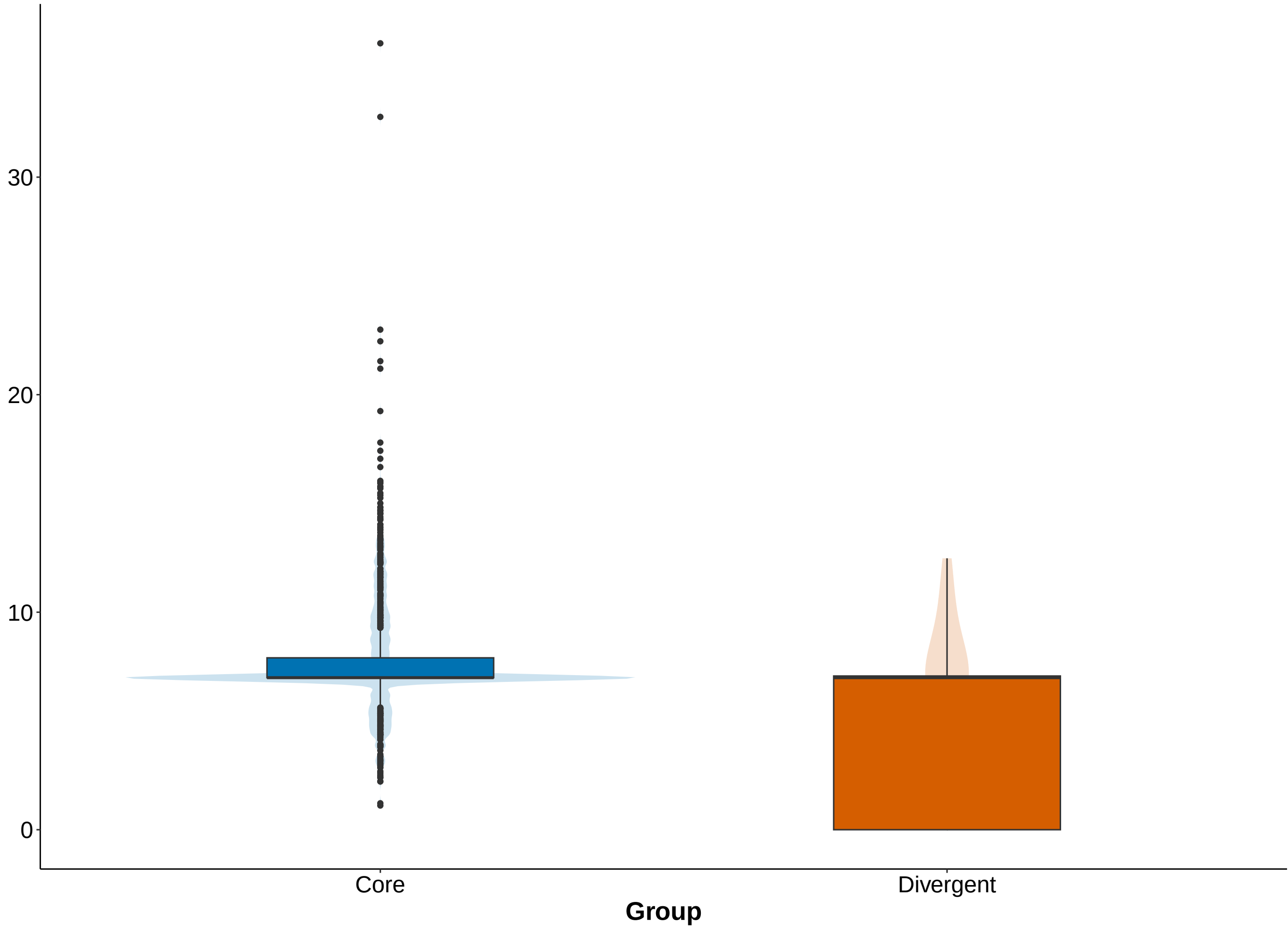
Metabolismo – . 3. Metabolismo Lipídico



Metabolismo – . 3. Metabolismo Lipídico

LipidPL_L1w

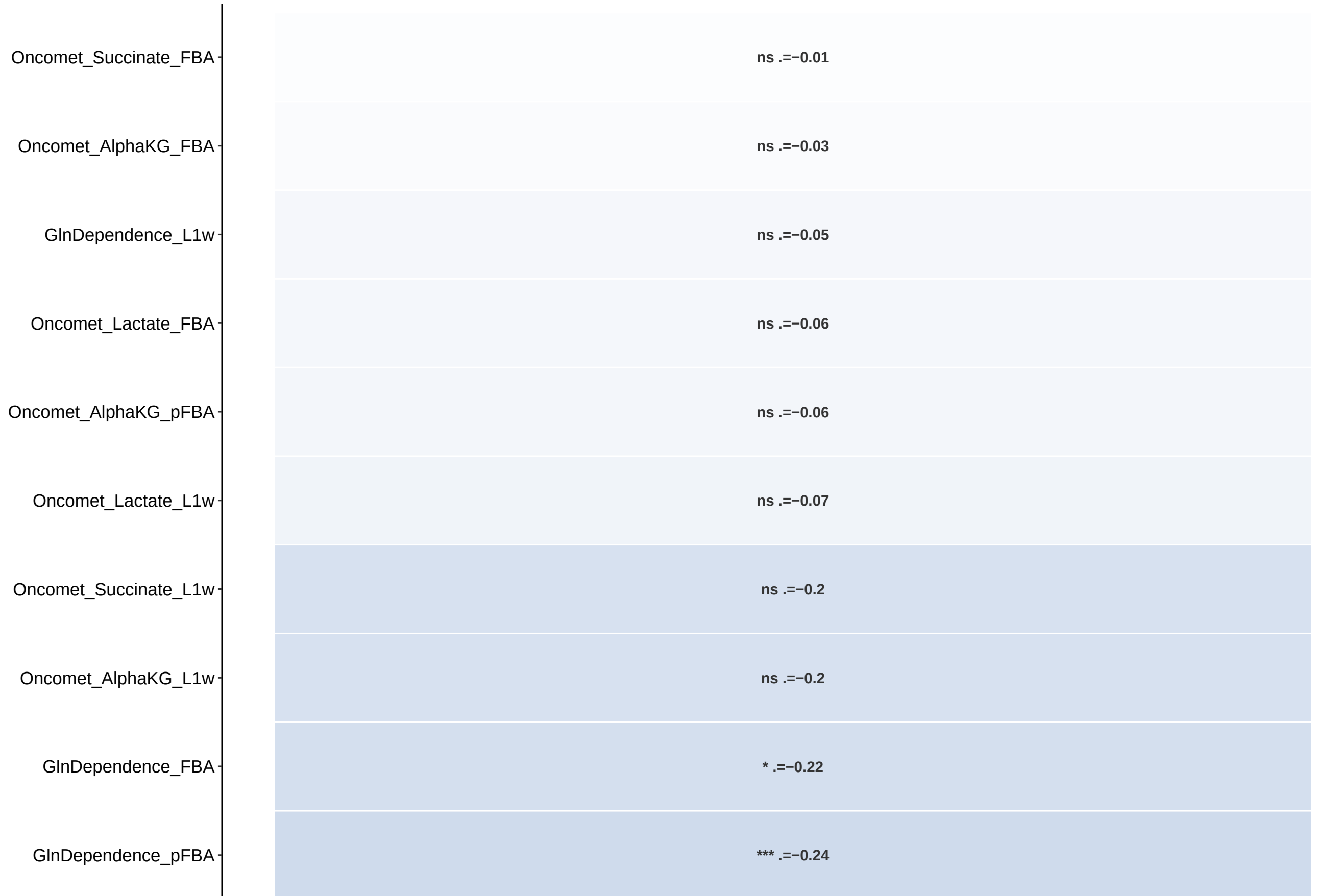
p-adj: 0.00242 | .=-0.26 (Pequeño) | .%: 0



Cliff's .: . 4. Dependencias Metabólicas Específicas

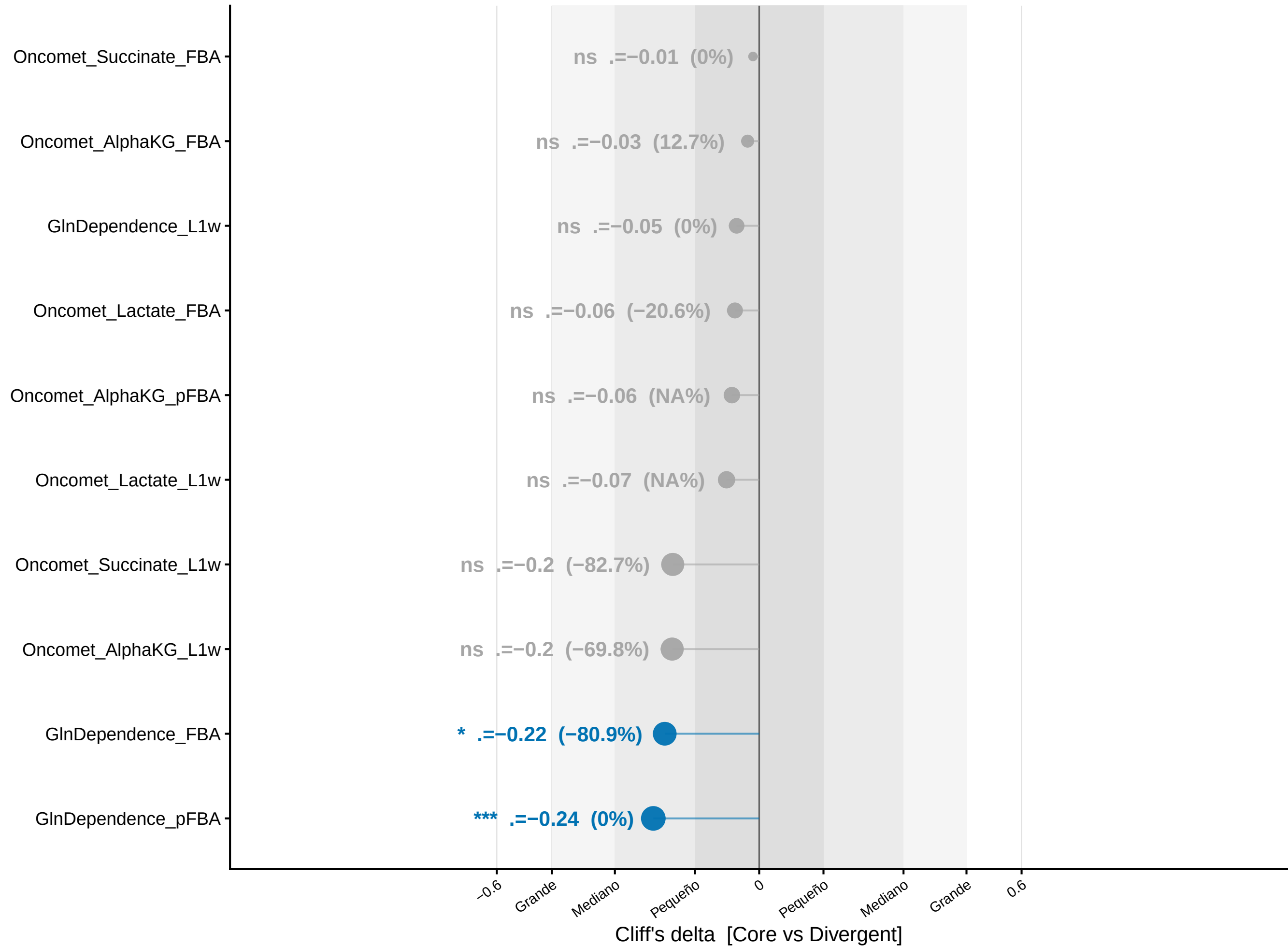
Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core



Cliff's . - . 4. Dependencias Metabólicas Específicas

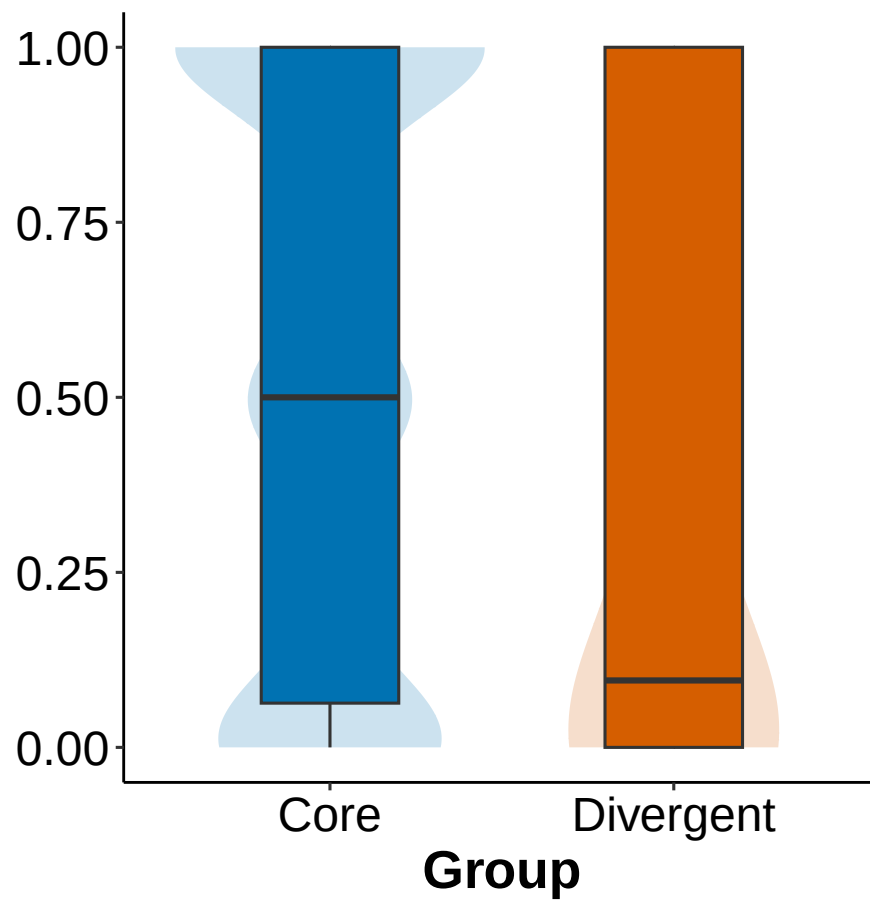
Cliff's . + .% mediana | azul = mayor en Core · naranja = mayor en Divergent



Metabolismo – 4. Dependencias Metabólicas Específicas

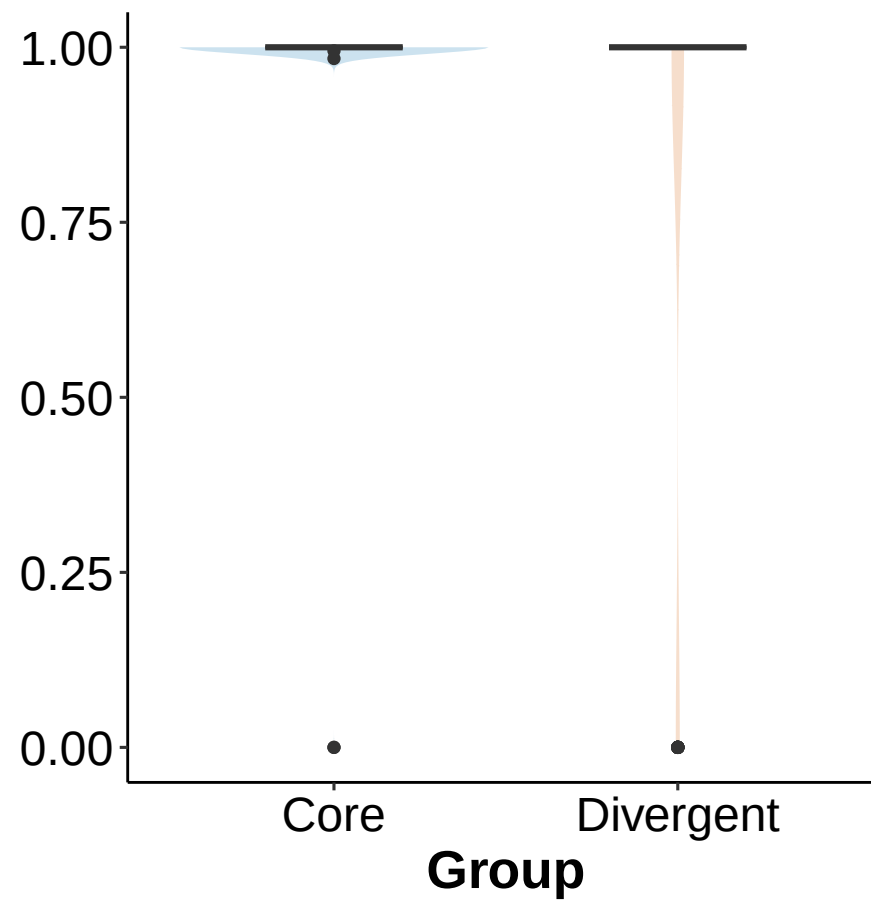
GlnDependence_FBA

p-adj: 0.0211 | .=-0.22 (Pequeño) | .%: -80.9



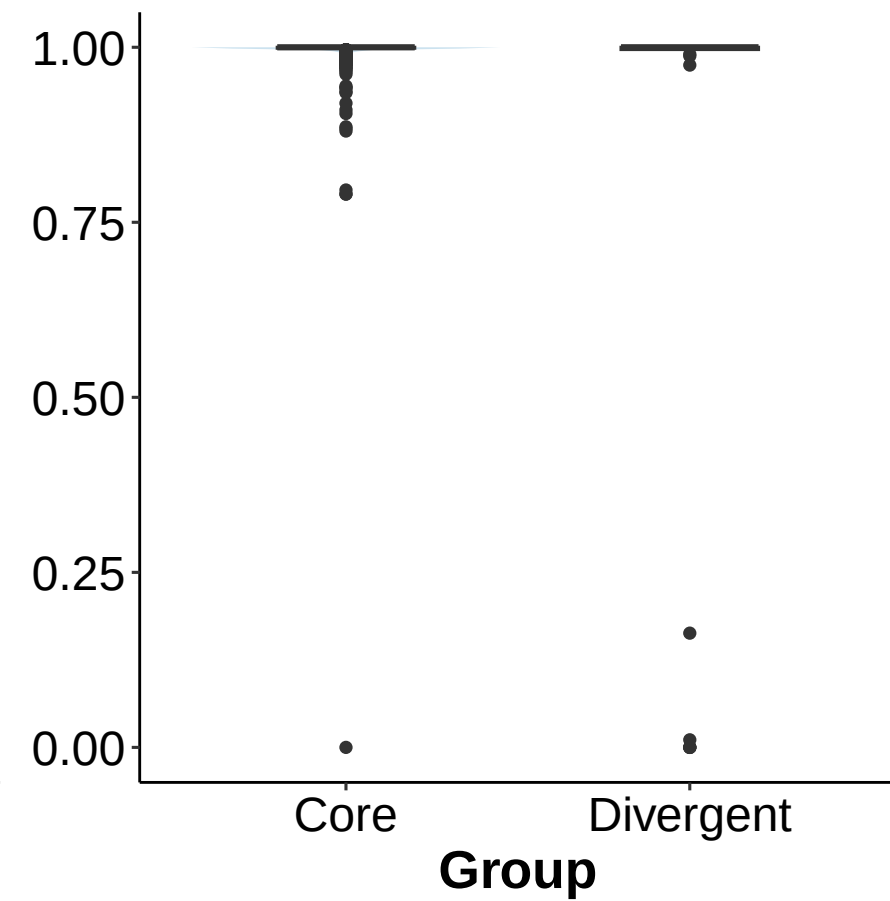
GlnDependence_pFBA

p-adj: 1.69e-44 | .=-0.24 (Pequeño) | .%: 0



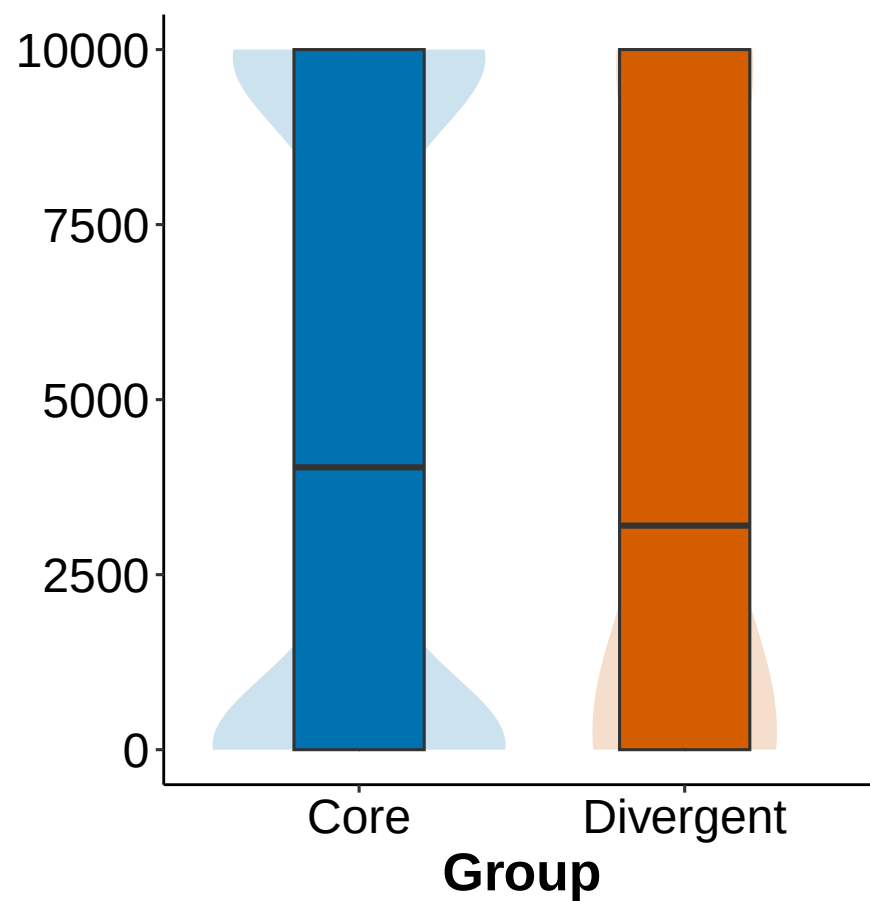
GlnDependence_L1w

p-adj: 0.705 | .=-0.05 (Negligible) | .%: 0



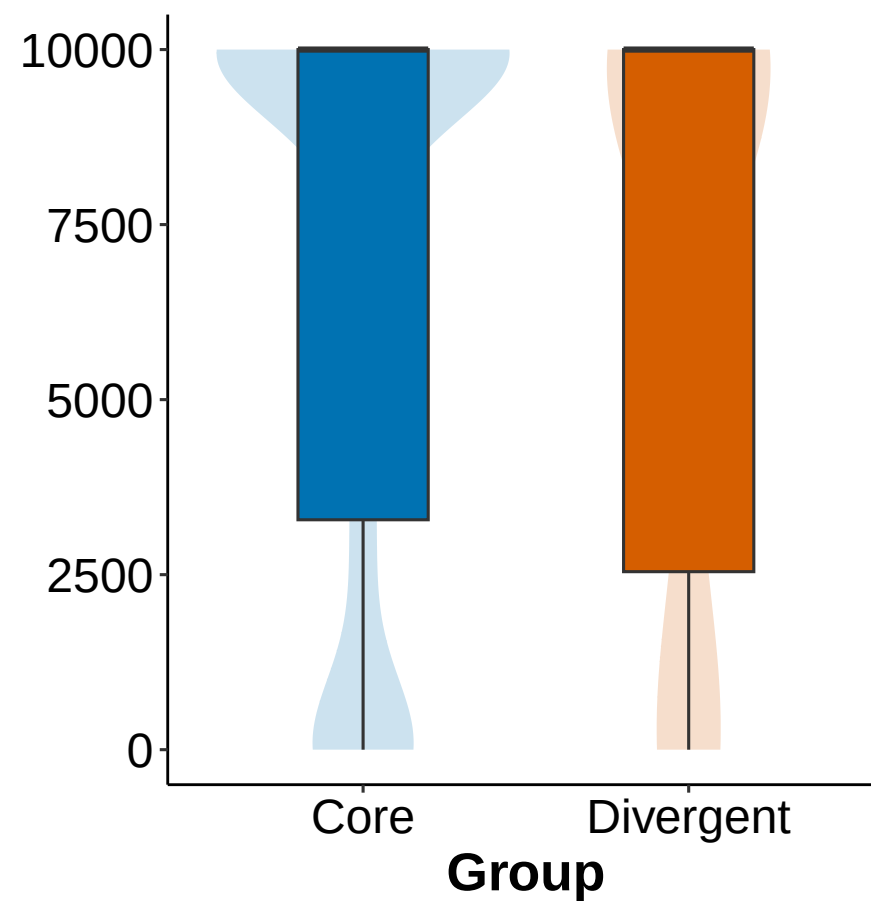
Oncomet_Lactate_FBA

p-adj: 0.68 | .=-0.06 (Negligible) | .%: -20.6



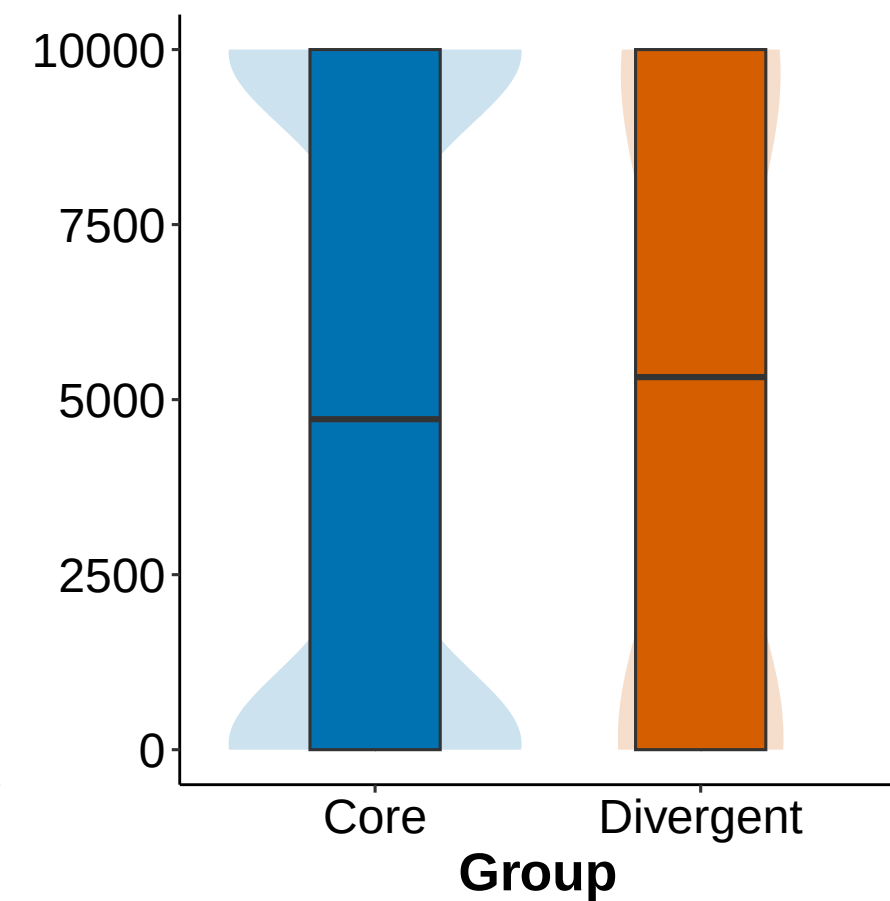
Oncomet_Succinate_FBA

p-adj: 0.922 | .=-0.01 (Negligible) | .%: 0



Oncomet_AlphaKG_FBA

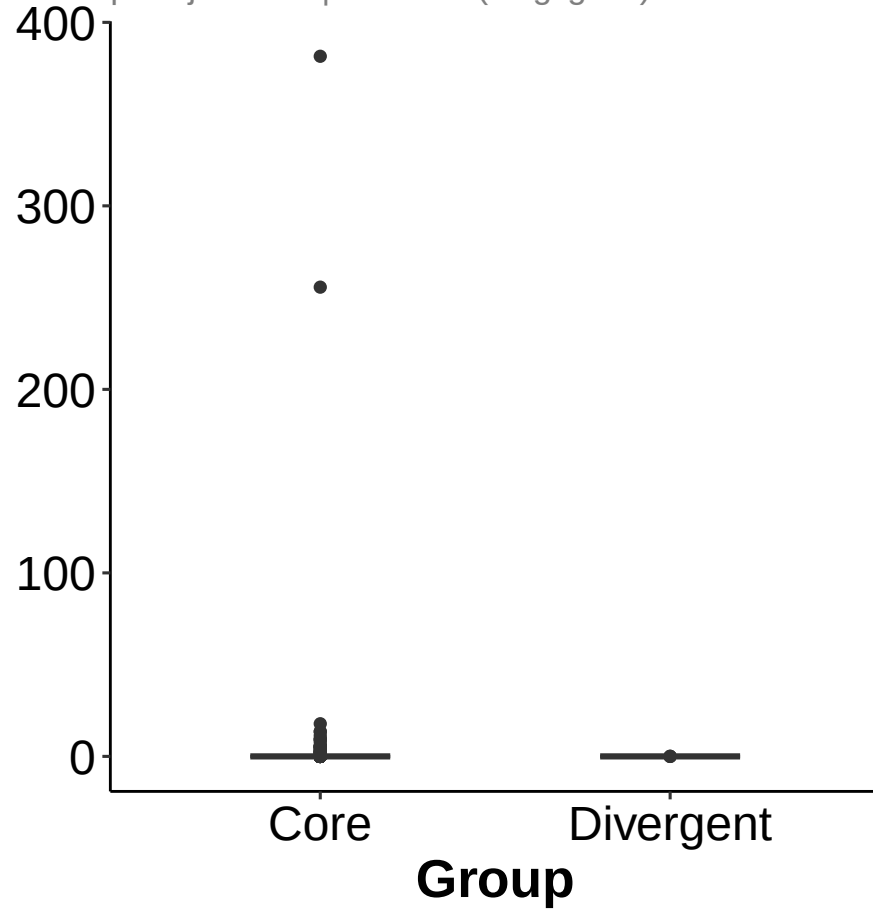
p-adj: 0.847 | .=-0.03 (Negligible) | .%: 12.7



Metabolismo – . 4. Dependencias Metabólicas Específicas

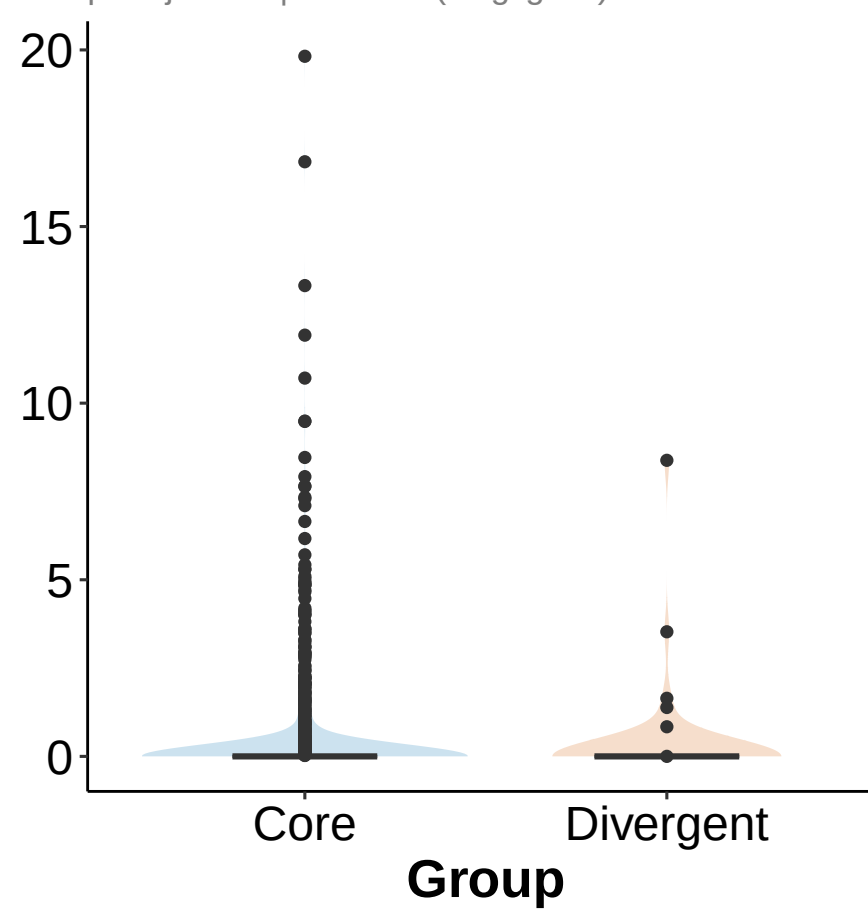
Oncomet_AlphaKG_pFBA

p-adj: 0.405 | .=-0.06 (Negligible)



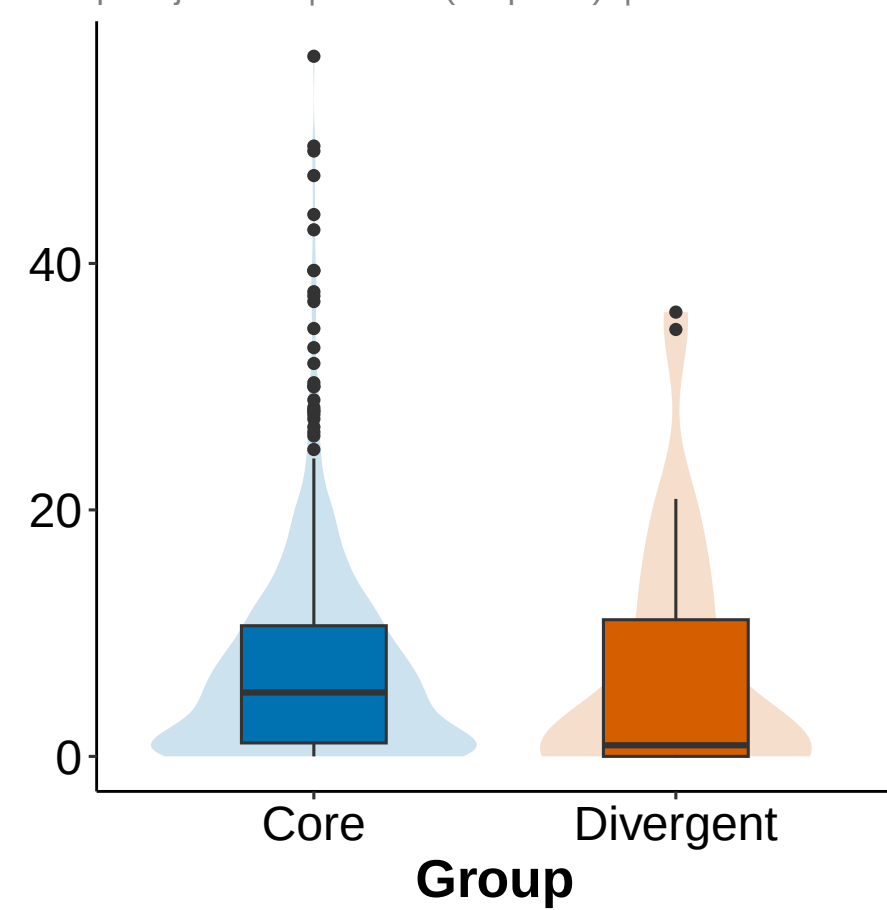
Oncomet_Lactate_L1w

p-adj: 0.32 | .=-0.07 (Negligible)



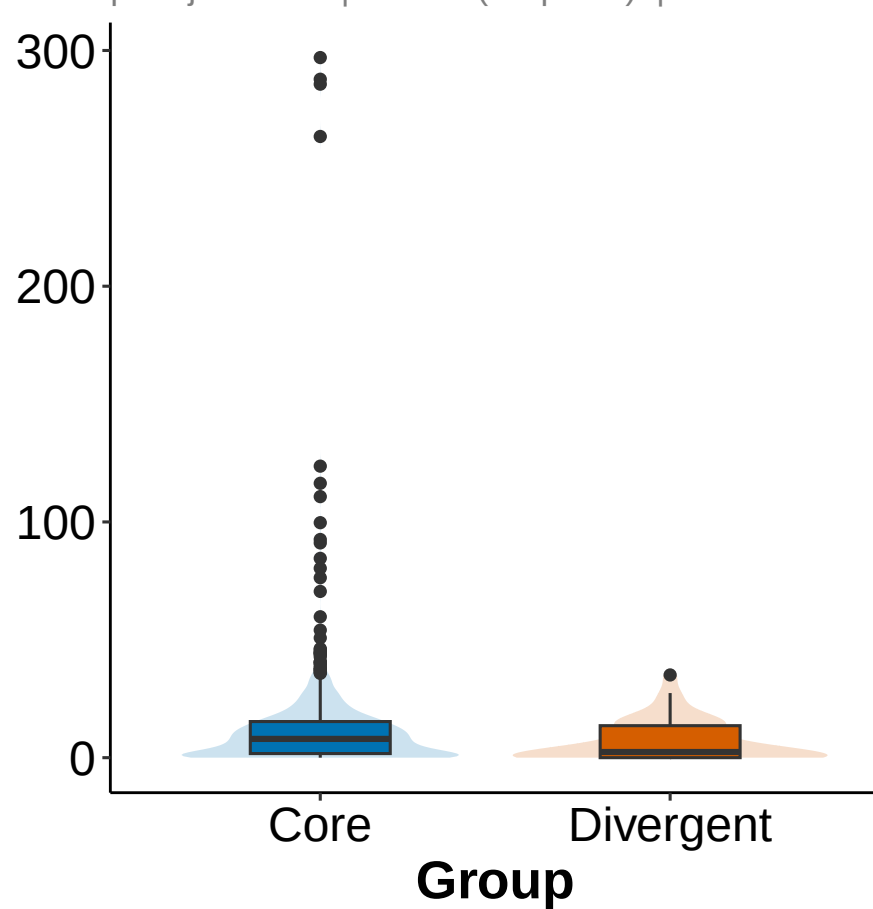
Oncomet_Succinate_L1w

p-adj: 0.122 | .=-0.2 (Pequeño) | .%: -82.7



Oncomet_AlphaKG_L1w

p-adj: 0.0701 | .=-0.2 (Pequeño) | .%: -69.8



Cliff's .: . 5. Reprogramación Sistémica (SA_*)

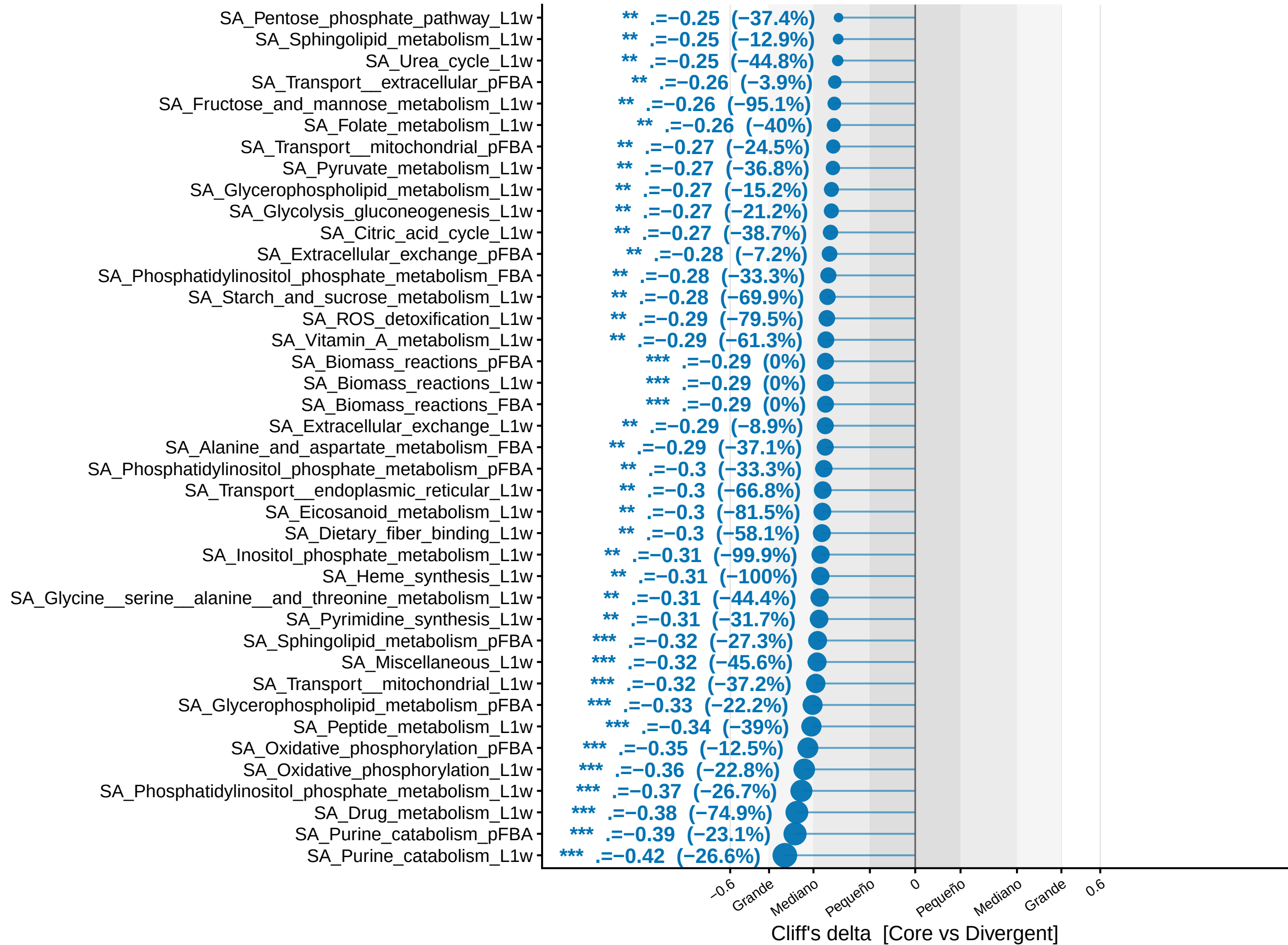
Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05



SA_Pentose_phosphate_pathway_L1w	**	.-0.25
SA_Sphingolipid_metabolism_L1w	**	.-0.25
SA_Urea_cycle_L1w	**	.-0.25
SA_Transport__extracellular_pFBA	**	.-0.26
SA_Fructose_and_mannose_metabolism_L1w	**	.-0.26
SA_Folate_metabolism_L1w	**	.-0.26
SA_Transport__mitochondrial_pFBA	**	.-0.27
SA_Pyruvate_metabolism_L1w	**	.-0.27
SA_Glycerophospholipid_metabolism_L1w	**	.-0.27
SA_Glycolysis_gluconeogenesis_L1w	**	.-0.27
SA_Citric_acid_cycle_L1w	**	.-0.27
SA_Extracellular_exchange_pFBA	**	.-0.28
SA_Phosphatidylinositol_phosphate_metabolism_FBA	**	.-0.28
SA_Starch_and_sucrose_metabolism_L1w	**	.-0.28
SA_ROS_detoxification_L1w	**	.-0.29
SA_Vitamin_A_metabolism_L1w	**	.-0.29
SA_Biomass_reactions_pFBA	***	.-0.29
SA_Biomass_reactions_L1w	***	.-0.29
SA_Biomass_reactions_FBA	***	.-0.29
SA_Extracellular_exchange_L1w	**	.-0.29
SA_Alanine_and_aspartate_metabolism_FBA	**	.-0.29
SA_Phosphatidylinositol_phosphate_metabolism_pFBA	**	.-0.3
SA_Transport__endoplasmic_reticular_L1w	**	.-0.3
SA_Eicosanoid_metabolism_L1w	**	.-0.3
SA_Dietary_fiber_binding_L1w	**	.-0.3
SA_Inositol_phosphate_metabolism_L1w	**	.-0.31
SA_Heme_synthesis_L1w	**	.-0.31
SA_Glycine__serine__alanine__and_threonine_metabolism_L1w	**	.-0.31
SA_Pyrimidine_synthesis_L1w	**	.-0.31
SA_Sphingolipid_metabolism_pFBA	***	.-0.32
SA_Miscellaneous_L1w	***	.-0.32
SA_Transport__mitochondrial_L1w	***	.-0.32
SA_Glycerophospholipid_metabolism_pFBA	***	.-0.33
SA_Peptide_metabolism_L1w	***	.-0.34
SA_Oxidative_phosphorylation_pFBA	***	.-0.35
SA_Oxidative_phosphorylation_L1w	***	.-0.36
SA_Phosphatidylinositol_phosphate_metabolism_L1w	***	.-0.37
SA_Drug_metabolism_L1w	***	.-0.38
SA_Purine_catabolism_pFBA	***	.-0.39
SA_Purine_catabolism_L1w	***	.-0.42

Cliff's . – . 5. Reprogramación Sistémica (SA_*)

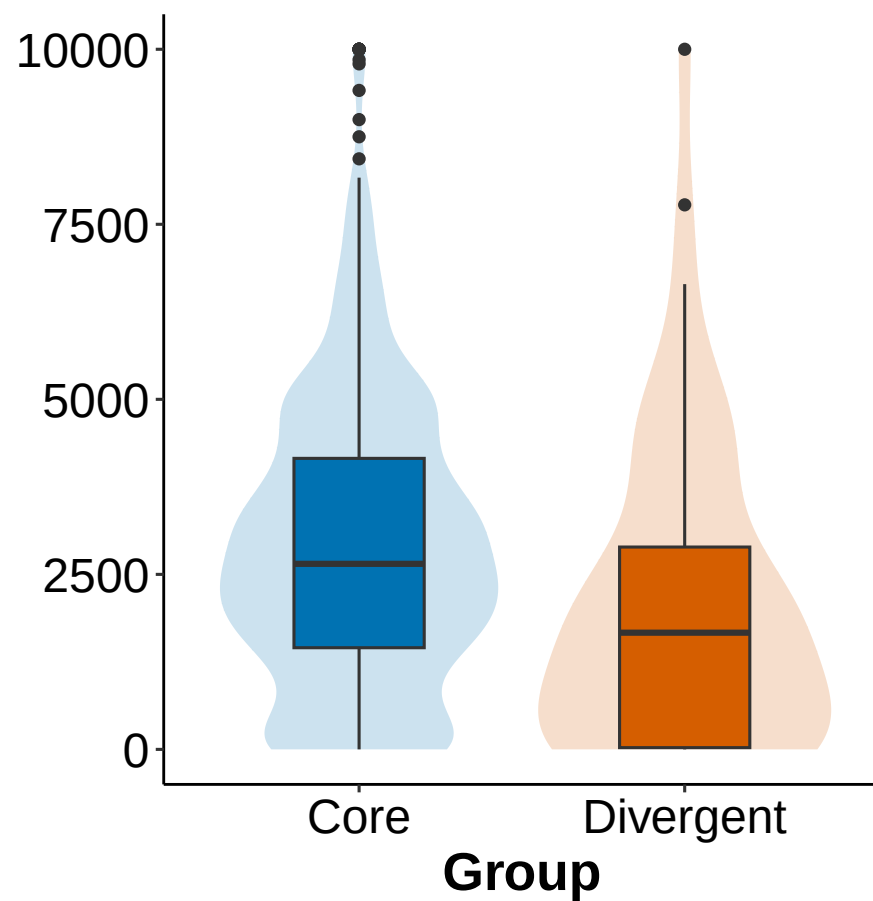
Cliff's . + .% mediana | azul = mayor en Core · naranja = mayor en Divergent



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

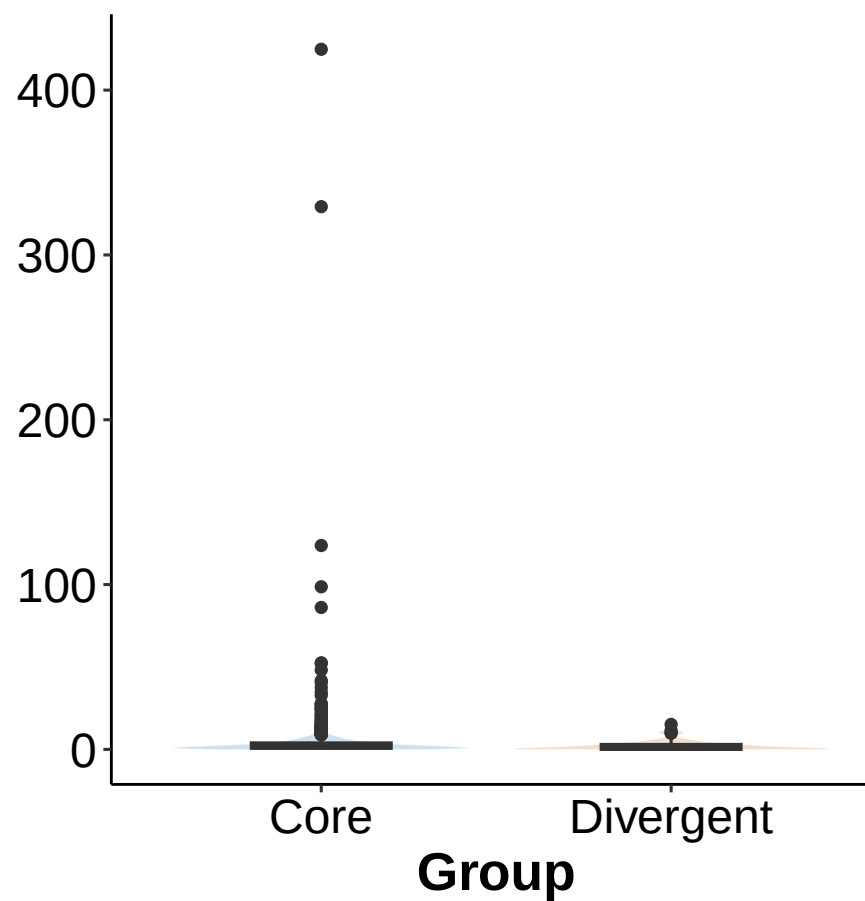
SA_Alanine_and_aspartate_metabolism

p-adj: 0.00199 | .=-0.29 (Pequeño) | .%: -37



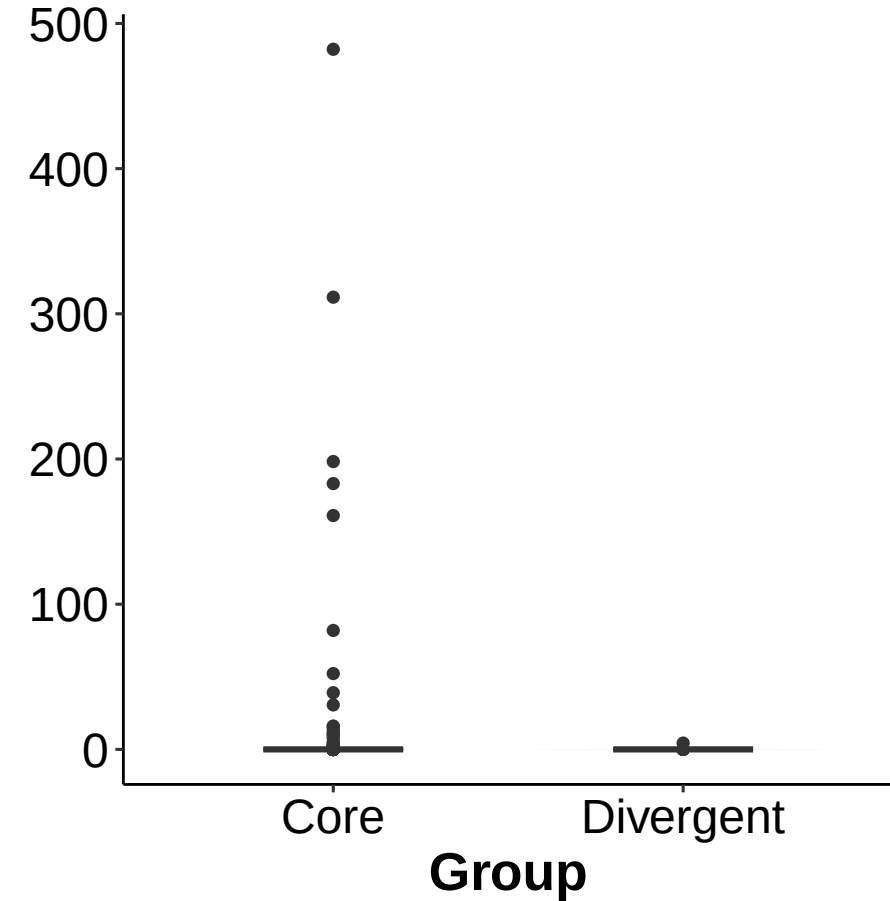
SA_Alanine_and_aspartate_metabolism

p-adj: 0.0226 | .=-0.22 (Pequeño) | .%: -35.6



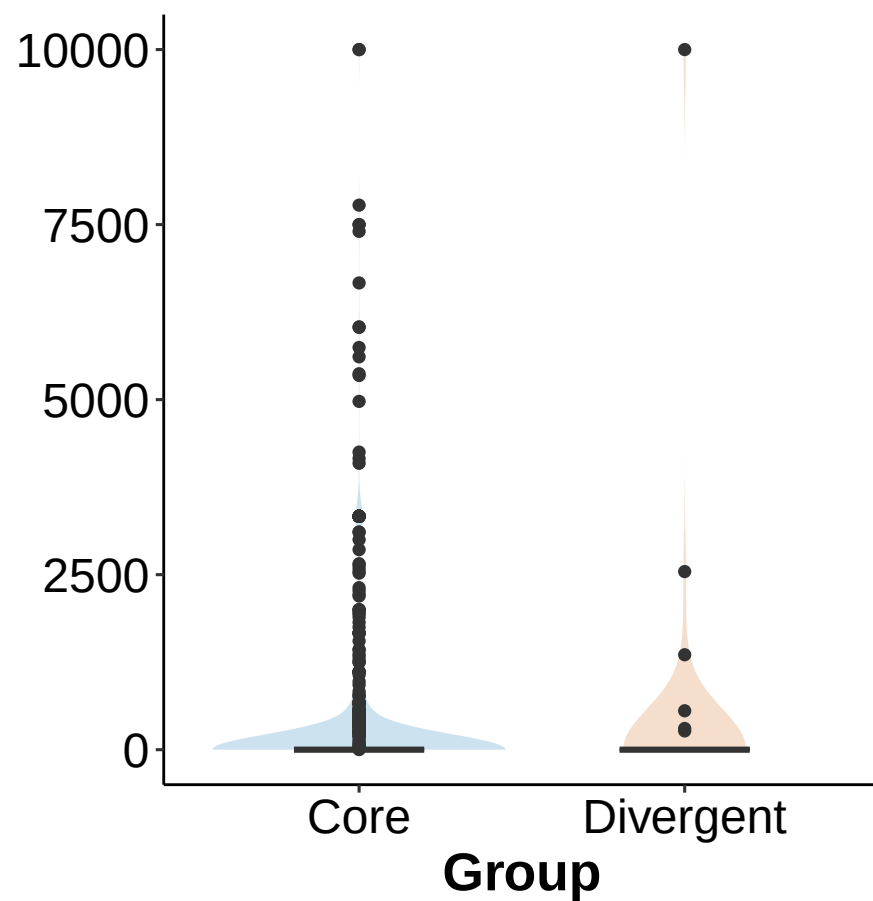
SA_Alanine_and_aspartate_metabolism

p-adj: 0.839 | .=-0.02 (Negligible)



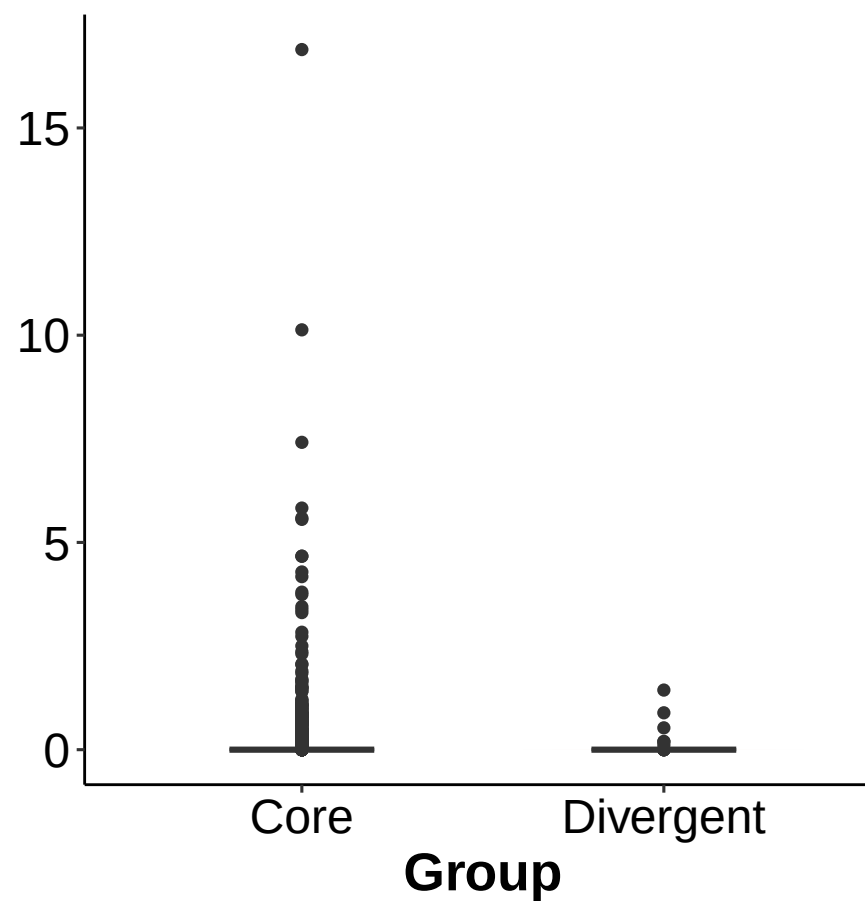
SA_Aminosugar_metabolism_FBA

p-adj: 0.578 | .=-0.05 (Negligible)



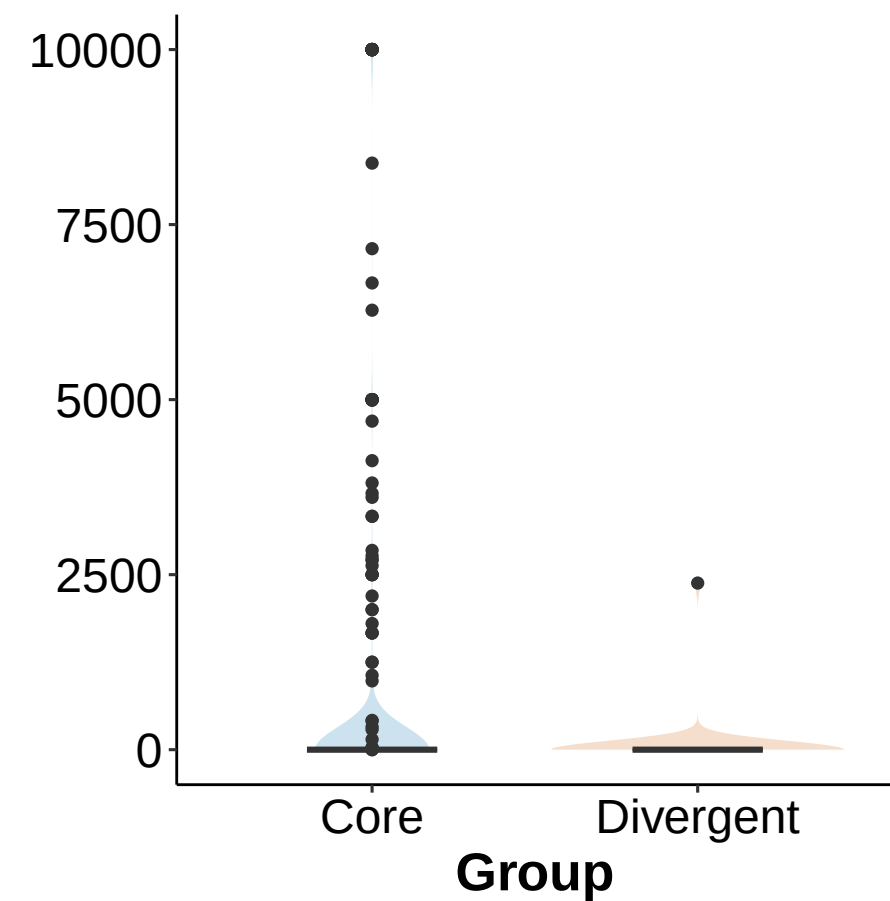
SA_Aminosugar_metabolism_L1w

p-adj: 0.0924 | .=-0.17 (Pequeño) | .%: -69.1



SA_Androgen_and_estrogen_synthesis

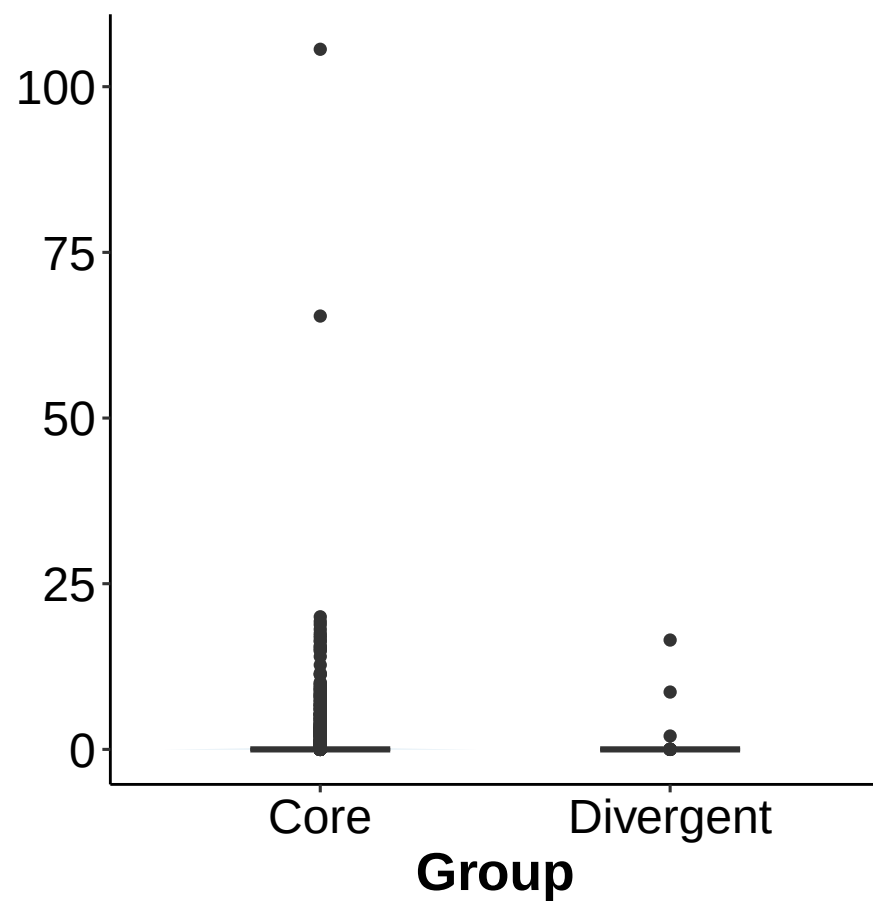
p-adj: 0.349 | .=-0.04 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

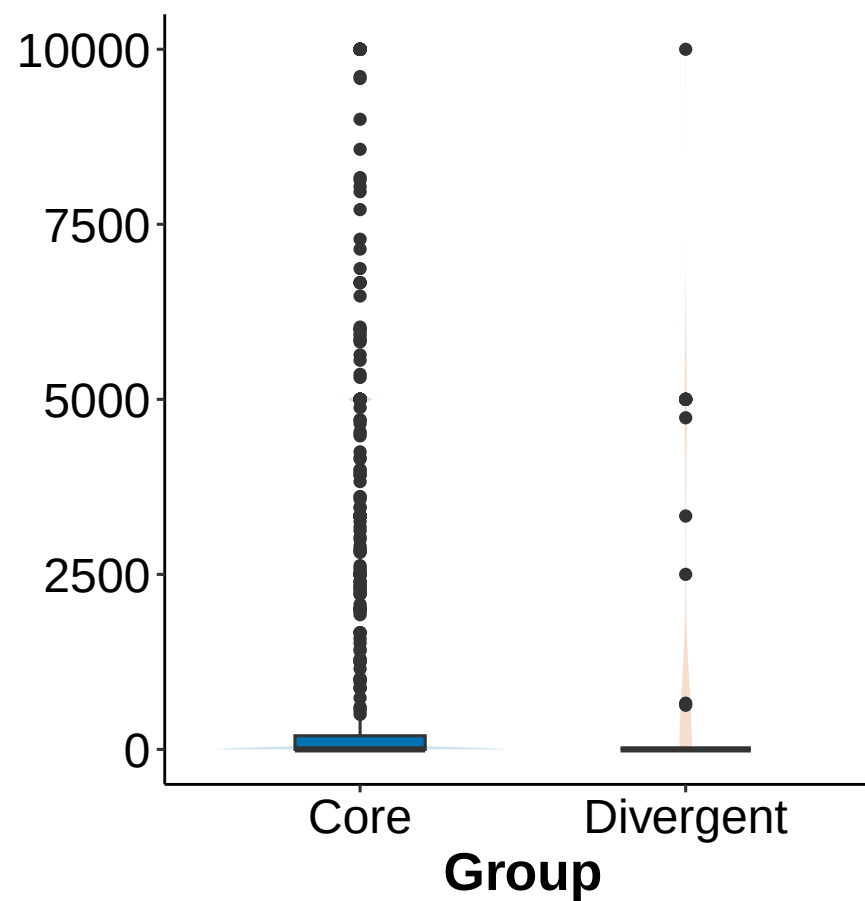
SA_Androgen_and_estrogen_synthe

p-adj: 0.251 | .=-0.1 (Negligible)



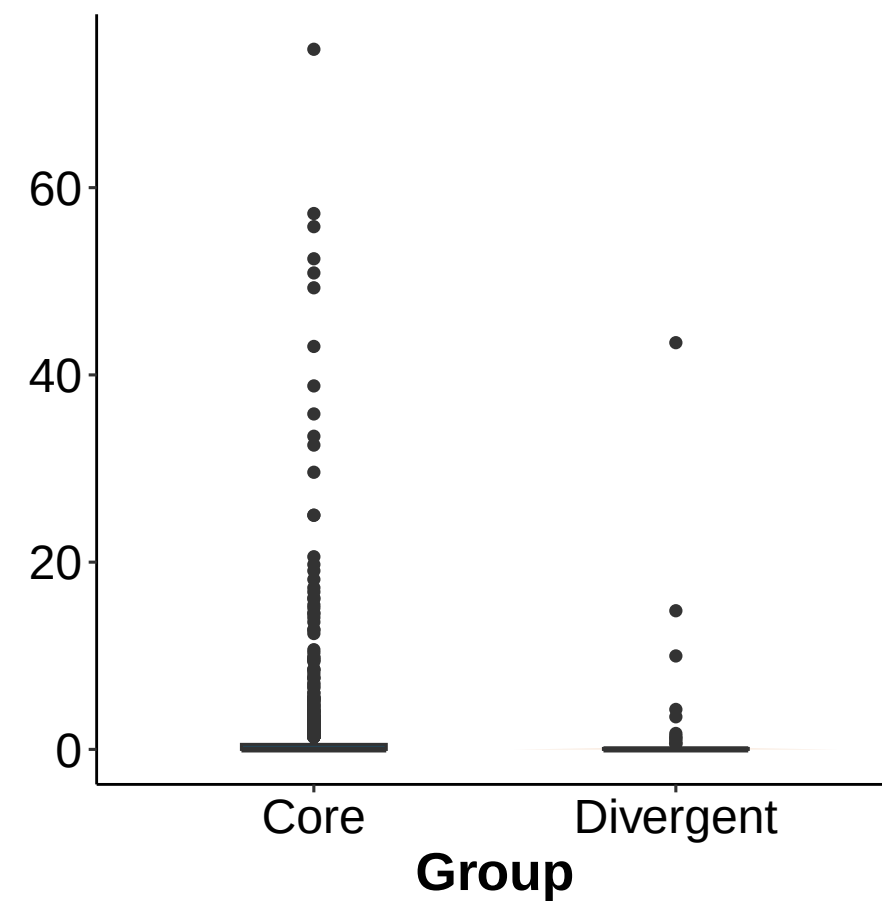
SA_Arachidonic_acid_metabolism

p-adj: 0.853 | .=-0.02 (Negligible)



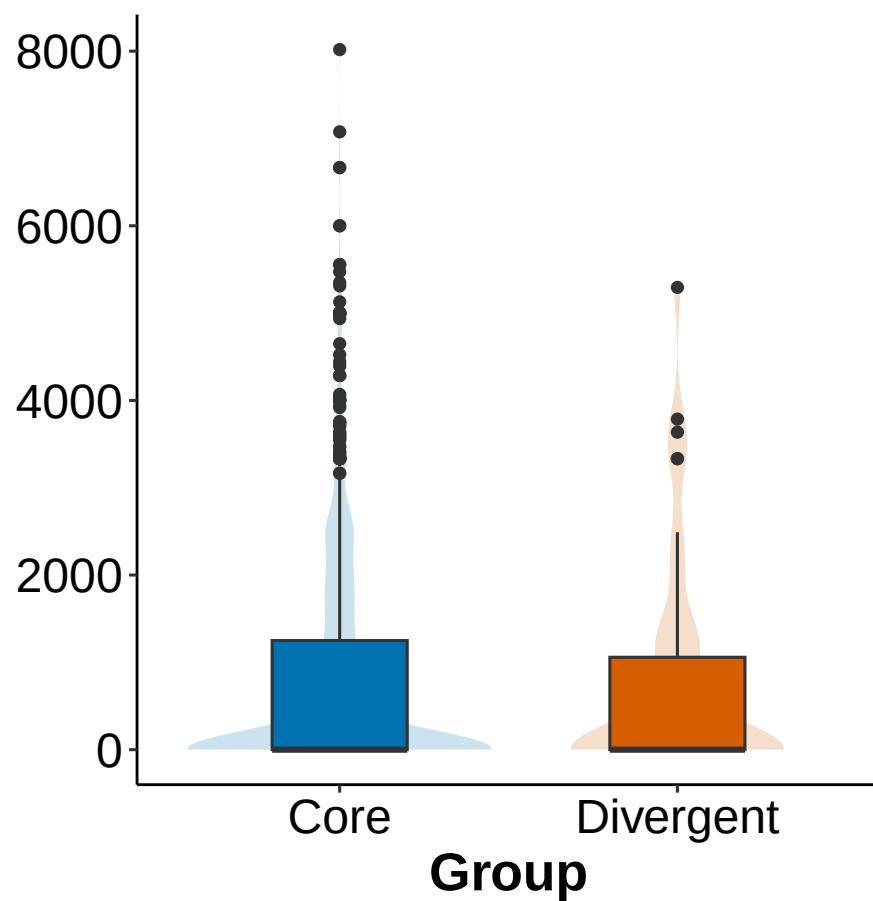
SA_Arachidonic_acid_metabolism_L1

p-adj: 0.243 | .=-0.12 (Negligible) | .%: -91.8



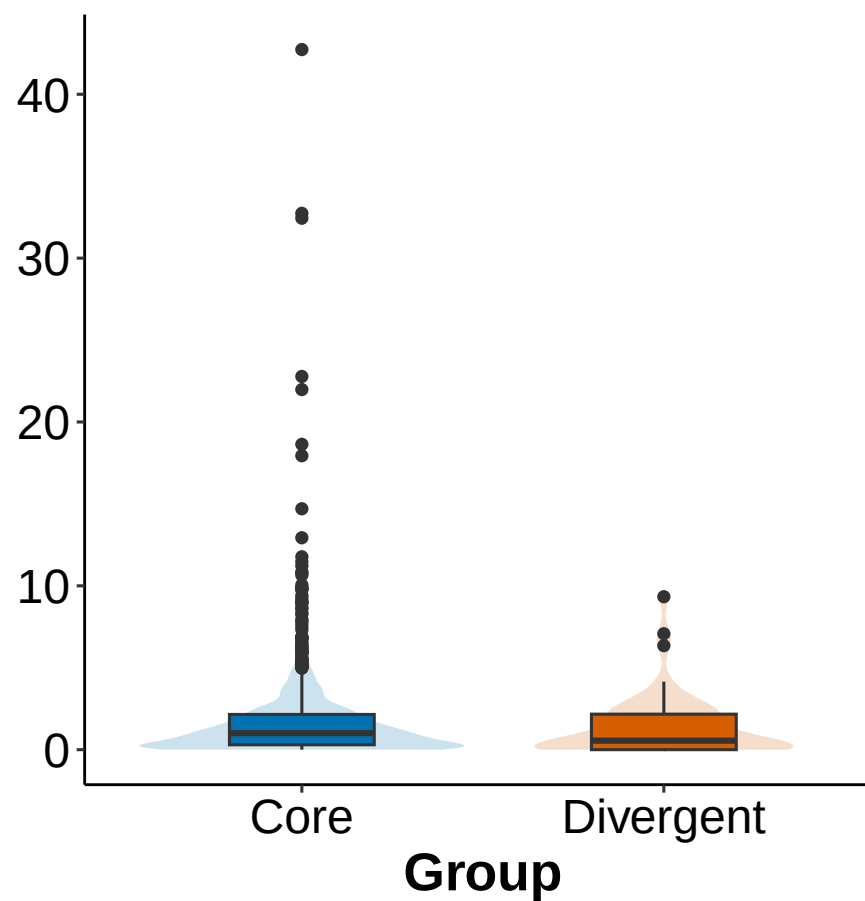
SA_Arginine_and_proline_metaboli

p-adj: 0.62 | .=-0.06 (Negligible)



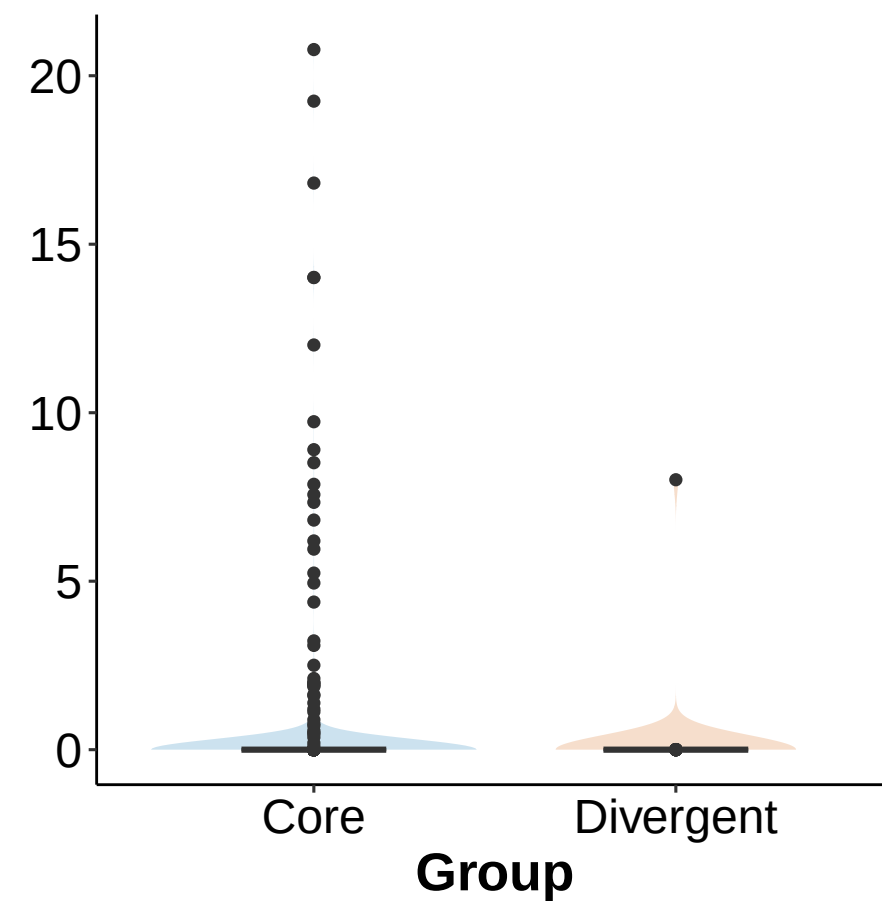
SA_Arginine_and_proline_metabolism

p-adj: 0.192 | .=-0.14 (Negligible) | .%: -45.2



SA_Arginine_and_proline_metabolism

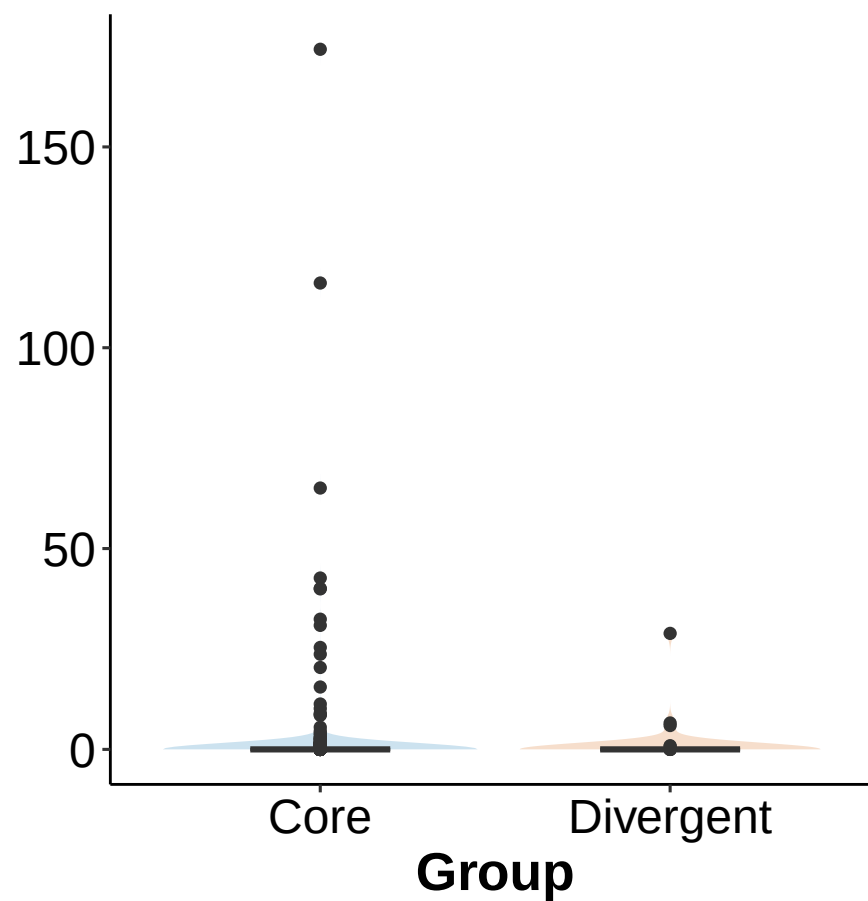
p-adj: 0.758 | .=0.03 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

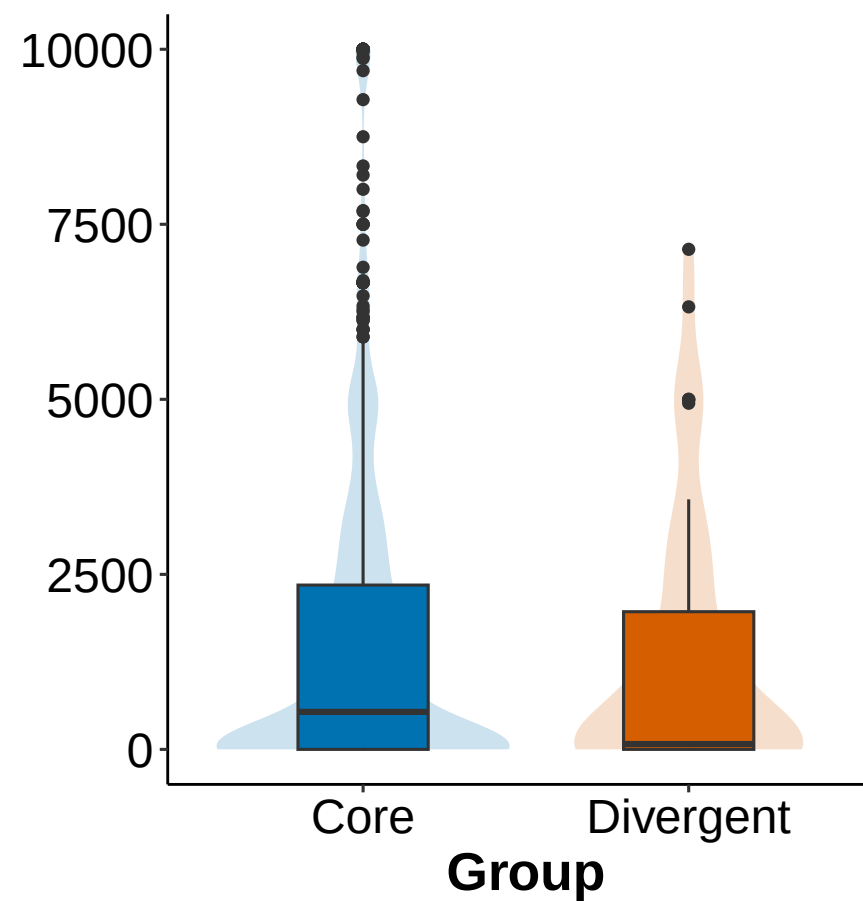
SA_Beta_Alanine_metabolism_L1w

p-adj: 0.72 | .:=0.03 (Negligible)



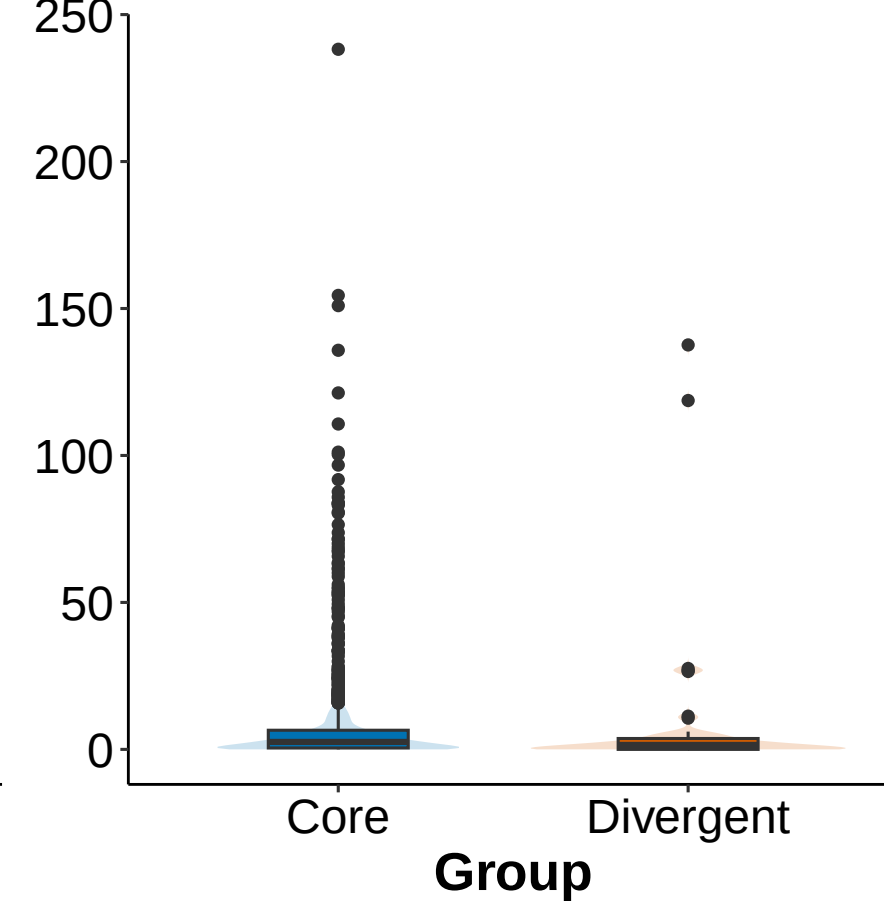
SA_Bile_acid_synthesis_FBA

p-adj: 0.56 | .:=−0.07 (Negligible) | .%: −85.7



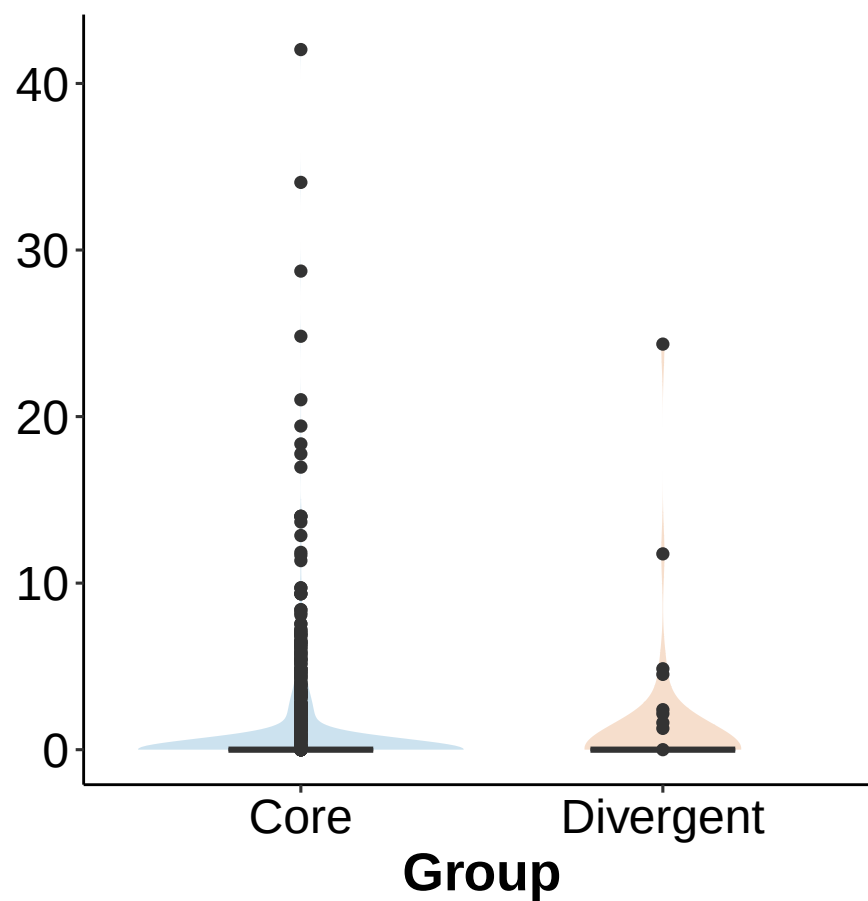
SA_Bile_acid_synthesis_L1w

p-adj: 0.0202 | .:=−0.22 (Pequeño) | .%: −40



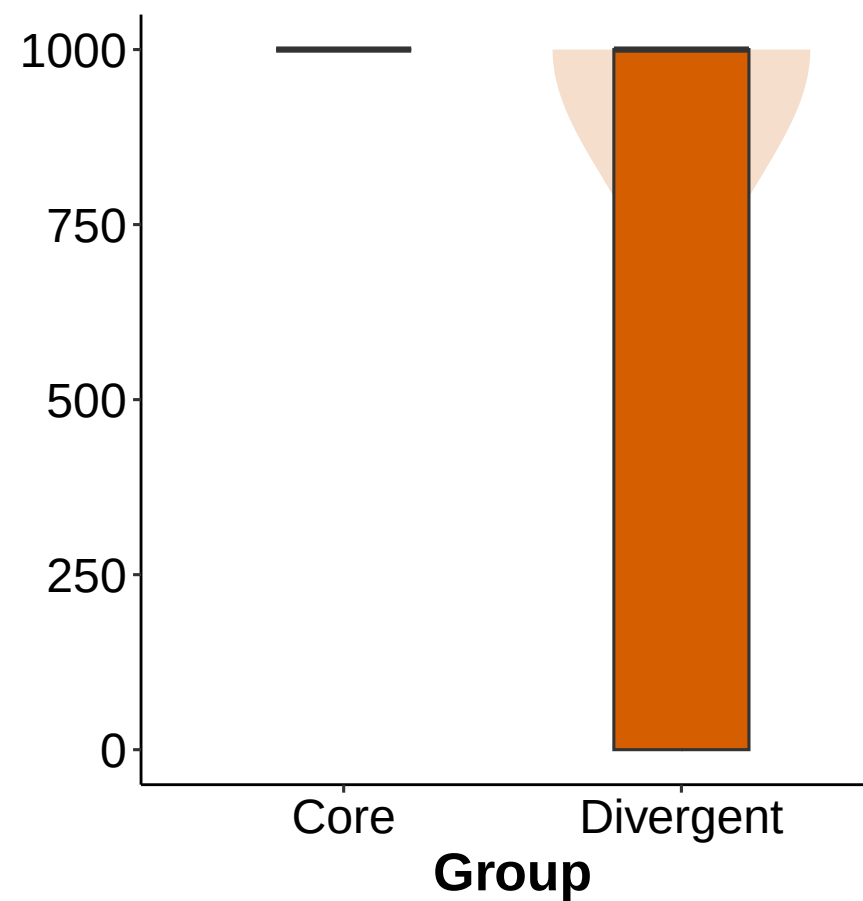
SA_Bile_acid_synthesis_pFBA

p-adj: 0.537 | .:=−0.06 (Negligible)



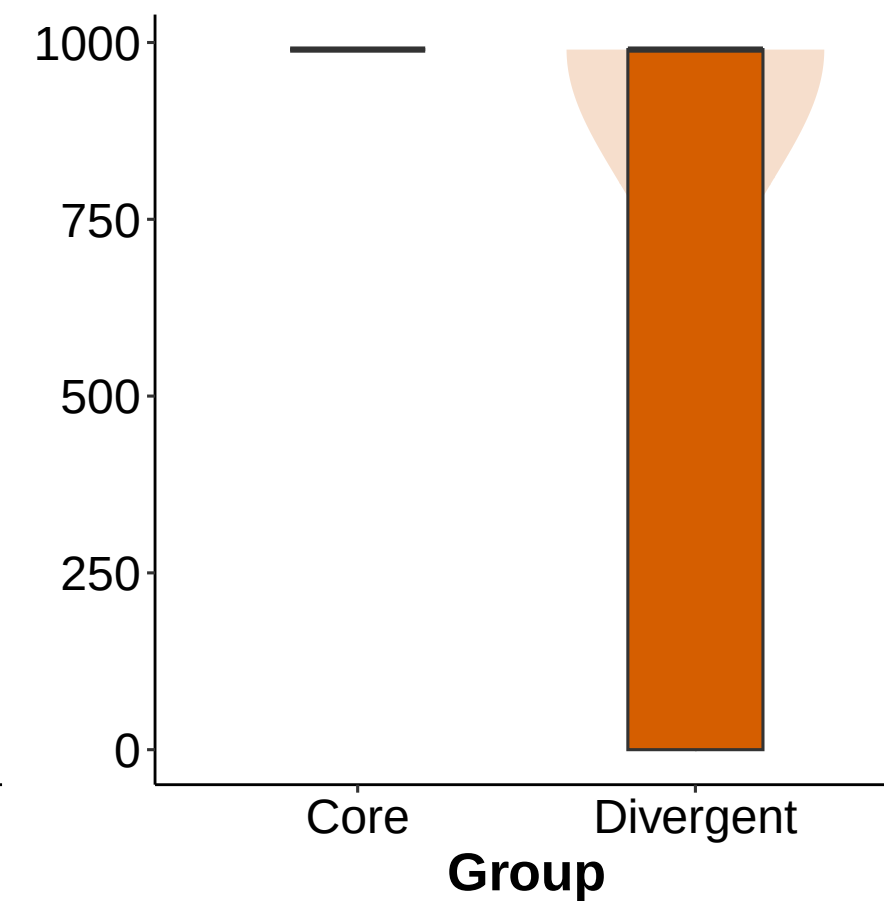
SA_Biomass_reactions_FBA

p-adj: 2.96e-71 | .:=−0.29 (Pequeño) | .%: 0



SA_Biomass_reactions_L1w

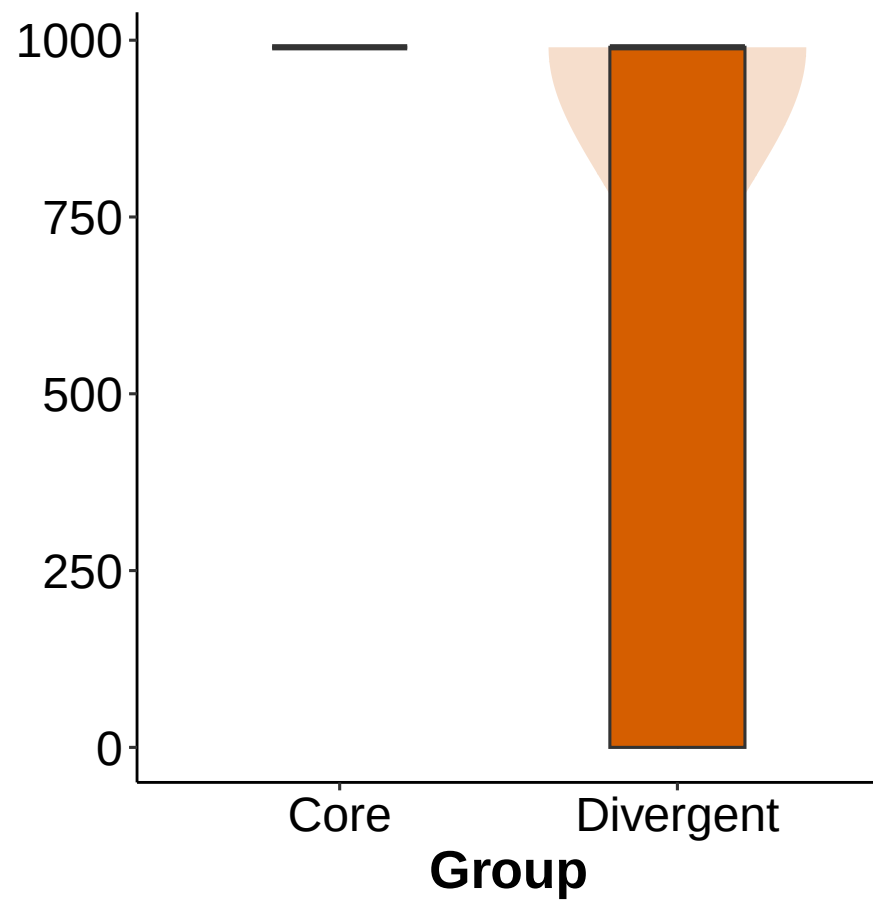
p-adj: 2.96e-71 | .:=−0.29 (Pequeño) | .%: 0



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

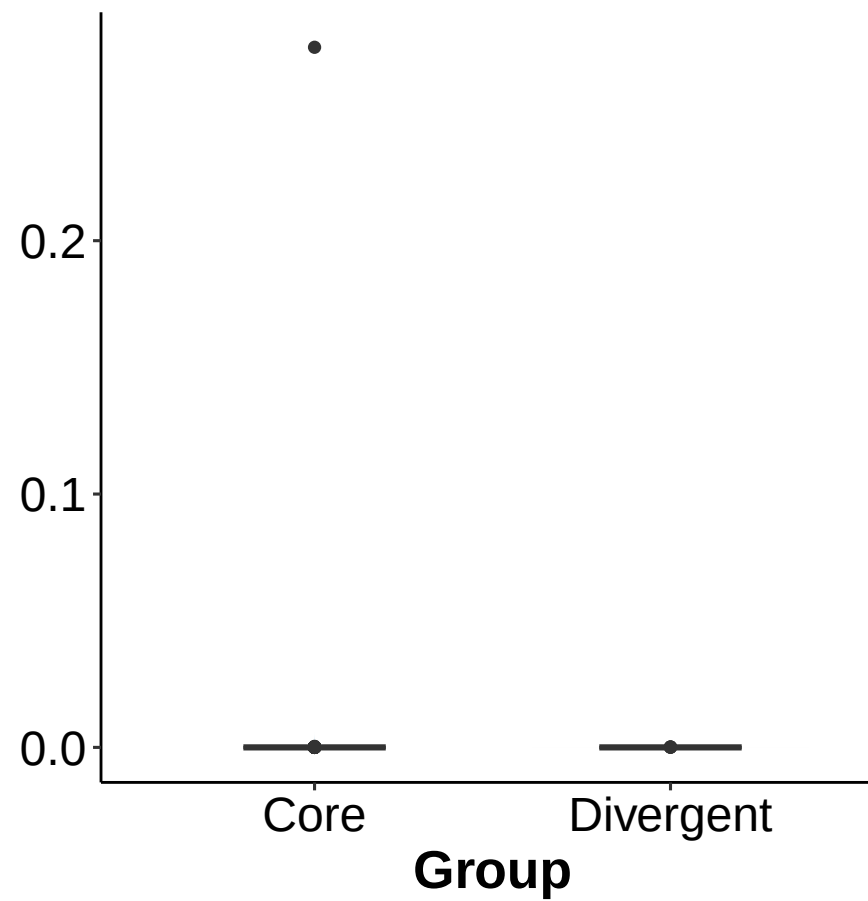
SA_Biomass_reactions_pFBA

p-adj: 2.96e-71 | .=-0.29 (Pequeño) | .%: 0



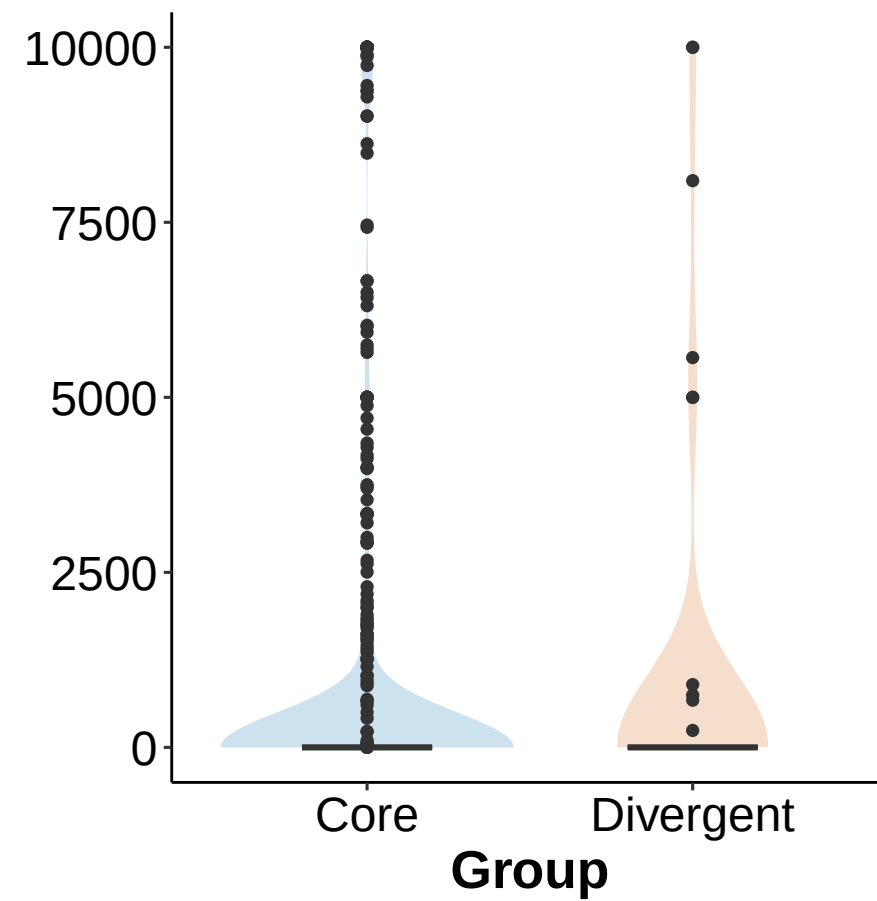
SA_Biotin_metabolism_L1w

p-adj: 0.731 | .=-0.02 (Negligible)



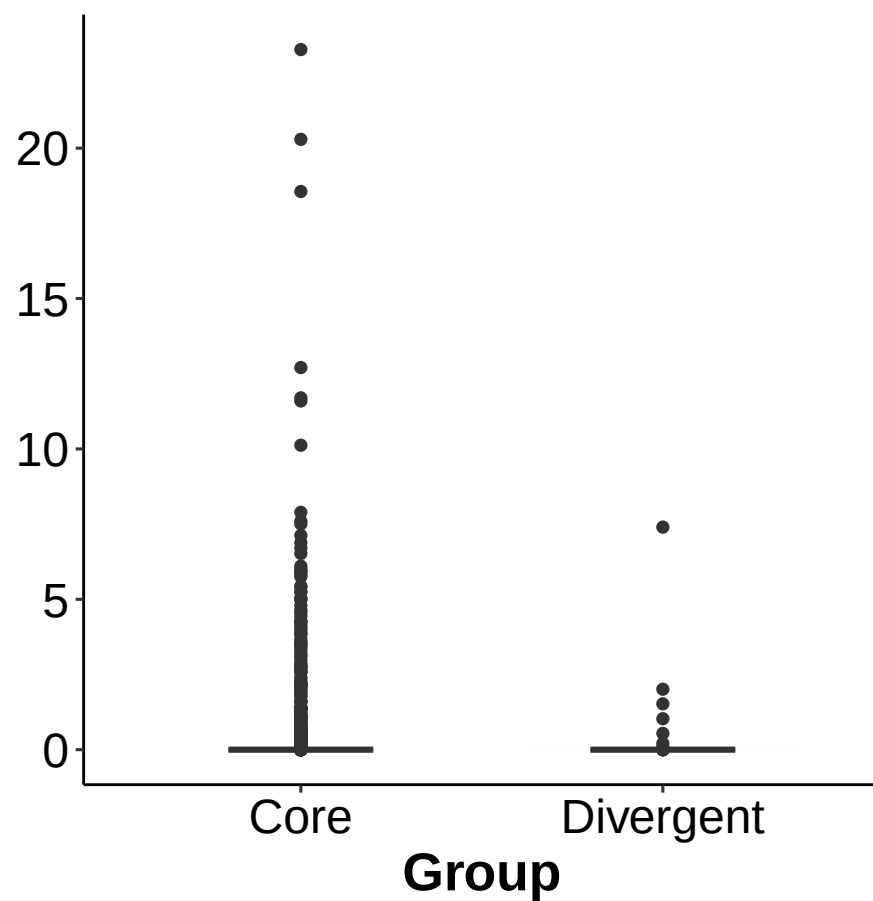
SA_Butanoate_metabolism_FBA

p-adj: 0.563 | .=-0.04 (Negligible)



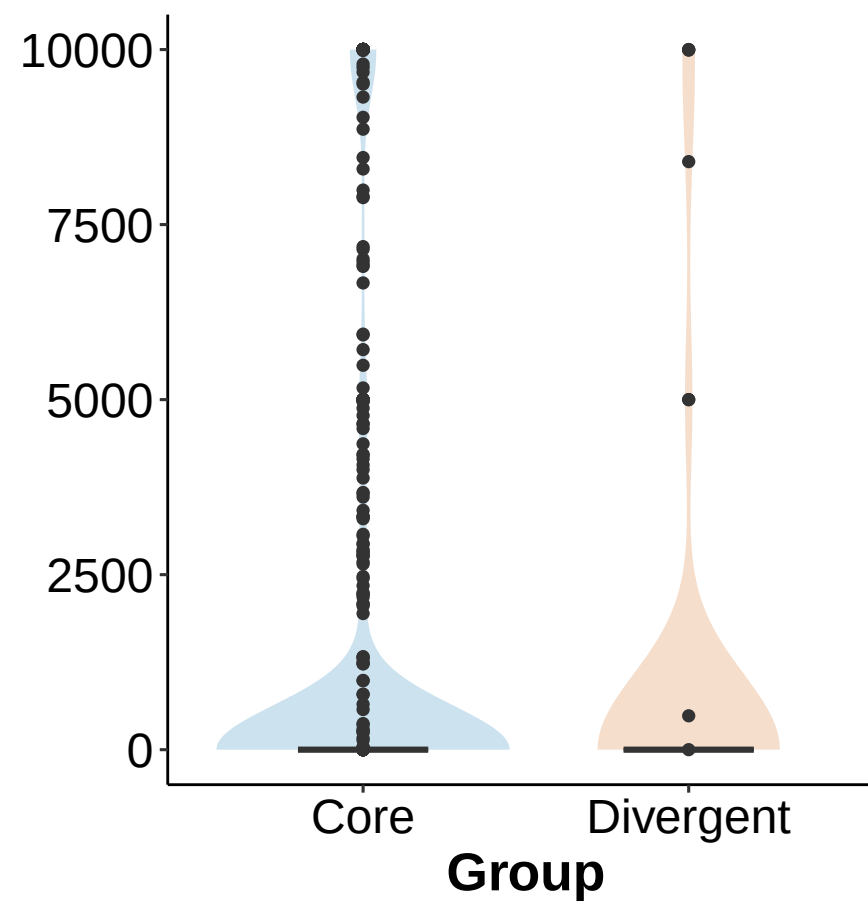
SA_Butanoate_metabolism_L1w

p-adj: 0.829 | .=-0.02 (Negligible)



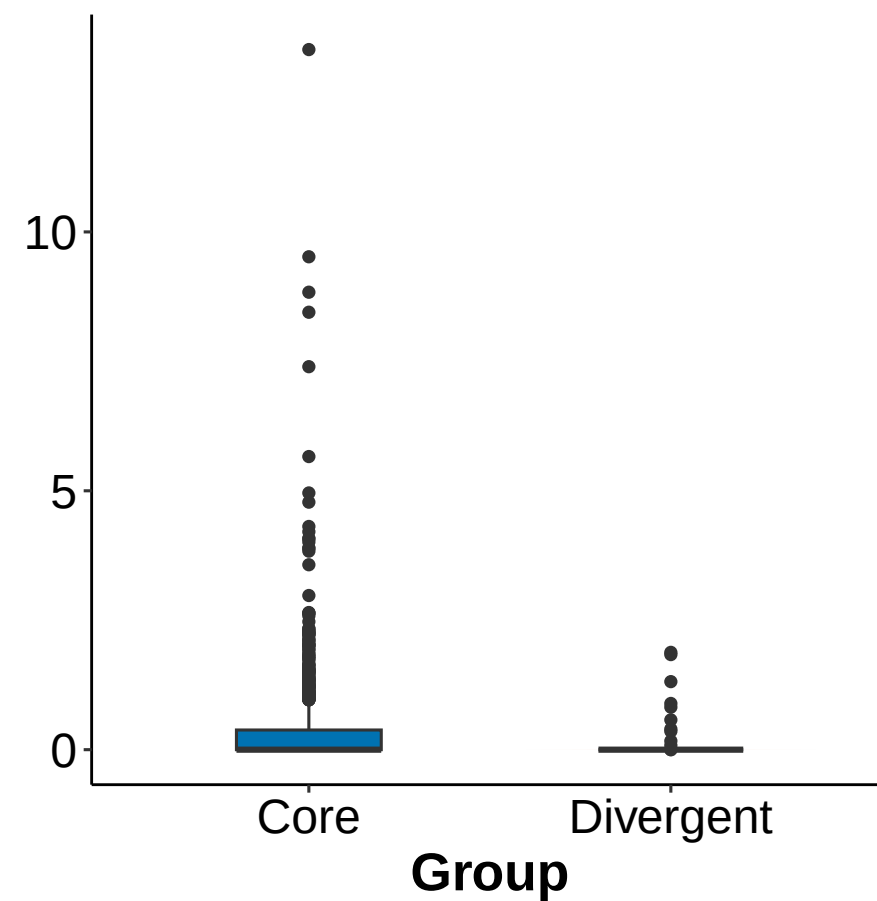
SA_C5_branched_dibasic_acid_m

p-adj: 0.67 | .=-0.04 (Negligible)



SA_C5_branched_dibasic_acid_metal

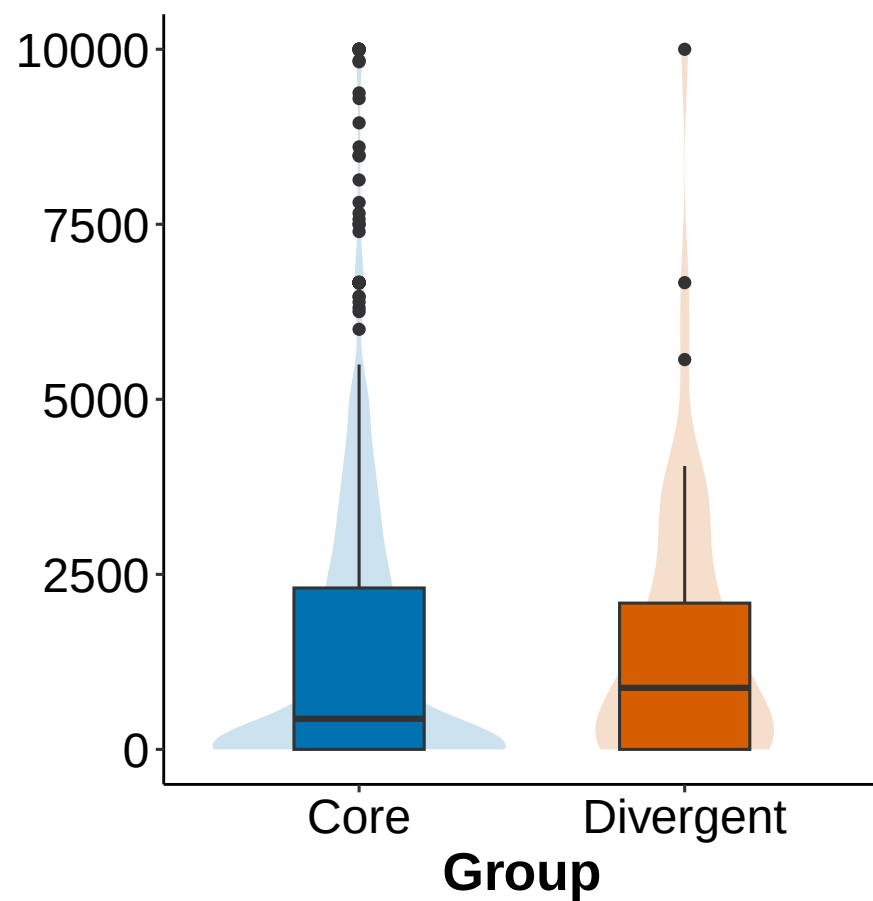
p-adj: 0.319 | .=-0.1 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

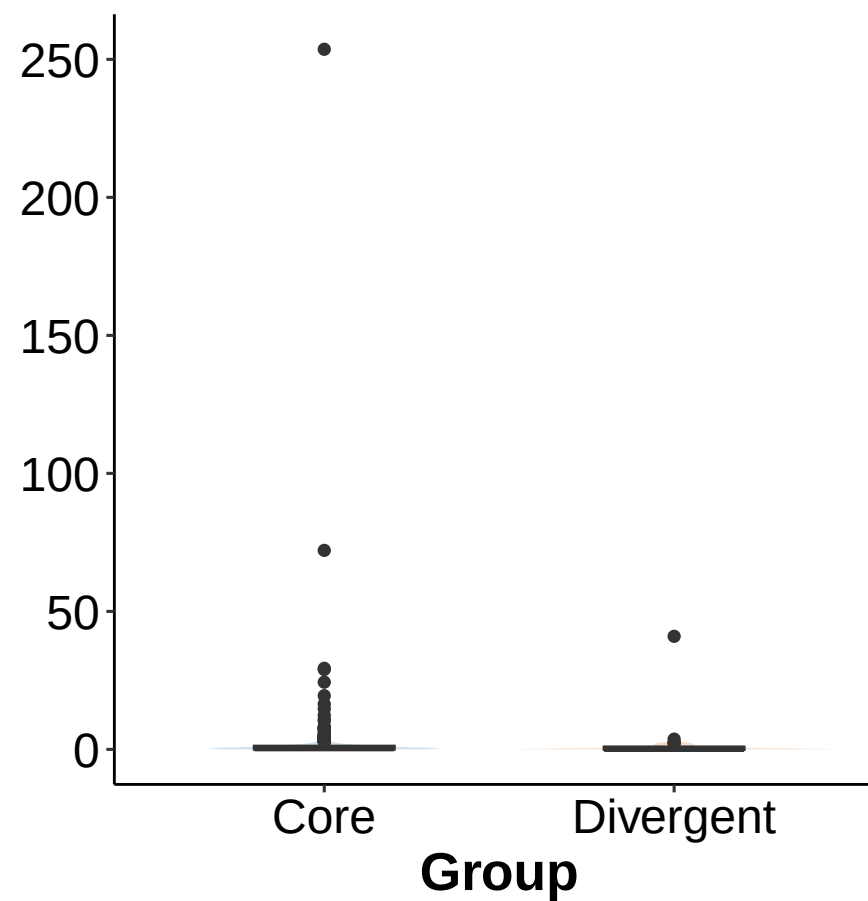
SA_Cholesterol_metabolism_FBA

p-adj: 0.72 | .:=0.05 (Negligible) | .%: 101



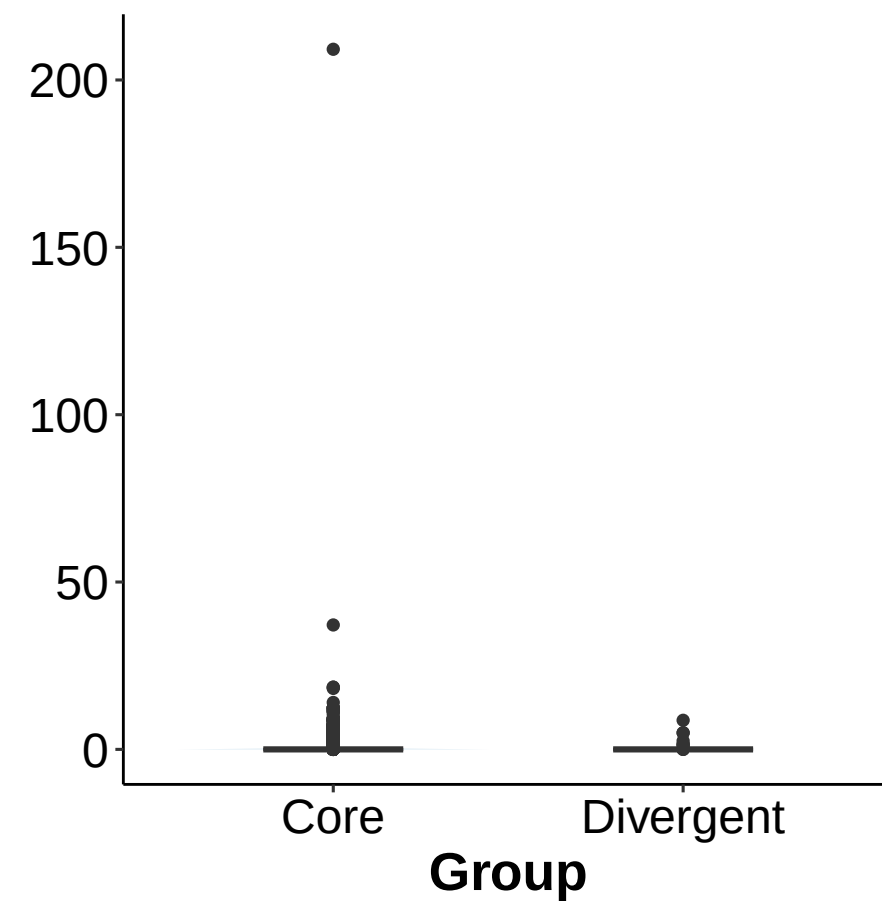
SA_Cholesterol_metabolism_L1w

p-adj: 0.0803 | .:=-0.18 (Pequeño) | .%: -54.9



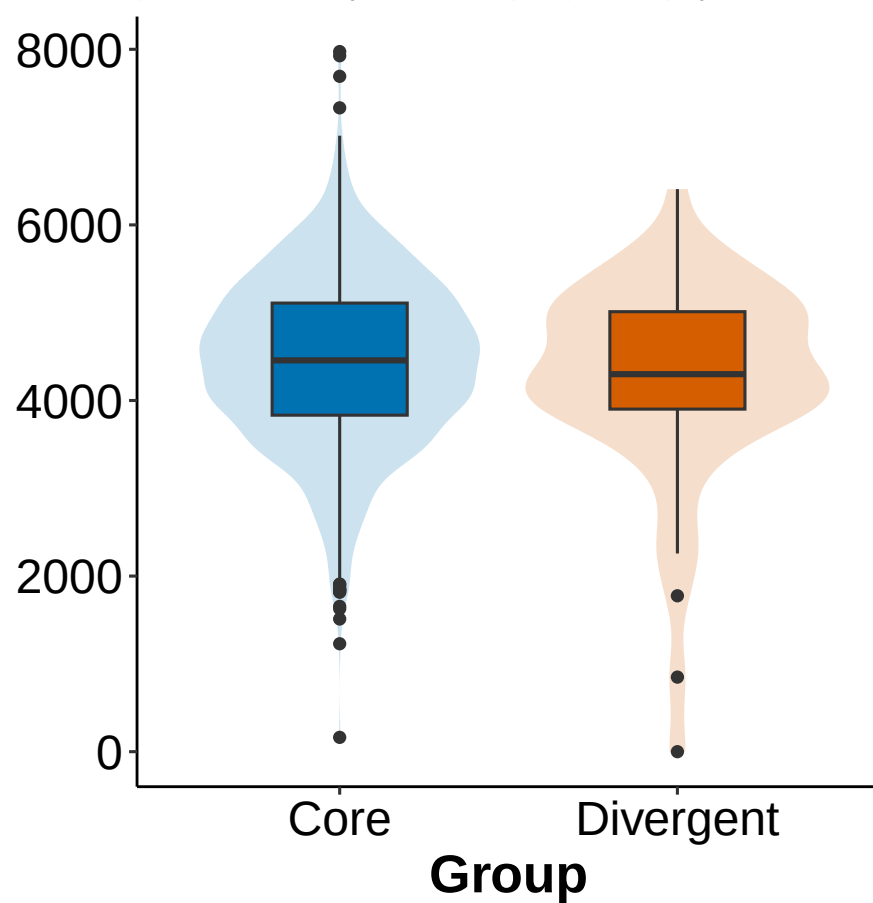
SA_Cholesterol_metabolism_pFBA

p-adj: 0.919 | .:=0.01 (Negligible)



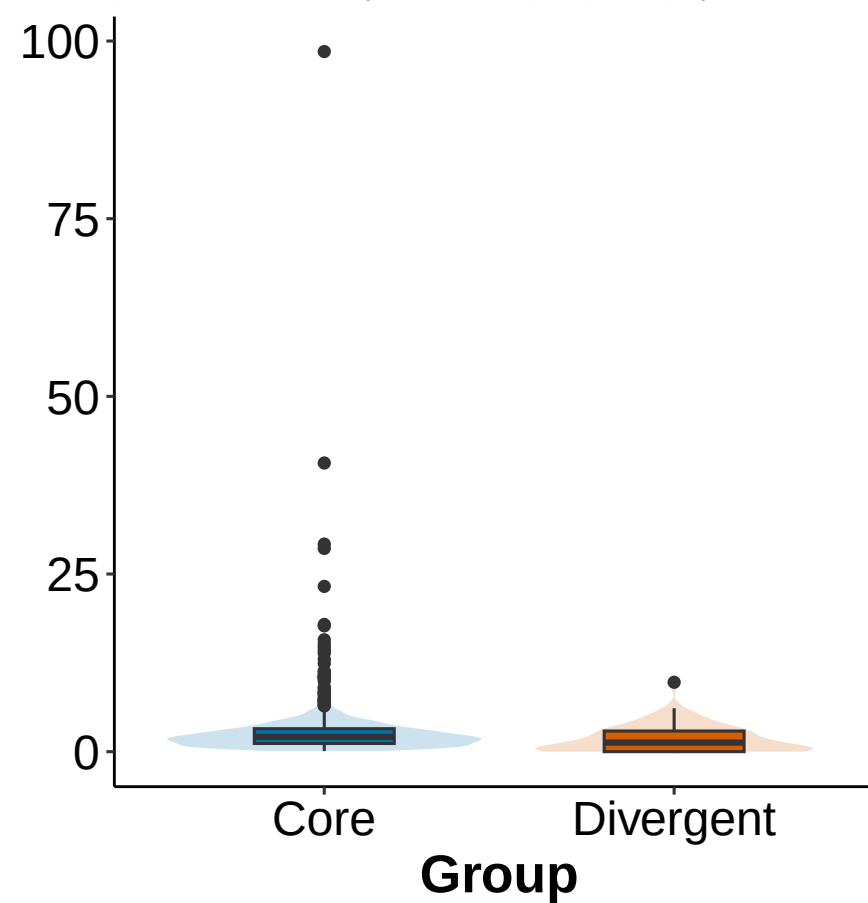
SA_Citric_acid_cycle_FBA

p-adj: 0.653 | .:=-0.06 (Negligible) | .%: -3.5



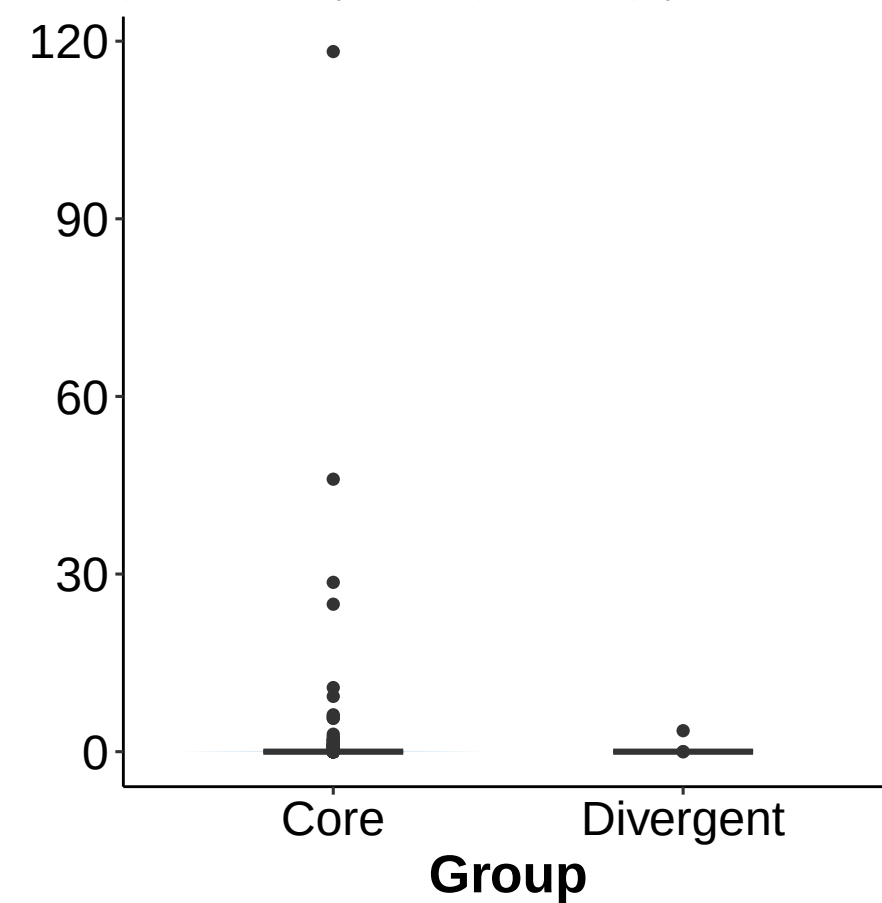
SA_Citric_acid_cycle_L1w

p-adj: 0.00357 | .:=-0.27 (Pequeño) | .%: -38.7



SA_Citric_acid_cycle_pFBA

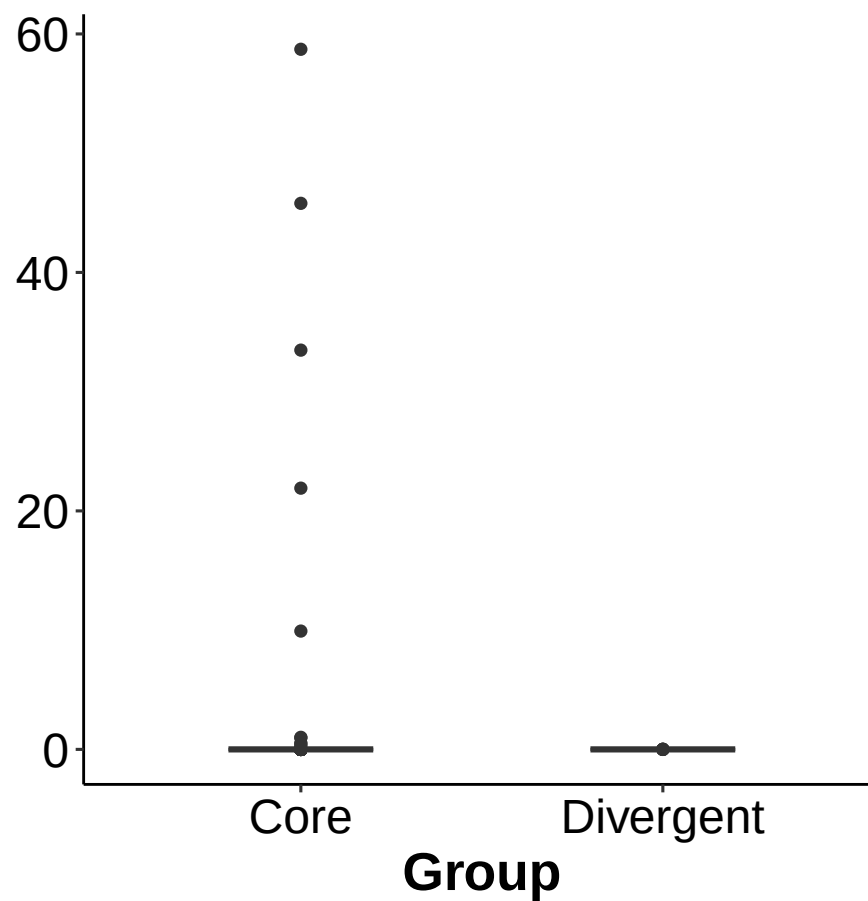
p-adj: 0.557 | .:=0.08 (Negligible) | .%: 22



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

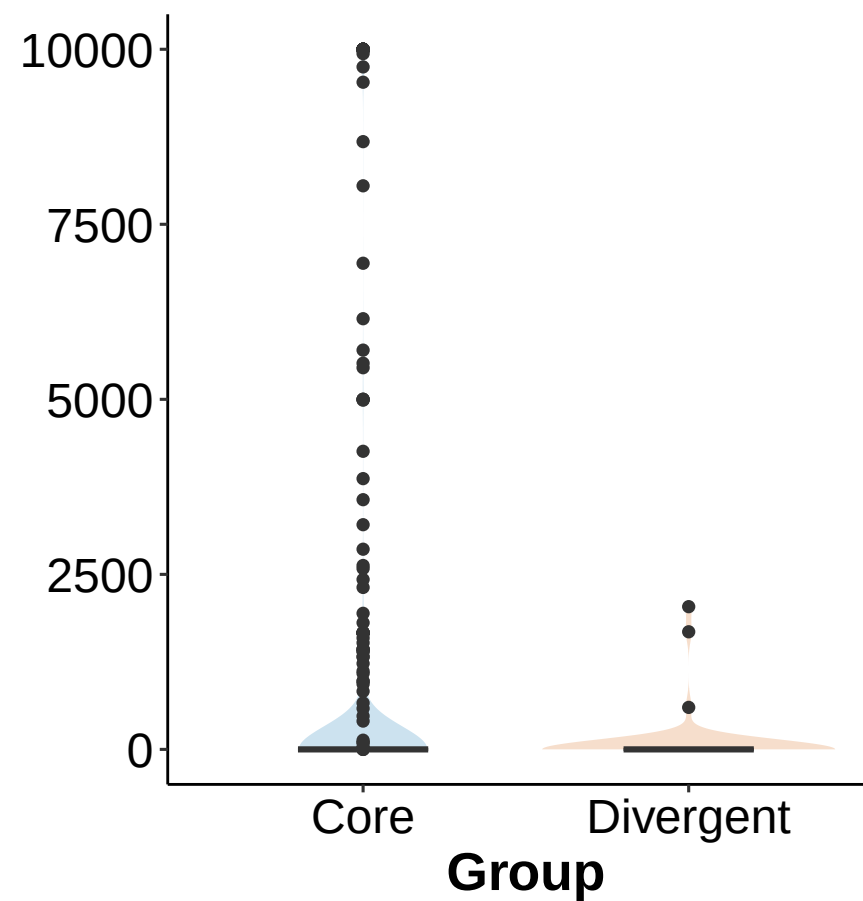
SA_CoA_catabolism_L1w

p-adj: 0.705 | .:=0.02 (Negligible)



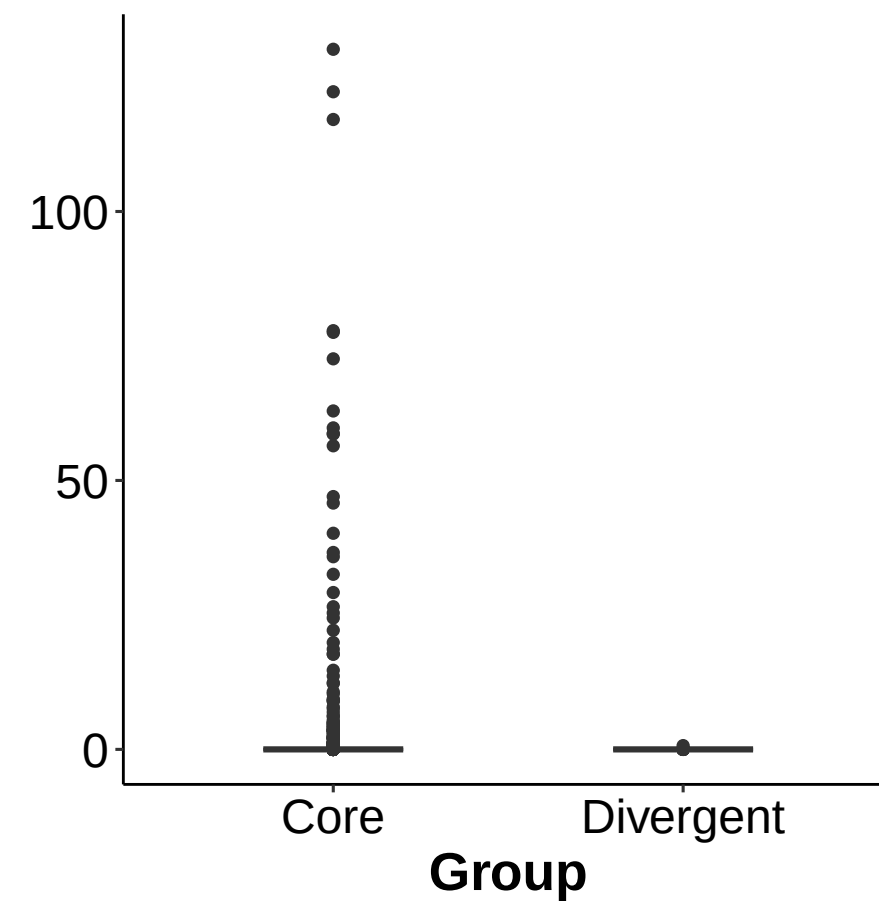
SA_CoA_synthesis_FBA

p-adj: 0.804 | .:=−0.02 (Negligible)



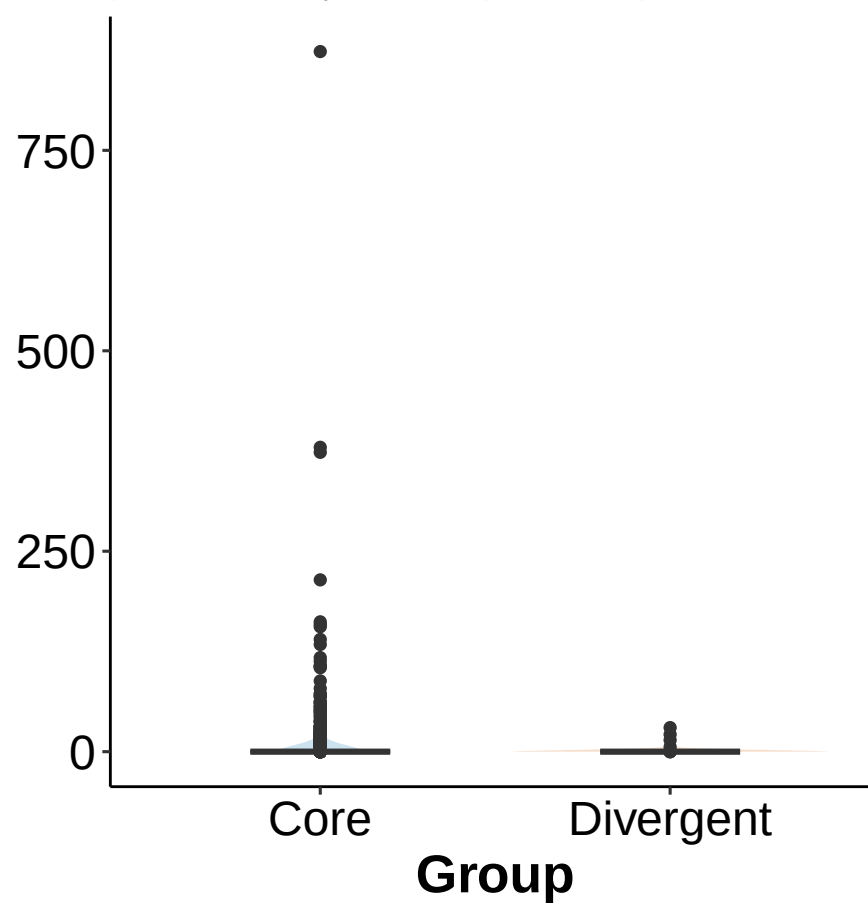
SA_CoA_synthesis_L1w

p-adj: 0.378 | .:=−0.09 (Negligible)



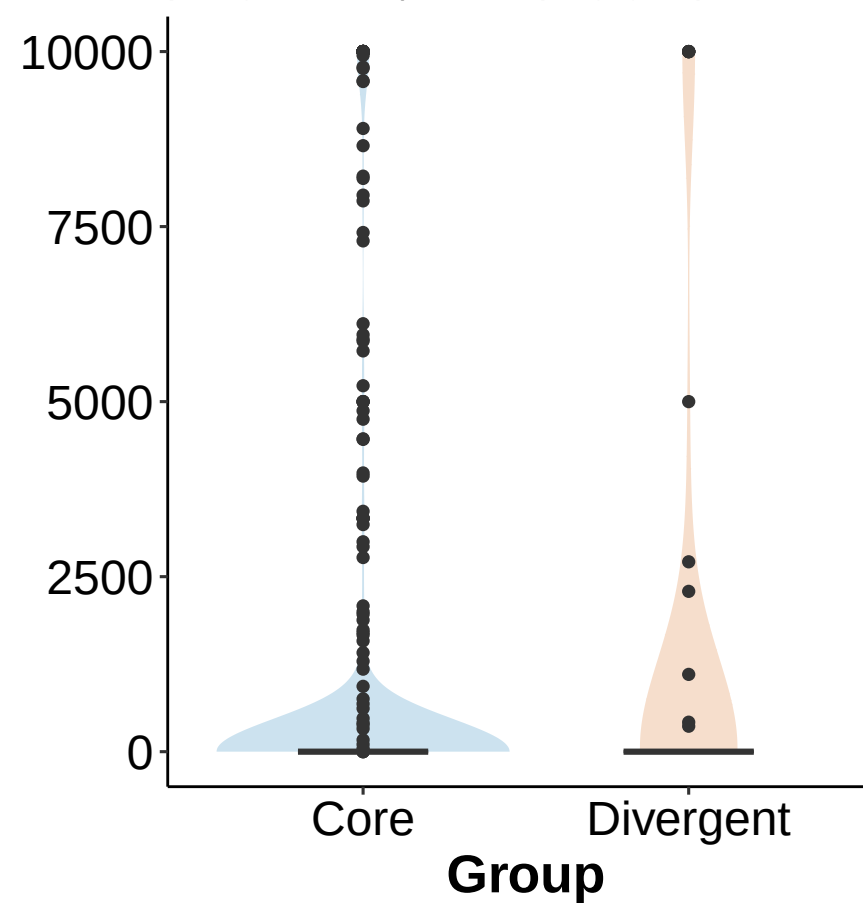
SA_Cytochrome_metabolism_L1w

p-adj: 0.845 | .:=−0.02 (Negligible)



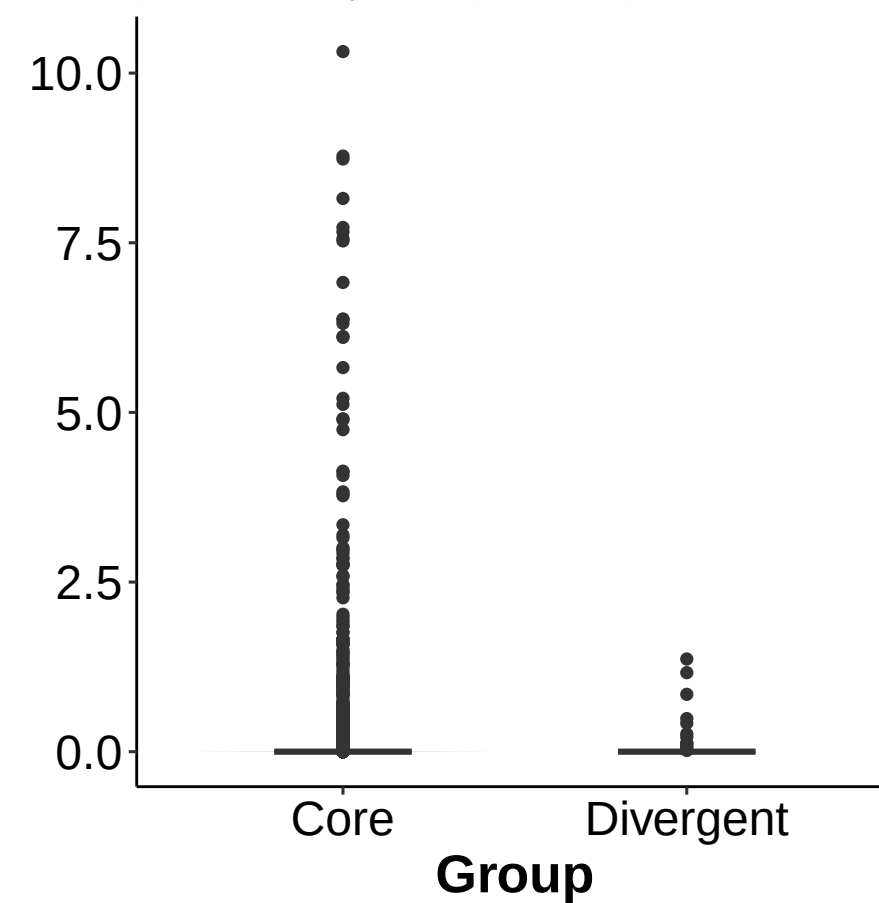
SA_D_alanine_metabolism_FBA

p-adj: 0.0109 | .:=0.13 (Negligible)



SA_D_alanine_metabolism_L1w

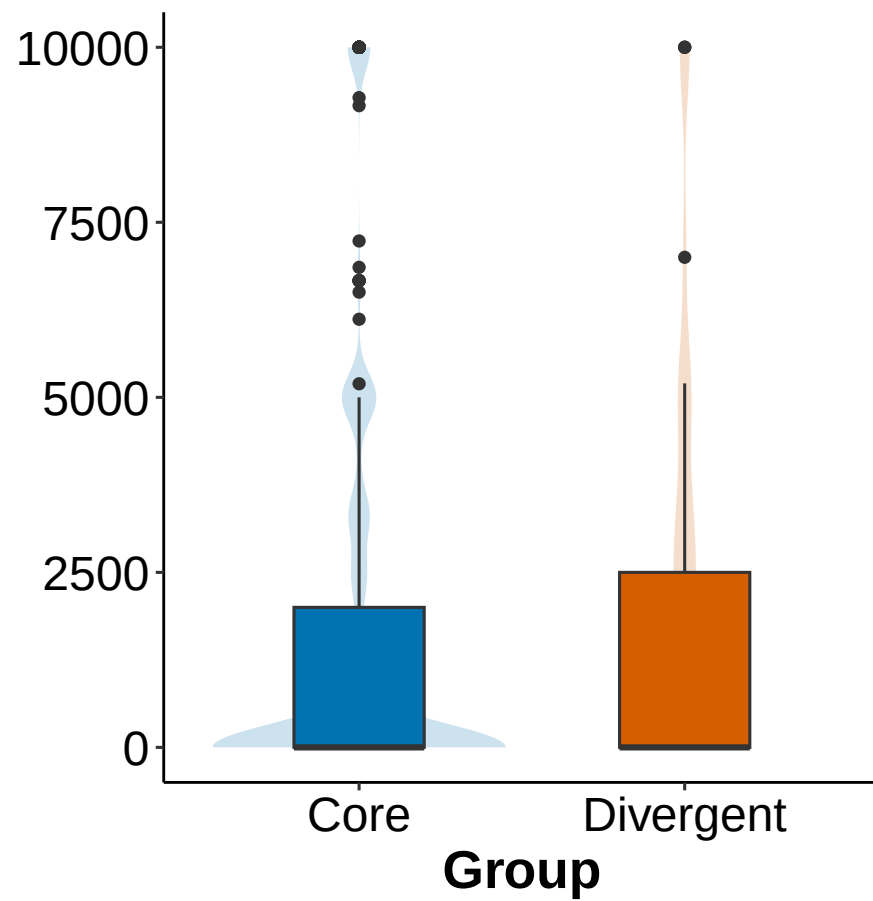
p-adj: 0.258 | .:=0.1 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

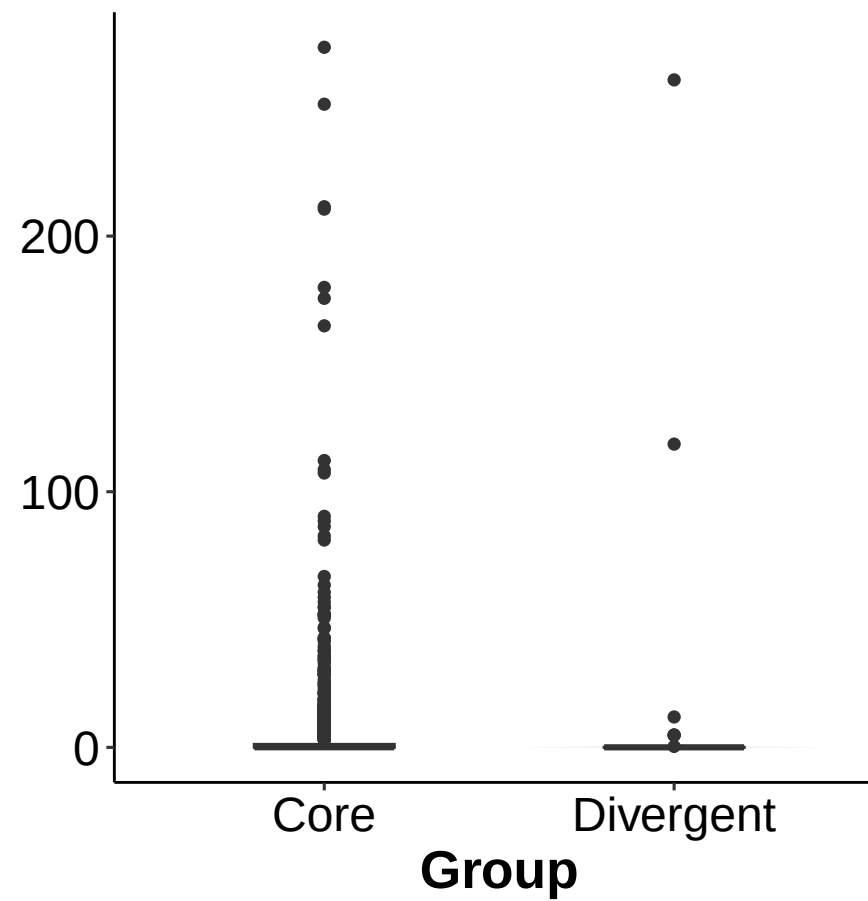
SA_Dietary_fiber_binding_FBA

p-adj: 0.731 | . = 0.04 (Negligible)



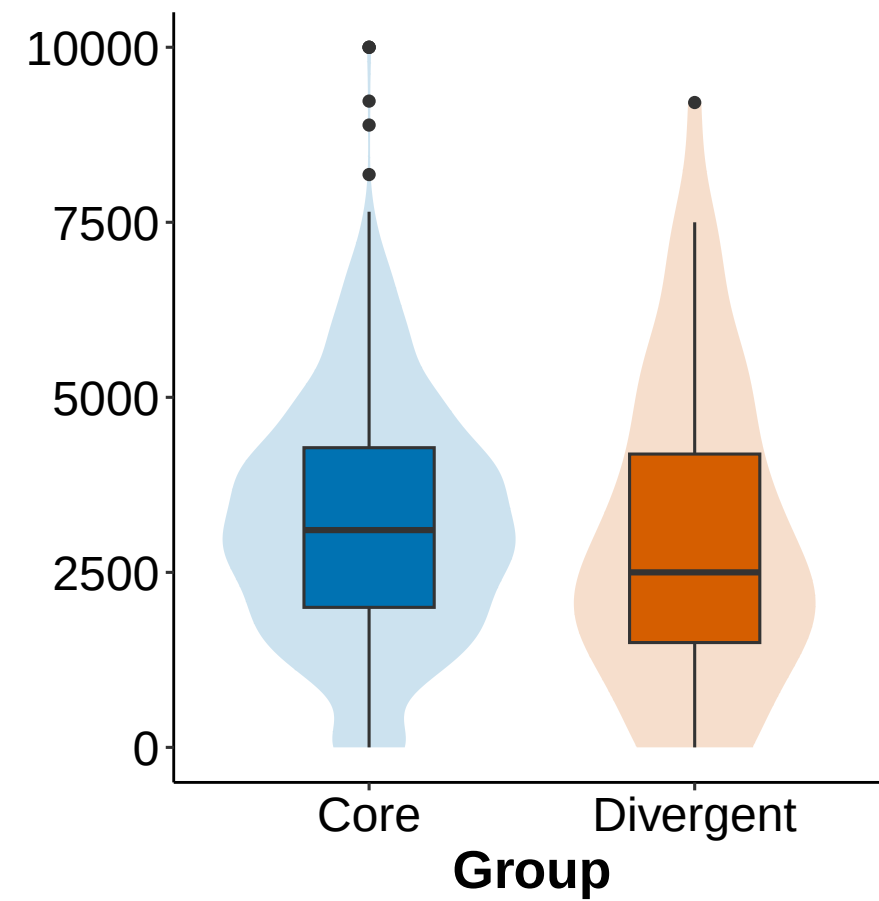
SA_Dietary_fiber_binding_L1w

p-adj: 0.00138 | . = -0.3 (Pequeño) | .%: -58.1



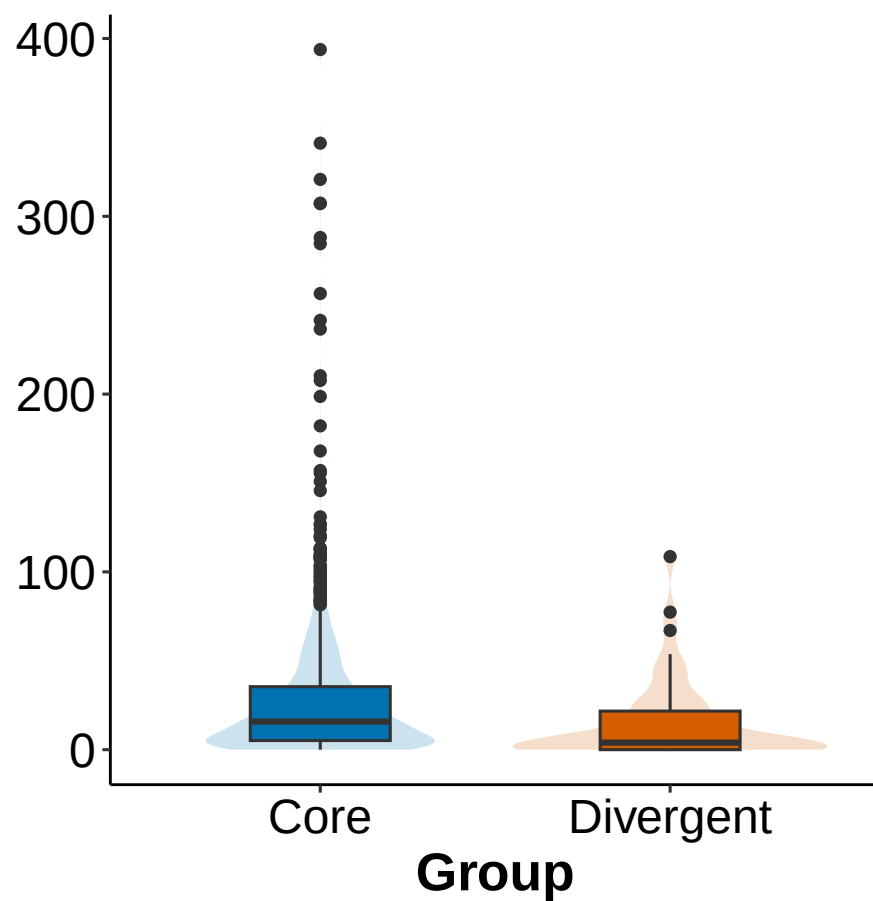
SA_Drug_metabolism_FBA

p-adj: 0.345 | . = -0.11 (Negligible) | .%: -19.4



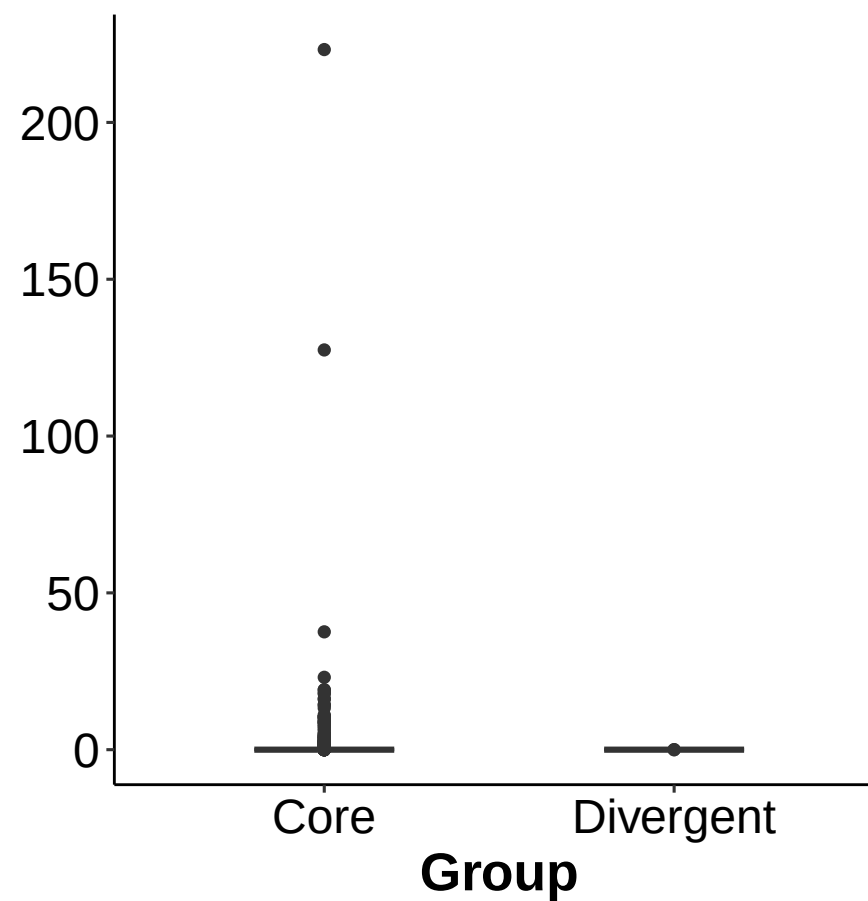
SA_Drug_metabolism_L1w

p-adj: 4.71e-05 | . = -0.38 (Mediano) | .%: -74.9



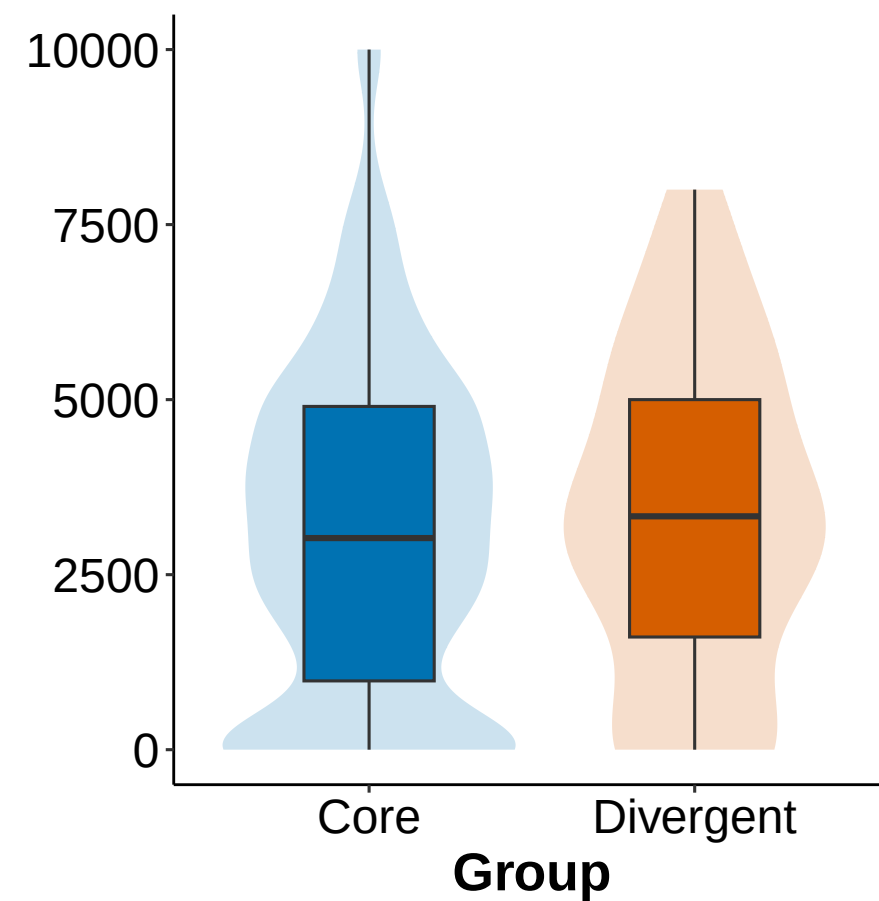
SA_Drug_metabolism_pFBA

p-adj: 0.109 | . = -0.09 (Negligible)



SA_Eicosanoid_metabolism_FBA

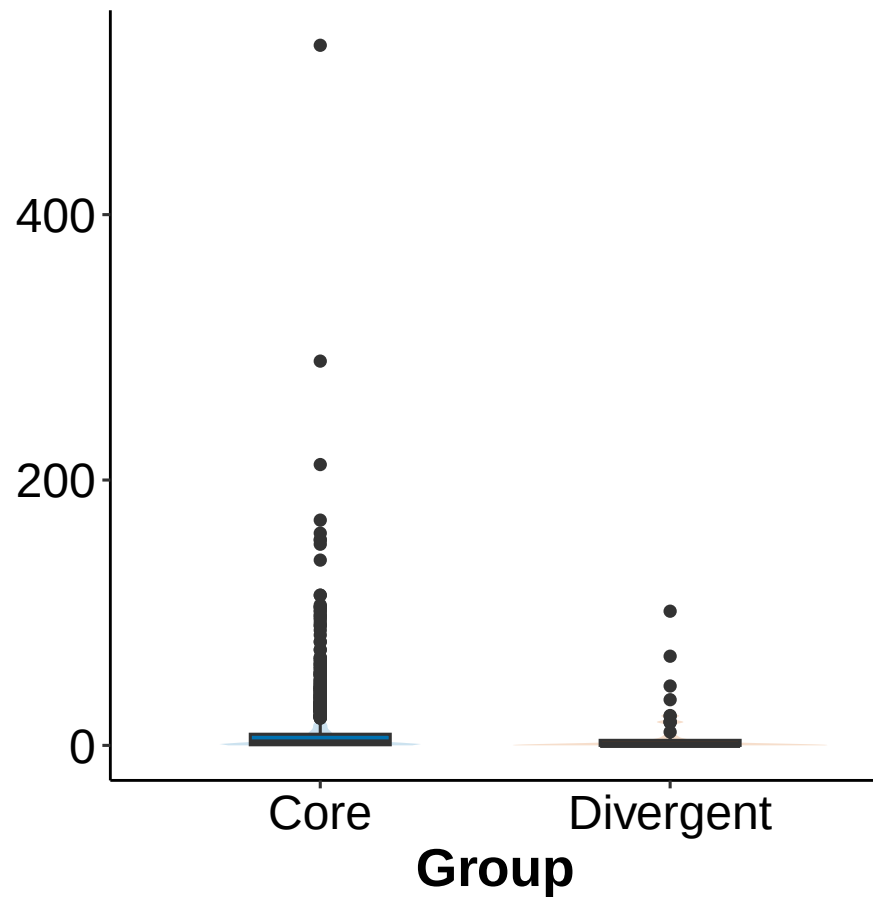
p-adj: 0.659 | . = 0.06 (Negligible) | .%: 10.3



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

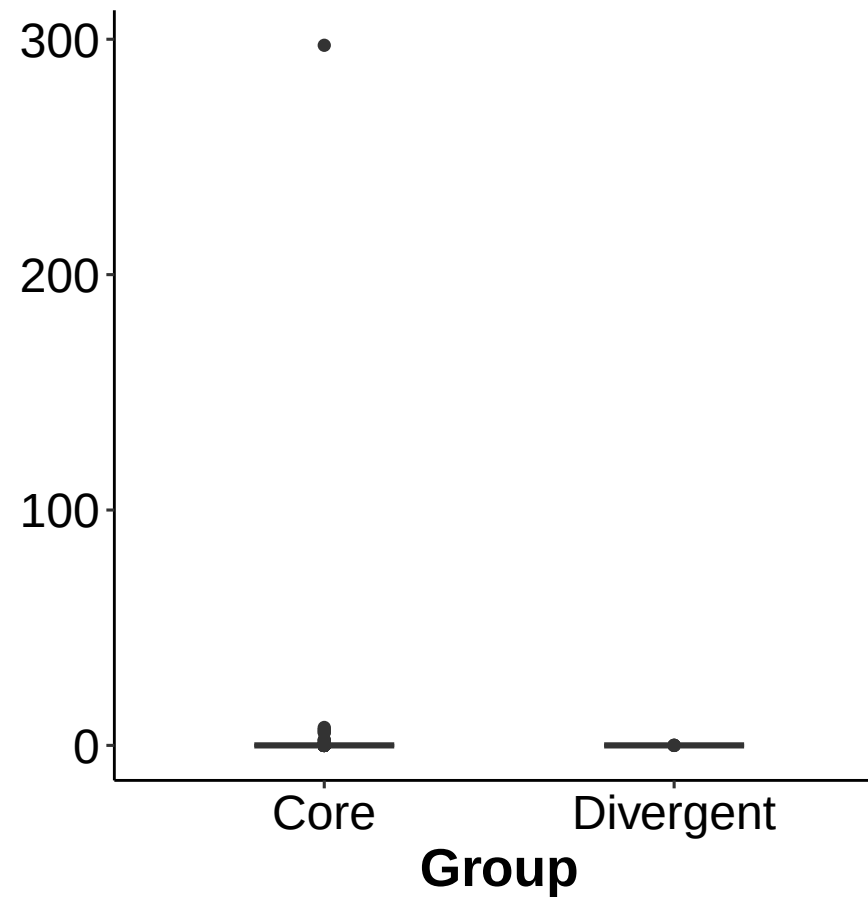
SA_Eicosanoid_metabolism_L1w

p-adj: 0.00143 | .=-0.3 (Pequeño) | .%: -81.5



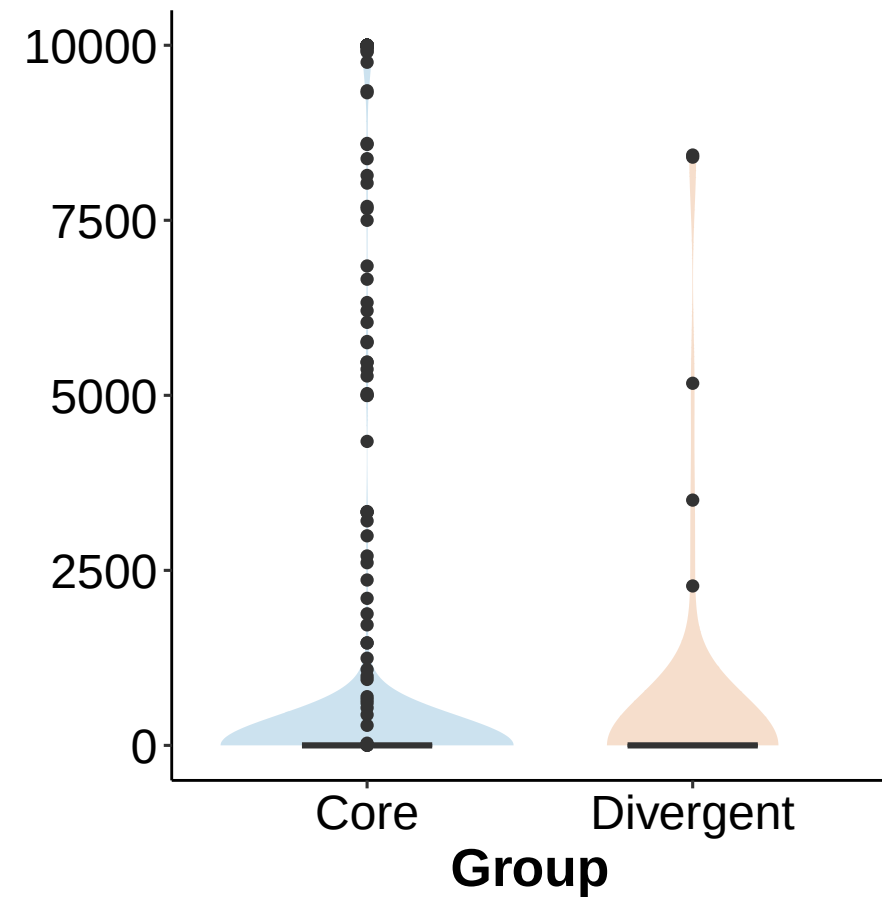
SA_Eicosanoid_metabolism_pFBA

p-adj: 0.787 | .=-0.02 (Negligible)



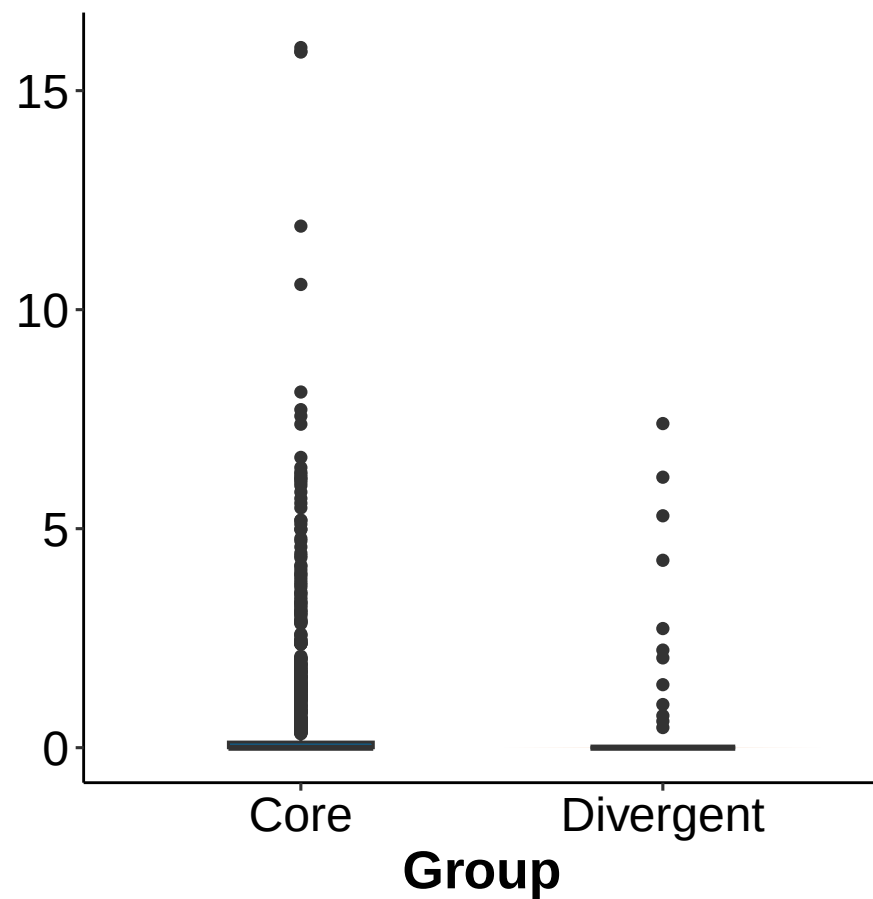
SA_Exchange_demand_reaction_L1w

p-adj: 0.962 | .=0 (Negligible)



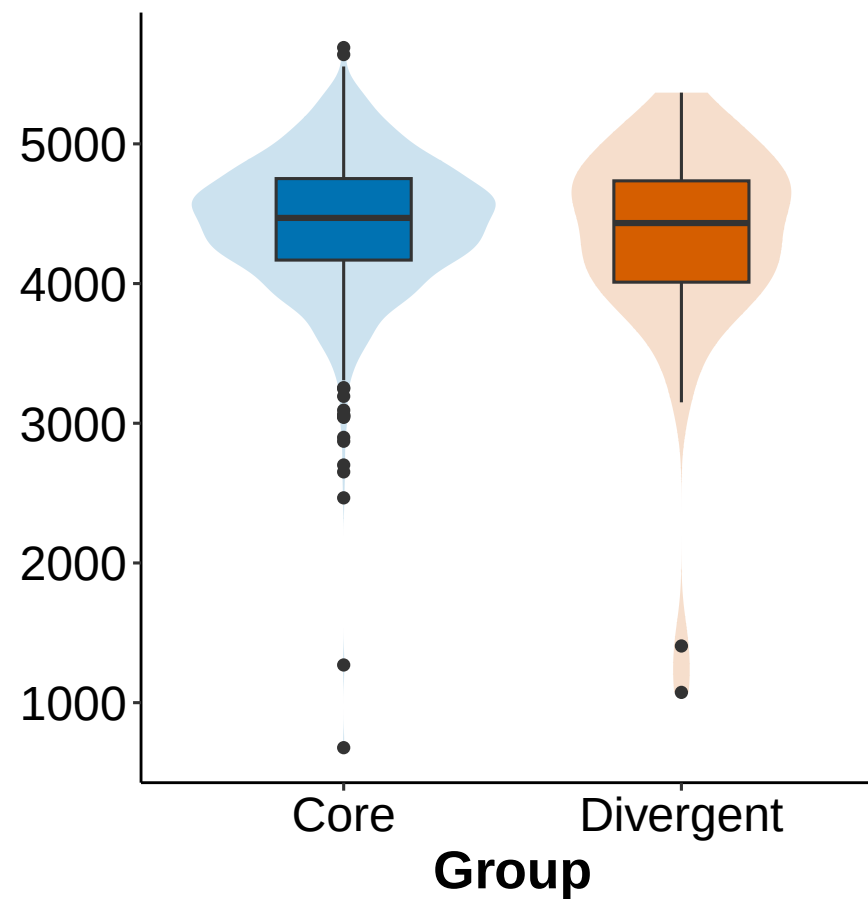
SA_Exchange_demand_reaction_L1w

p-adj: 0.653 | .=0.05 (Negligible)



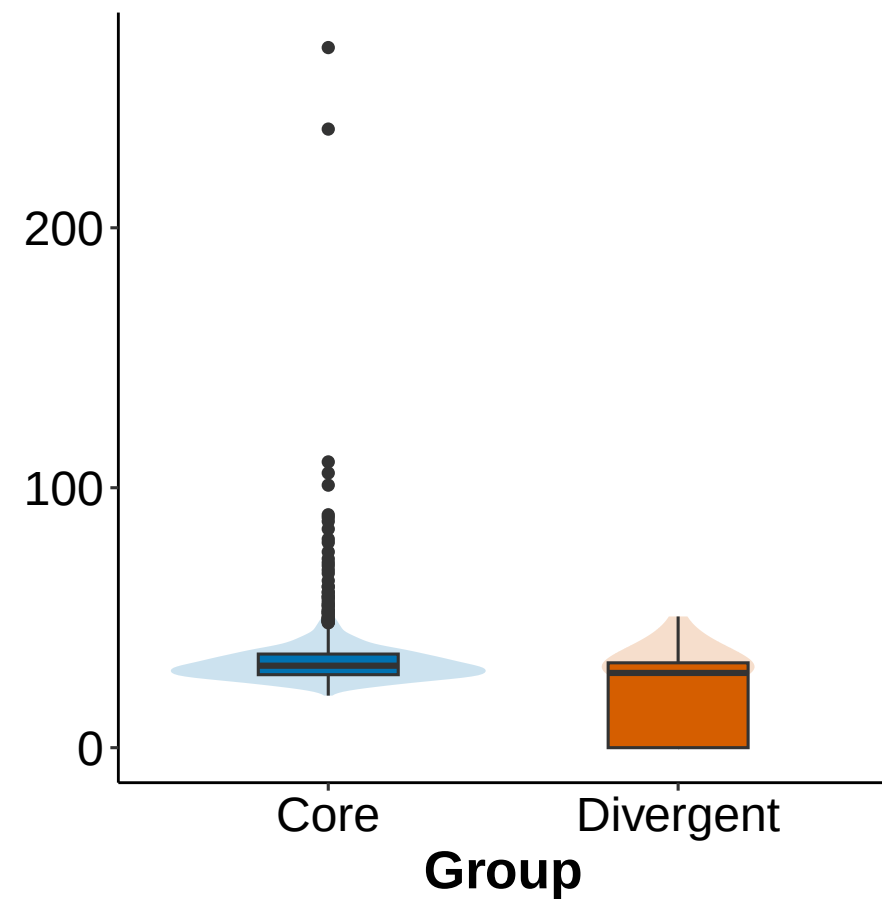
SA_Extracellular_exchange_FBA

p-adj: 0.69 | .=-0.06 (Negligible) | .%: -0.8



SA_Extracellular_exchange_L1w

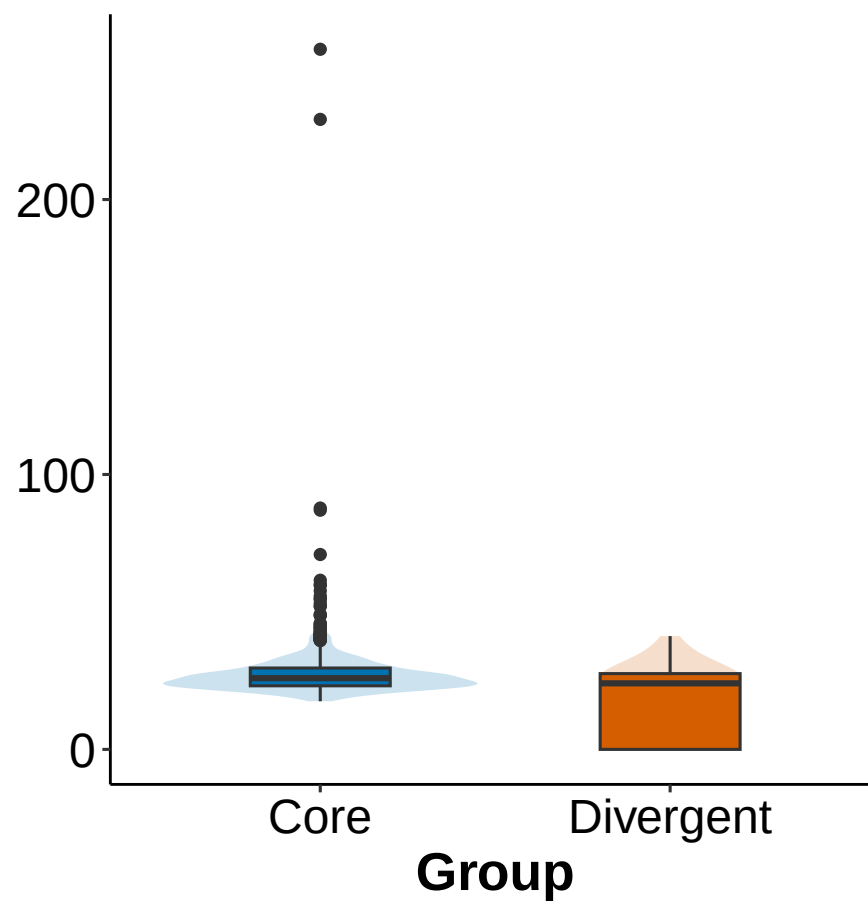
p-adj: 0.00199 | .=-0.29 (Pequeño) | .%: -8.9



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

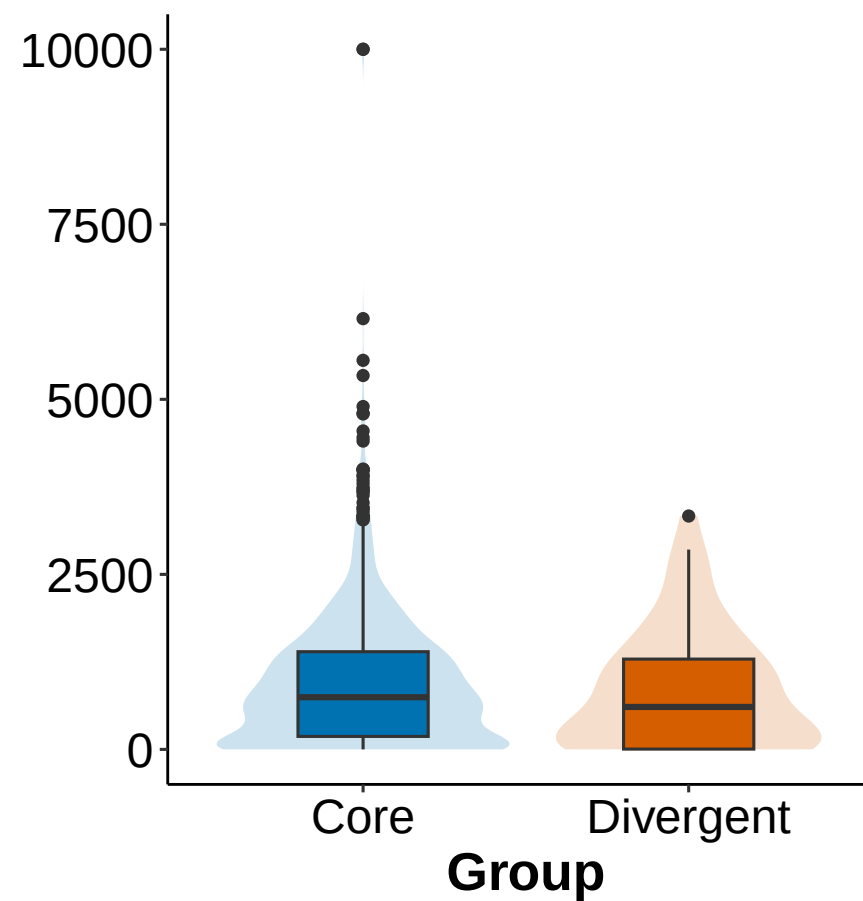
SA_Extracellular_exchange_pFBA

p-adj: 0.00317 | .=-0.28 (Pequeño) | .%: -7.2



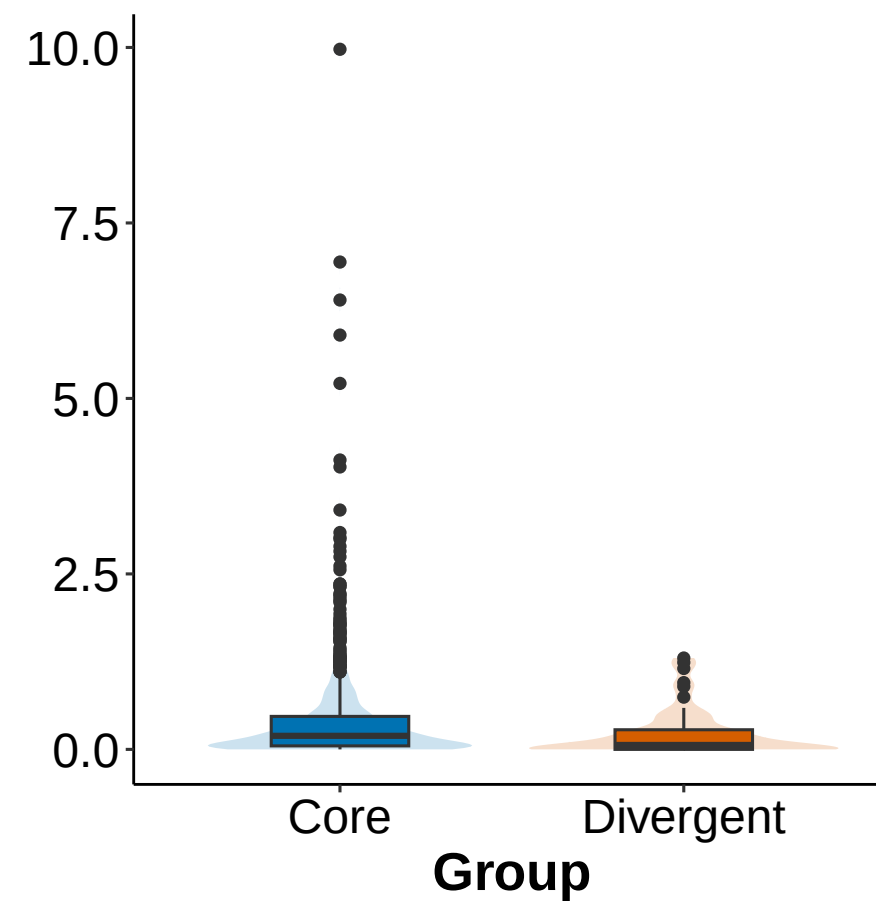
SA_Fatty_acid_oxidation_FBA

p-adj: 0.62 | .=-0.07 (Negligible) | .%: -18.7



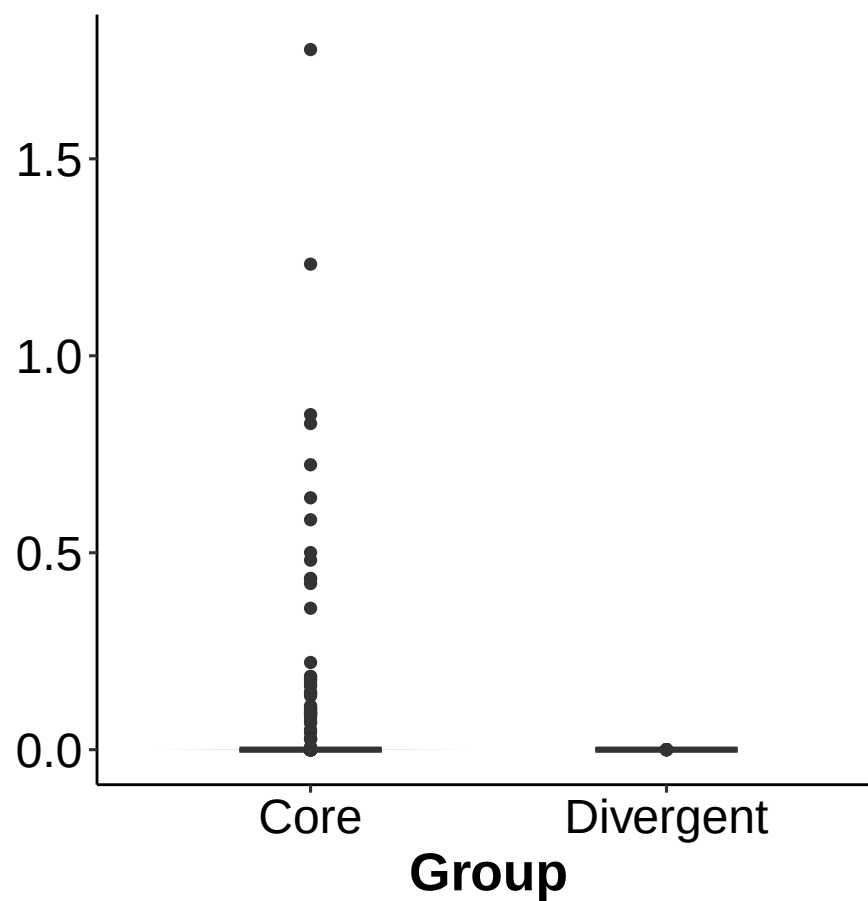
SA_Fatty_acid_oxidation_L1w

p-adj: 0.017 | .=-0.23 (Pequeño) | .%: -67.4



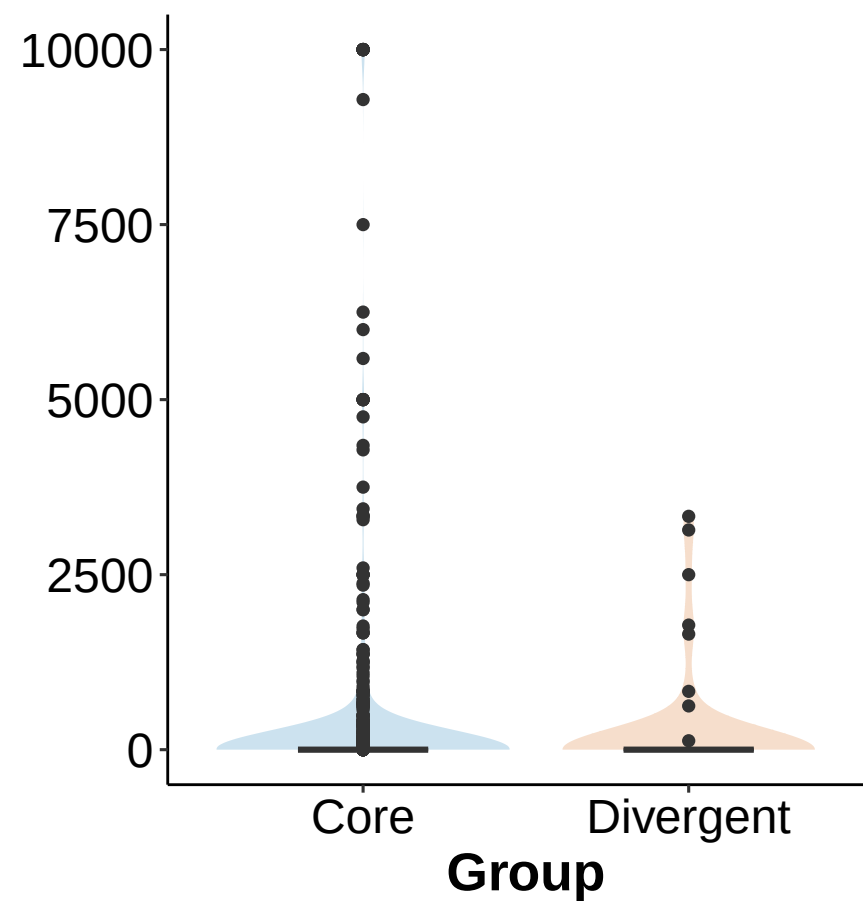
SA_Fatty_acid_oxidation_pFBA

p-adj: 0.817 | .=0.03 (Negligible) | .%: 23.4



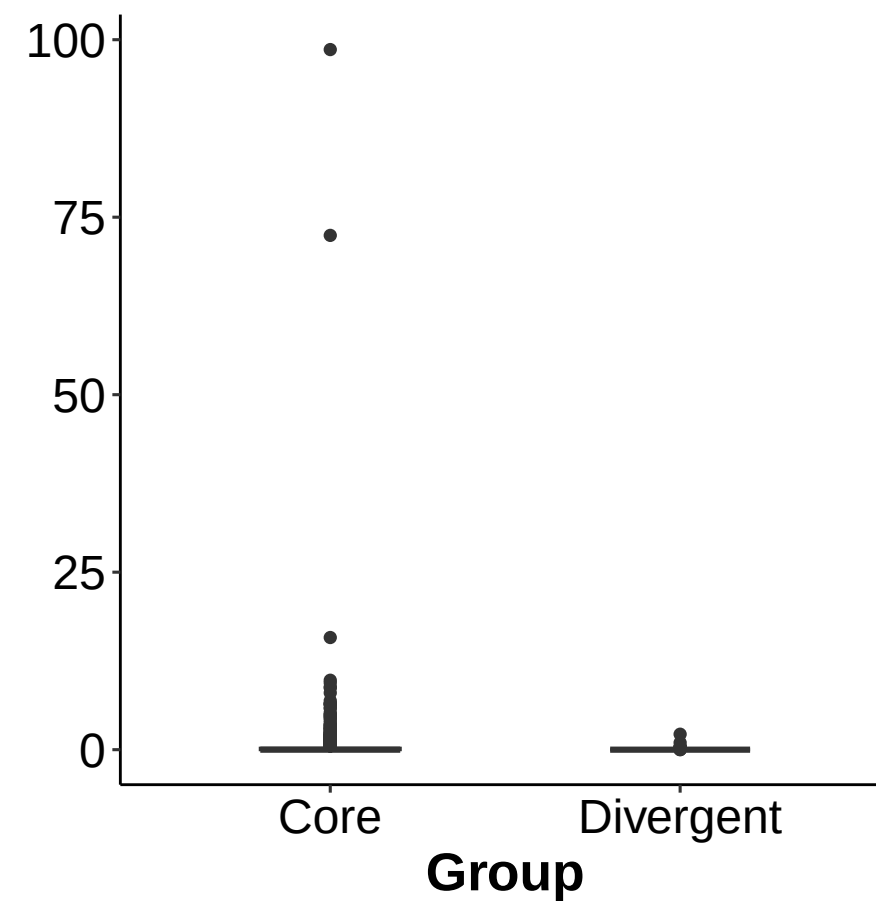
SA_Fatty_acid_synthesis_FBA

p-adj: 0.962 | .=0 (Negligible)



SA_Fatty_acid_synthesis_L1w

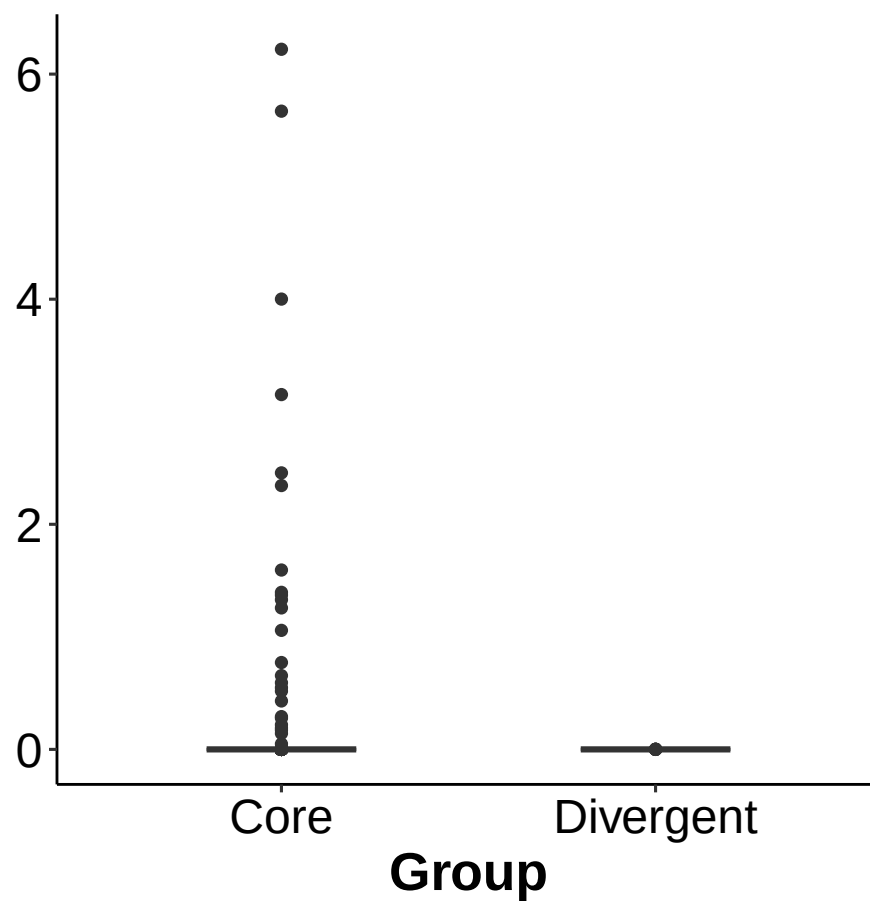
p-adj: 0.119 | .=-0.16 (Pequeño) | .%: -45.1



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

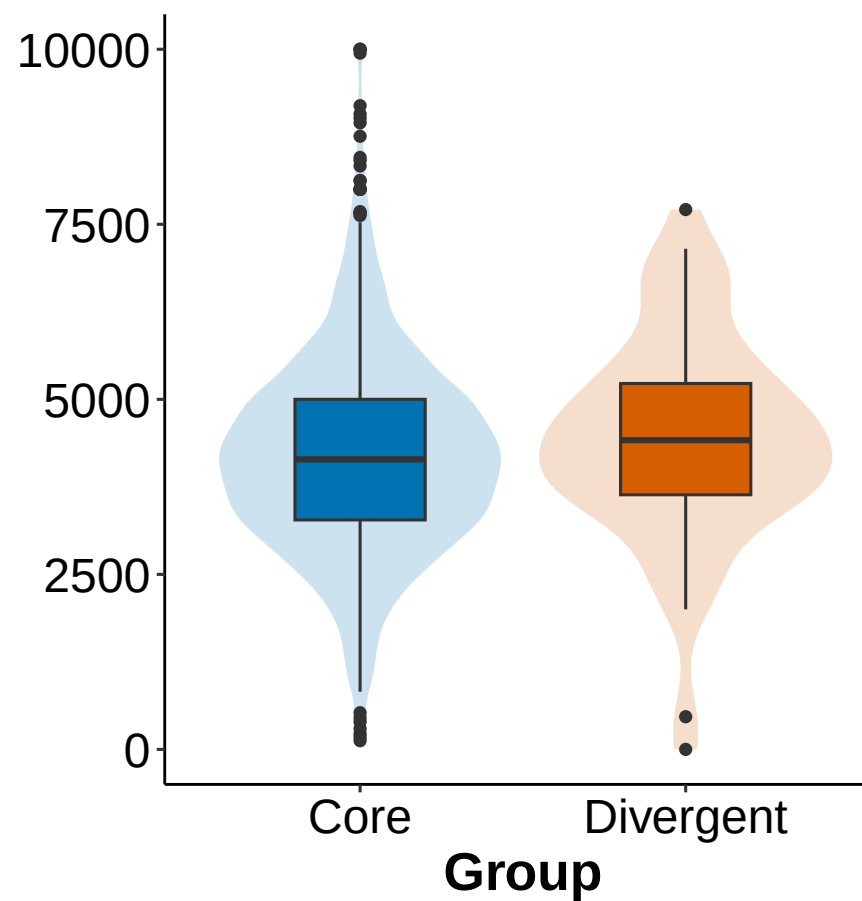
SA_Fatty_acid_synthesis_pFBA

p-adj: 0.432 | .=-0.06 (Negligible)



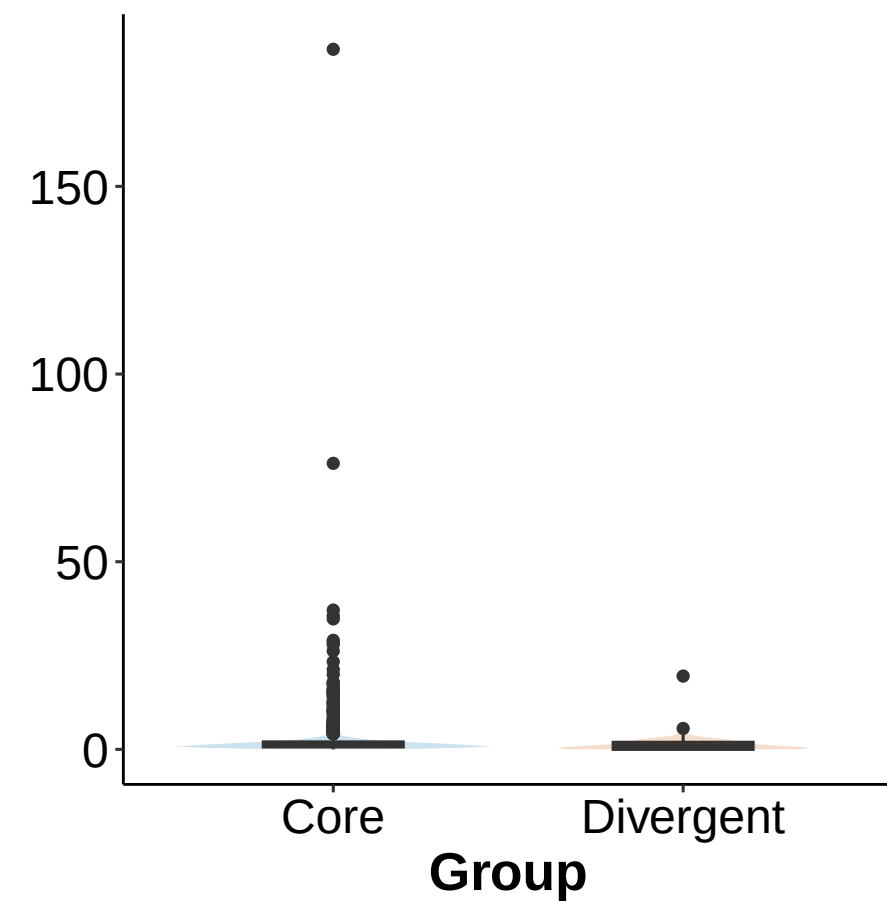
SA_Folate_metabolism_FBA

p-adj: 0.319 | .=-0.11 (Negligible) | .%: 6.6



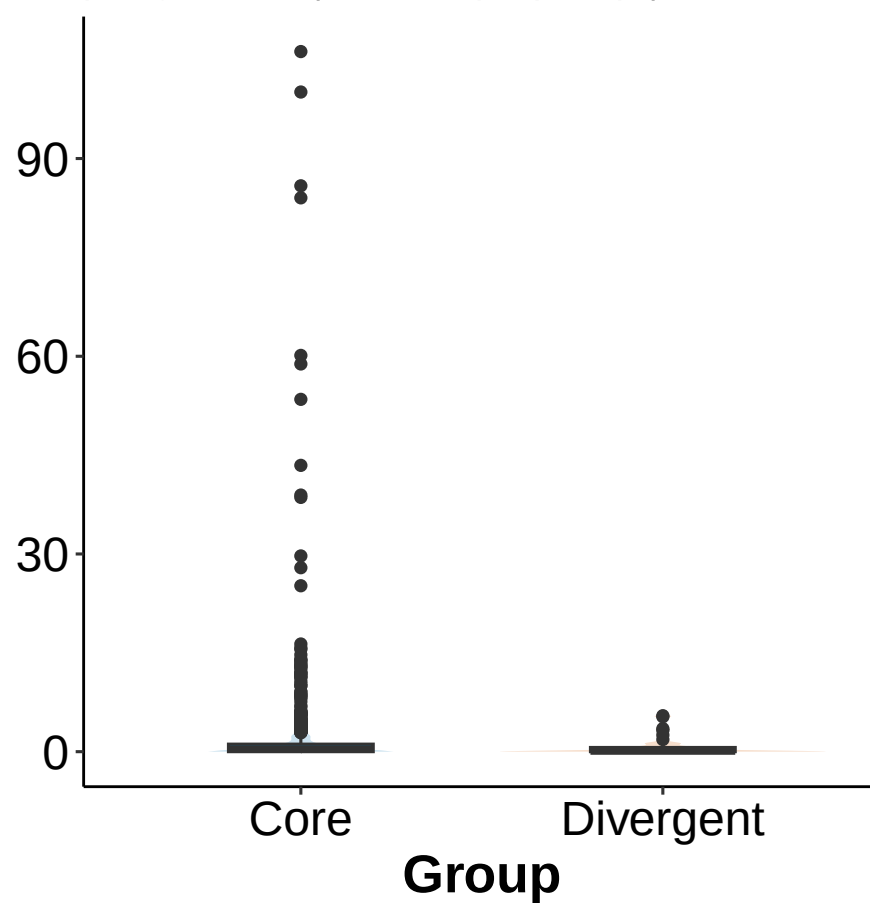
SA_Folate_metabolism_L1w

p-adj: 0.00507 | .=-0.26 (Pequeño) | .%: -40



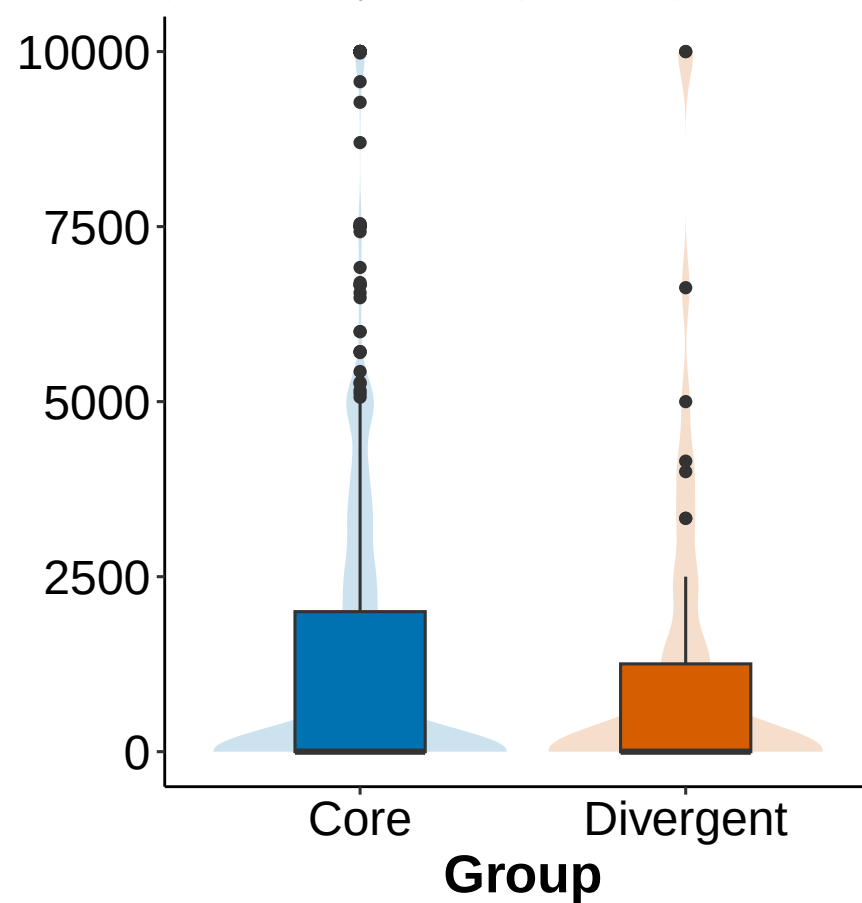
SA_Folate_metabolism_pFBA

p-adj: 0.0925 | .=-0.17 (Pequeño) | .%: -99.9



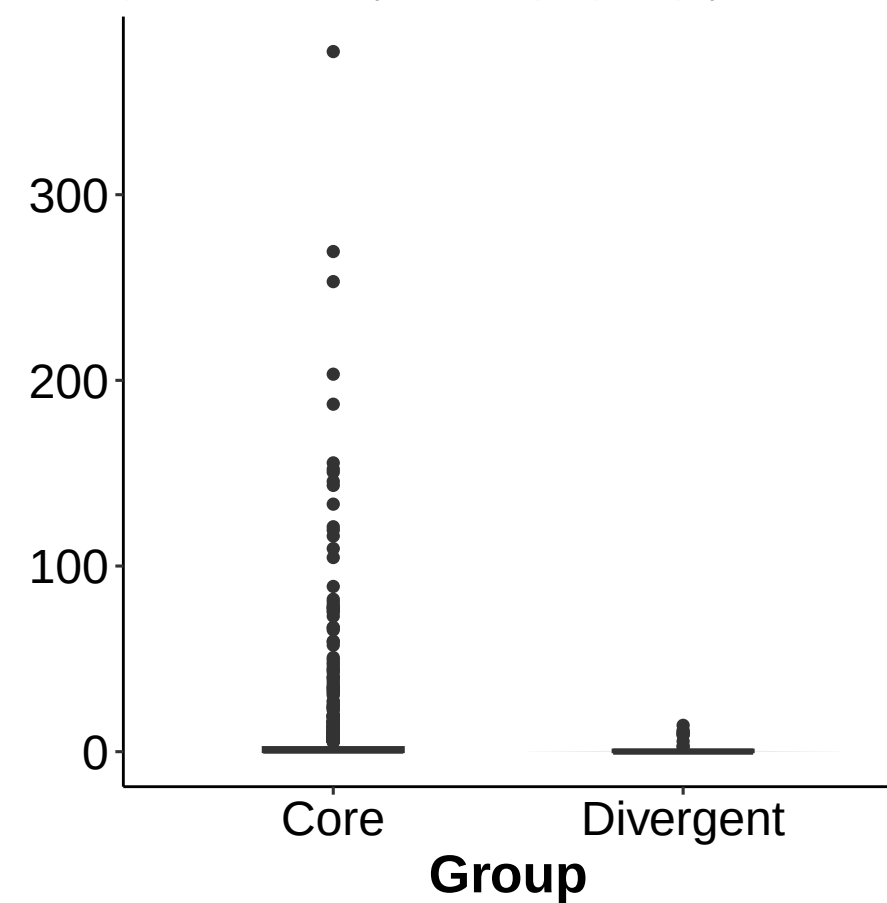
SA_Fructose_and_mannose_metabo

p-adj: 0.72 | .=-0.05 (Negligible)



SA_Fructose_and_mannose_metabo

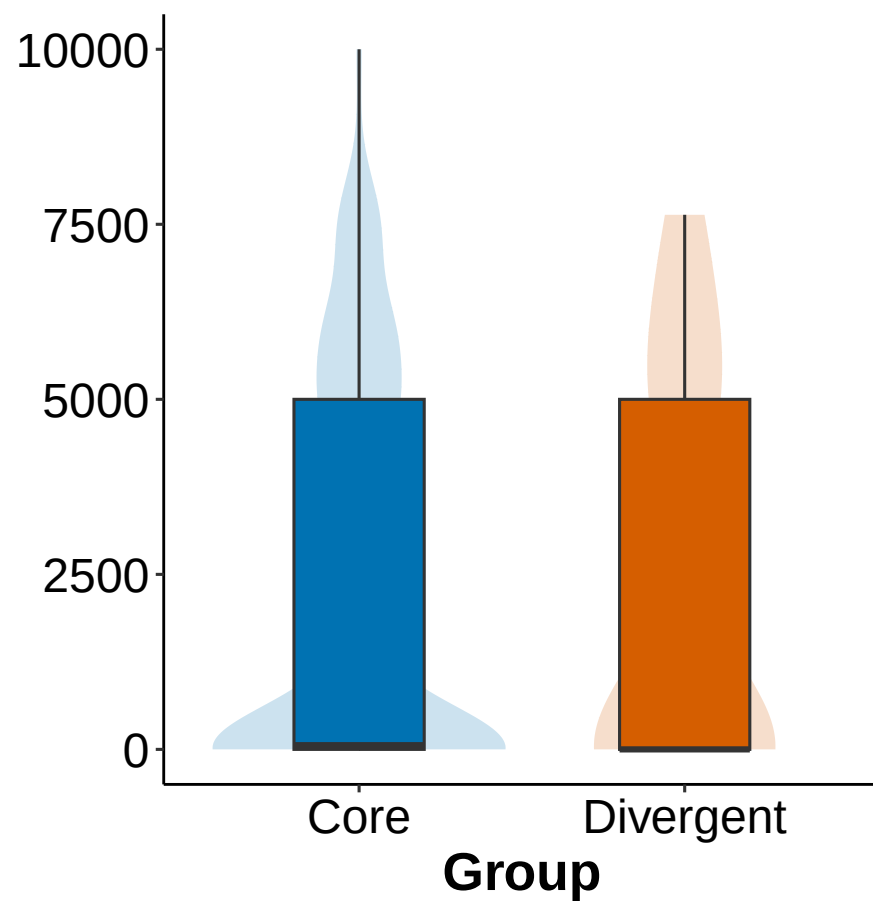
p-adj: 0.00535 | .=-0.26 (Pequeño) | .%: -95.1



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

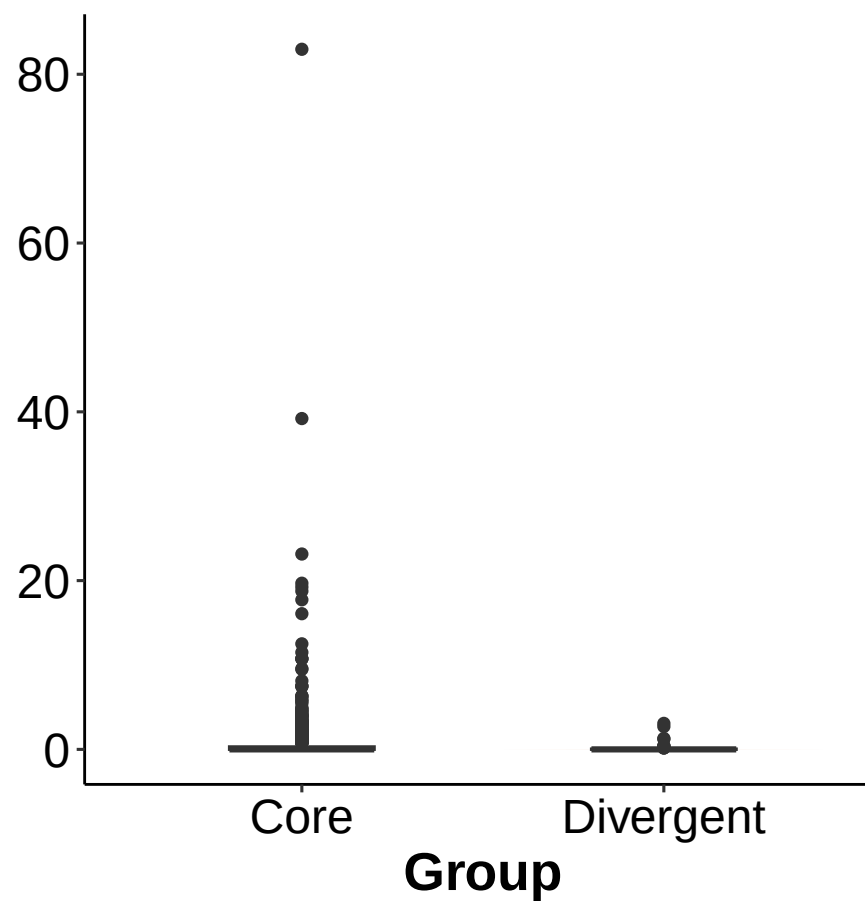
SA_Galactose_metabolism_FBA

p-adj: 0.728 | .=-0.05 (Negligible) | .%: -100



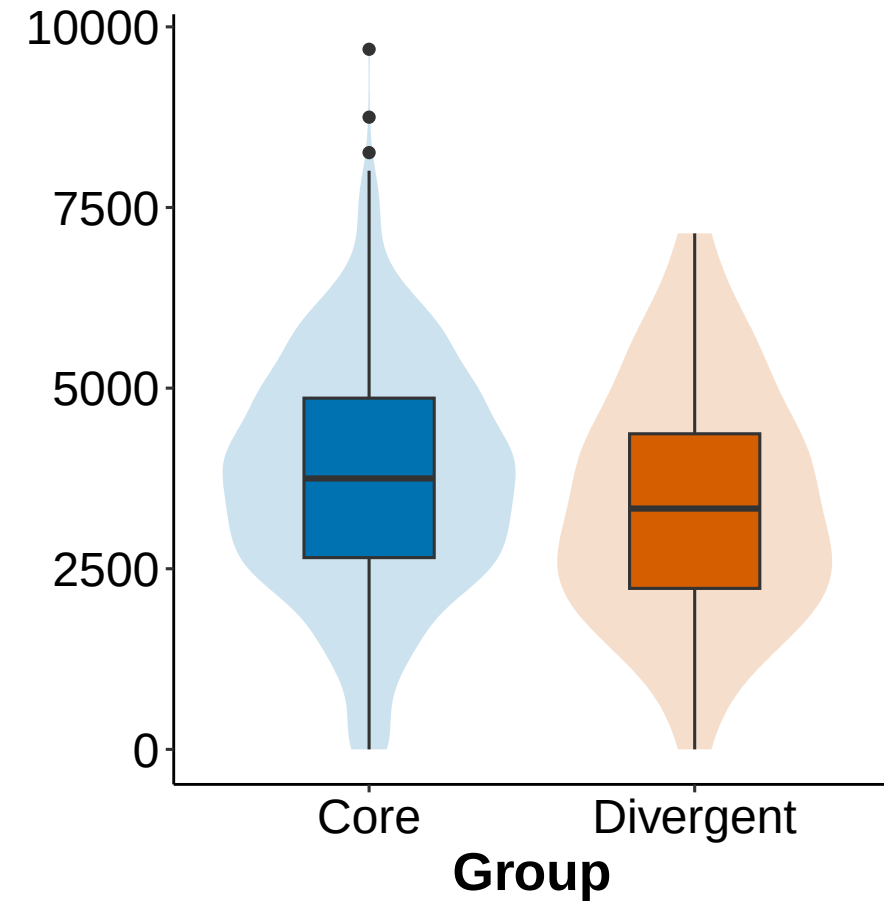
SA_Galactose_metabolism_L1w

p-adj: 0.345 | .=-0.11 (Negligible) | .%: -66.9



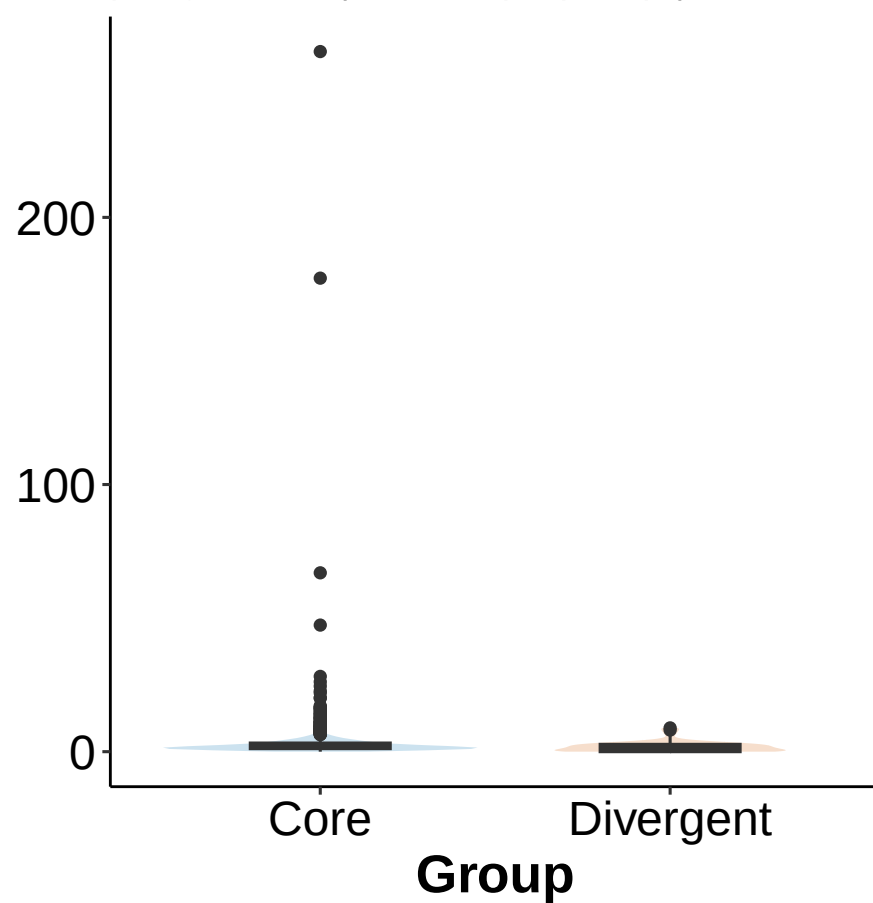
SA_Glutamate_metabolism_FBA

p-adj: 0.192 | .=-0.14 (Negligible) | .%: -11.1



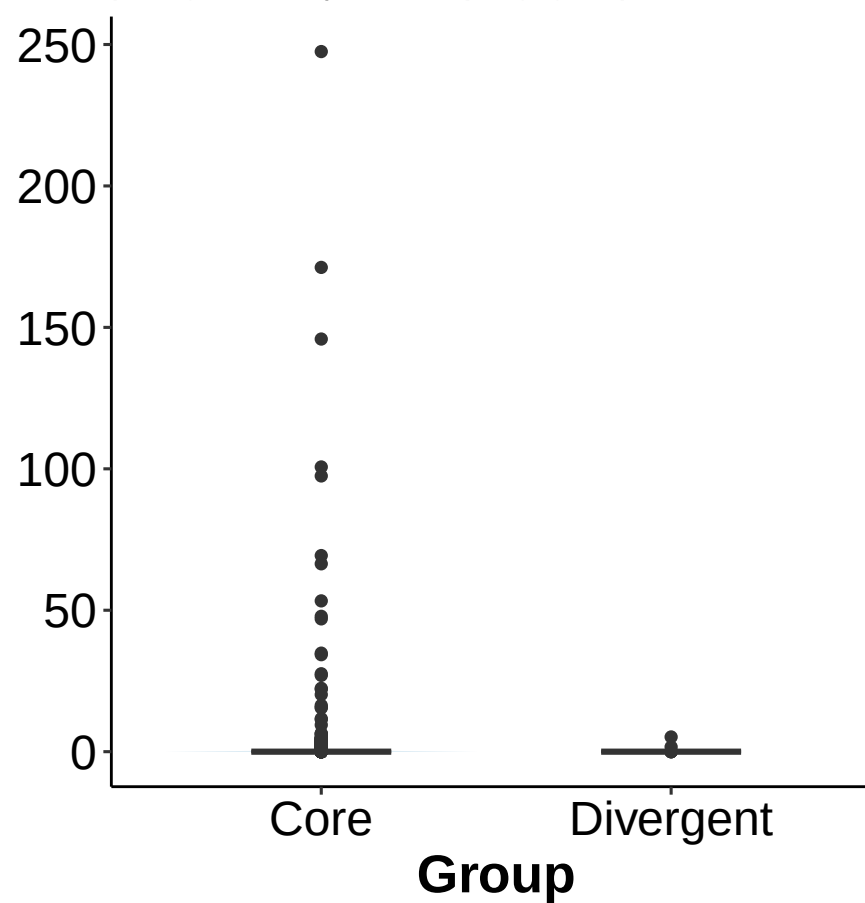
SA_Glutamate_metabolism_L1w

p-adj: 0.0211 | .=-0.22 (Pequeño) | .%: -22.3



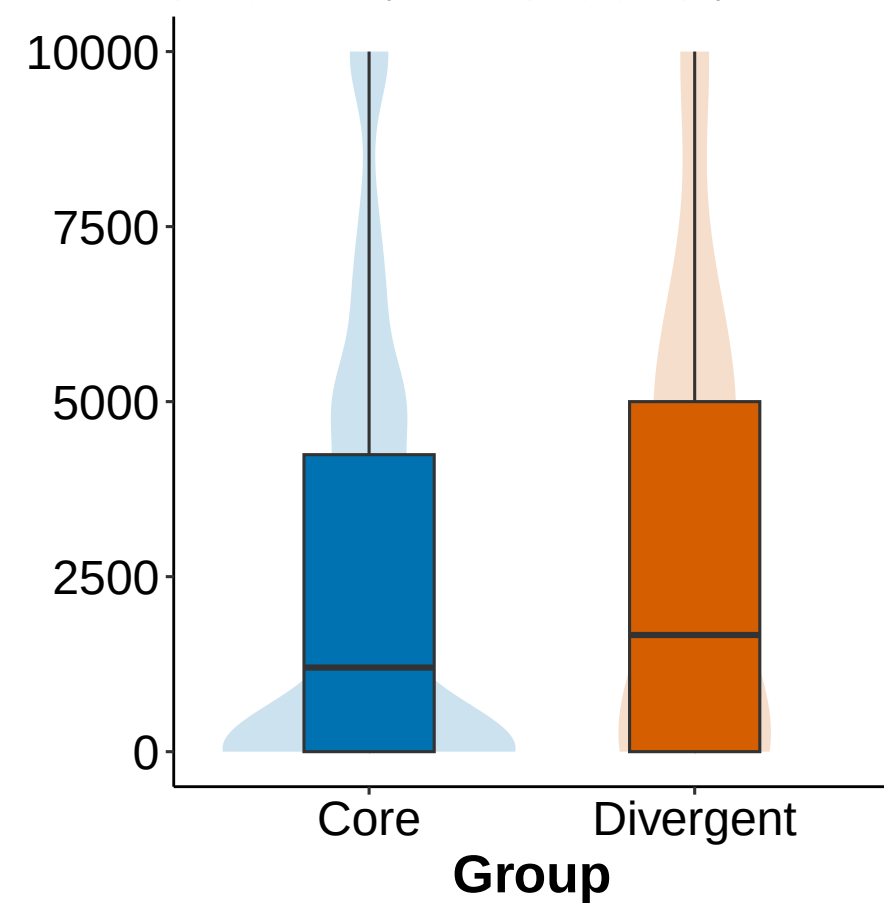
SA_Glutamate_metabolism_pFBA

p-adj: 0.922 | .=0.01 (Negligible)

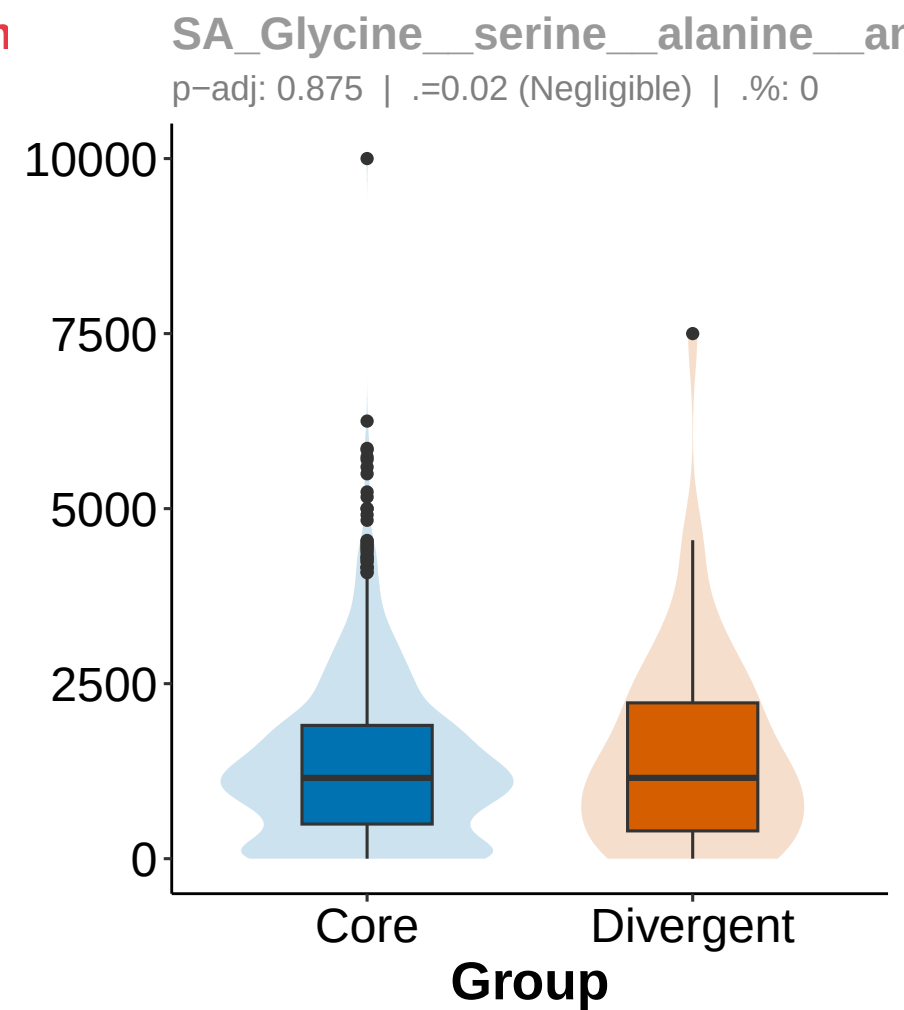
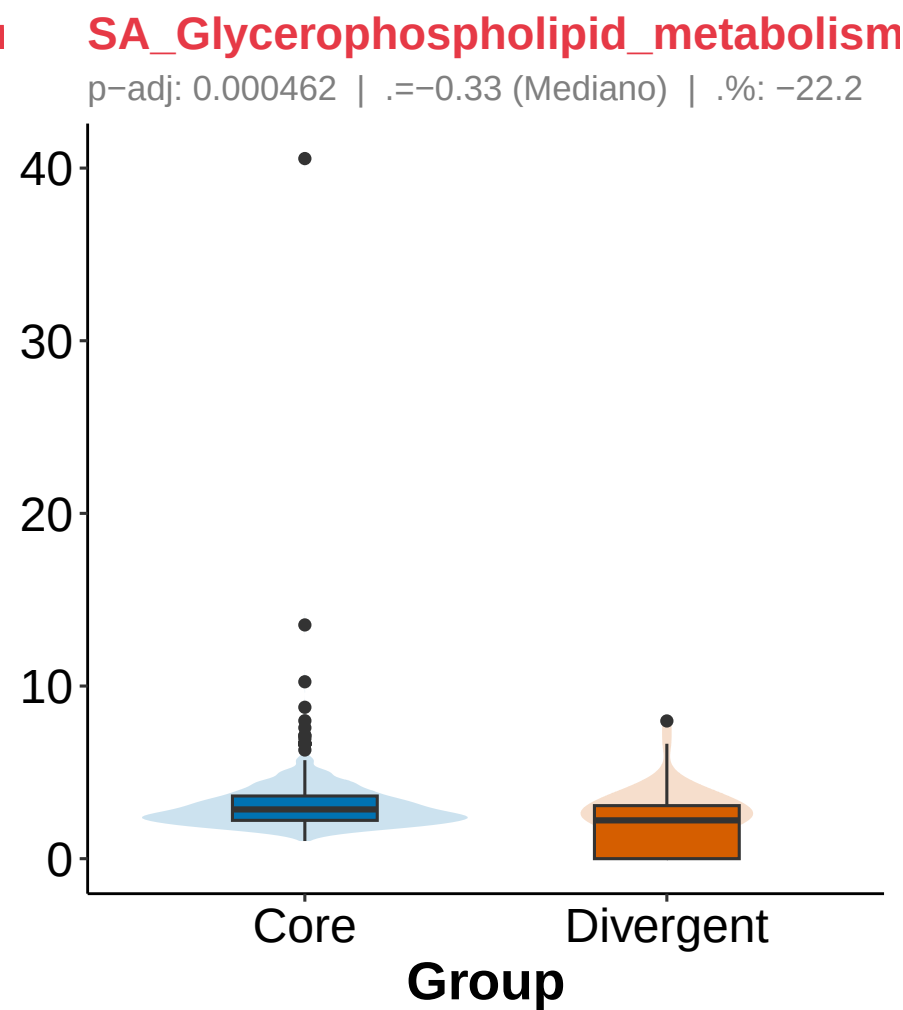
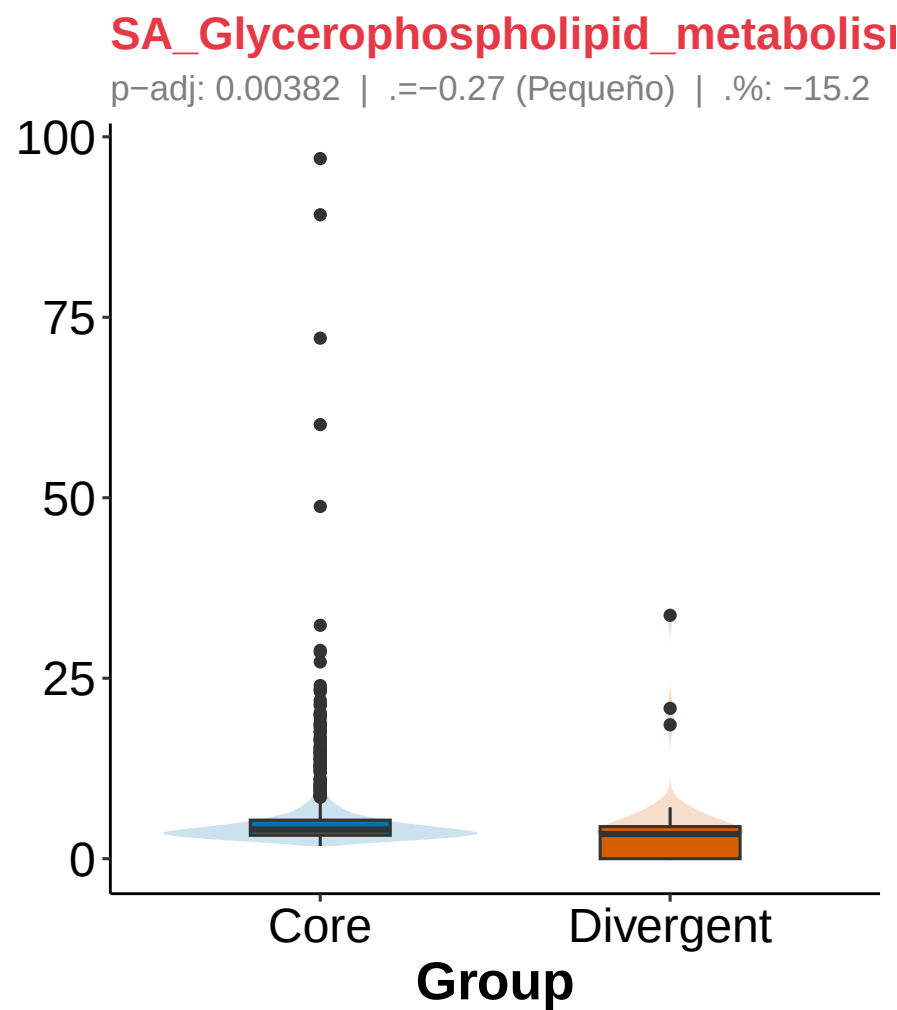
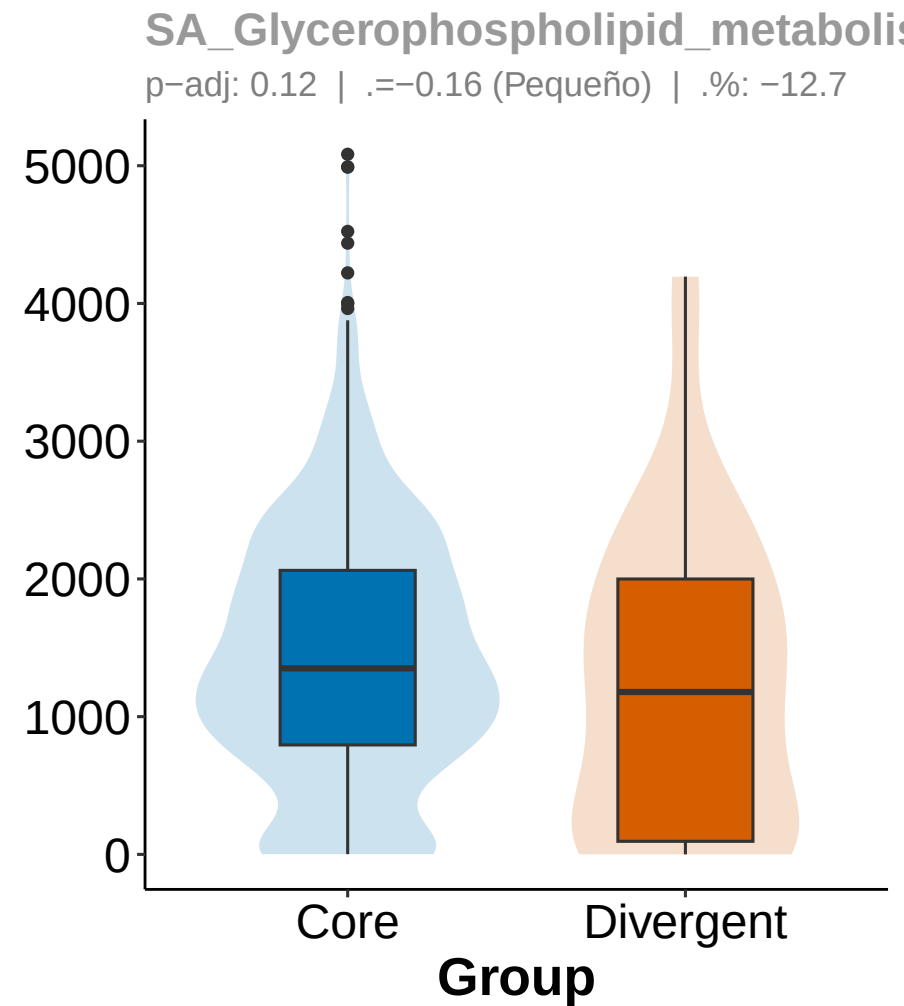
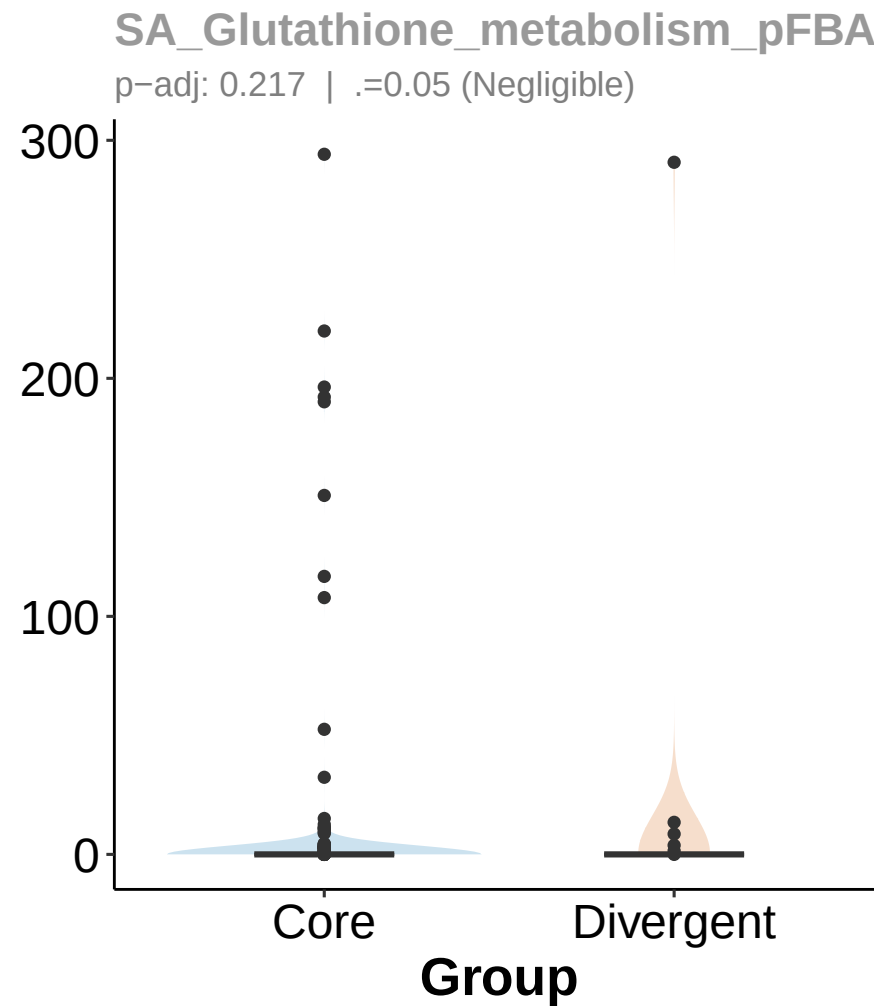
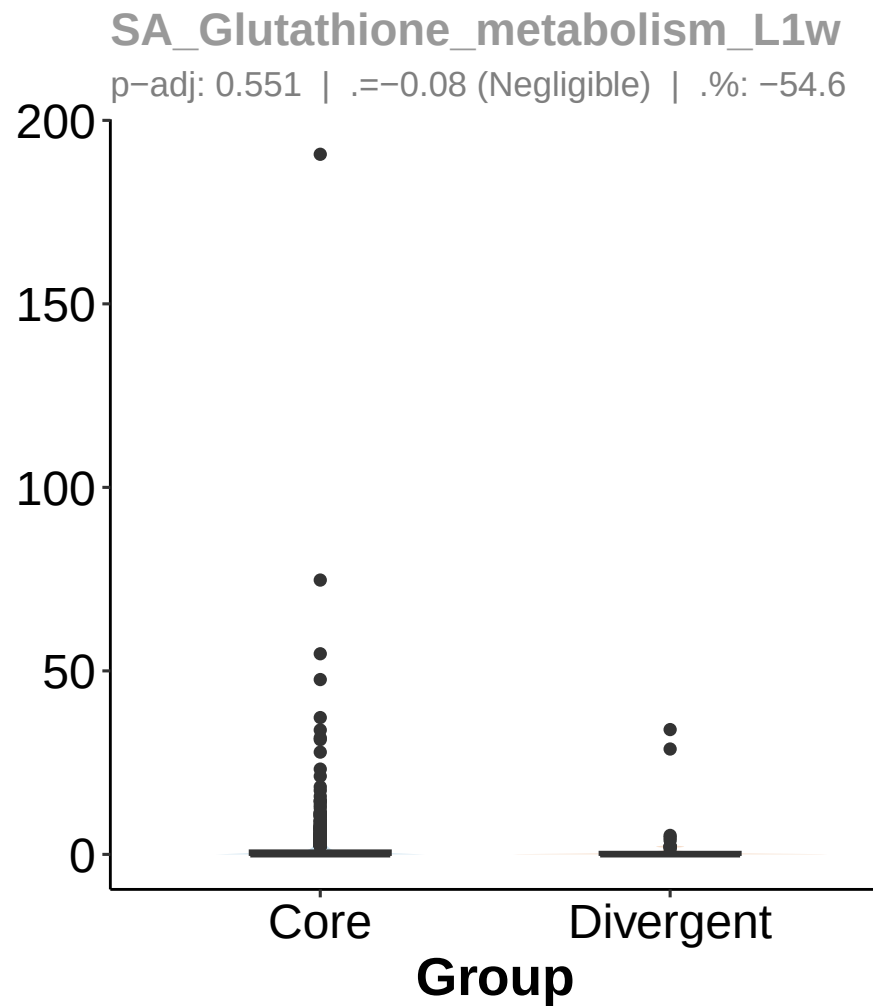


SA_Glutathione_metabolism_FBA

p-adj: 0.537 | .=0.08 (Negligible) | .%: 38.7



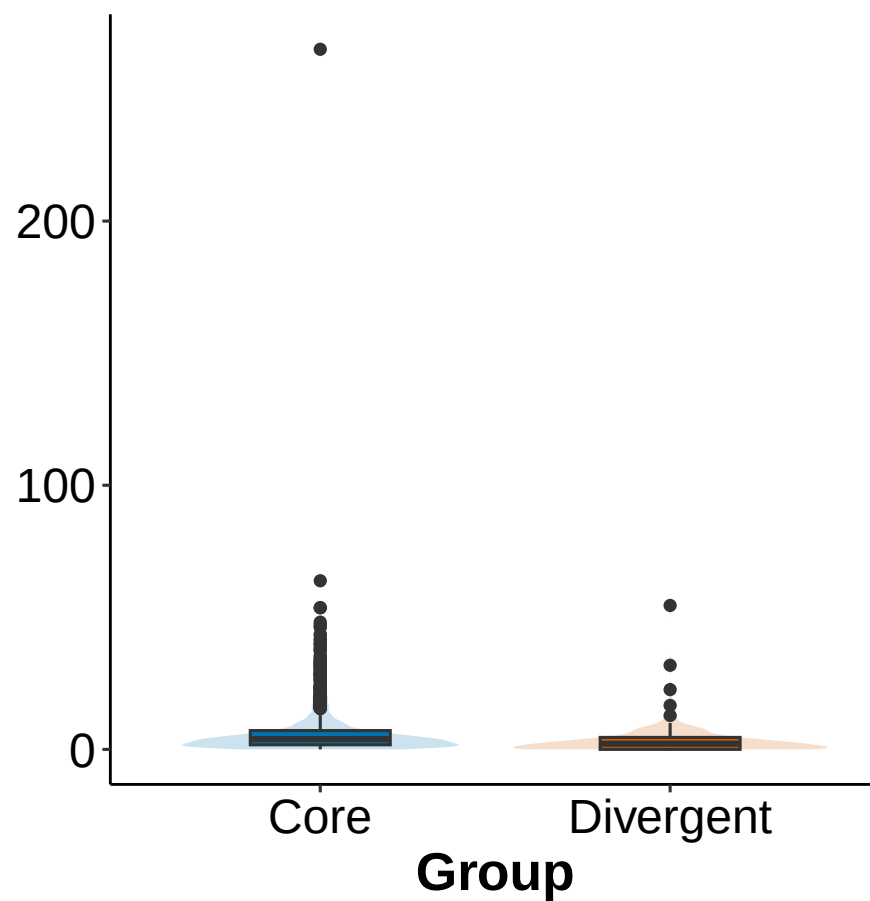
Metabolismo – . 5. Reprogramación Sistémica (SA_*)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

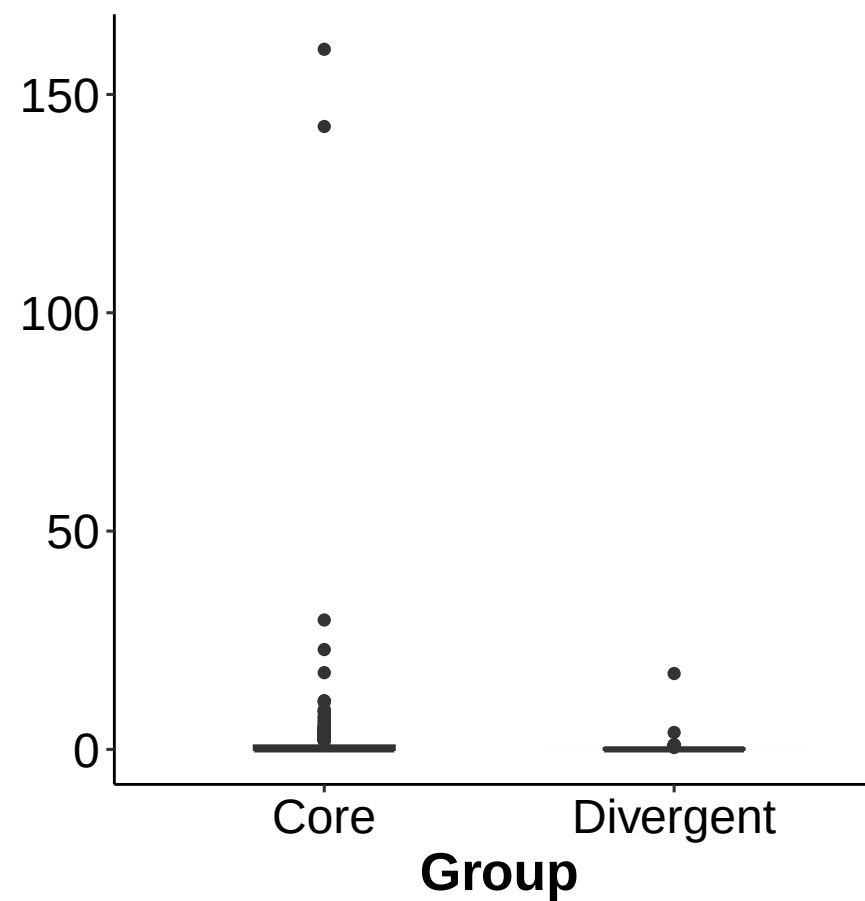
SA_Glycine_serine_alanine_and

p-adj: 0.00117 | .=-0.31 (Pequeño) | .%: -44.4



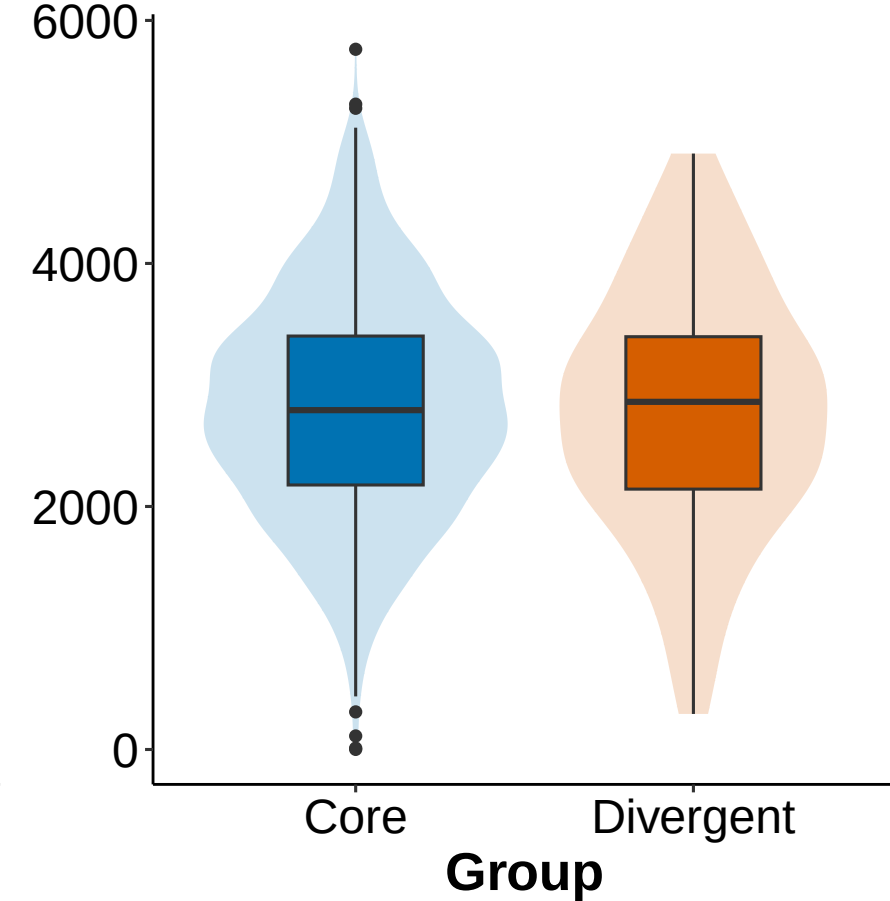
SA_Glycine_serine_alanine_and

p-adj: 0.0211 | .=-0.21 (Pequeño) | .%: -100



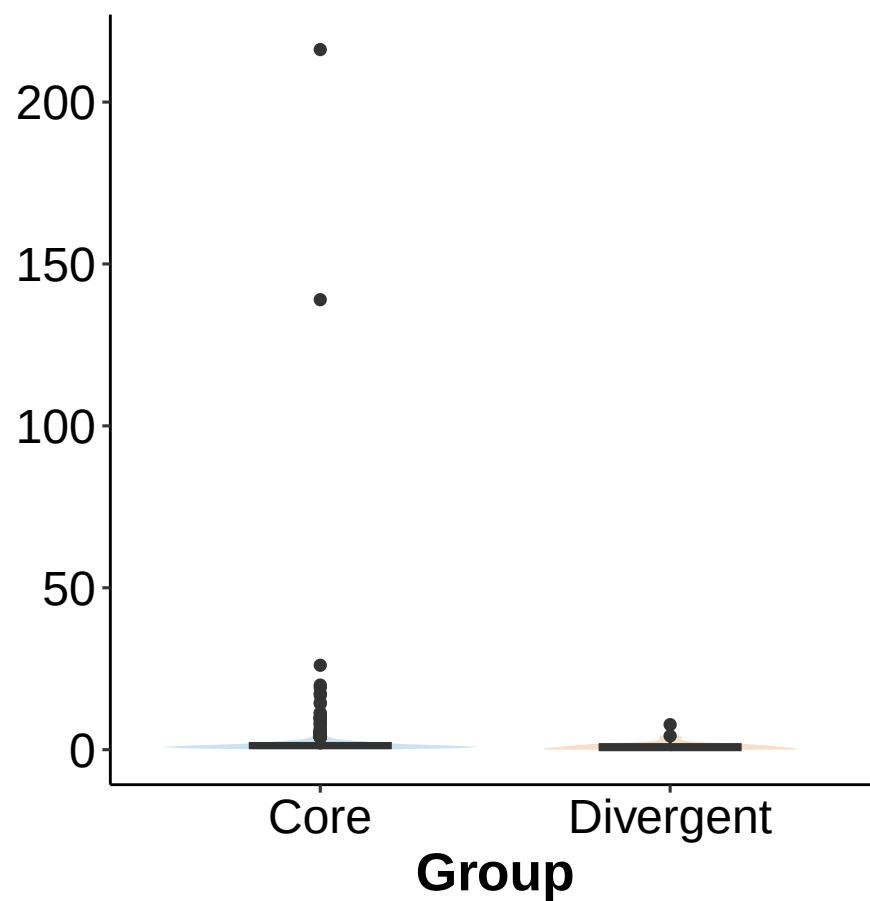
SA_Glycolysis_gluconeogenesis_F

p-adj: 0.922 | .=-0.01 (Negligible) | .%: 2.5



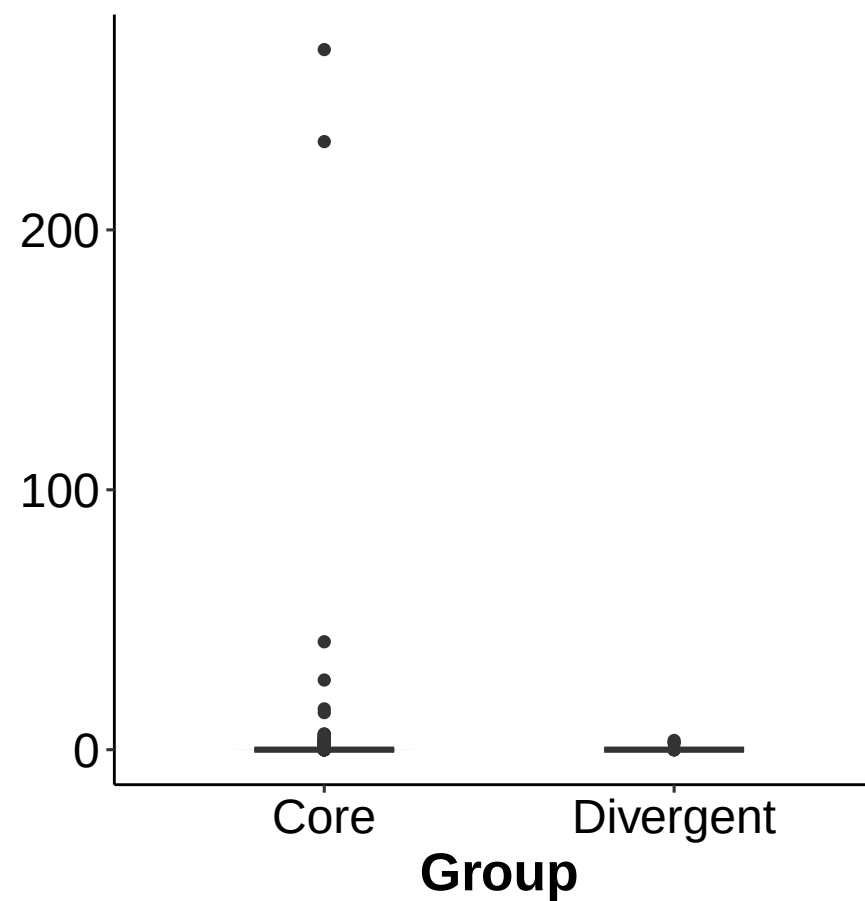
SA_Glycolysis_gluconeogenesis_L1

p-adj: 0.00382 | .=-0.27 (Pequeño) | .%: -21.2



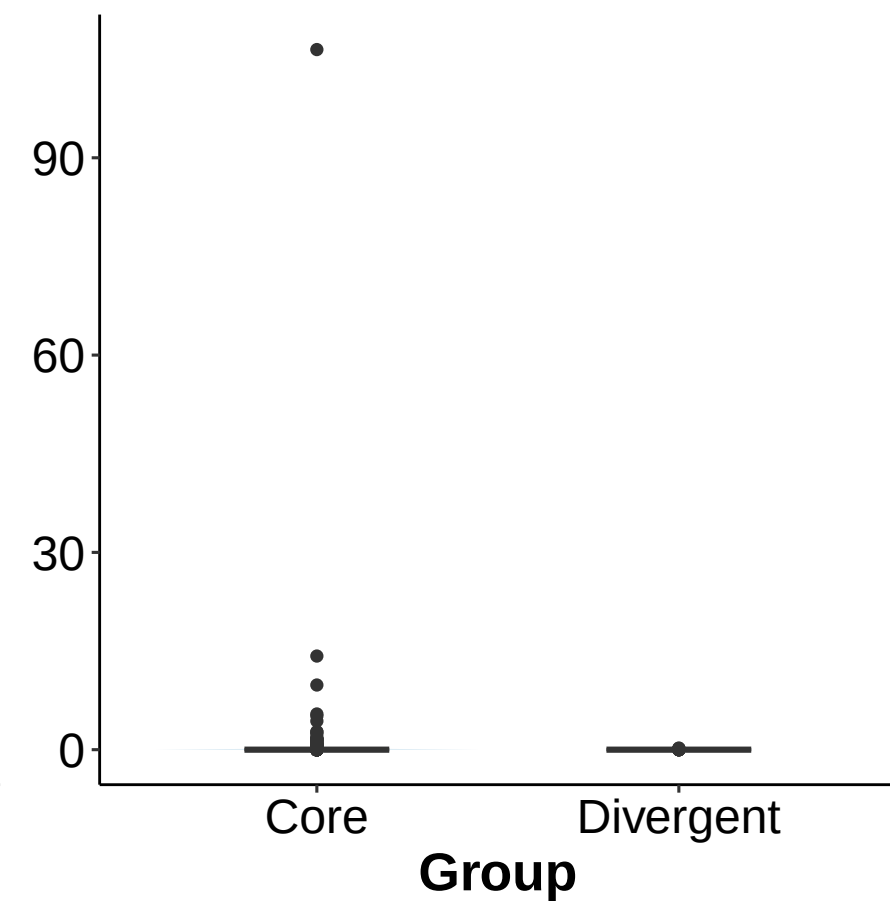
SA_Glycolysis_gluconeogenesis_pF

p-adj: 0.563 | .=-0.07 (Negligible) | .%: -42.9



SA_Glycosphingolipid_metabolism_L

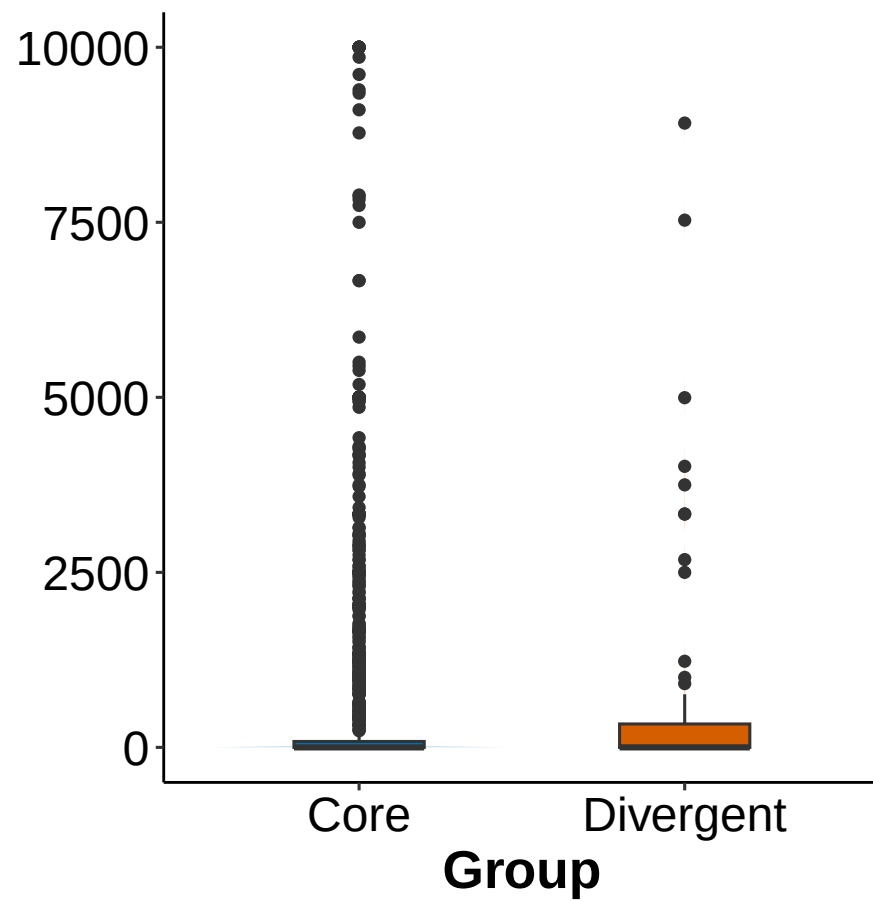
p-adj: 0.849 | .=-0.02 (Negligible) | .%: -100



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

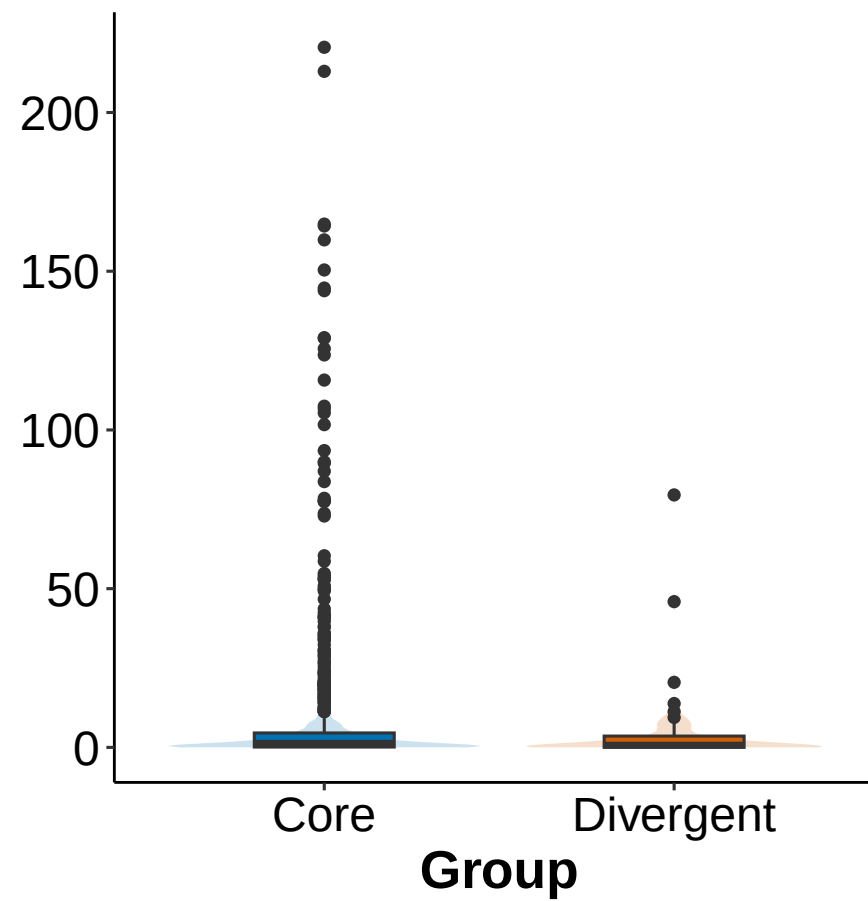
SA_Glyoxylate_and_dicarboxylate

p-adj: 0.759 | . = 0.03 (Negligible)



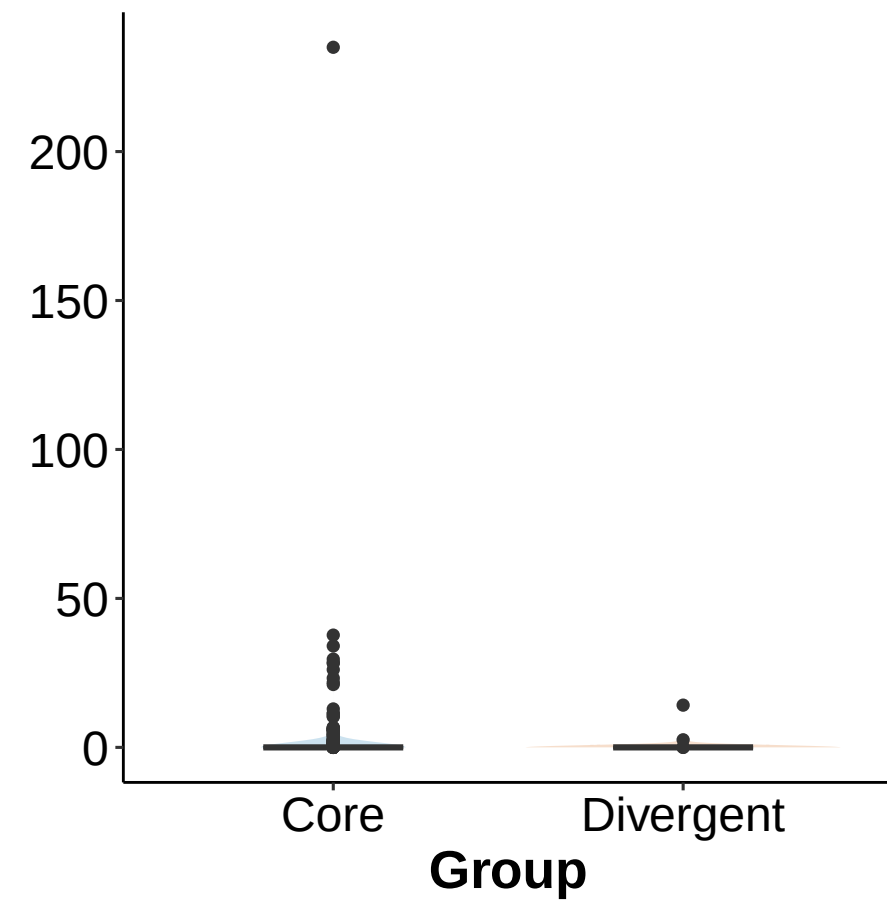
SA_Glyoxylate_and_dicarboxylate_n

p-adj: 0.301 | . = -0.11 (Negligible) | .%: -42.6



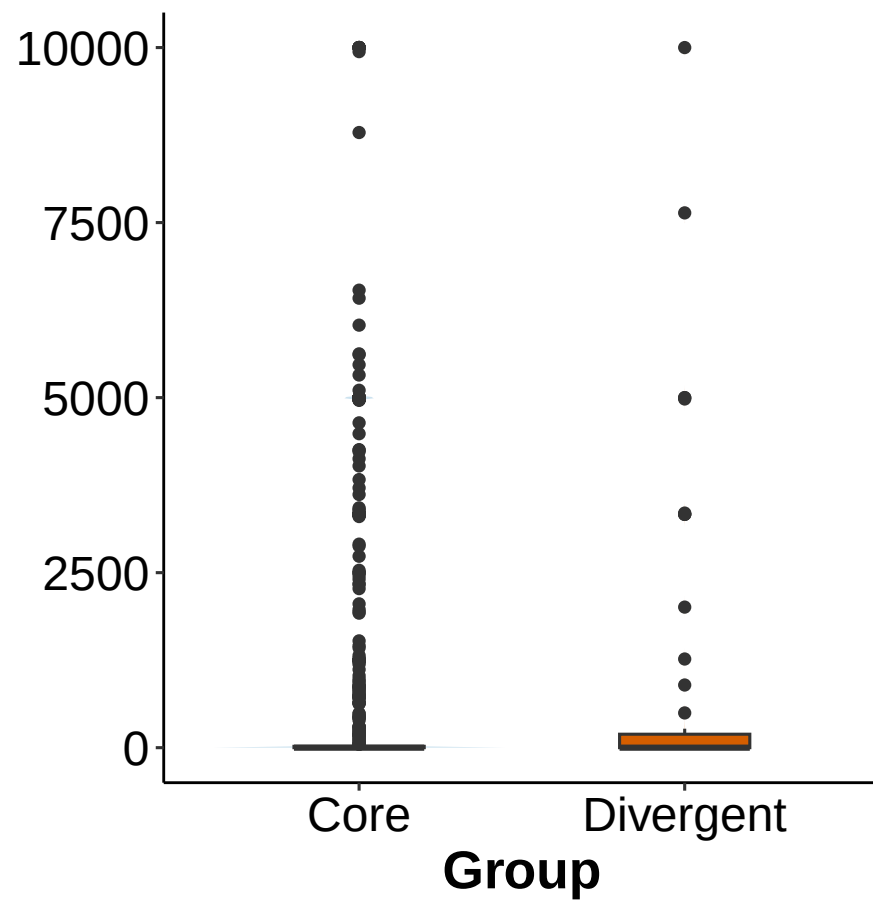
SA_Glyoxylate_and_dicarboxylate_n

p-adj: 0.931 | . = 0 (Negligible)



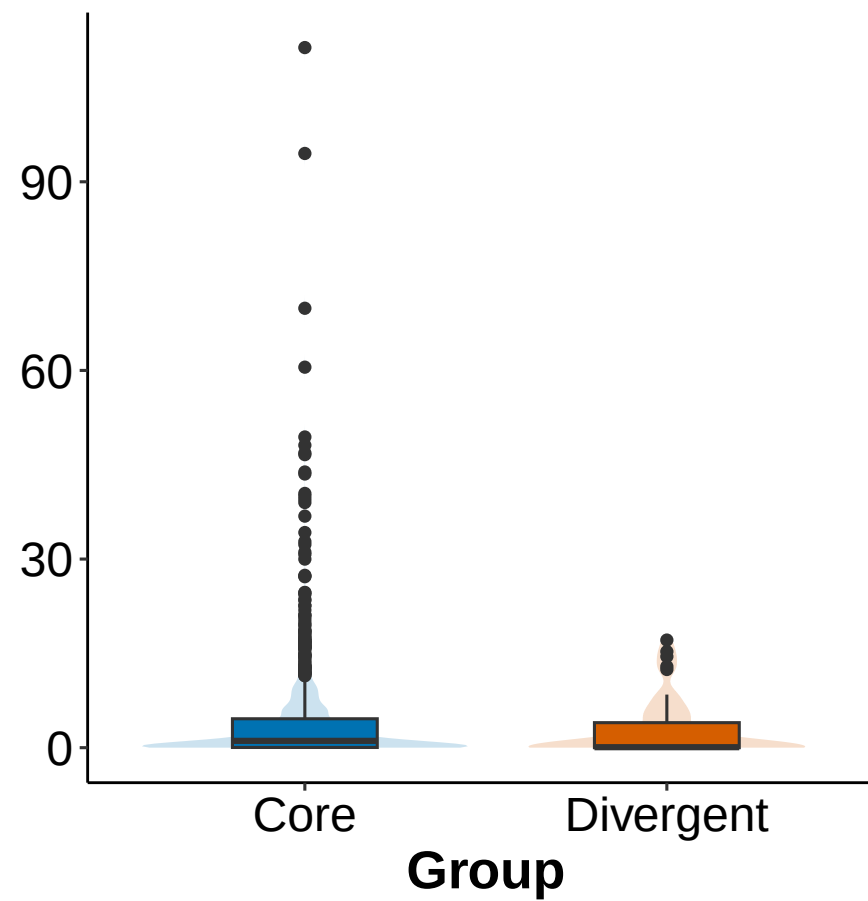
SA_Heme_degradation_FBA

p-adj: 0.804 | . = 0.03 (Negligible)



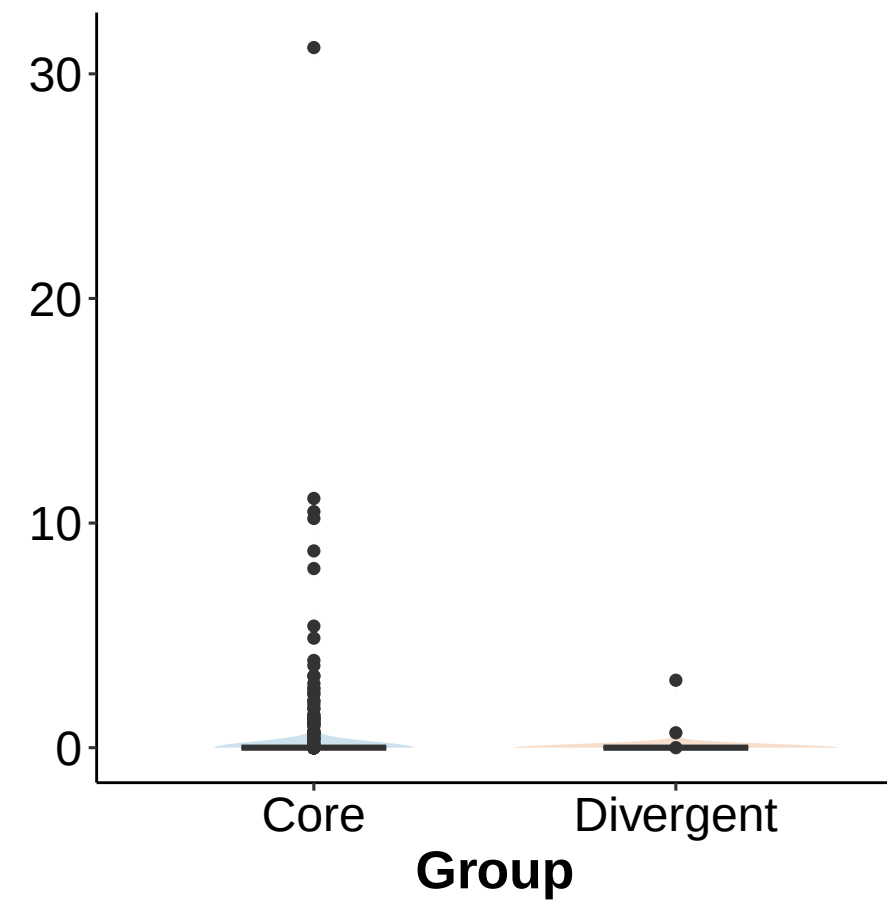
SA_Heme_degradation_L1w

p-adj: 0.127 | . = -0.16 (Pequeño) | .%: -92.9



SA_Heme_degradation_pFBA

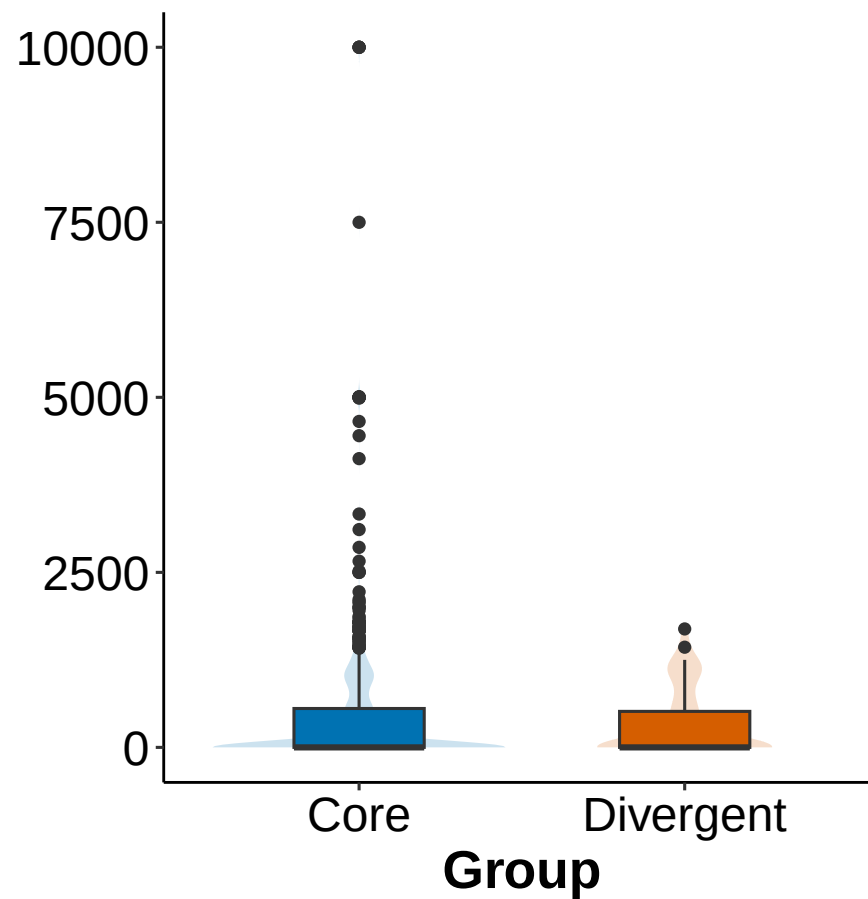
p-adj: 0.136 | . = -0.09 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

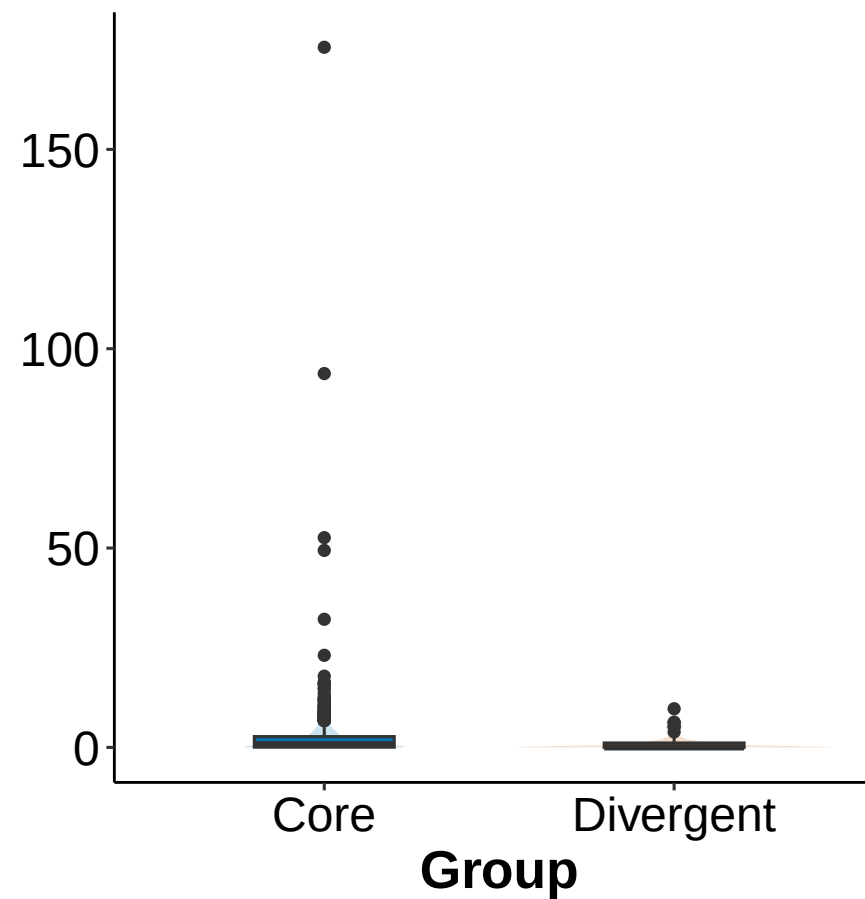
SA_Heme_synthesis_FBA

p-adj: 0.847 | .:=0.02 (Negligible)



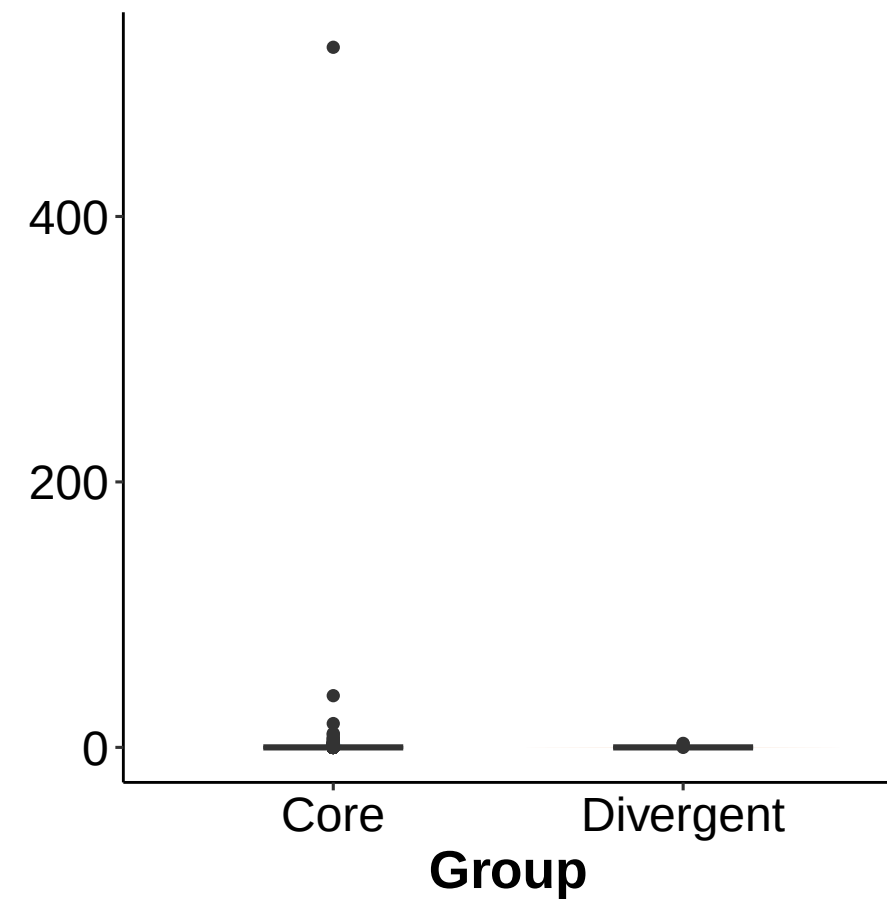
SA_Heme_synthesis_L1w

p-adj: 0.00124 | .=-0.31 (Pequeño) | .%: -100



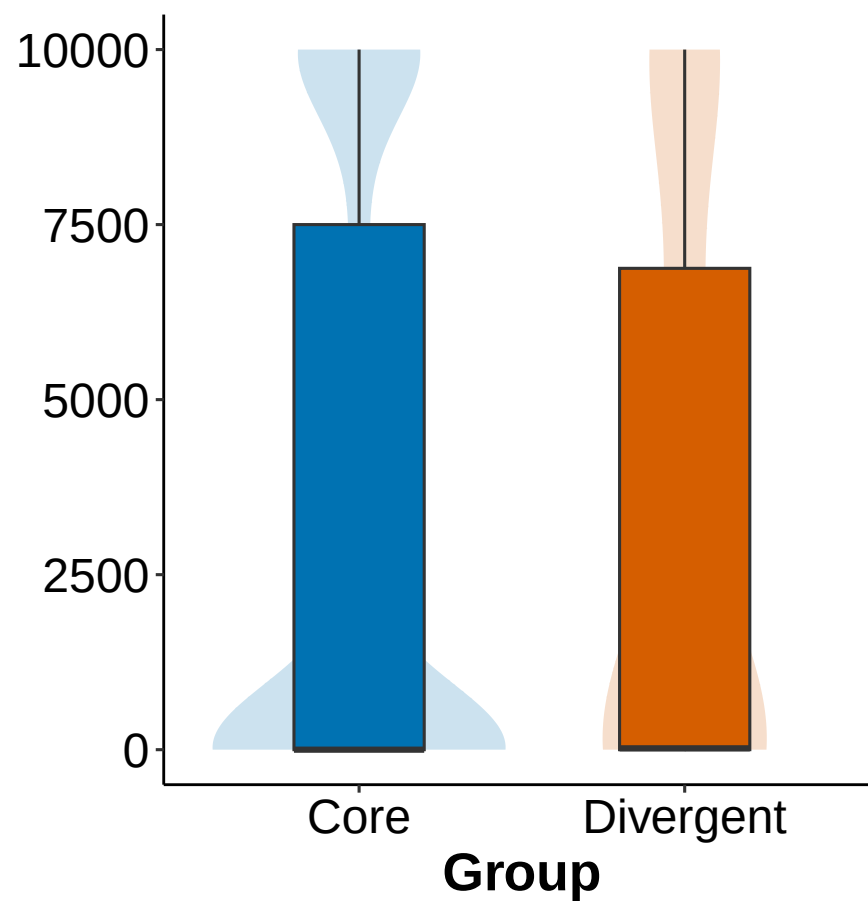
SA_Heme_synthesis_pFBA

p-adj: 0.119 | .=-0.1 (Negligible)



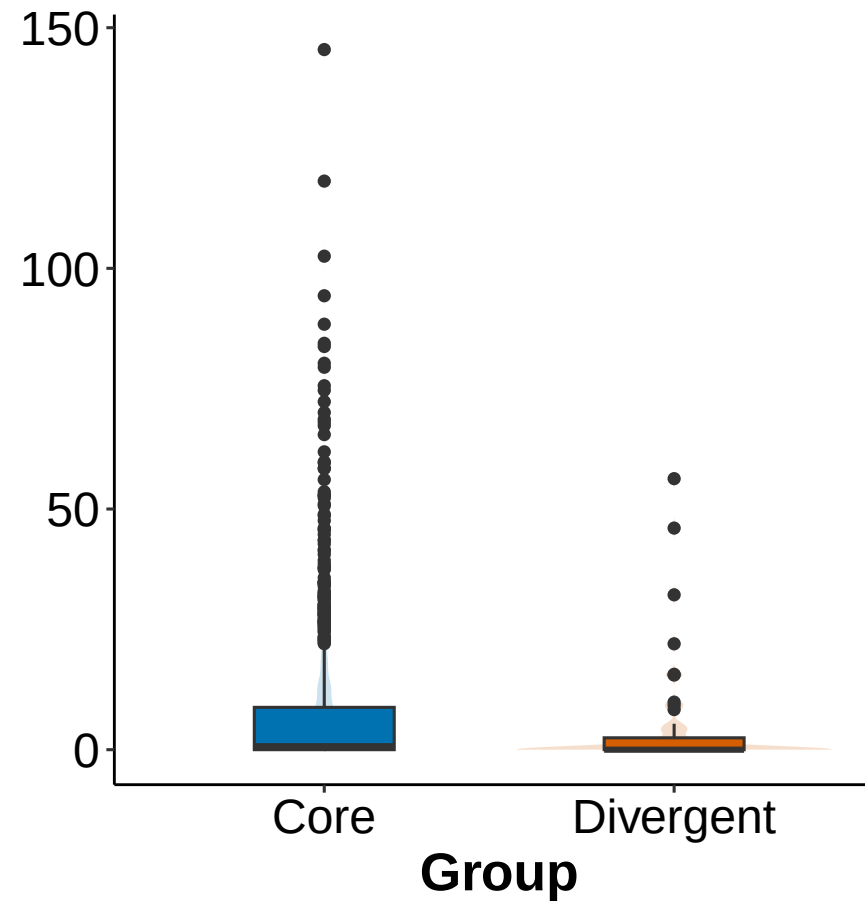
SA_Histidine_metabolism_FBA

p-adj: 0.847 | .:=0.02 (Negligible)



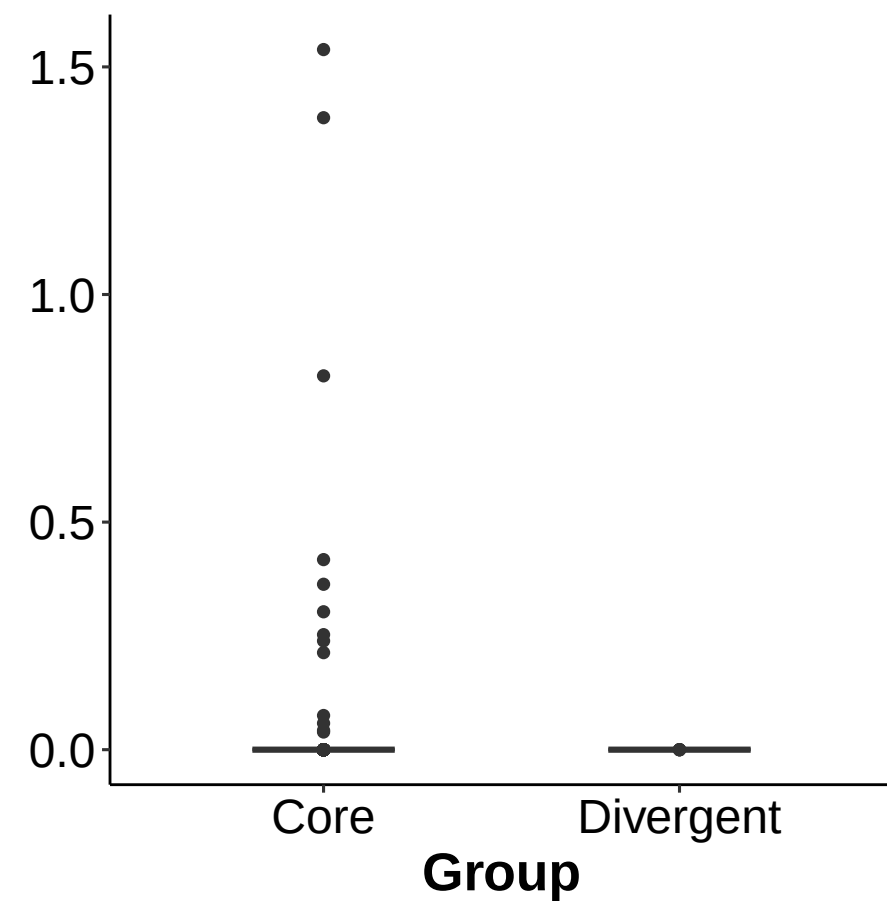
SA_Histidine_metabolism_L1w

p-adj: 0.537 | .=-0.08 (Negligible) | .%: -100



SA_Hyaluronan_metabolism_L1w

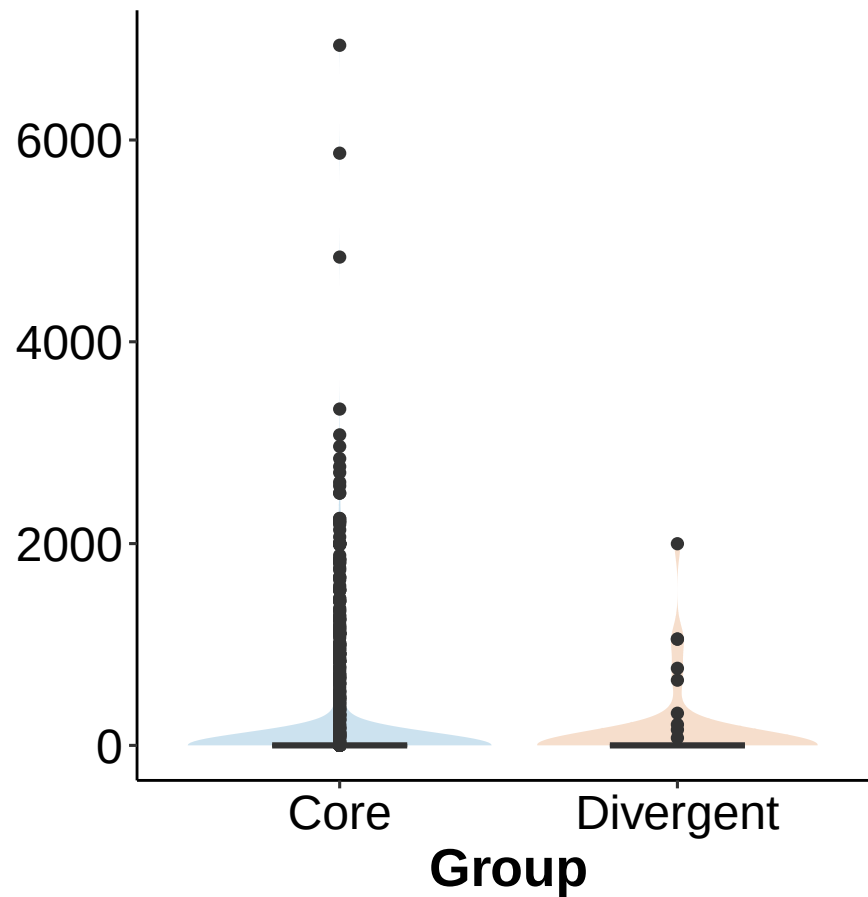
p-adj: 0.759 | .=-0.02 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

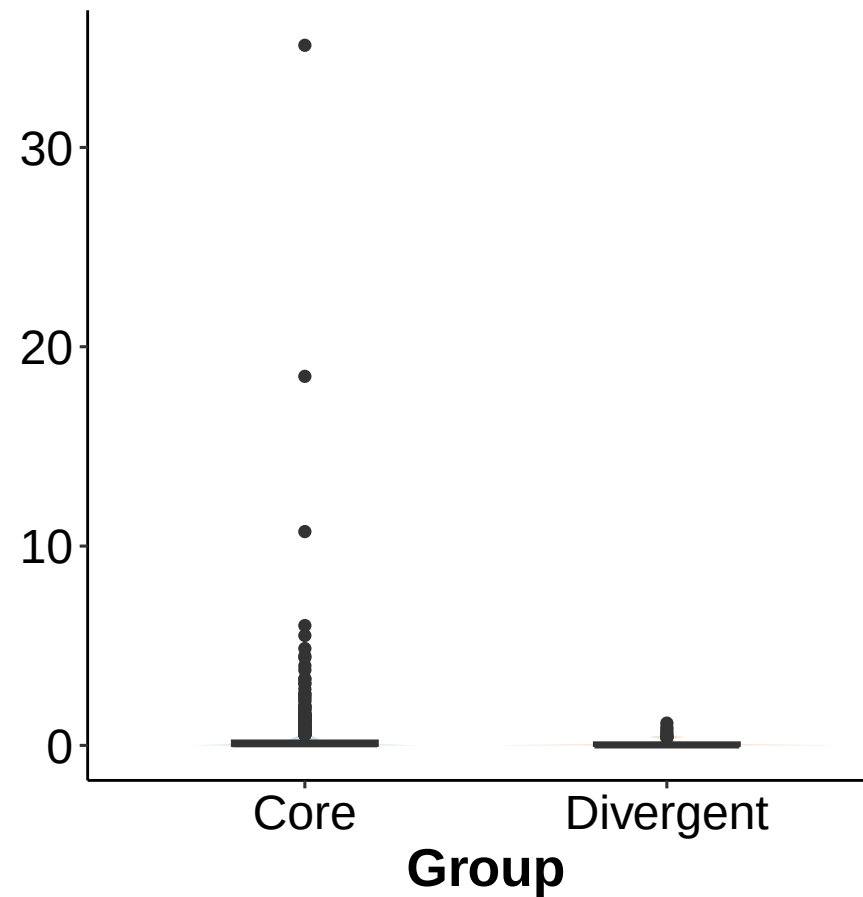
SA_Inositol_phosphate_metabolism

p-adj: 0.56 | .=-0.06 (Negligible)



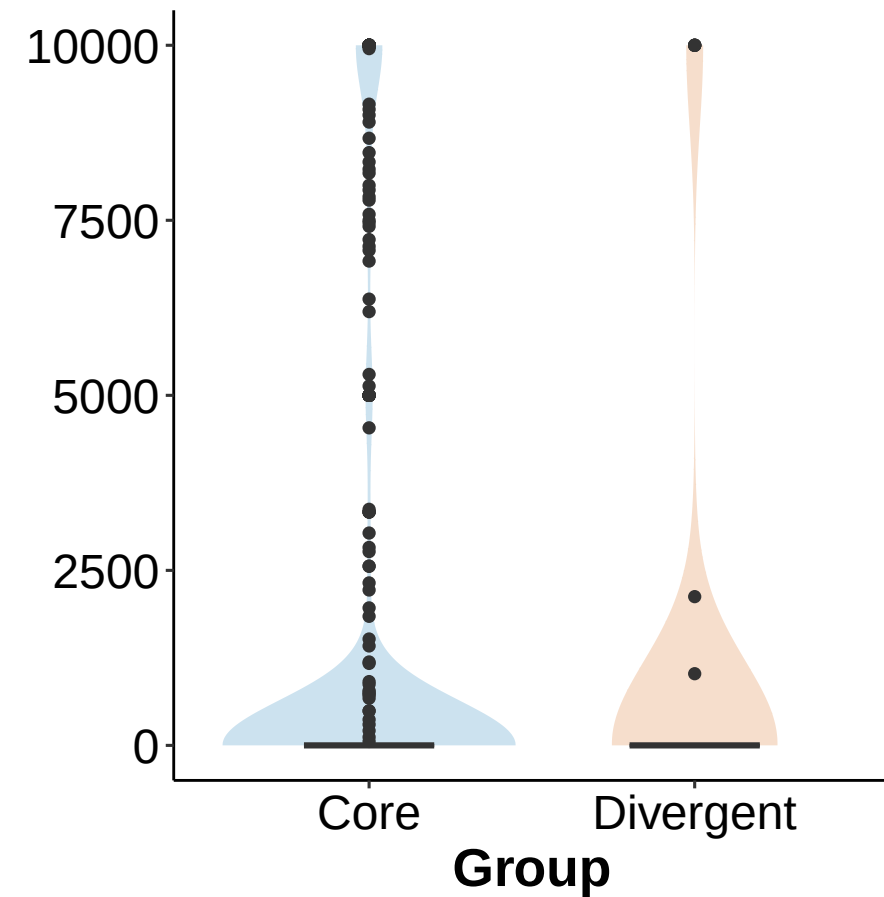
SA_Inositol_phosphate_metabolism_I

p-adj: 0.00124 | .=-0.31 (Pequeño) | .%: -99.9



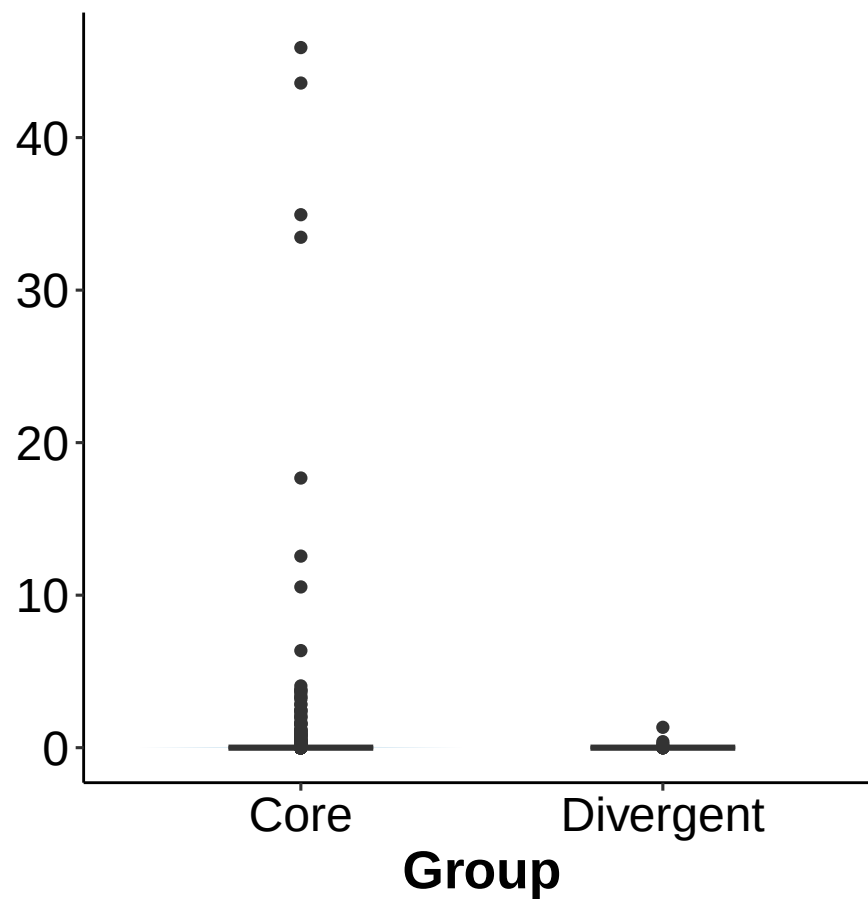
SA_Leukotriene_metabolism_FBA

p-adj: 0.839 | .=-0.02 (Negligible)



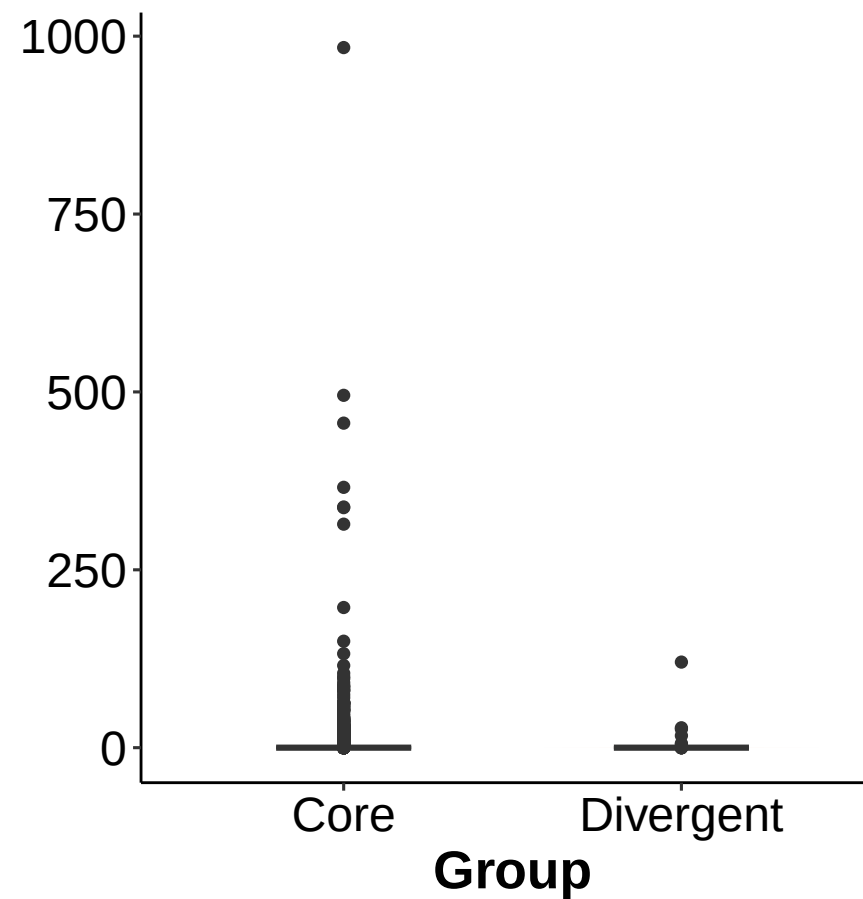
SA_Leukotriene_metabolism_L1w

p-adj: 0.735 | .=-0.04 (Negligible)



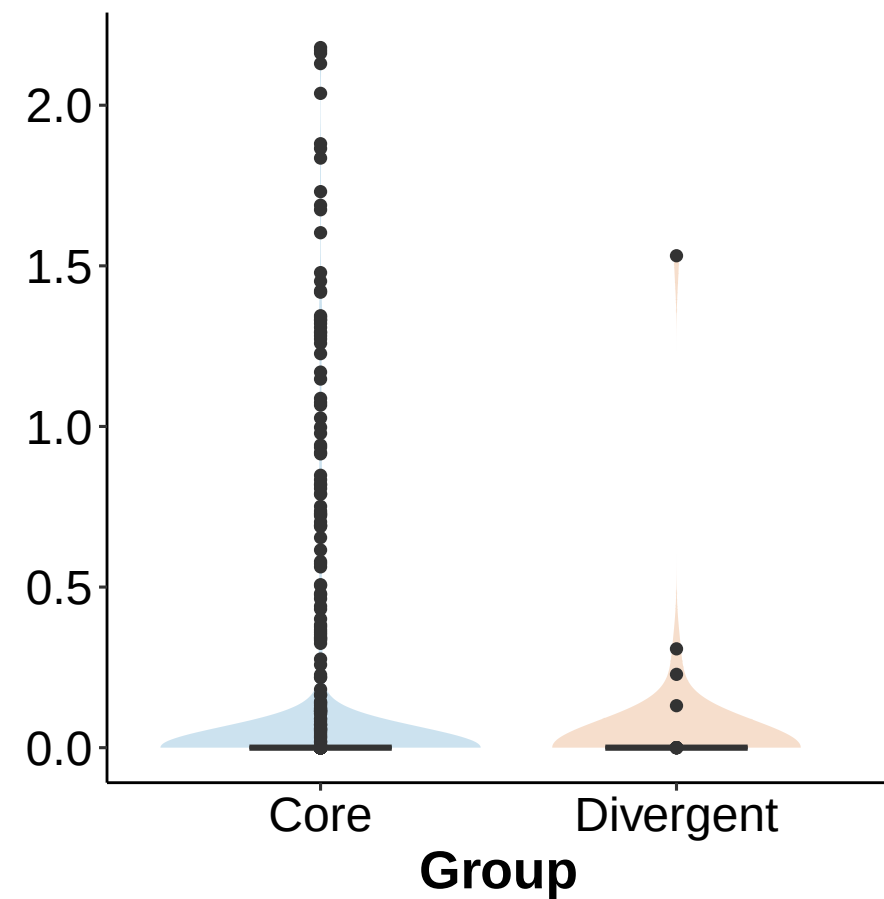
SA_Limonene_and_pinene_degrad

p-adj: 0.731 | .=-0.04 (Negligible)



SA_Linoleate_metabolism_L1w

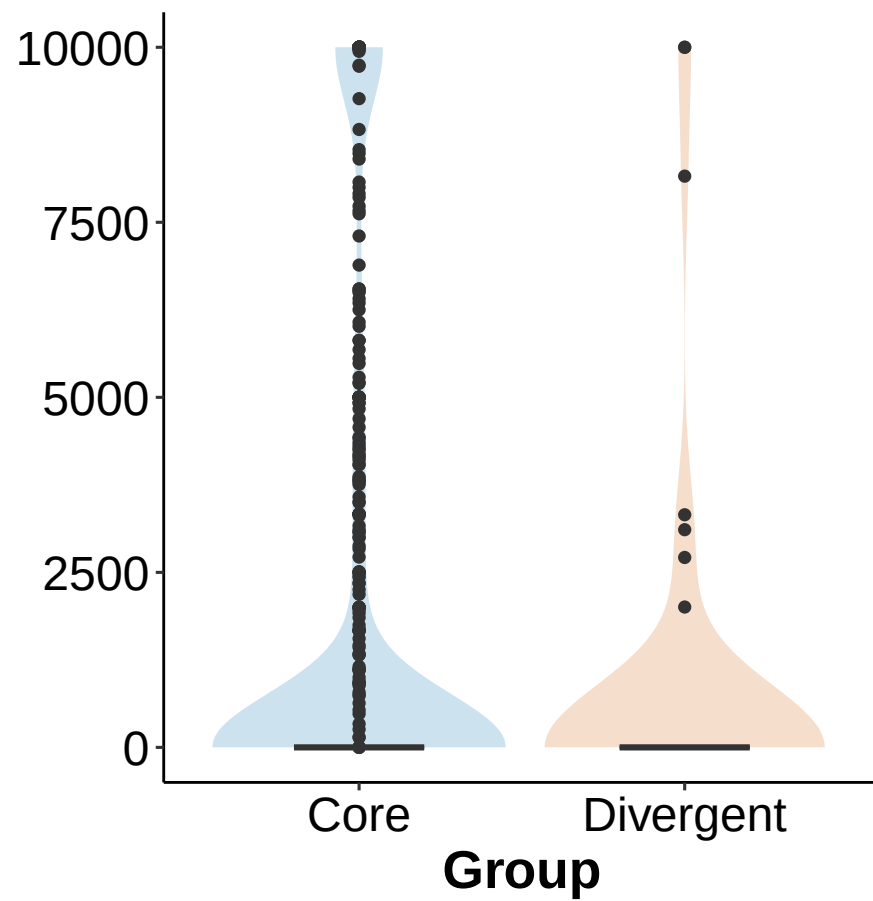
p-adj: 0.666 | .=-0.05 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

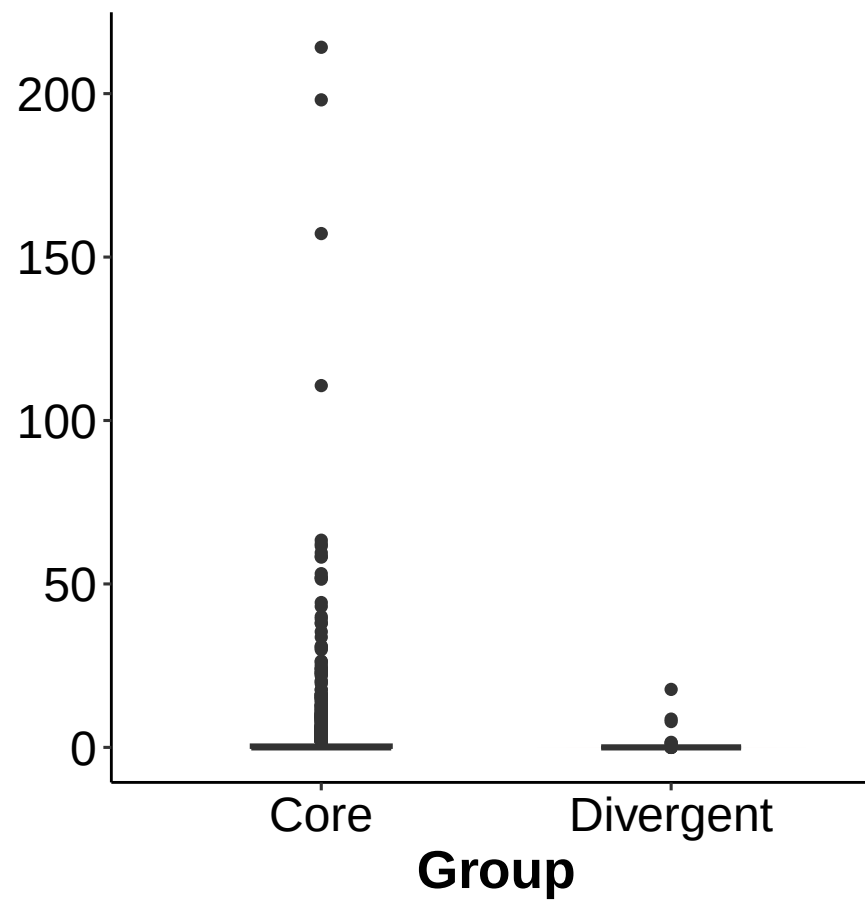
SA_Lysine_metabolism_FBA

p-adj: 0.0989 | .=-0.13 (Negligible)



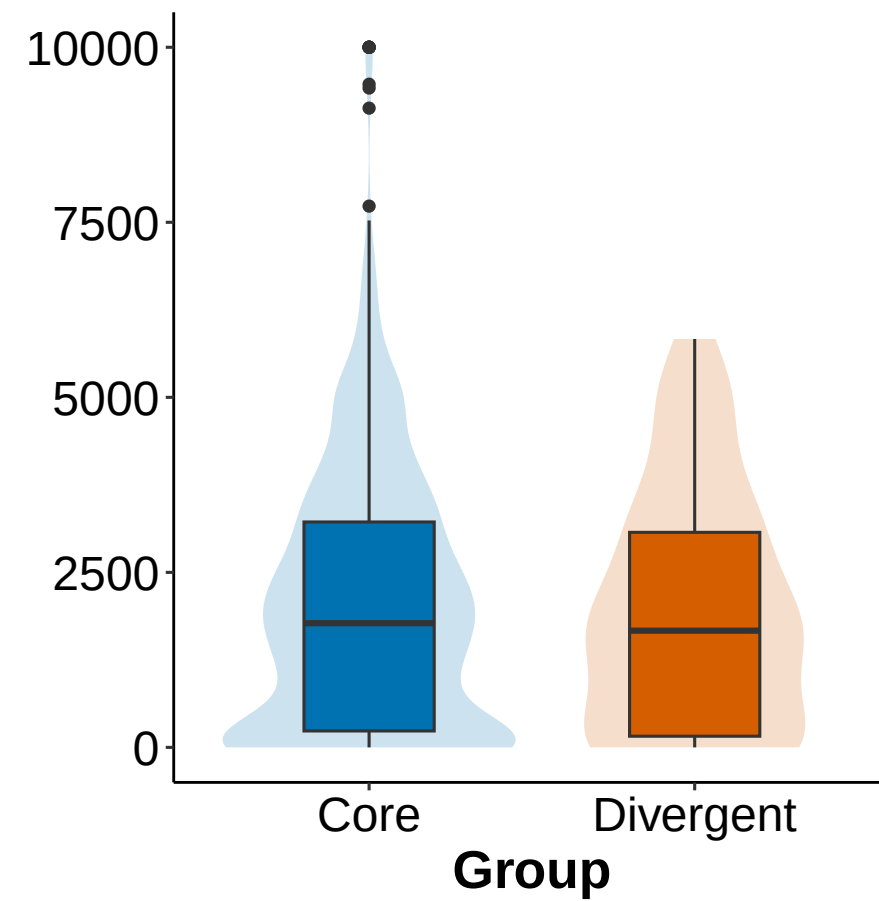
SA_Lysine_metabolism_L1w

p-adj: 0.061 | .=-0.16 (Pequeño)



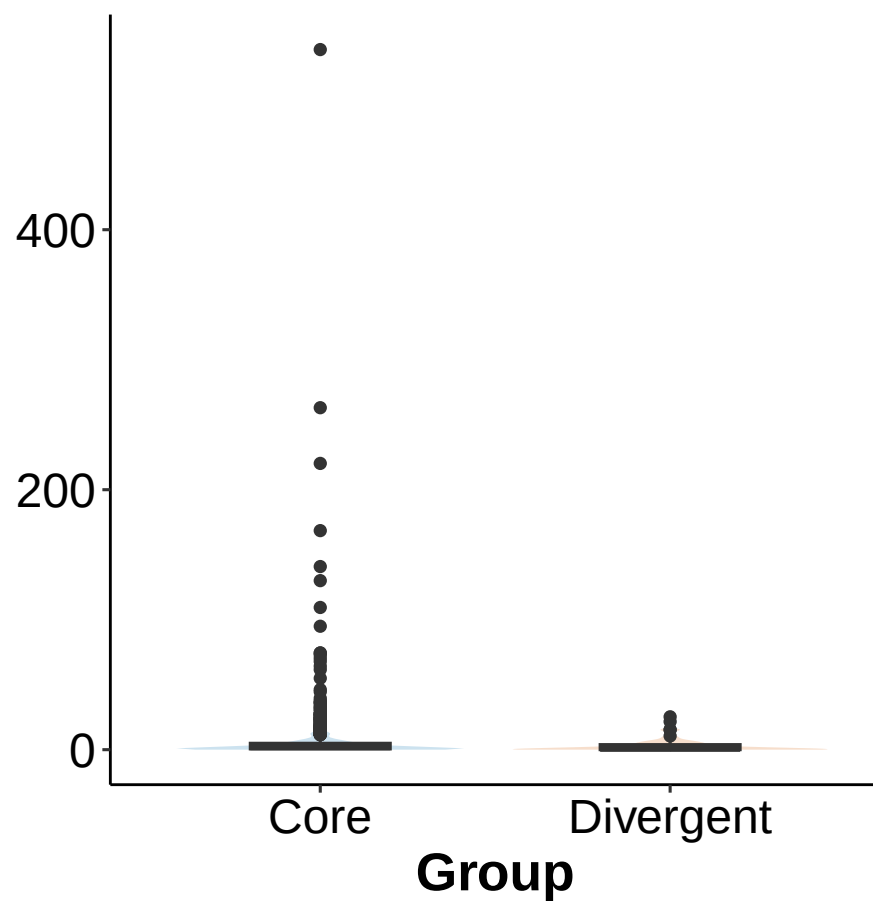
SA_Methionine_and_cysteine_me

p-adj: 0.852 | .=-0.02 (Negligible) | .%: -6



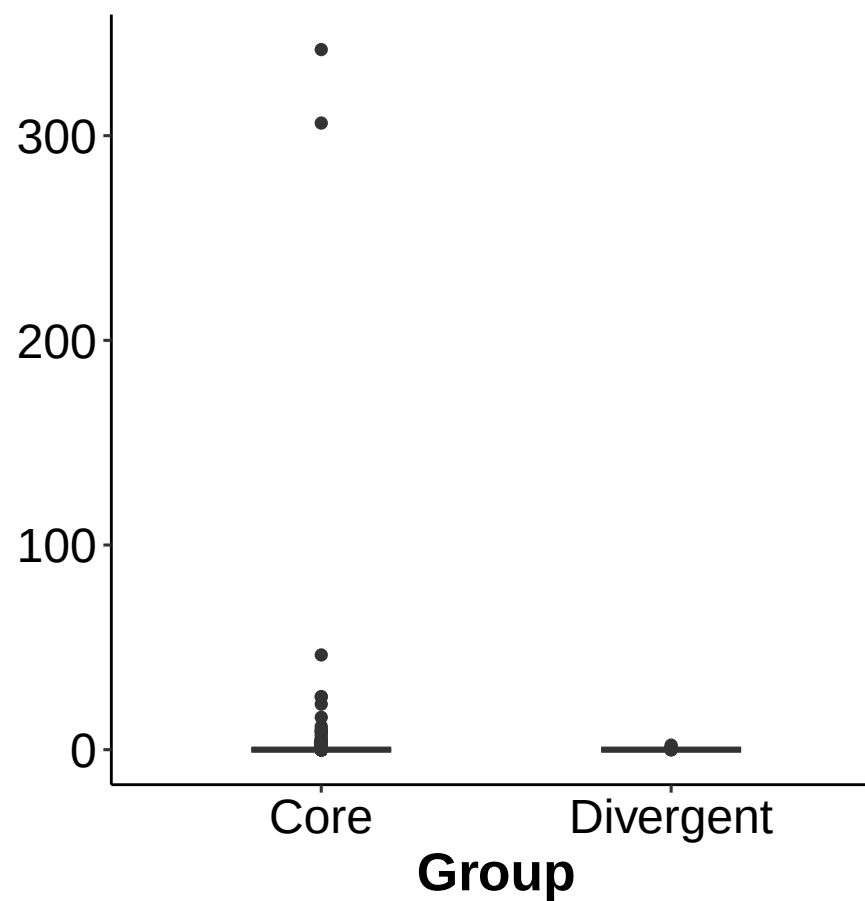
SA_Methionine_and_cysteine_metak

p-adj: 0.0245 | .=-0.22 (Pequeño) | .%: -61.4



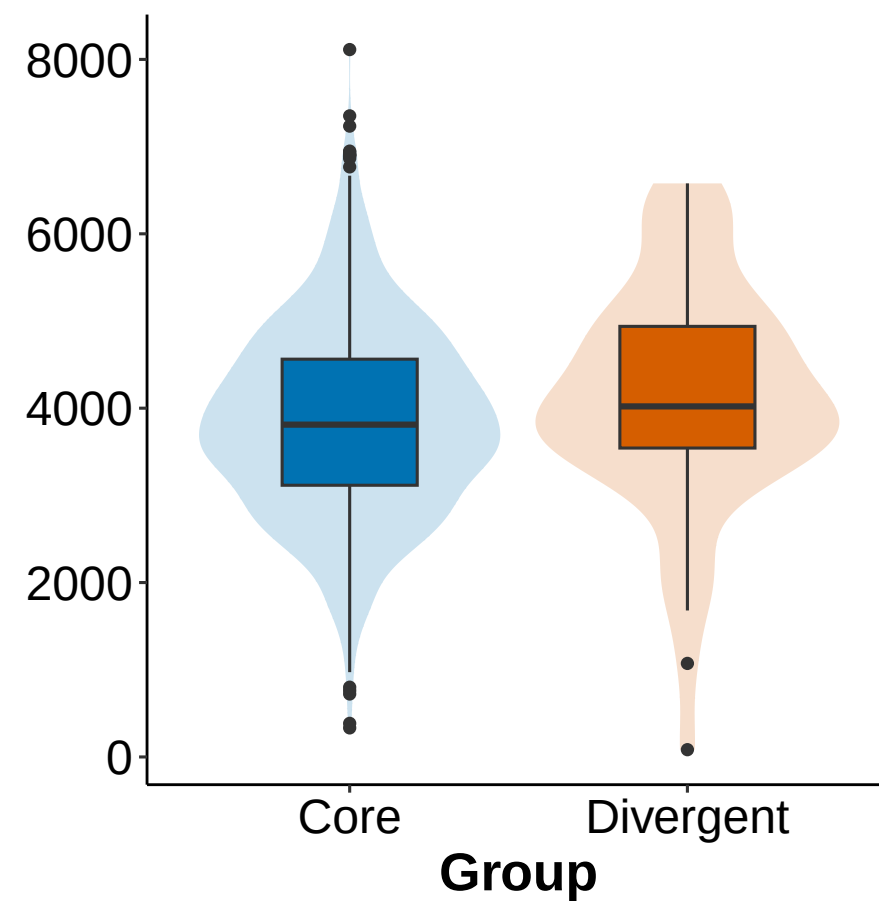
SA_Methionine_and_cysteine_metak

p-adj: 0.241 | .=0.1 (Negligible)



SA_Miscellaneous_FBA

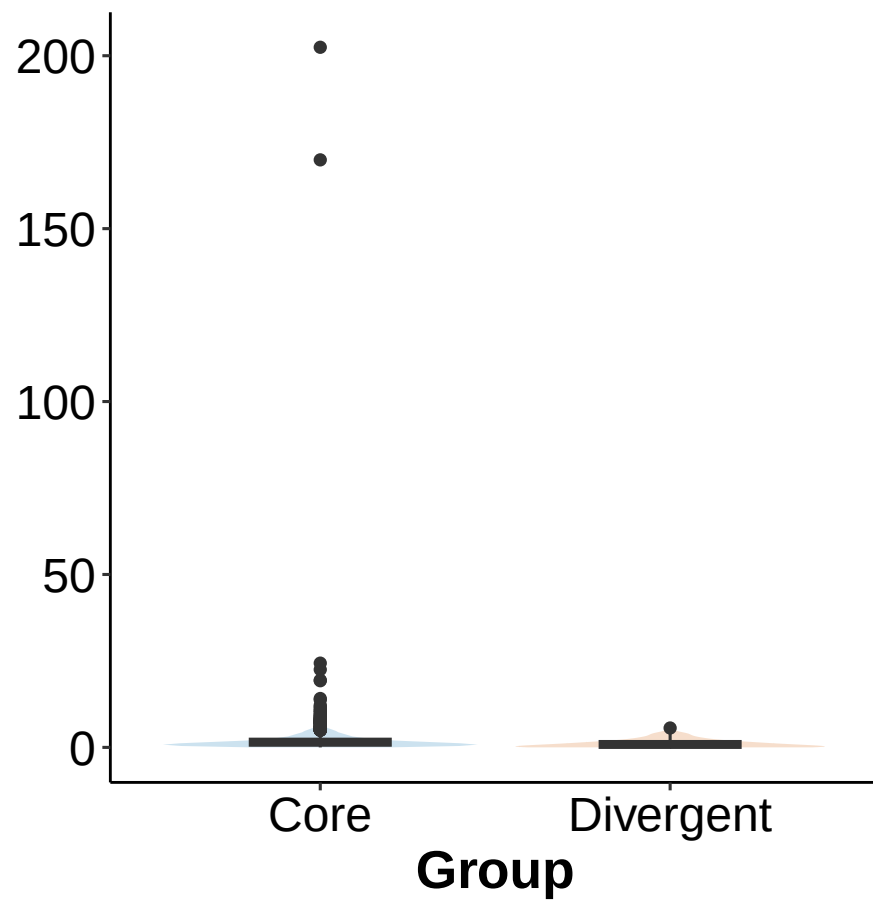
p-adj: 0.131 | .=0.16 (Pequeño) | .%: 5.5



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

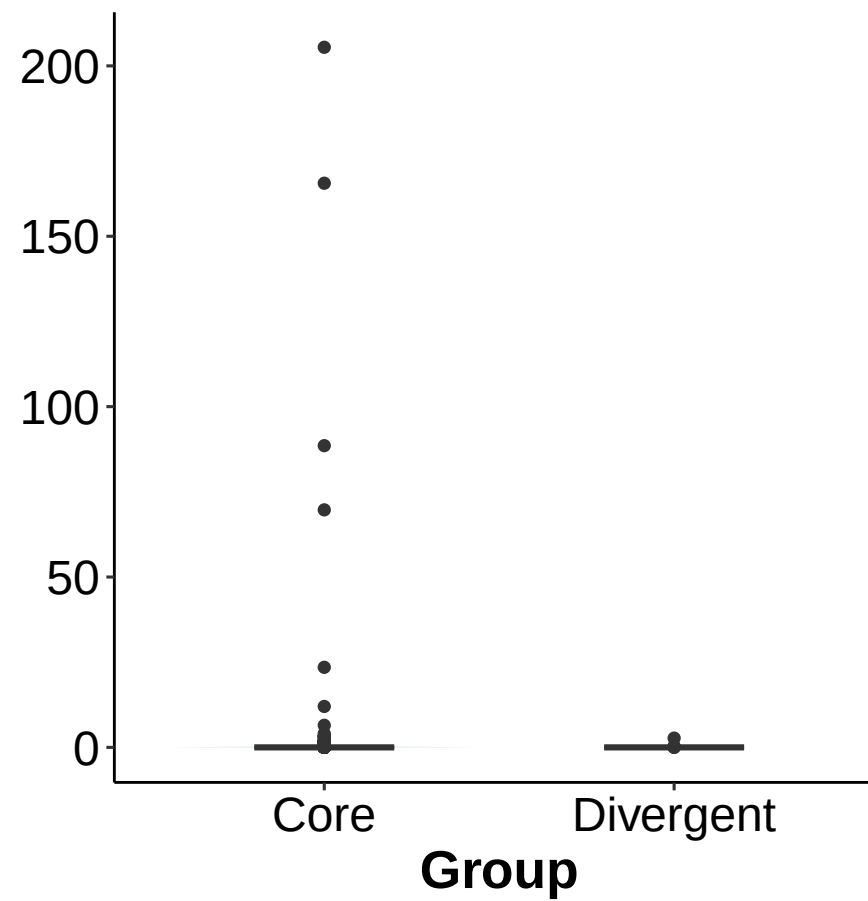
SA_Miscellaneous_L1w

p-adj: 0.000878 | .=-0.32 (Pequeño) | .%: -45.6



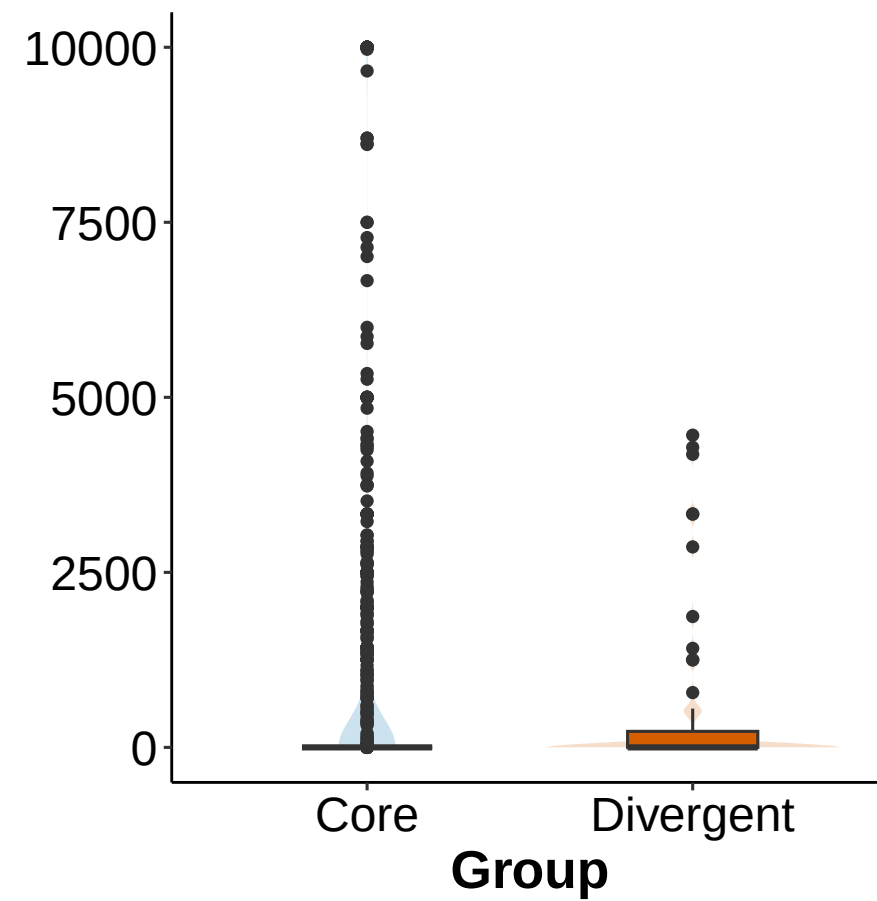
SA_Miscellaneous_pFBA

p-adj: 0.537 | .=-0.08 (Negligible) | .%: -11.7



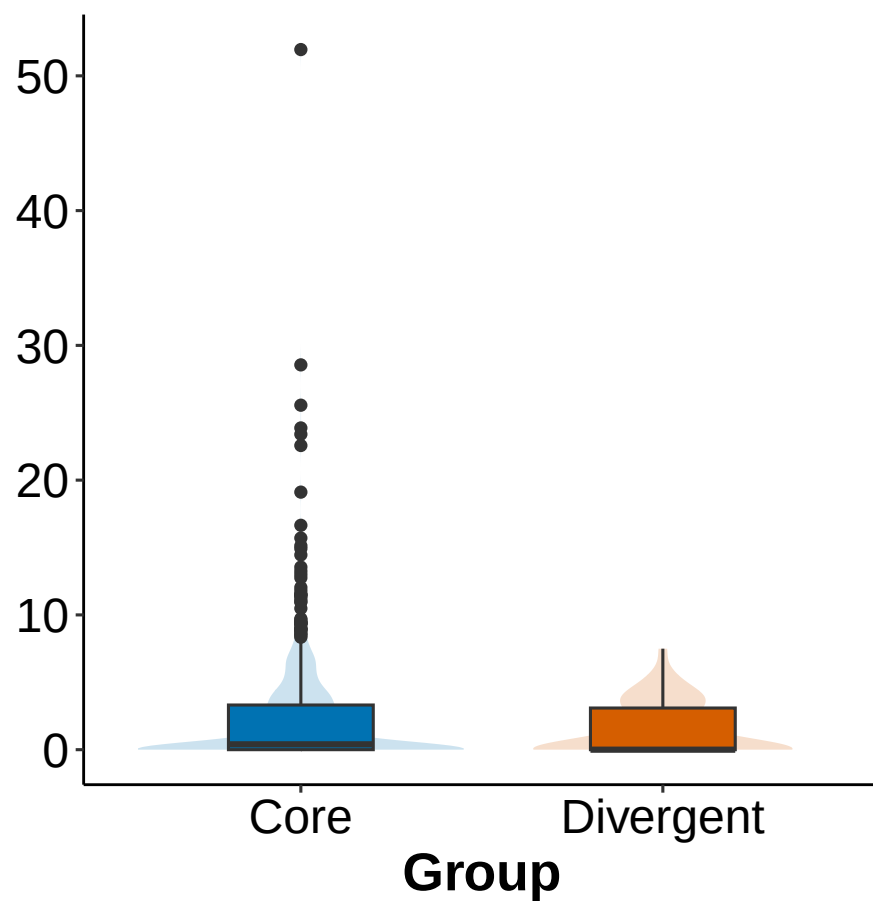
SA_NAD_metabolism_FBA

p-adj: 0.719 | .=0.04 (Negligible)



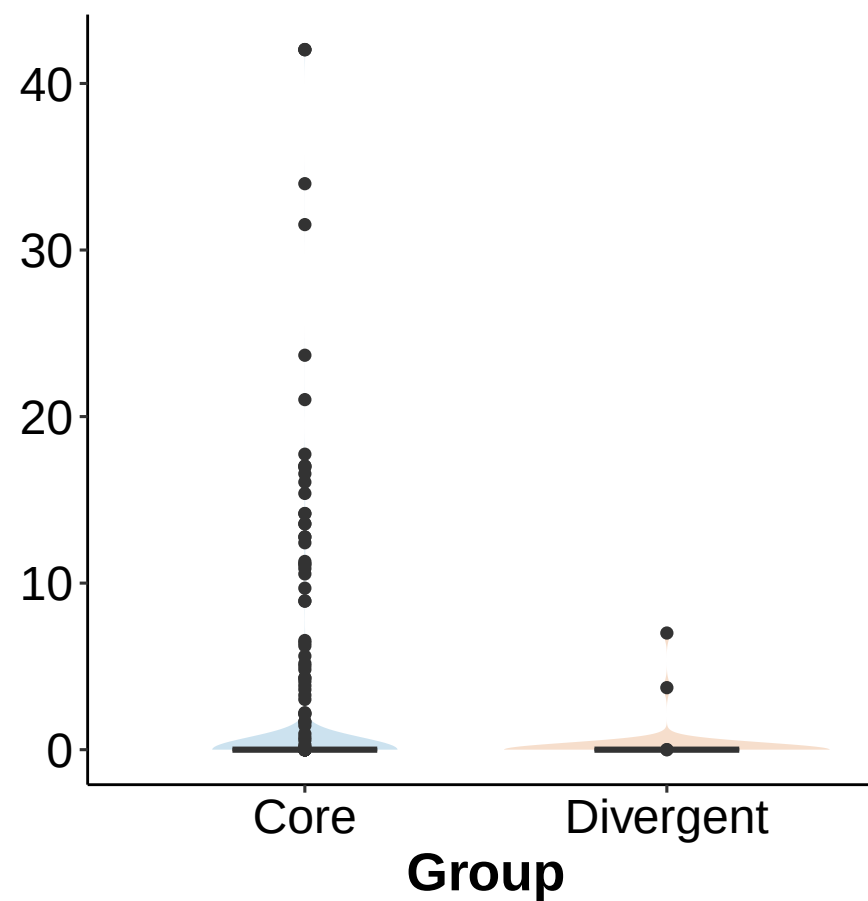
SA_NAD_metabolism_L1w

p-adj: 0.839 | .=-0.03 (Negligible) | .%: -99.8



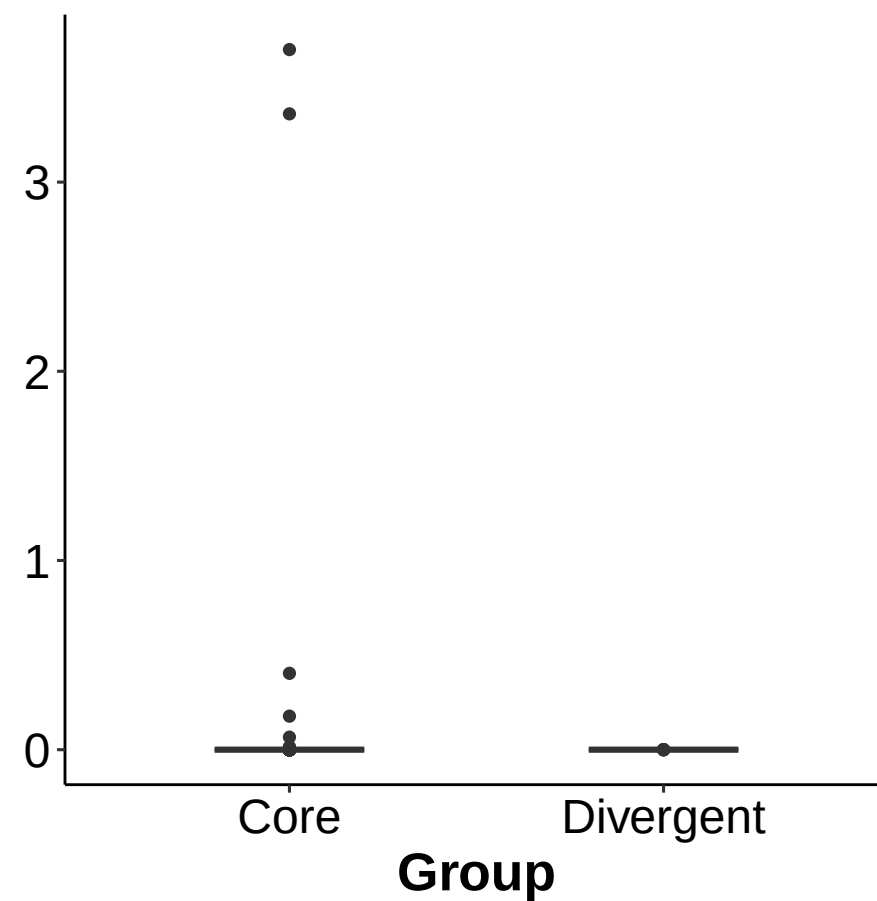
SA_NAD_metabolism_pFBA

p-adj: 0.663 | .=-0.03 (Negligible)



SA_N_glycan_synthesis_L1w

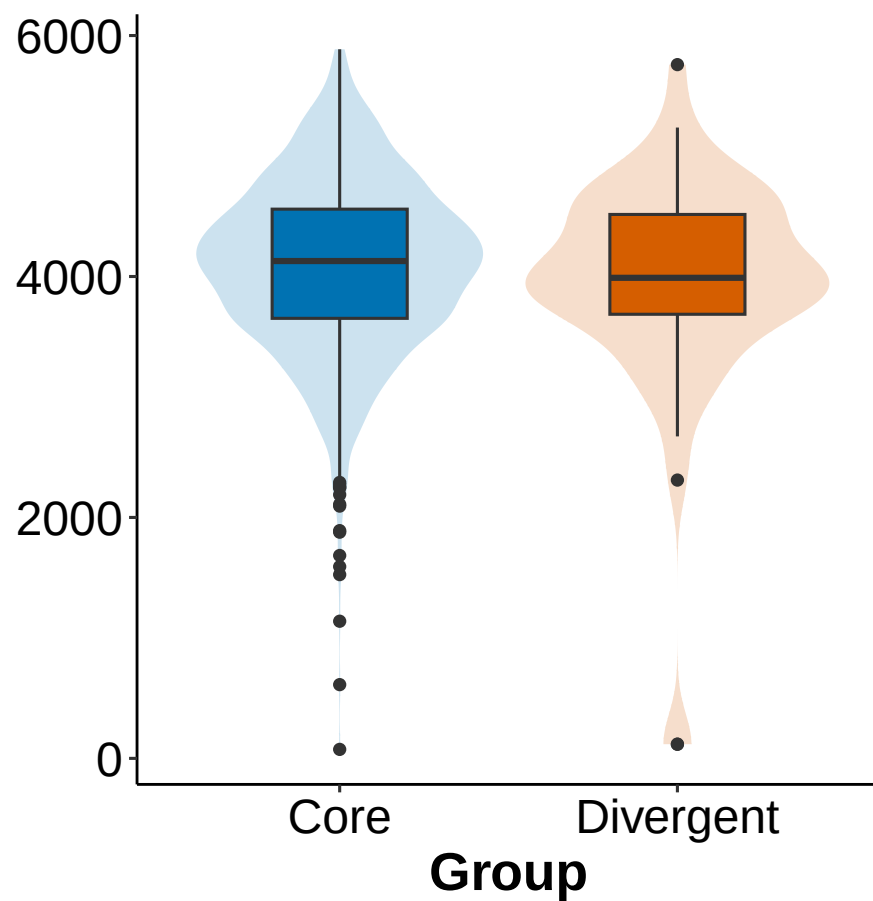
p-adj: 0.804 | .=0.02 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

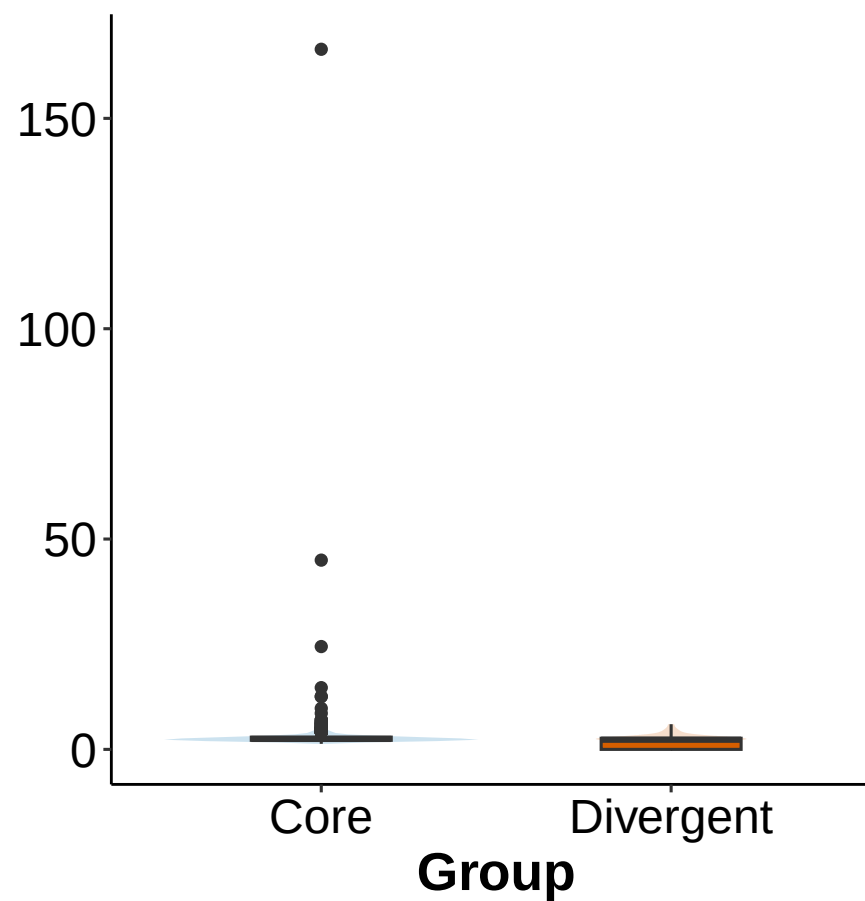
SA_Nucleotide_interconversion_FB

p-adj: 0.572 | .=-0.07 (Negligible) | .%: -3.4



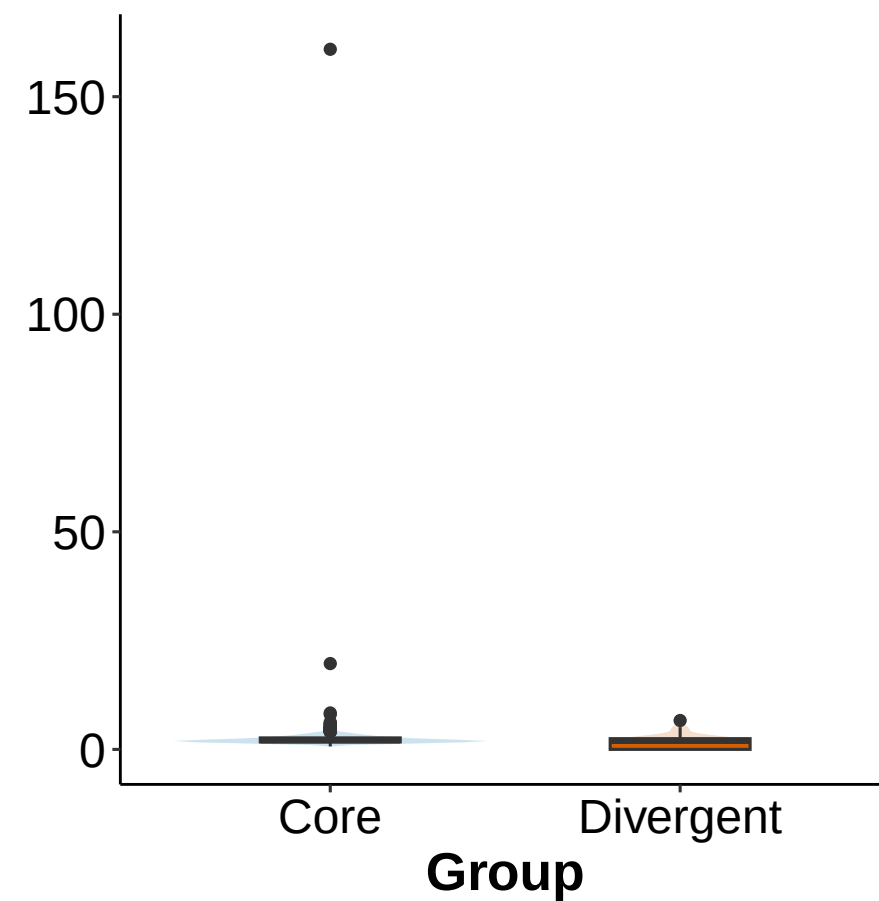
SA_Nucleotide_interconversion_L1w

p-adj: 0.017 | .=-0.23 (Pequeño) | .%: -7.7



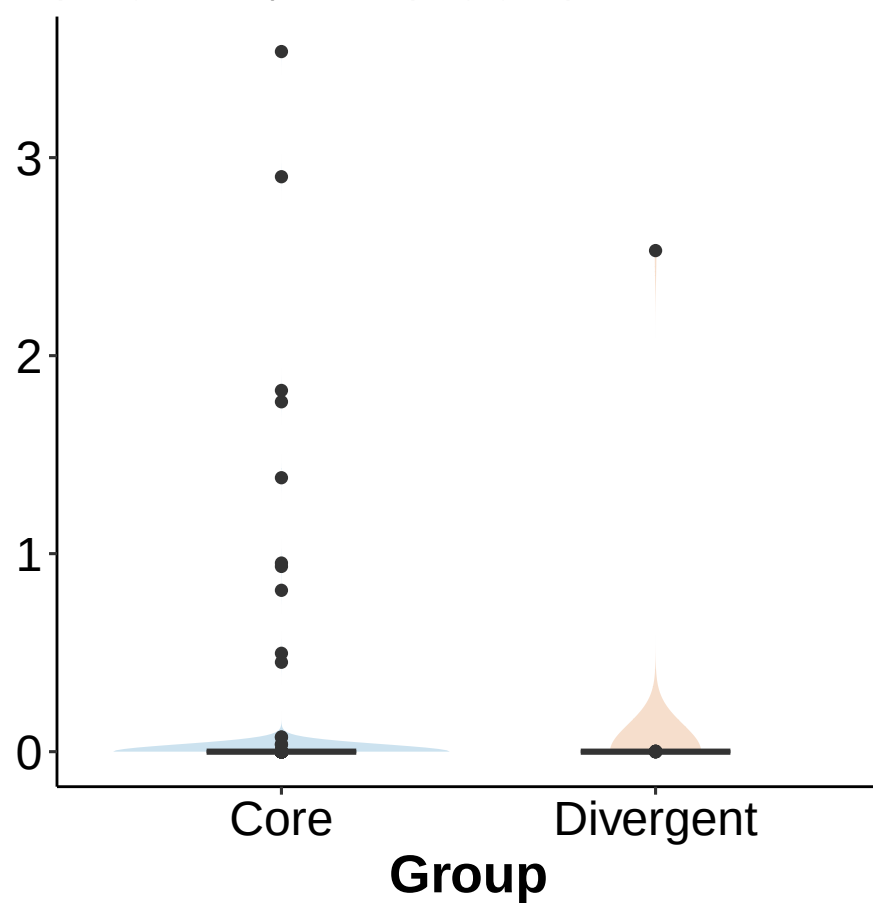
SA_Nucleotide_interconversion_pFB

p-adj: 0.0525 | .=-0.19 (Pequeño) | .%: -5.5



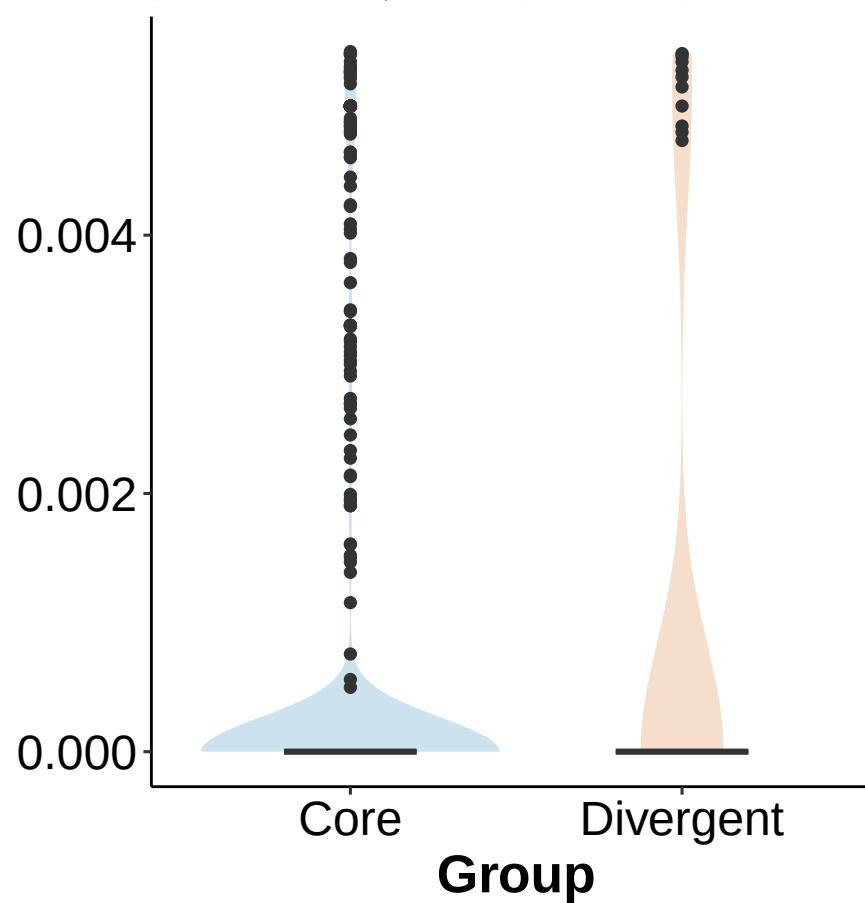
SA_Nucleotide_metabolism_L1w

p-adj: 0.145 | .=0.09 (Negligible)



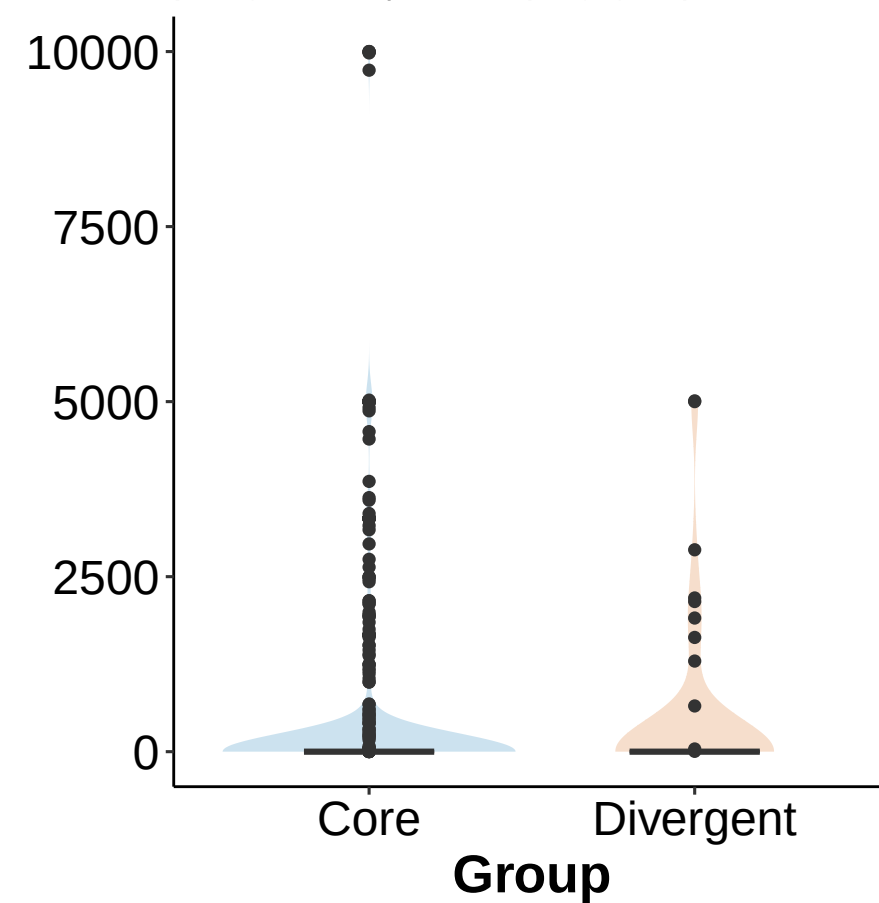
SA_Nucleotide_metabolism_pFBA

p-adj: 0.0162 | .=0.12 (Negligible)



SA_Nucleotide_salvage_pathway_pFBA

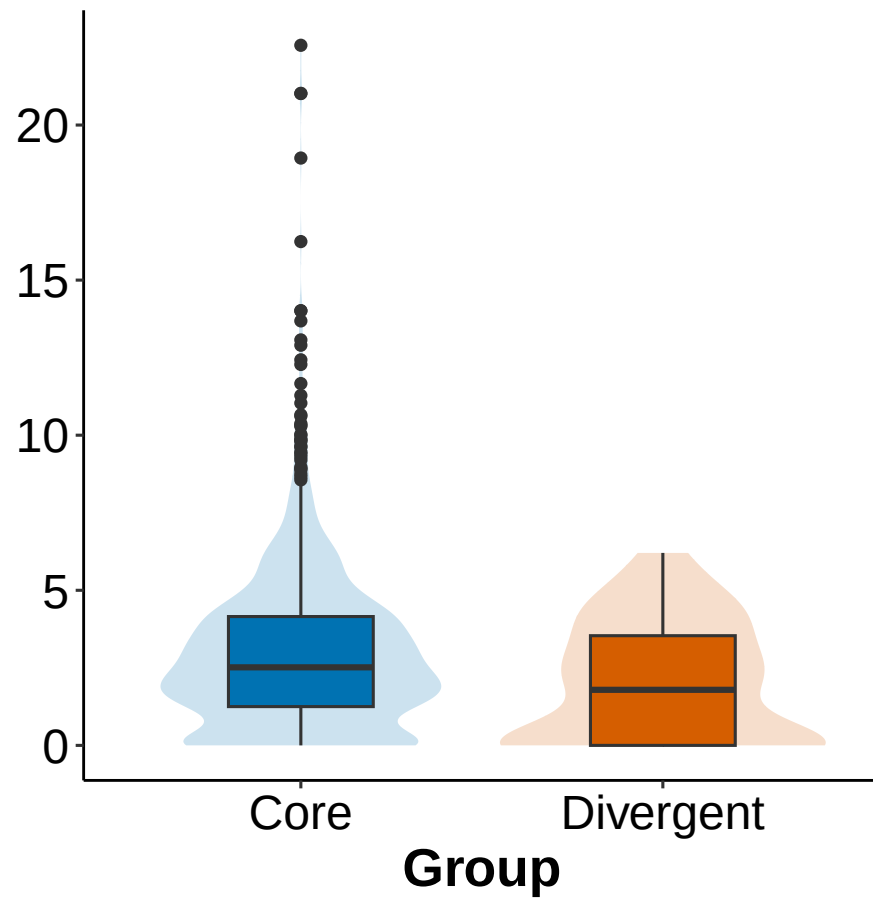
p-adj: 0.917 | .=0.01 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

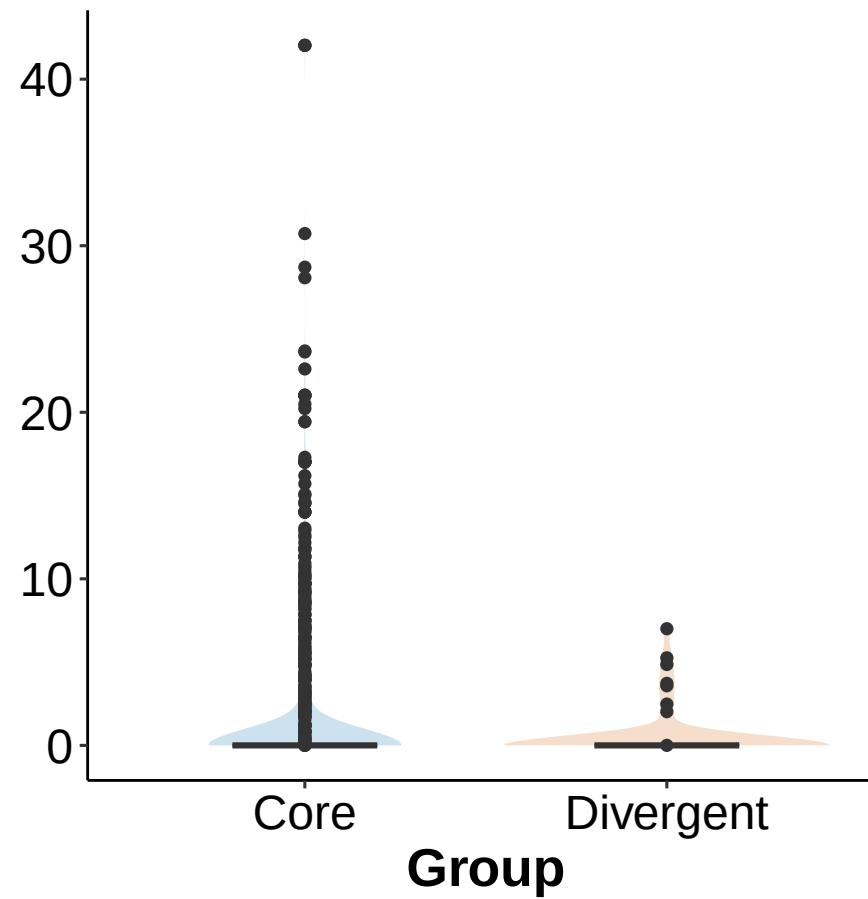
SA_Nucleotide_salvage_pathway_L1v

p-adj: 0.05 | .=-0.19 (Pequeño) | .%: -28.8



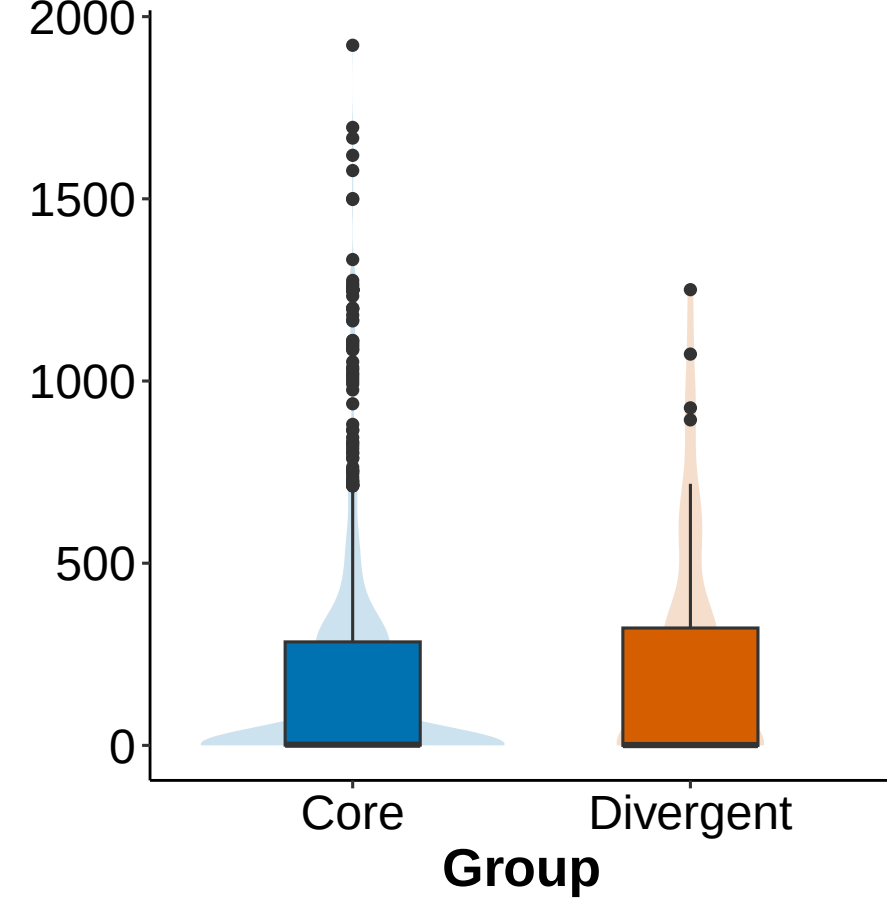
SA_Nucleotide_salvage_pathway_pFE

p-adj: 0.616 | .=-0.05 (Negligible)



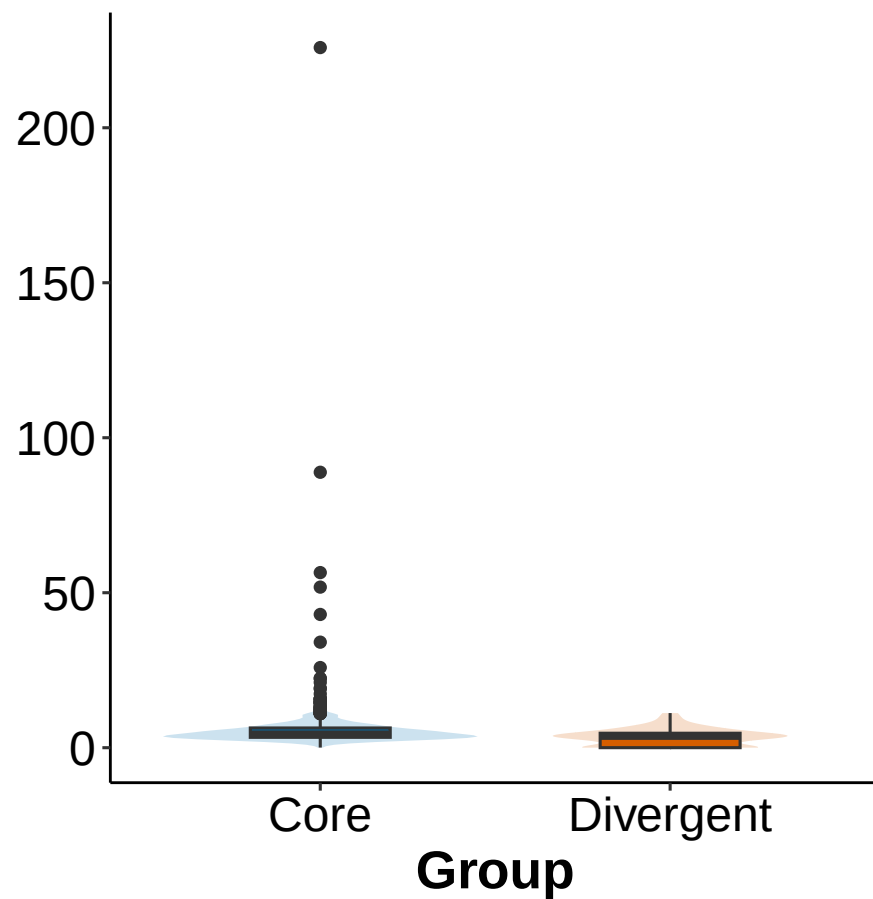
SA_Oxidative_phosphorylation_FB

p-adj: 0.829 | .=-0.03 (Negligible) | .%: -52.7



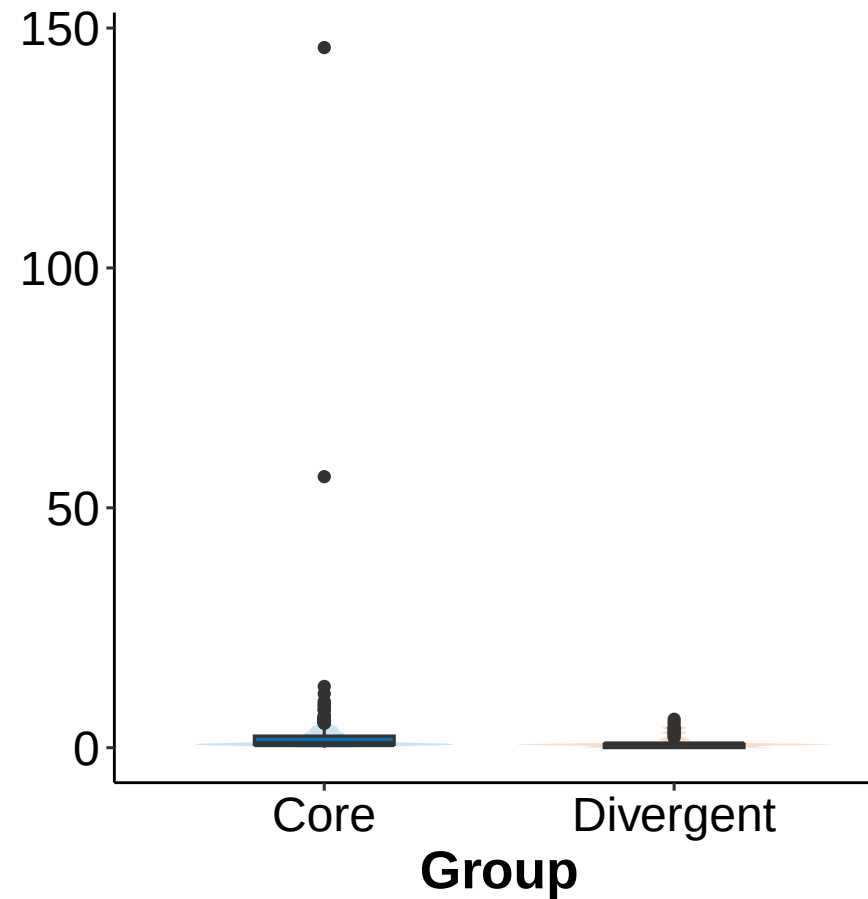
SA_Oxidative_phosphorylation_L1w

p-adj: 0.000162 | .=-0.36 (Mediano) | .%: -22.8



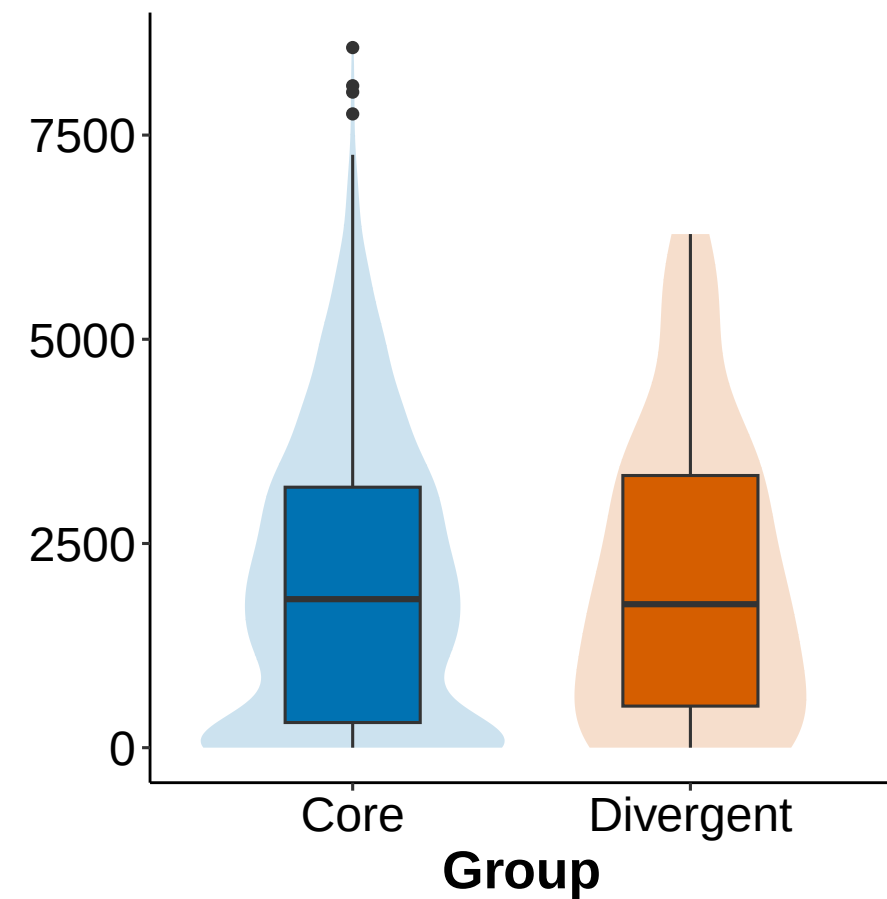
SA_Oxidative_phosphorylation_pFB

p-adj: 0.000264 | .=-0.35 (Mediano) | .%: -12.5



SA_Pentose_phosphate_pathway_L1w

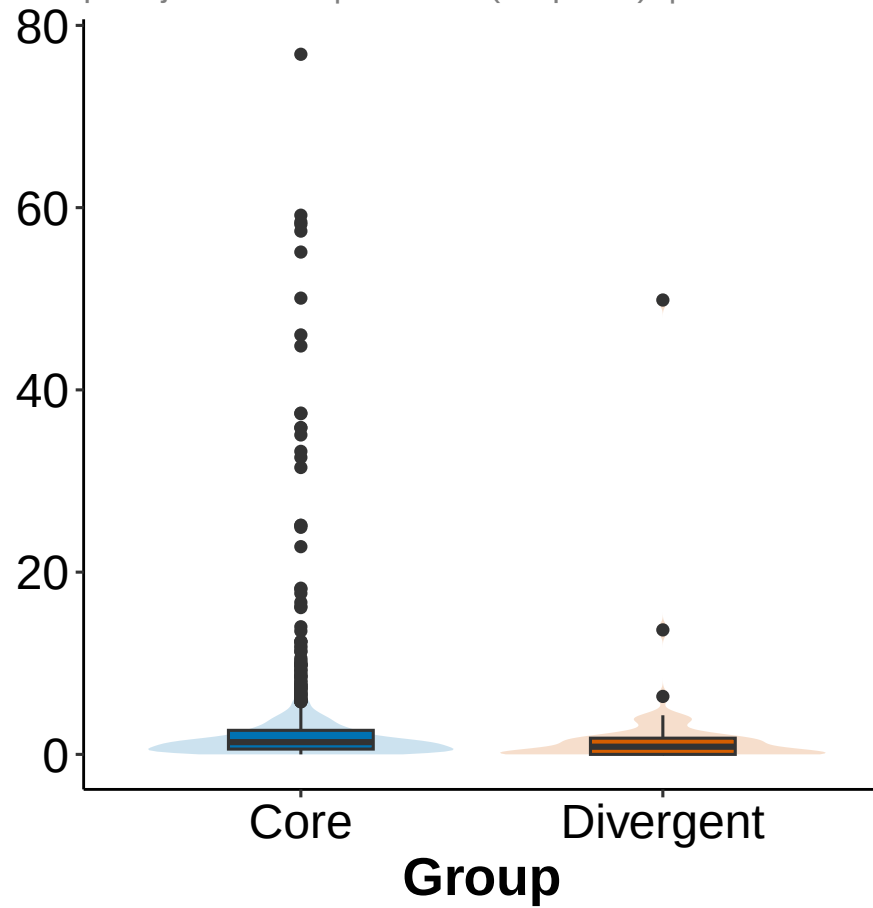
p-adj: 0.969 | .=0 (Negligible) | .%: -3.4



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

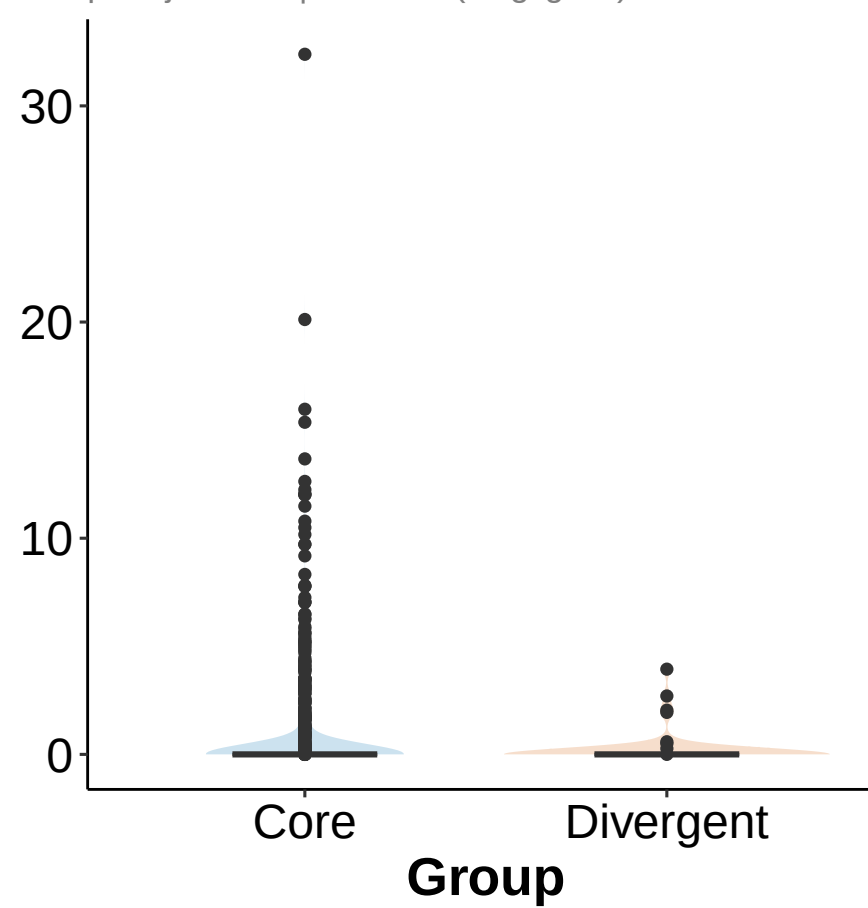
SA_Pentose_phosphate_pathway_L1w

p-adj: 0.00891 | .=-0.25 (Pequeño) | .%: -37.4



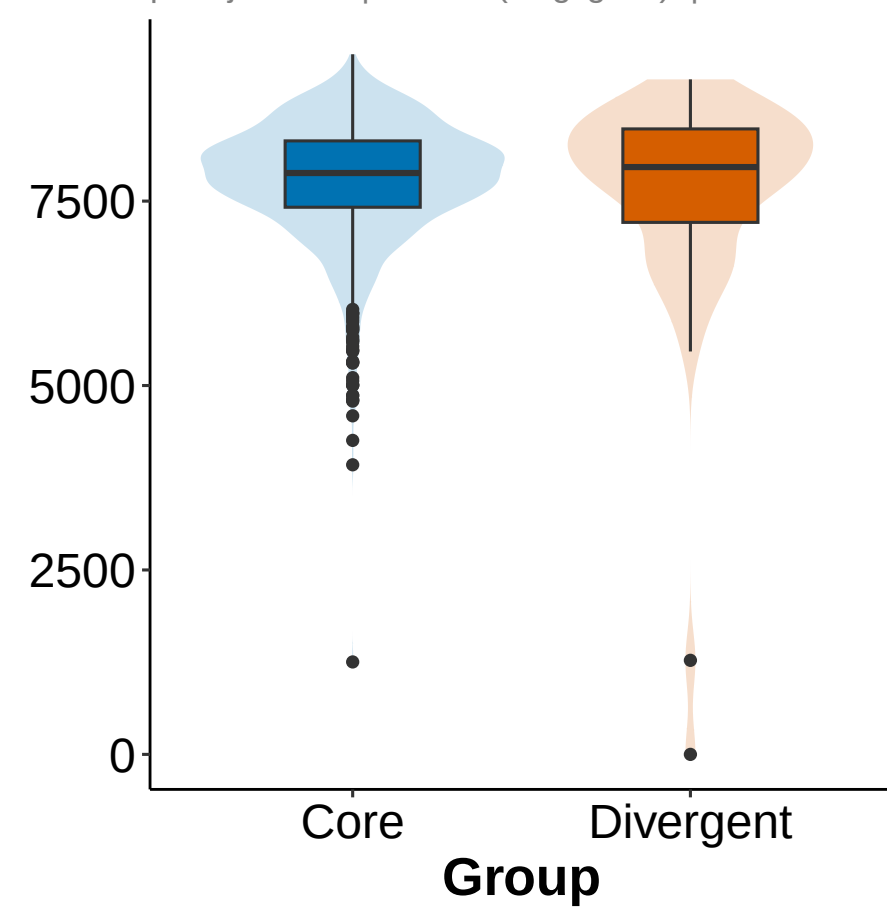
SA_Pentose_phosphate_pathway_pFI

p-adj: 0.618 | .=-0.05 (Negligible)



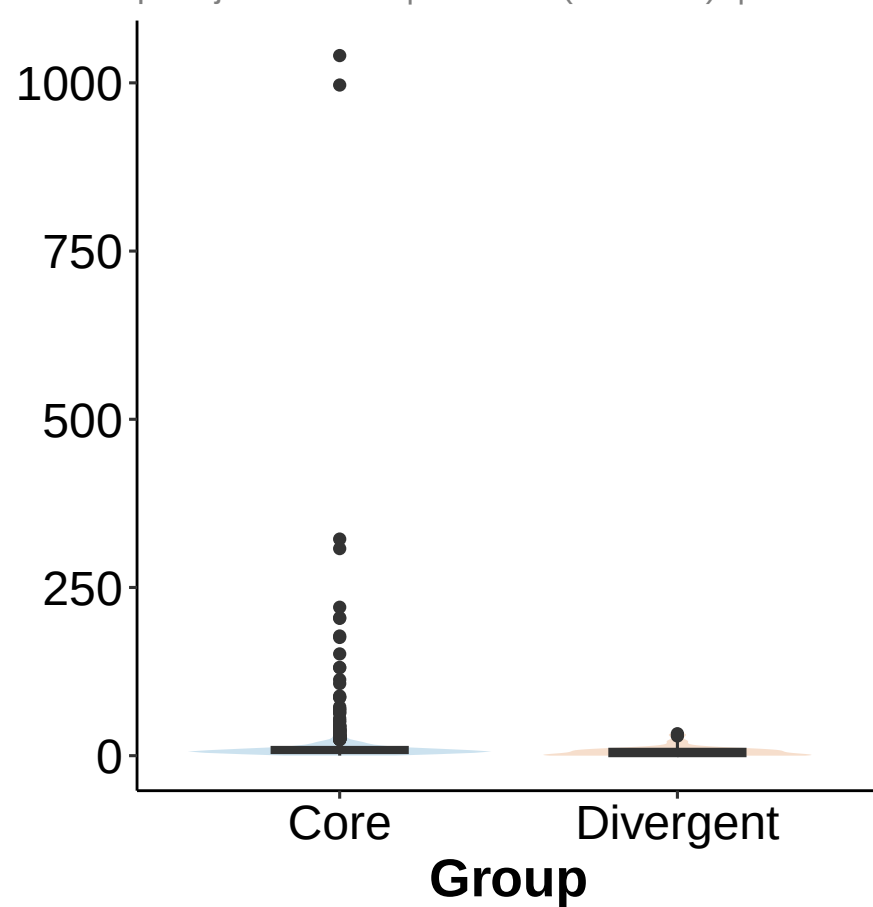
SA_Peptide_metabolism_FBA

p-adj: 0.838 | .=-0.03 (Negligible) | .%: 1



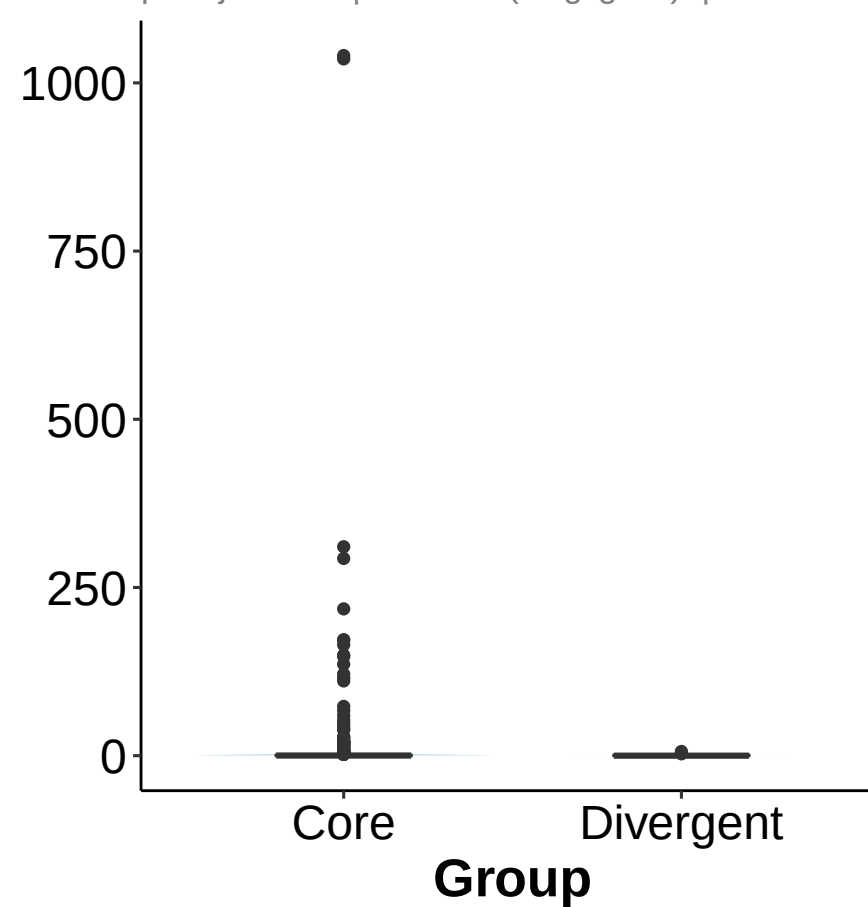
SA_Peptide_metabolism_L1w

p-adj: 0.000406 | .=-0.34 (Mediano) | .%: -39



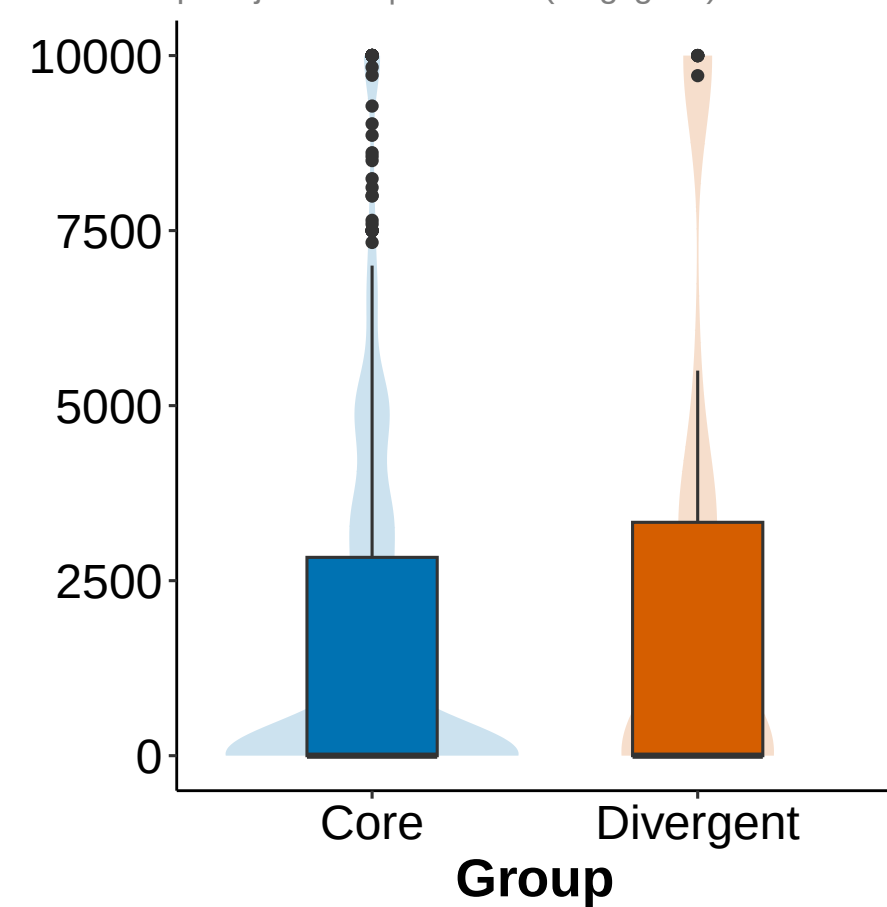
SA_Peptide_metabolism_pFBA

p-adj: 0.537 | .=-0.08 (Negligible) | .%: -99.9



SA_Phenylalanine_metabolism_FBA

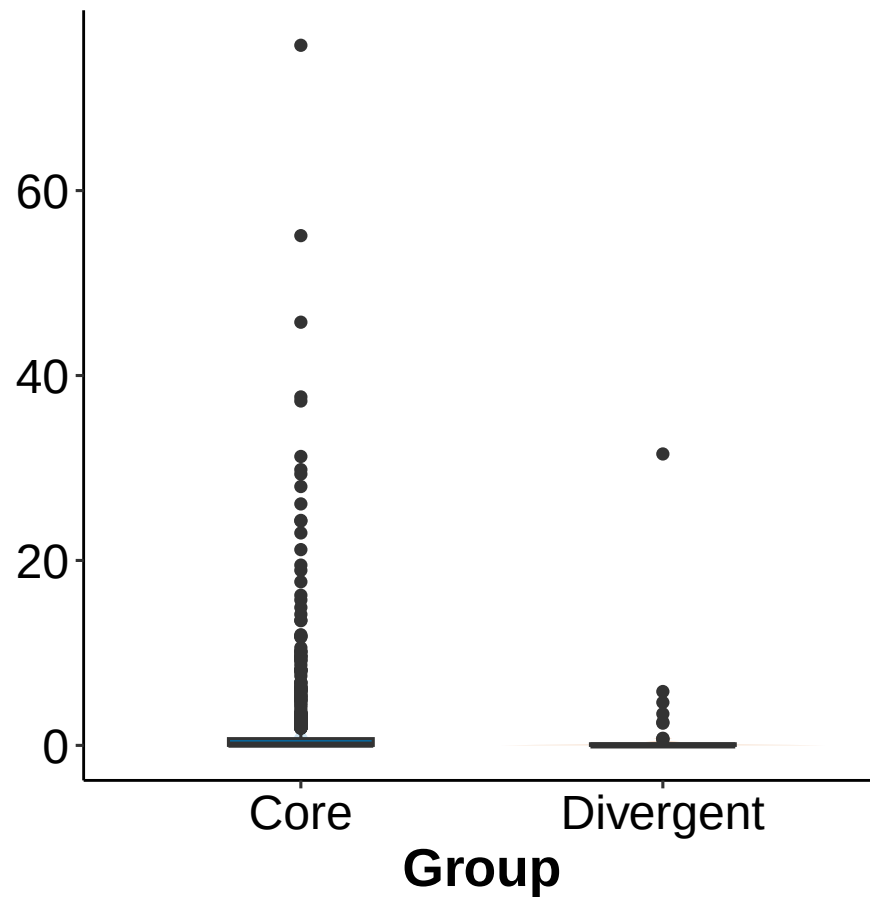
p-adj: 0.902 | .=-0.02 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

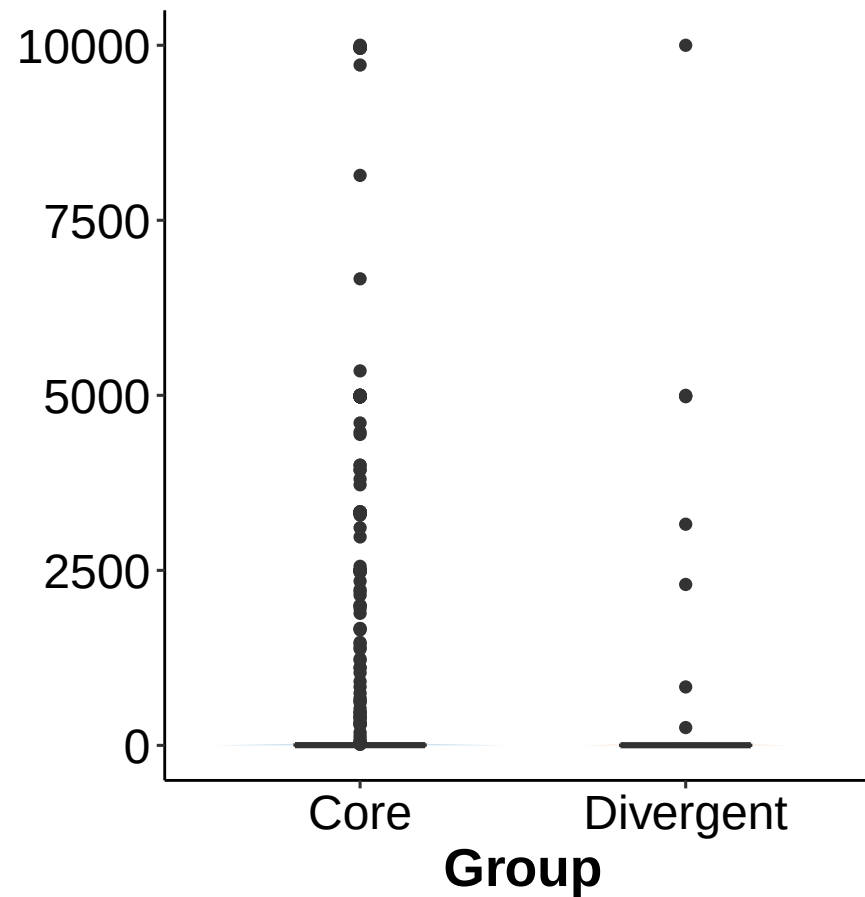
SA_Phenylalanine_metabolism_L1w

p-adj: 0.0654 | .=-0.18 (Pequeño) | .%: -99.9



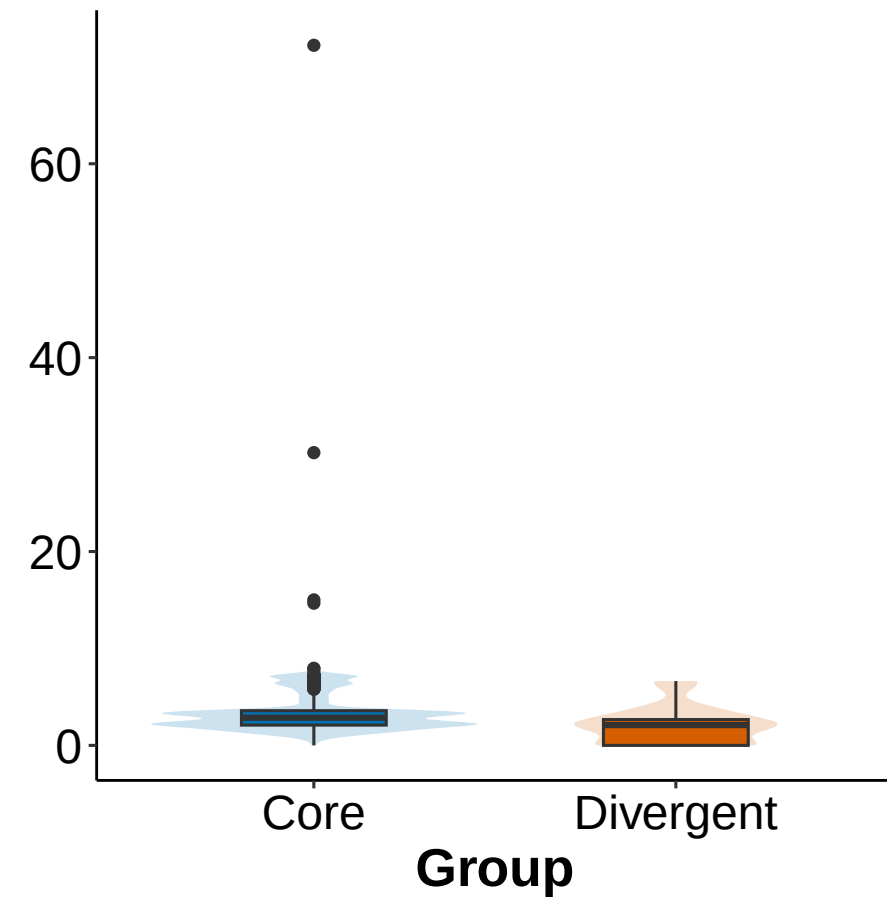
SA_Phosphatidylinositol_phosphatidylcholine

p-adj: 0.00226 | .=-0.28 (Pequeño) | .%: -33



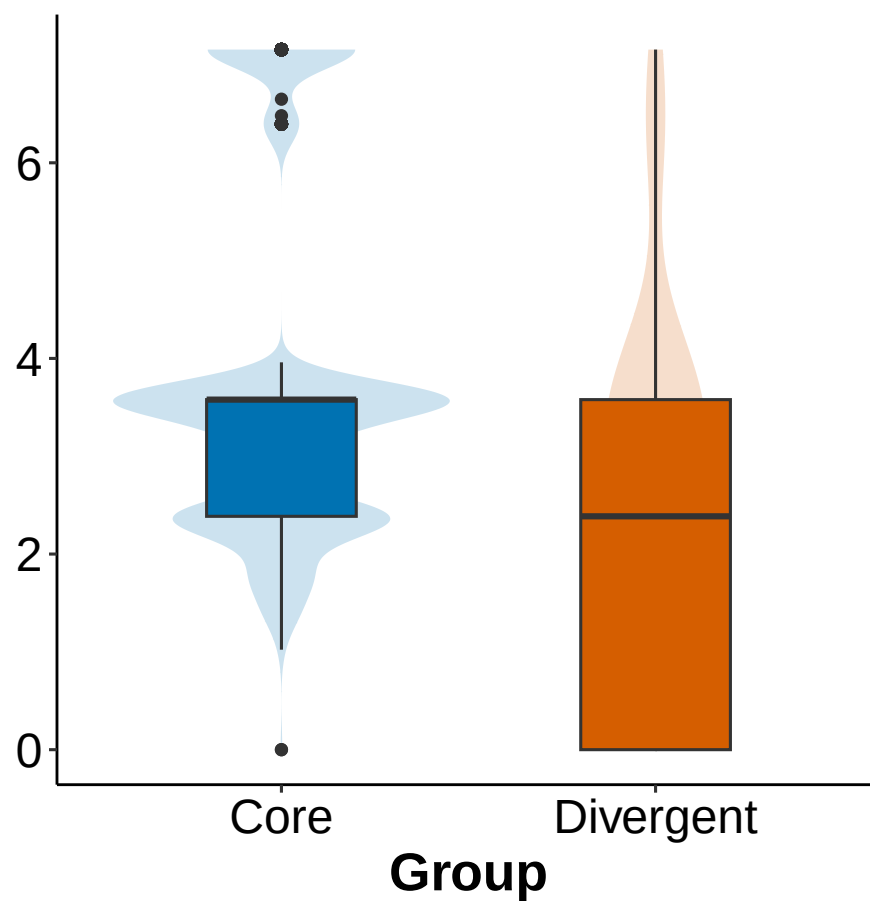
SA_Phosphatidylinositol_phosphate_n

p-adj: 0.000103 | .=-0.37 (Mediano) | .%: -26.7



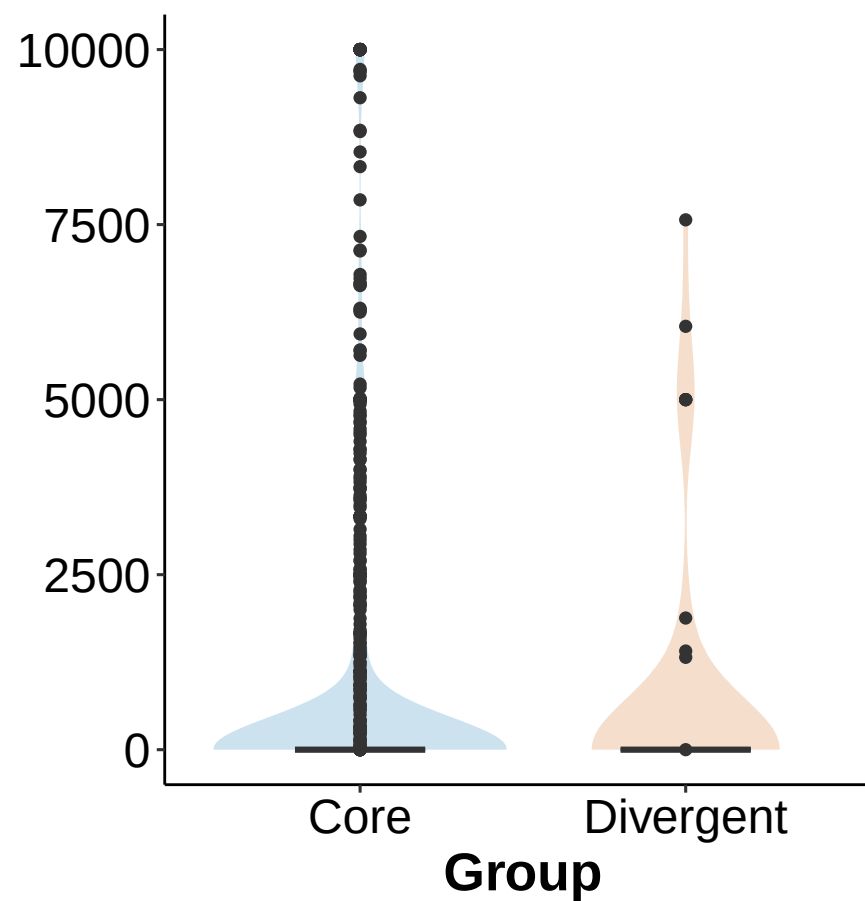
SA_Phosphatidylinositol_phosphate_n

p-adj: 0.00128 | .=-0.3 (Pequeño) | .%: -33.3



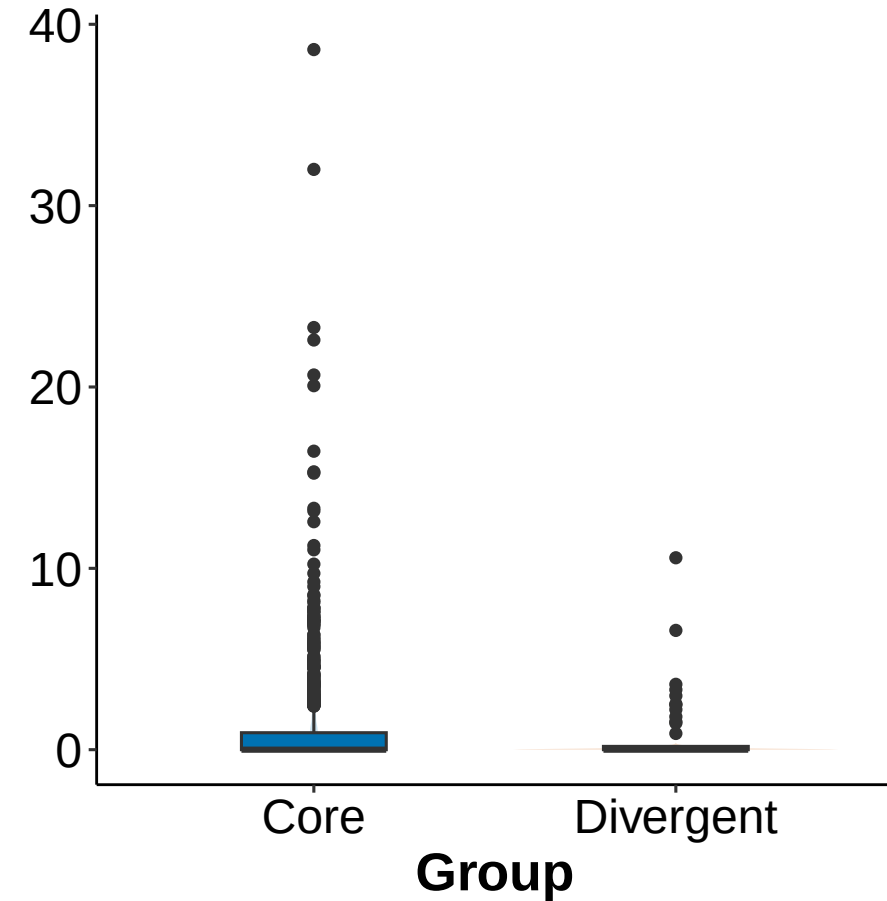
SA_Propanoate_metabolism_FBA

p-adj: 0.746 | .=-0.03 (Negligible)



SA_Propanoate_metabolism_L1w

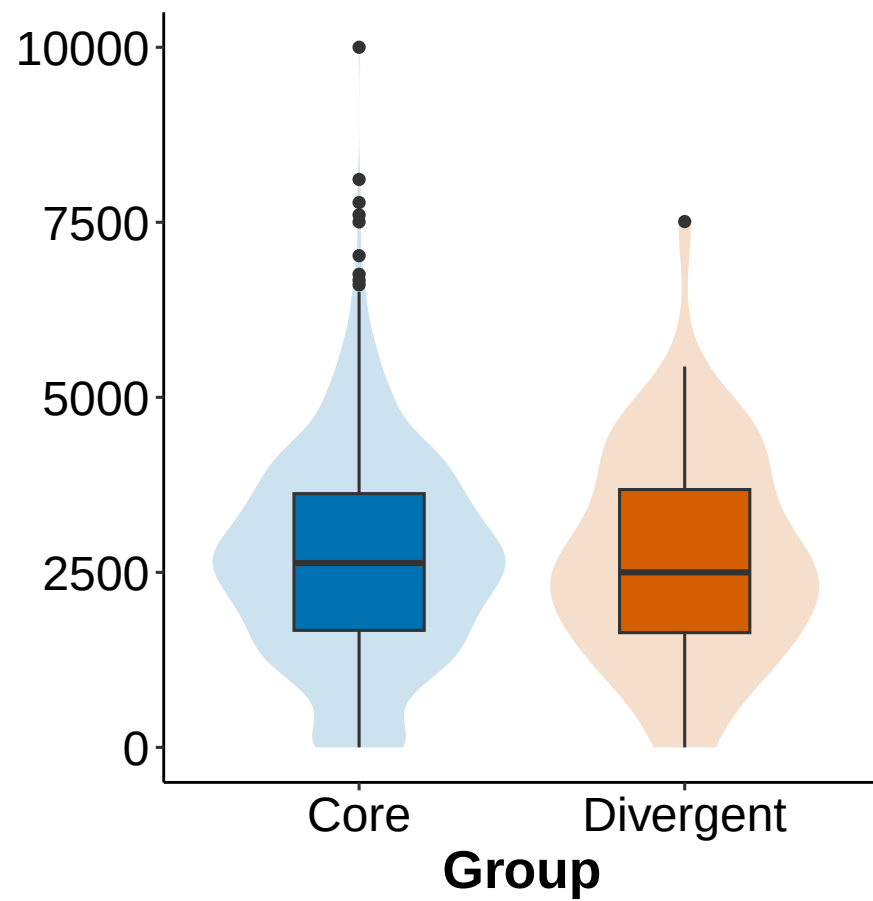
p-adj: 0.737 | .=-0.05 (Negligible) | .%: 134191076



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

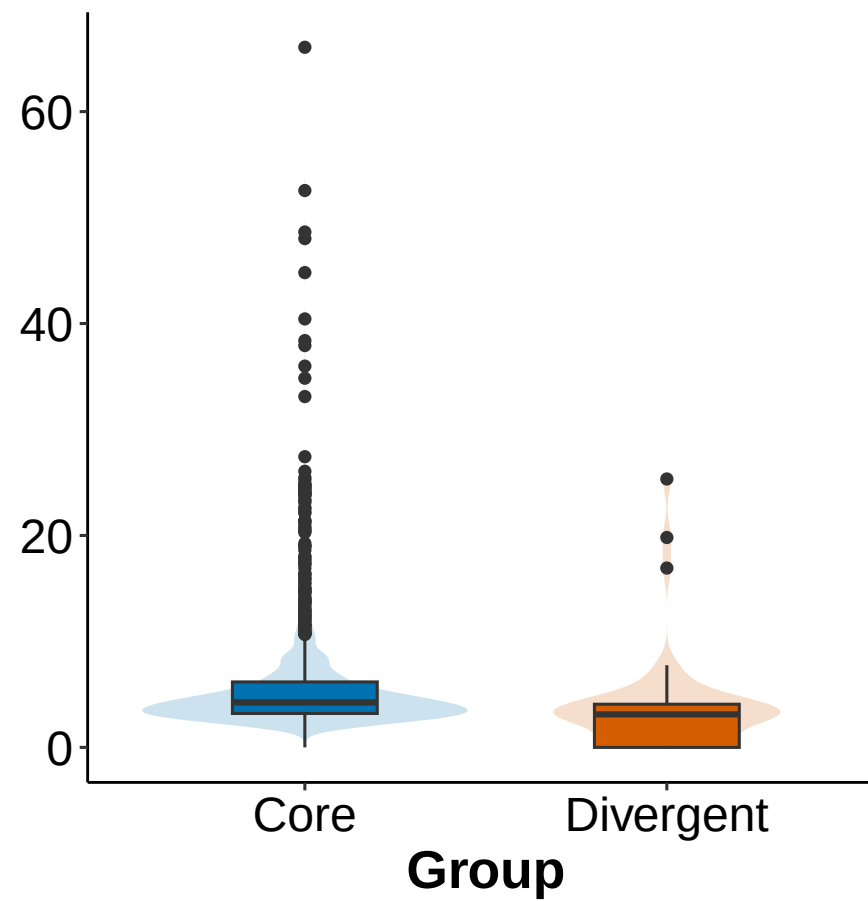
SA_Purine_catabolism_FBA

p-adj: 0.951 | .=-0.01 (Negligible) | .%: -5



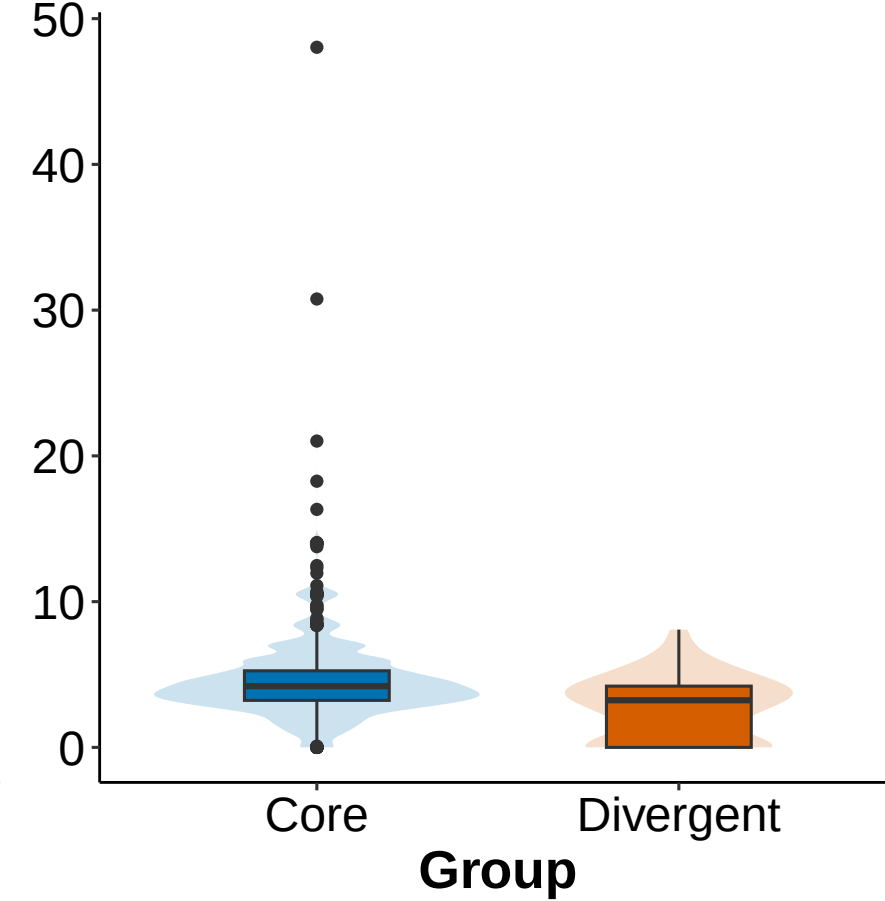
SA_Purine_catabolism_L1w

p-adj: 4.83e-06 | .=-0.42 (Mediano) | .%: -26.6



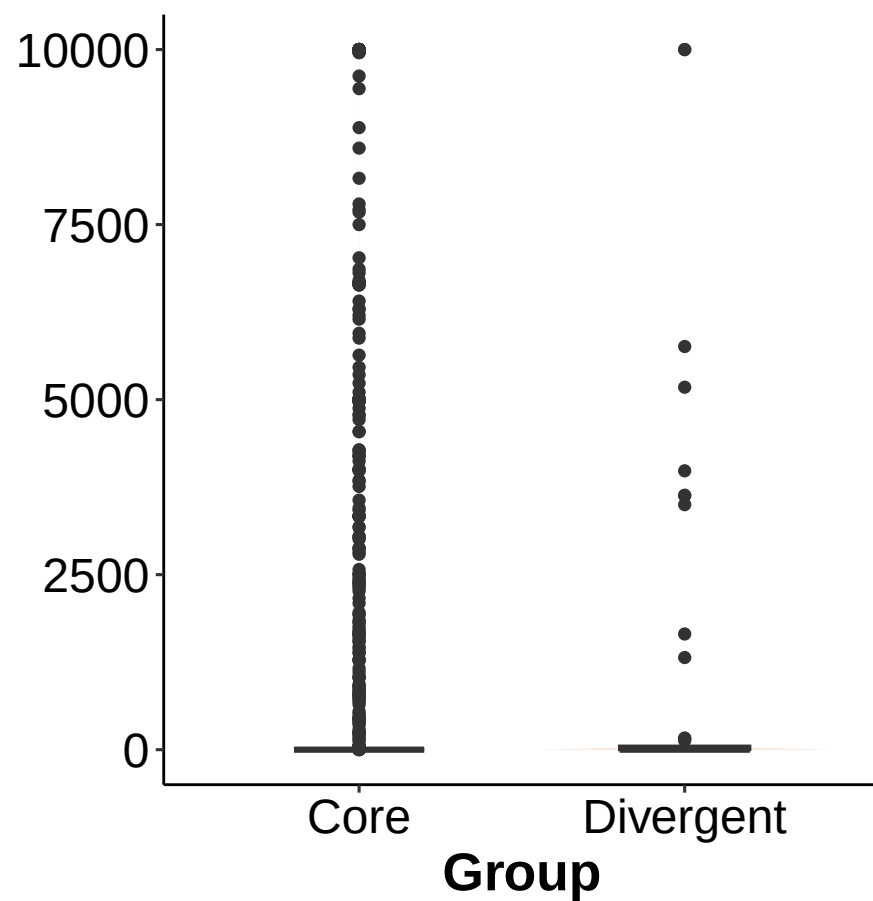
SA_Purine_catabolism_pFBA

p-adj: 3.6e-05 | .=-0.39 (Mediano) | .%: -23.1



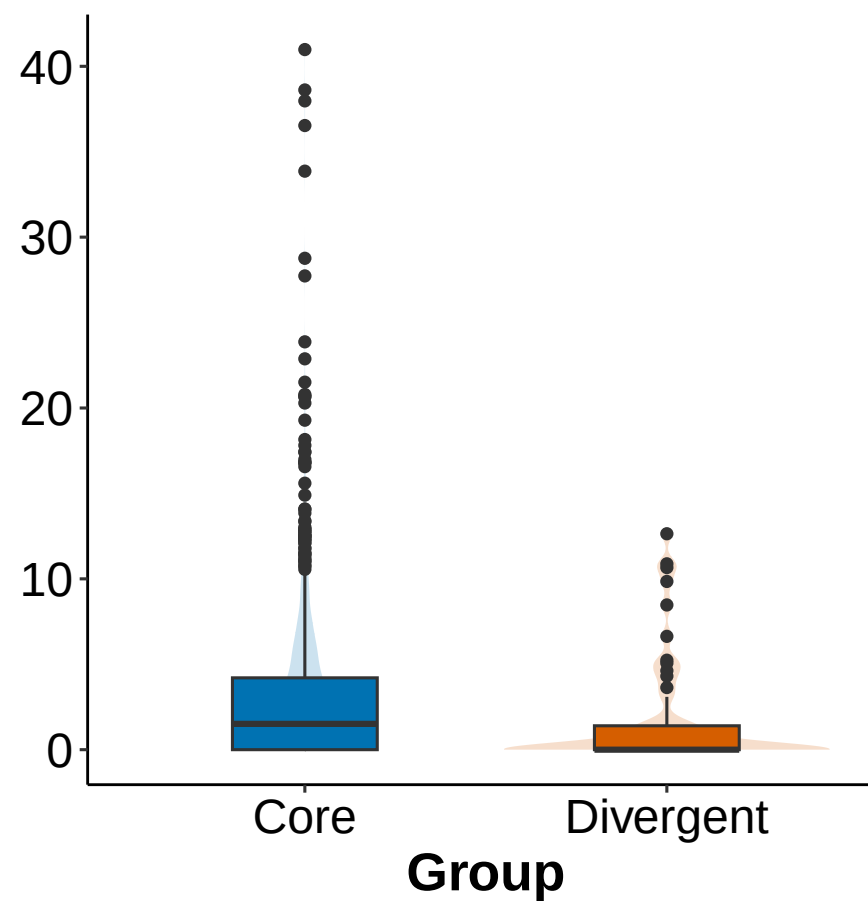
SA_Purine_synthesis_FBA

p-adj: 0.688 | .=-0.04 (Negligible)



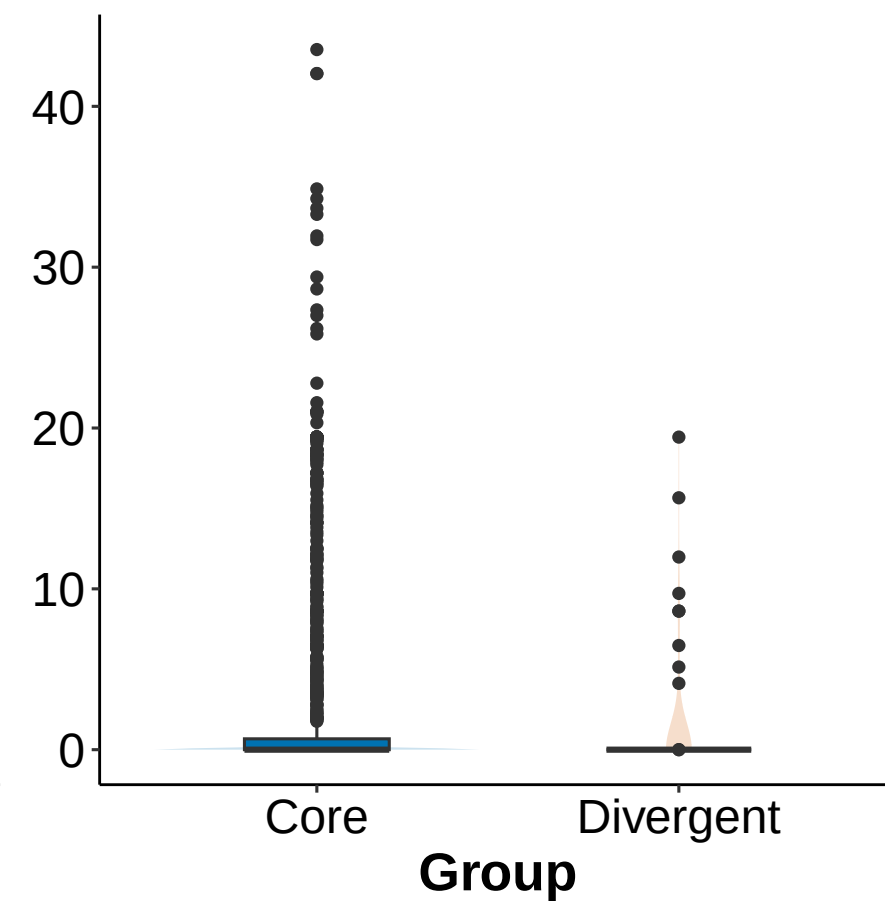
SA_Purine_synthesis_L1w

p-adj: 0.0544 | .=-0.19 (Pequeño) | .%: -100



SA_Purine_synthesis_pFBA

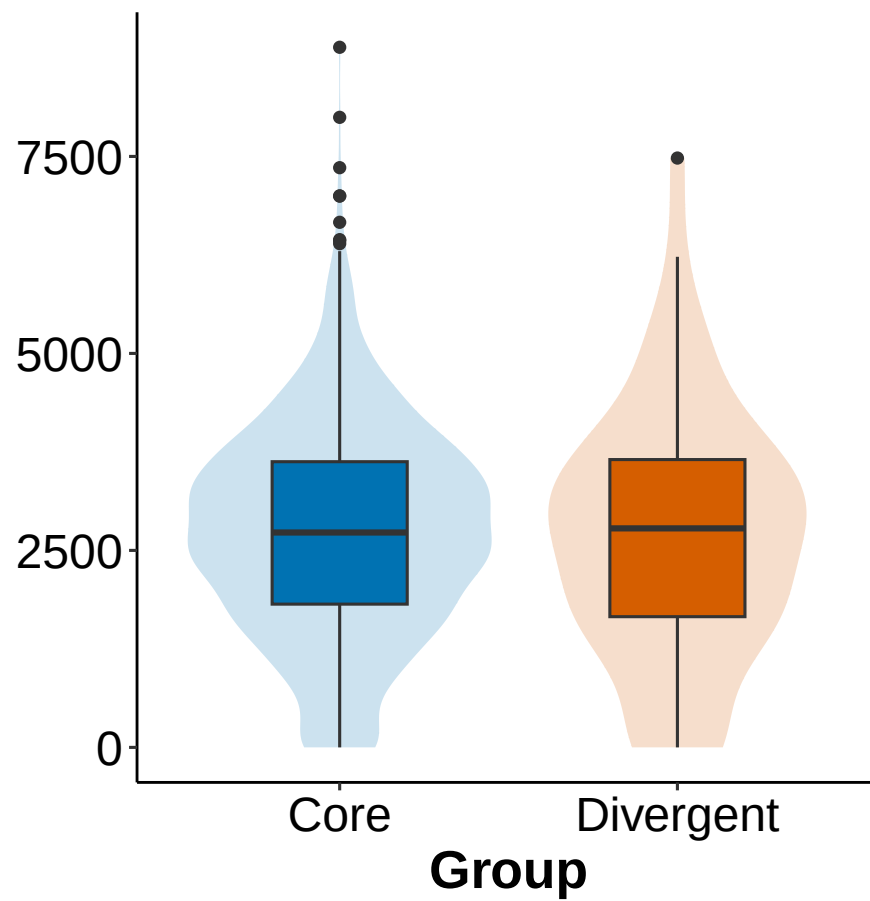
p-adj: 0.359 | .=-0.08 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

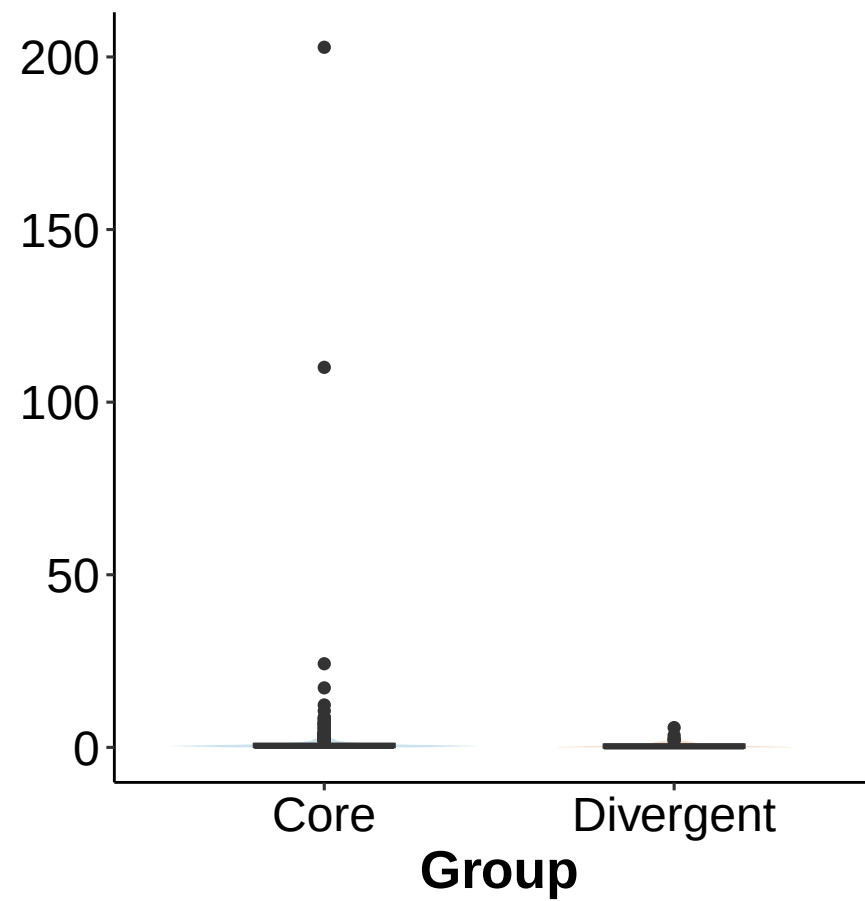
SA_Pyrimidine_catabolism_FBA

p-adj: 0.789 | .=-0.04 (Negligible) | .%: 1.9



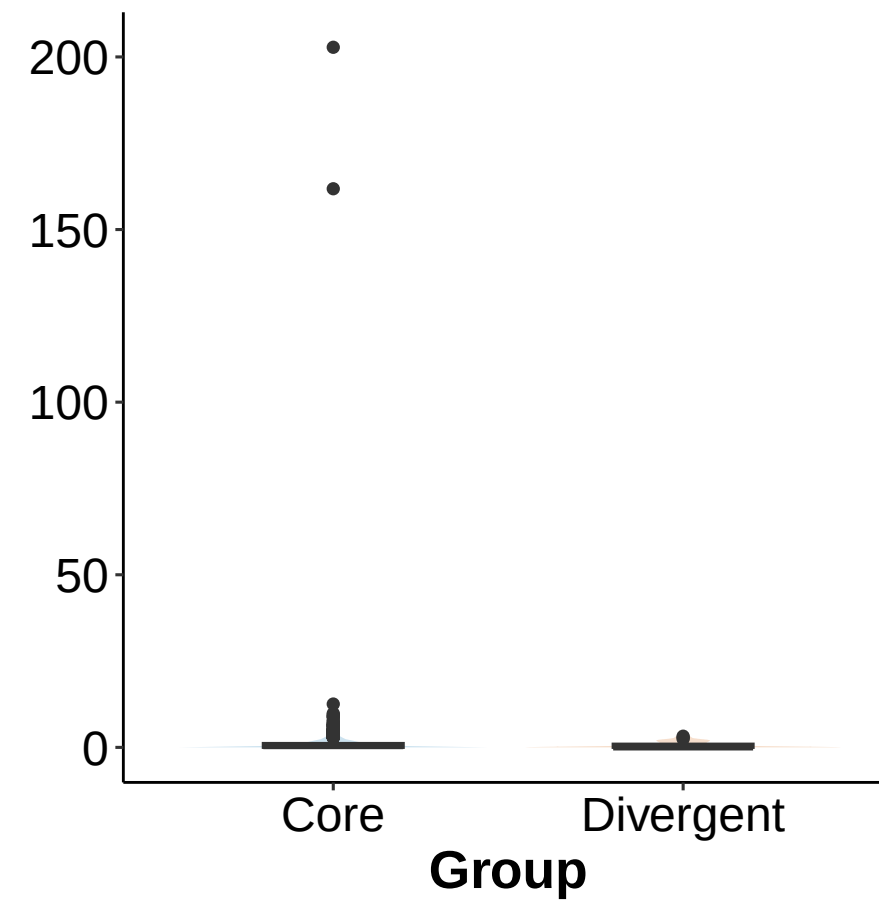
SA_Pyrimidine_catabolism_L1w

p-adj: 0.0278 | .=-0.21 (Pequeño) | .%: -35.9



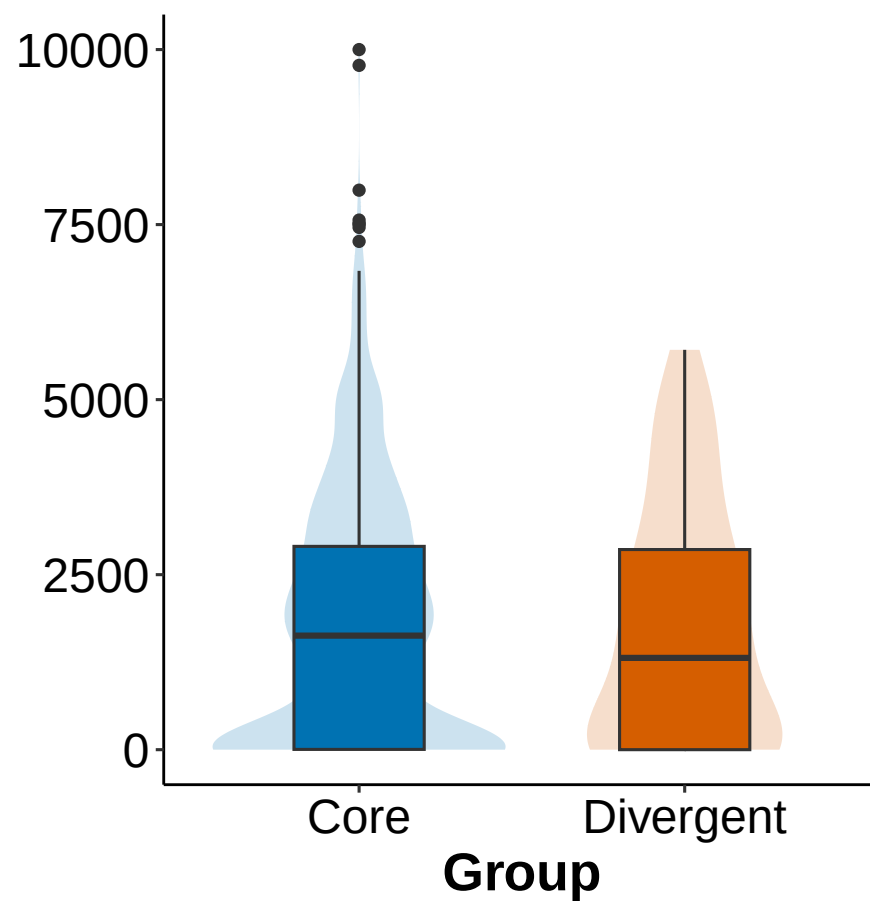
SA_Pyrimidine_catabolism_pFBA

p-adj: 0.227 | .=-0.12 (Negligible) | .%: -100



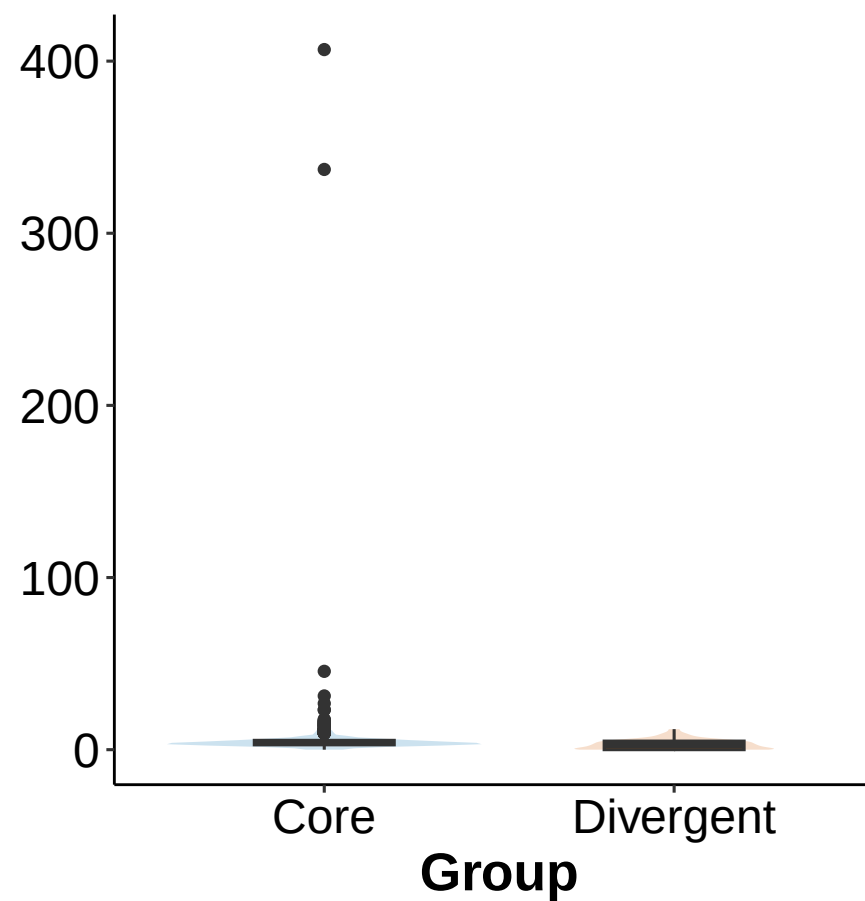
SA_Pyrimidine_synthesis_FBA

p-adj: 0.838 | .=-0.03 (Negligible) | .%: -19.5



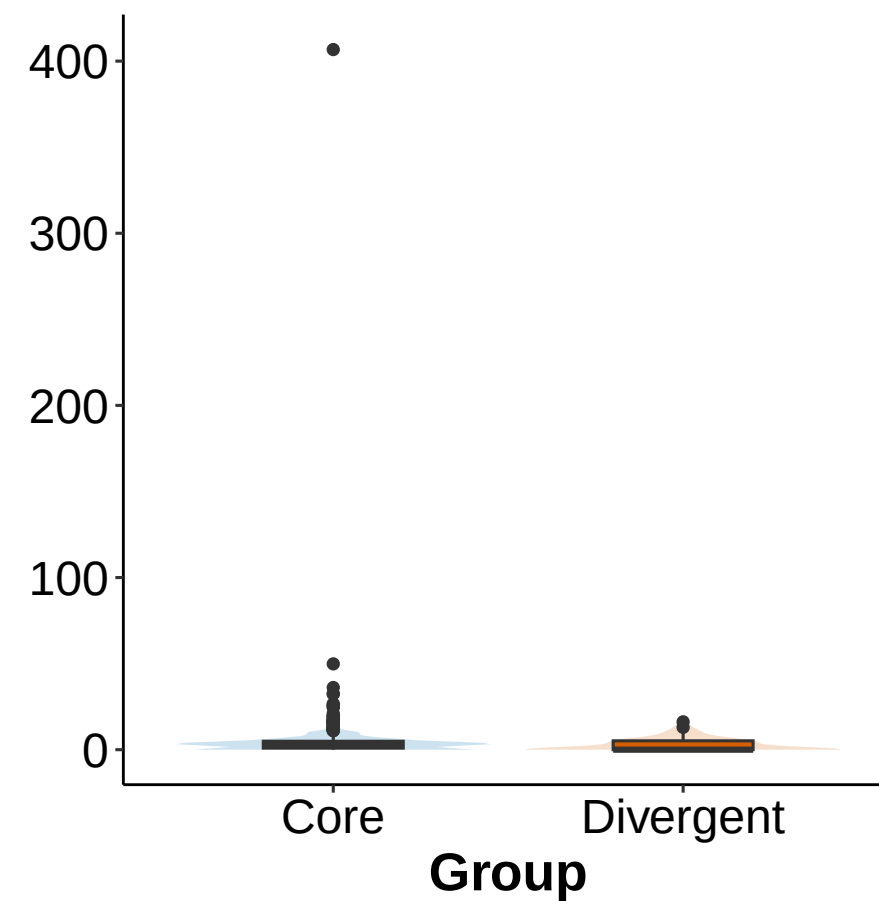
SA_Pyrimidine_synthesis_L1w

p-adj: 0.0011 | .=-0.31 (Pequeño) | .%: -31.7



SA_Pyrimidine_synthesis_pFBA

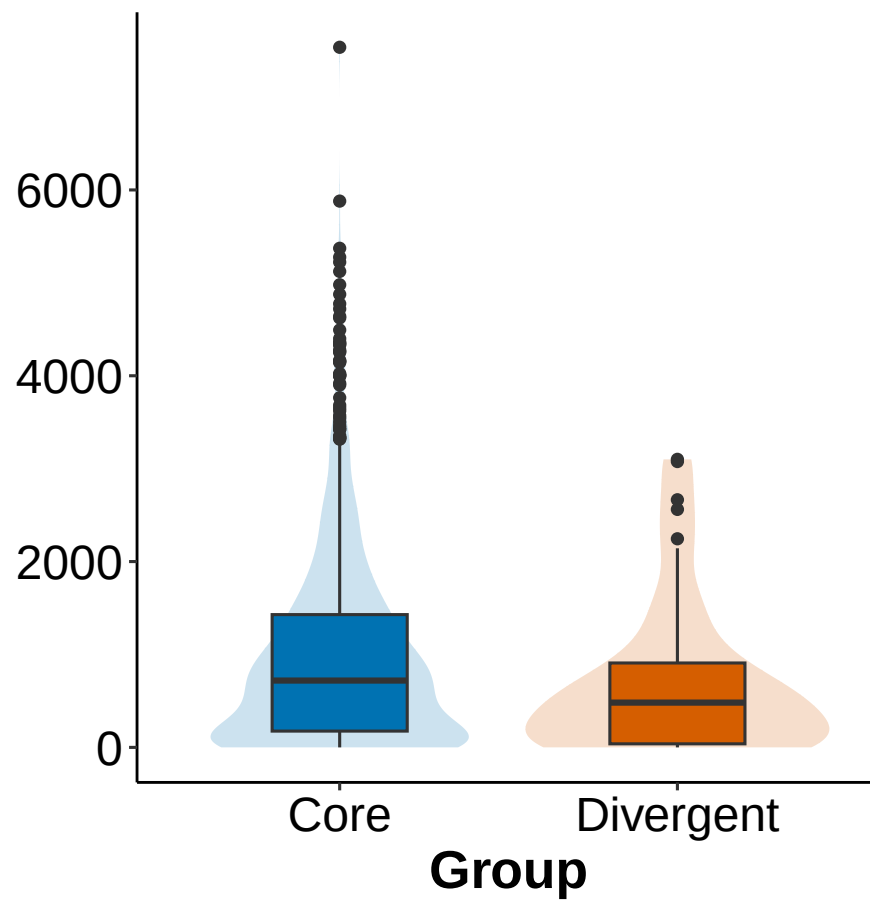
p-adj: 0.0354 | .=-0.2 (Pequeño) | .%: -100



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

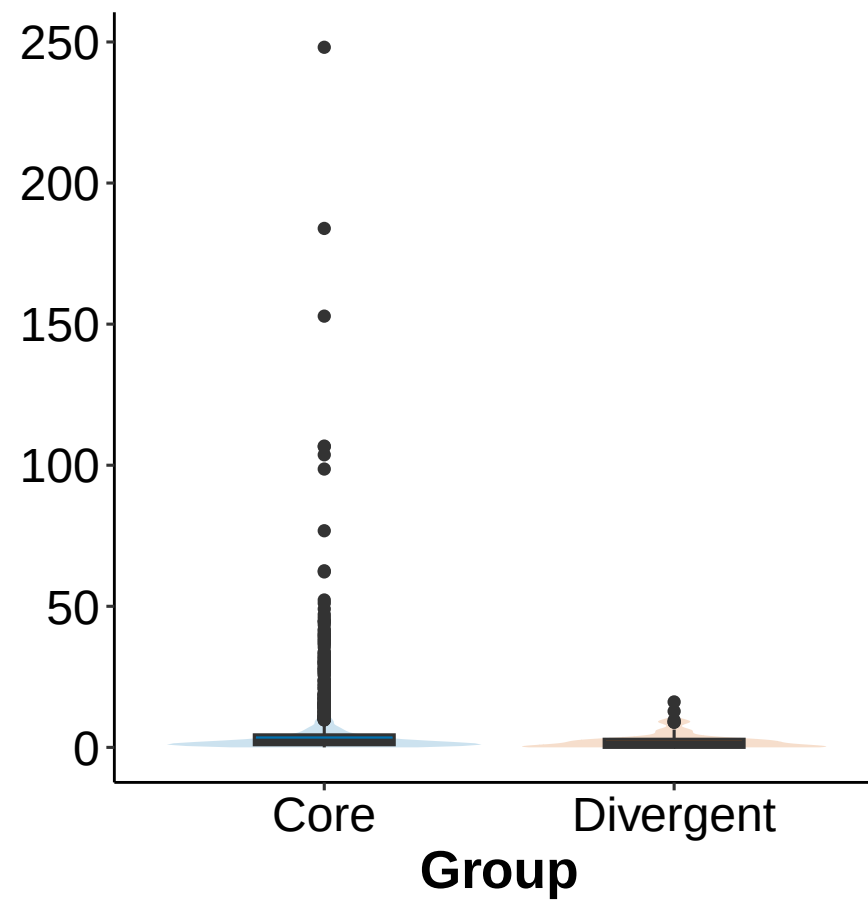
SA_Pyruvate_metabolism_FBA

p-adj: 0.112 | .=-0.16 (Pequeño) | .%: -33



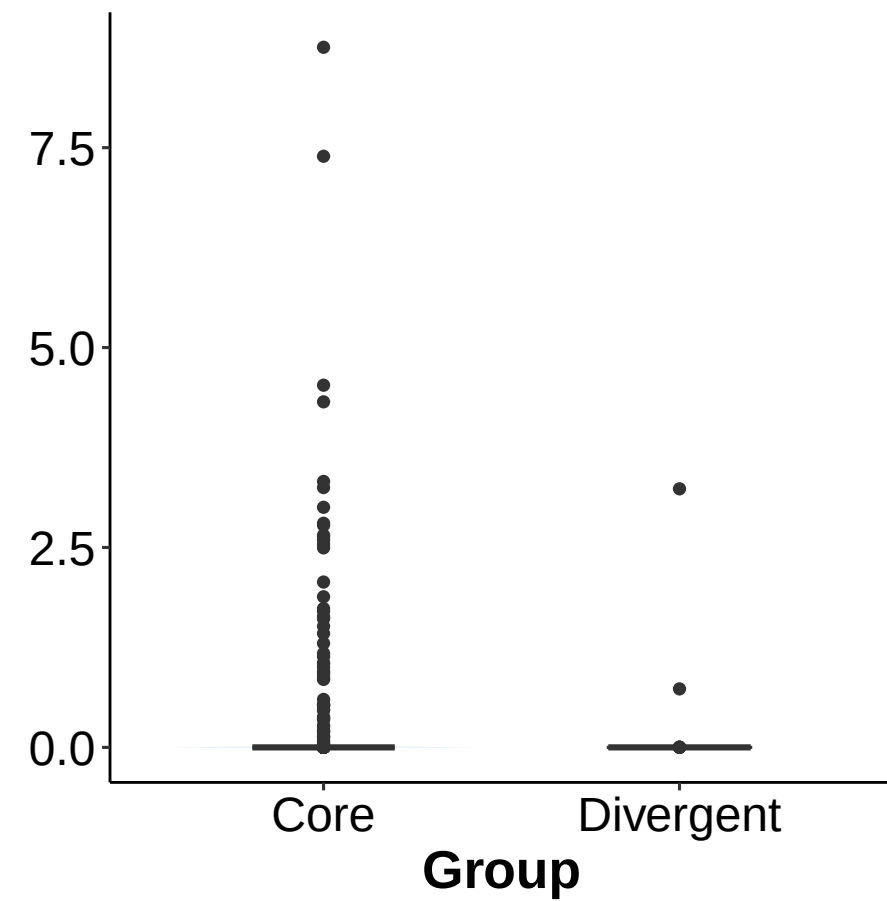
SA_Pyruvate_metabolism_L1w

p-adj: 0.00464 | .=-0.27 (Pequeño) | .%: -36.8



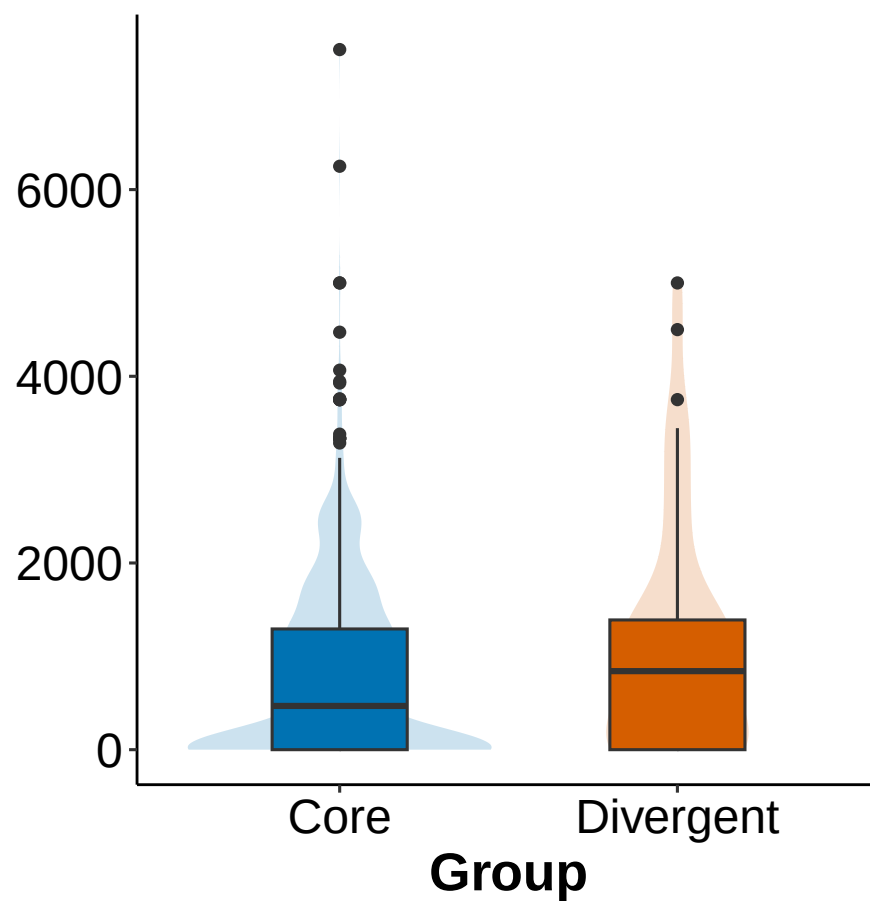
SA_Pyruvate_metabolism_pFBA

p-adj: 0.829 | .-=0.03 (Negligible) | .%: -29.6



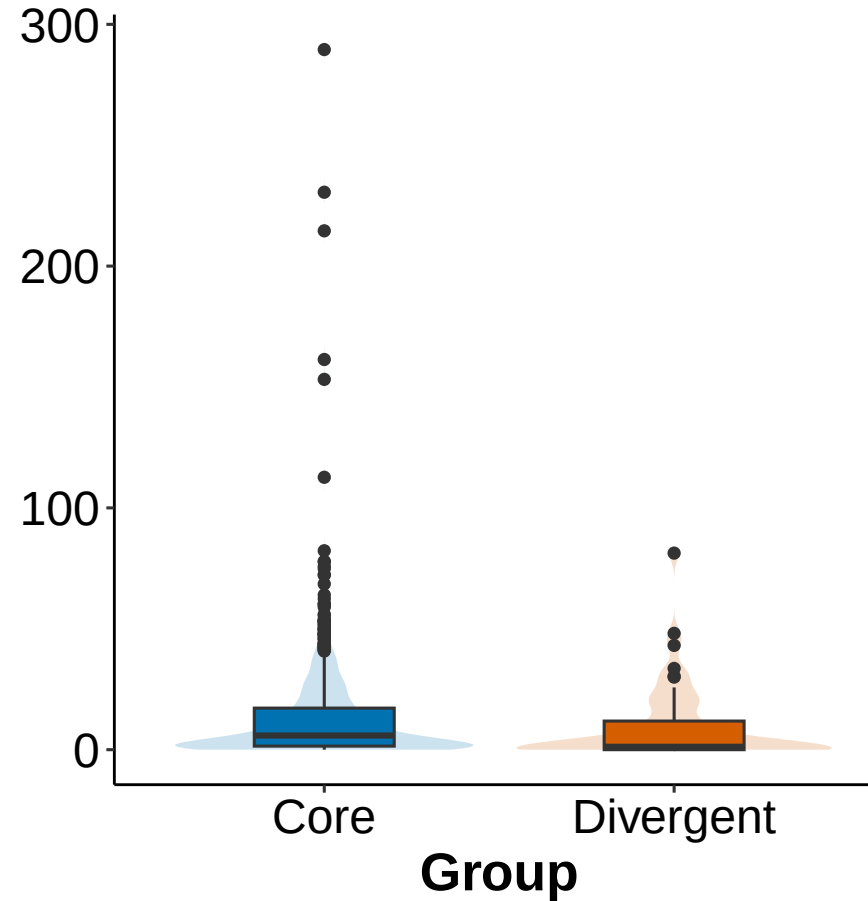
SA_ROS_detoxification_FBA

p-adj: 0.227 | .-=0.13 (Negligible) | .%: 79.5



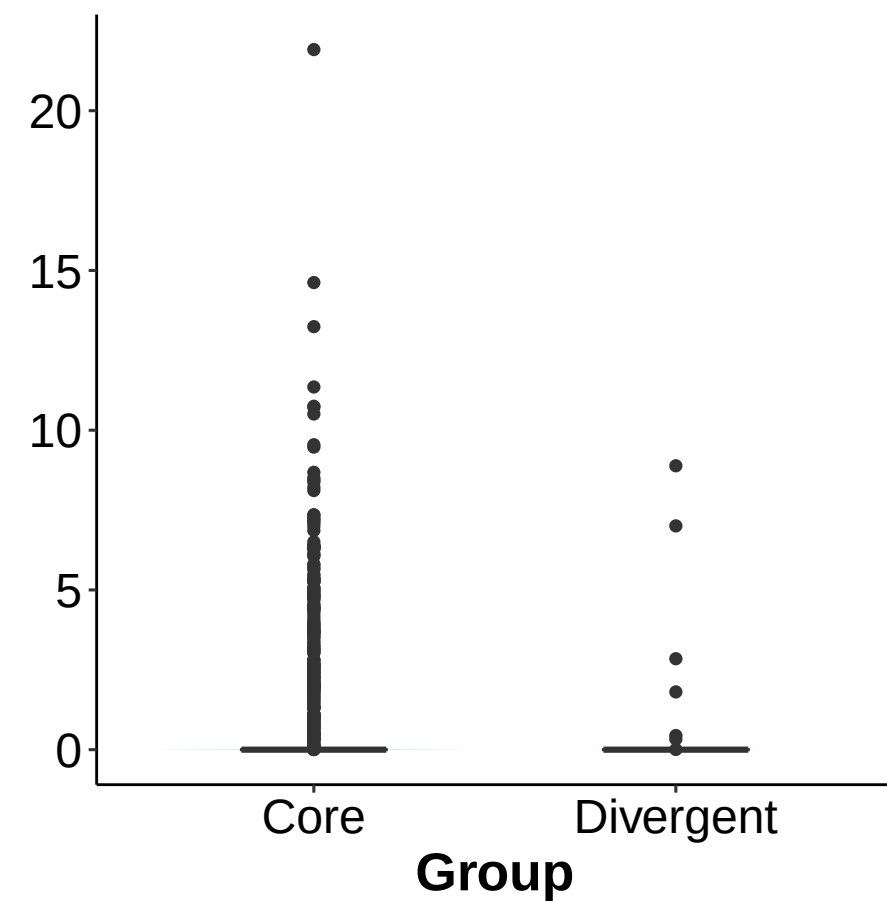
SA_ROS_detoxification_L1w

p-adj: 0.00234 | .=-0.29 (Pequeño) | .%: -79.5



SA_ROS_detoxification_pFBA

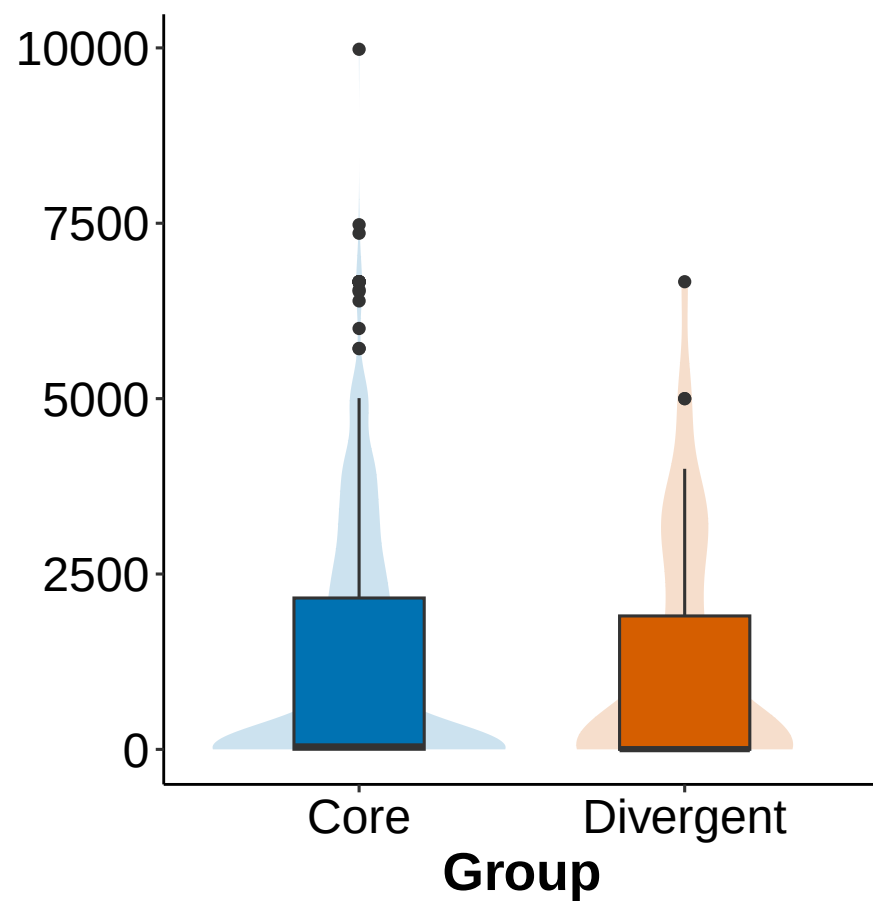
p-adj: 0.179 | .-=0.14 (Negligible) | .%: -30.6



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

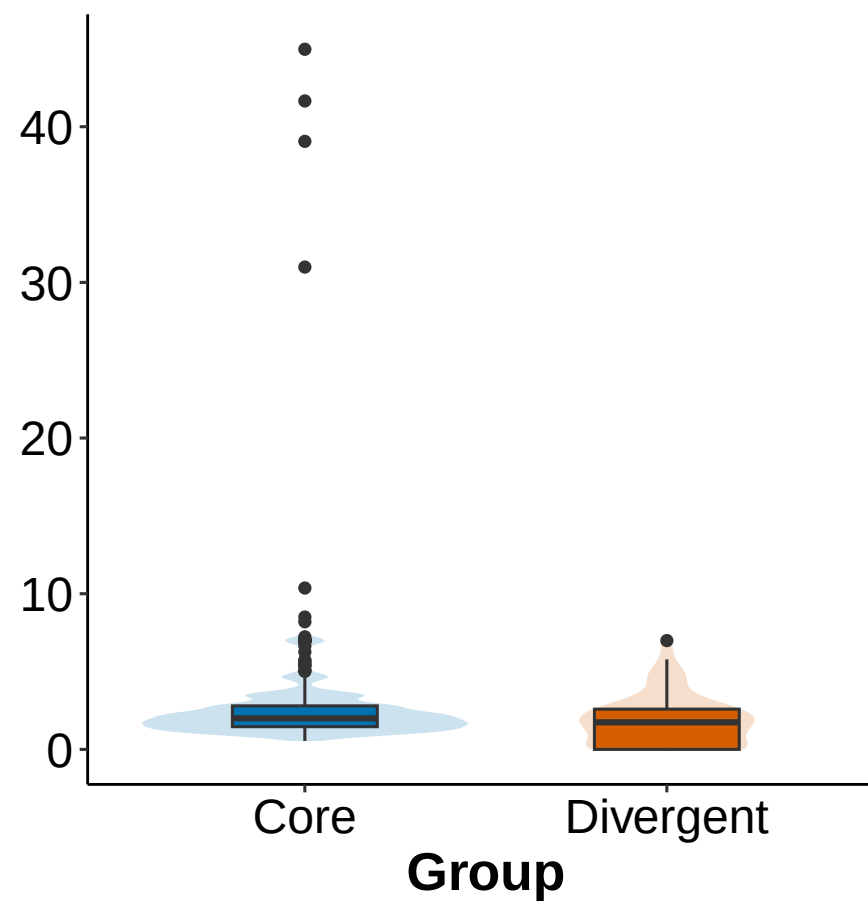
SA_Sphingolipid_metabolism_FB

p-adj: 0.0989 | .=-0.17 (Pequeño) | .%: -91.6



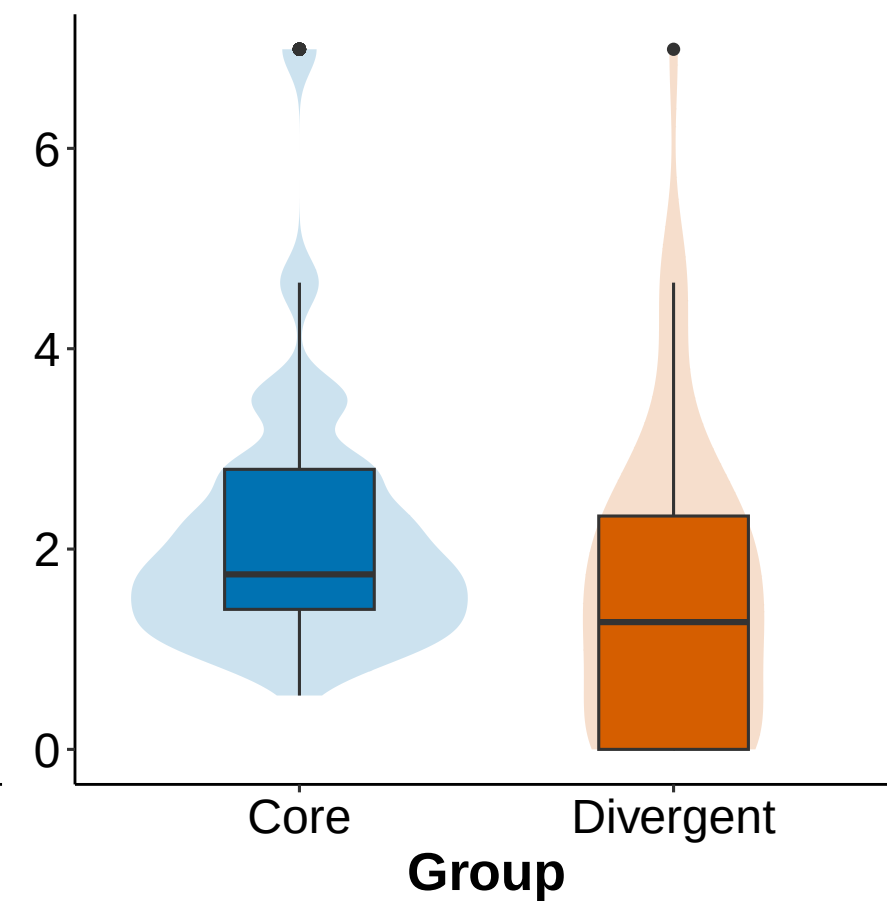
SA_Sphingolipid_metabolism_L1w

p-adj: 0.00864 | .=-0.25 (Pequeño) | .%: -12.9



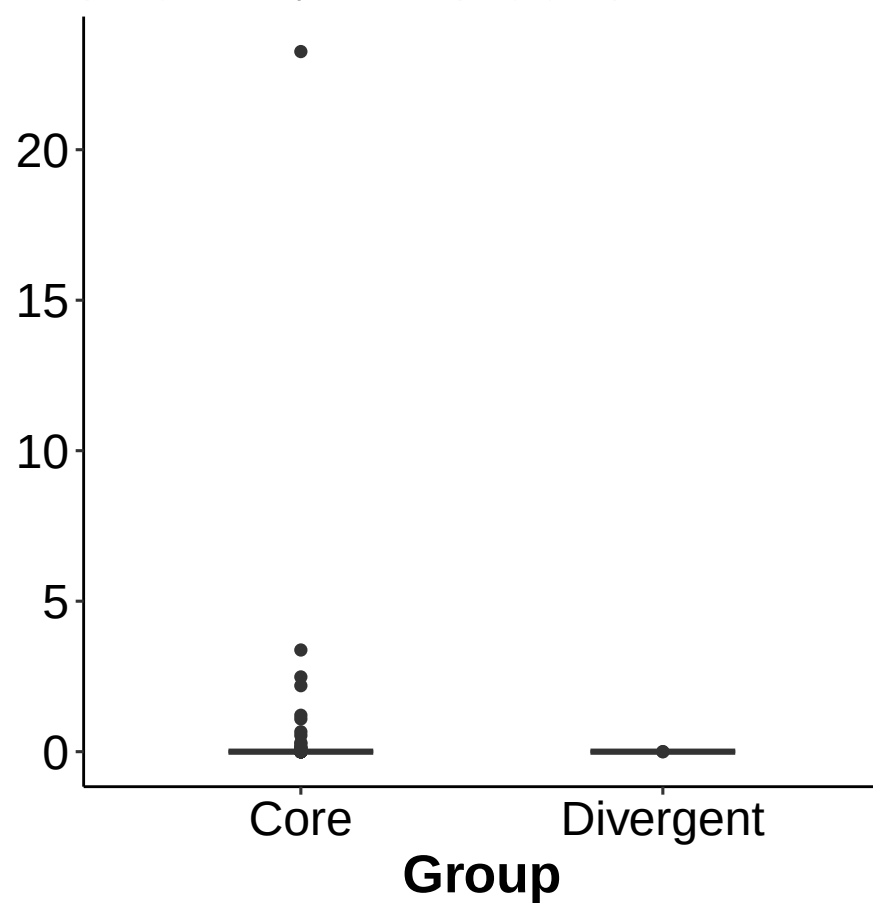
SA_Sphingolipid_metabolism_pFBA

p-adj: 0.000878 | .=-0.32 (Pequeño) | .%: -27.3



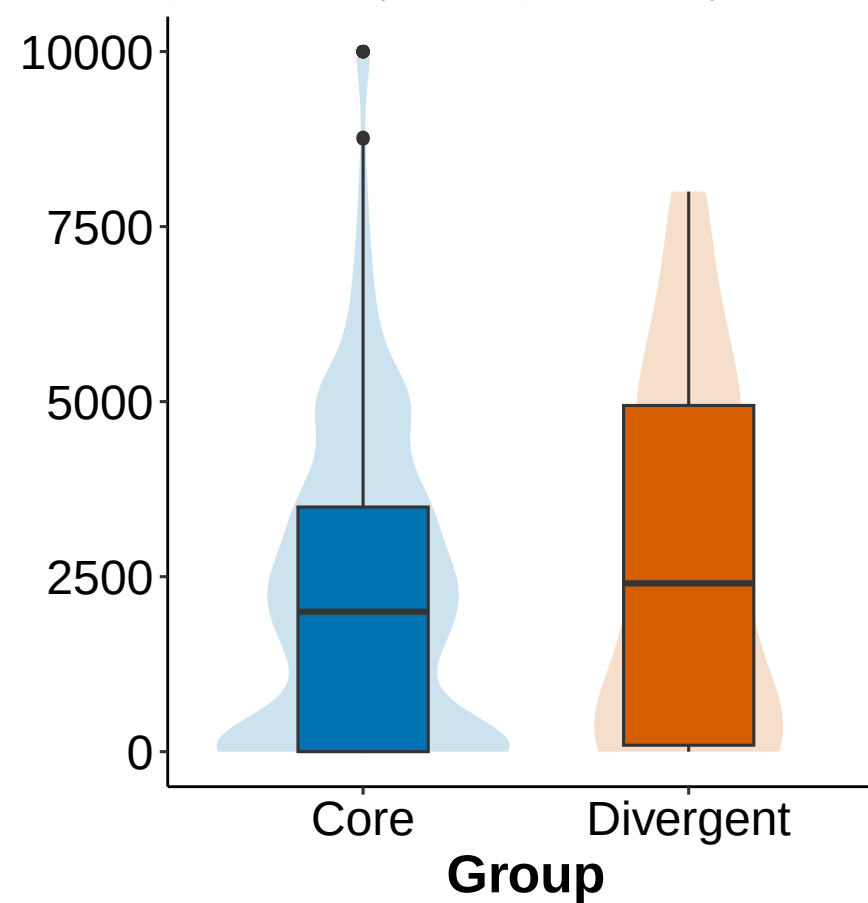
SA_Squalene_and_cholesterol_synth

p-adj: 0.804 | .=-0.01 (Negligible)



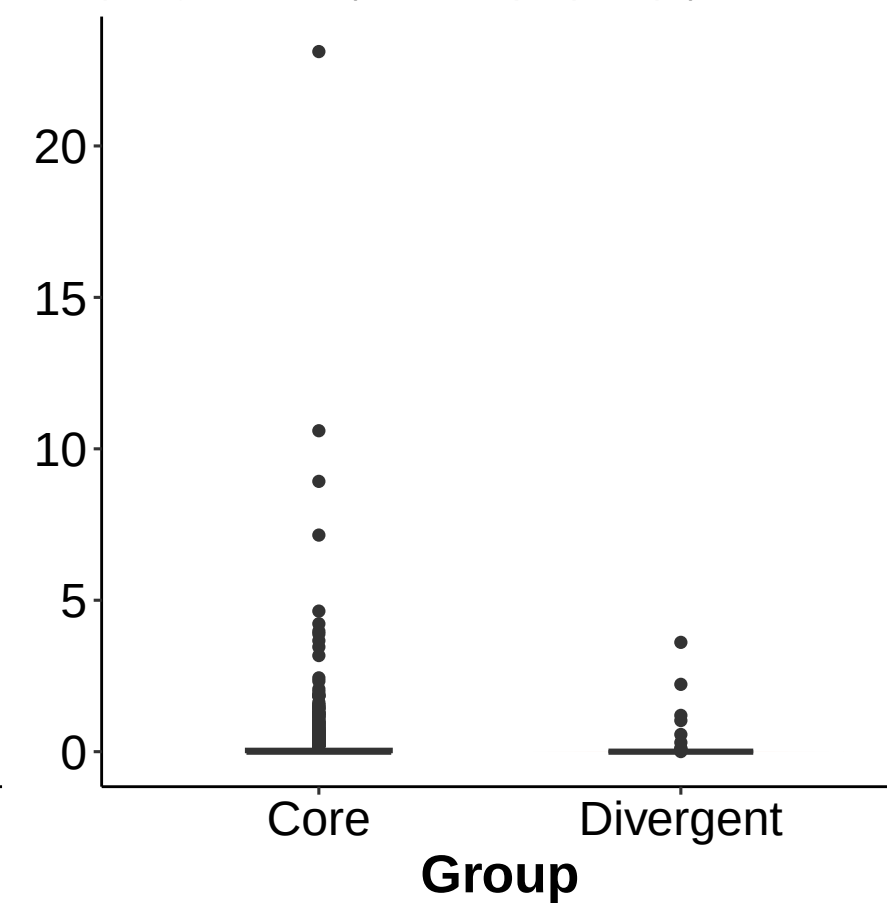
SA_Starch_and_sucrose_metabol

p-adj: 0.653 | .=-0.06 (Negligible) | .%: 20.2



SA_Starch_and_sucrose_metabolism

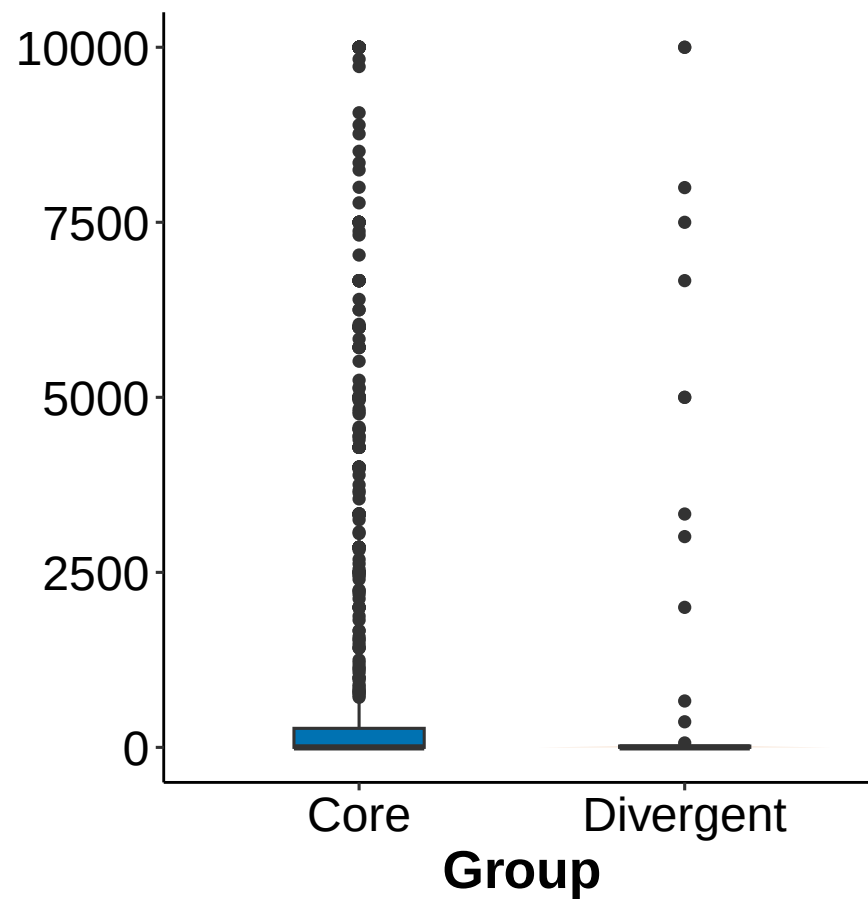
p-adj: 0.00242 | .=-0.28 (Pequeño) | .%: -69.9



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

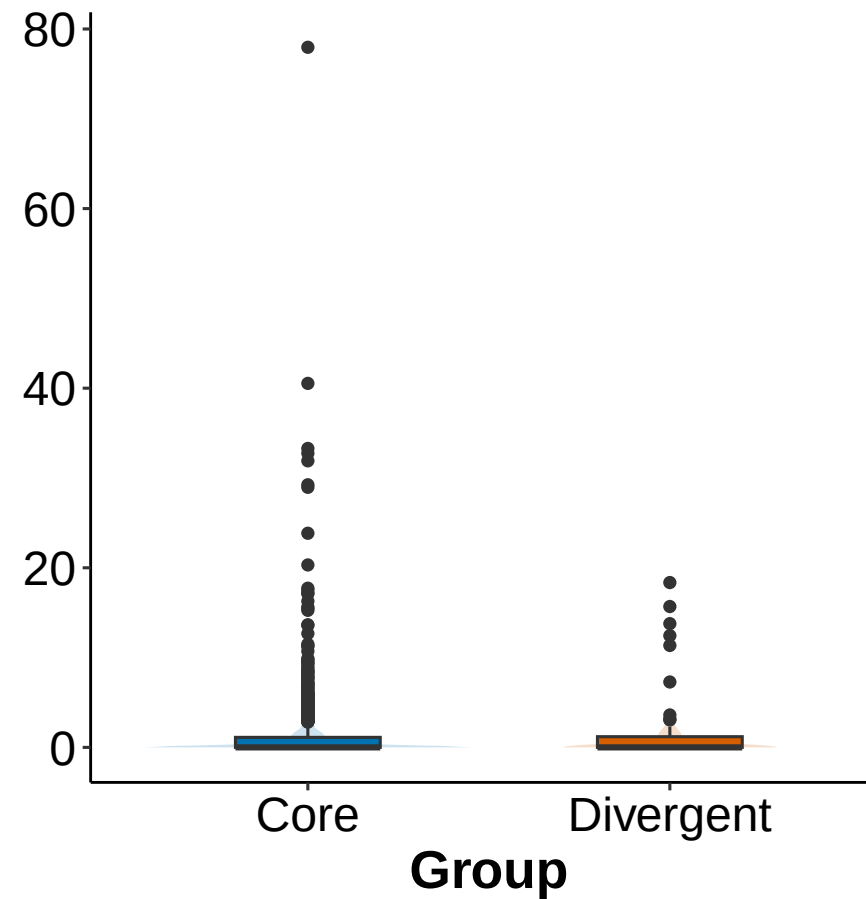
SA_Steroid_metabolism_FBA

p-adj: 0.839 | .=-0.02 (Negligible)



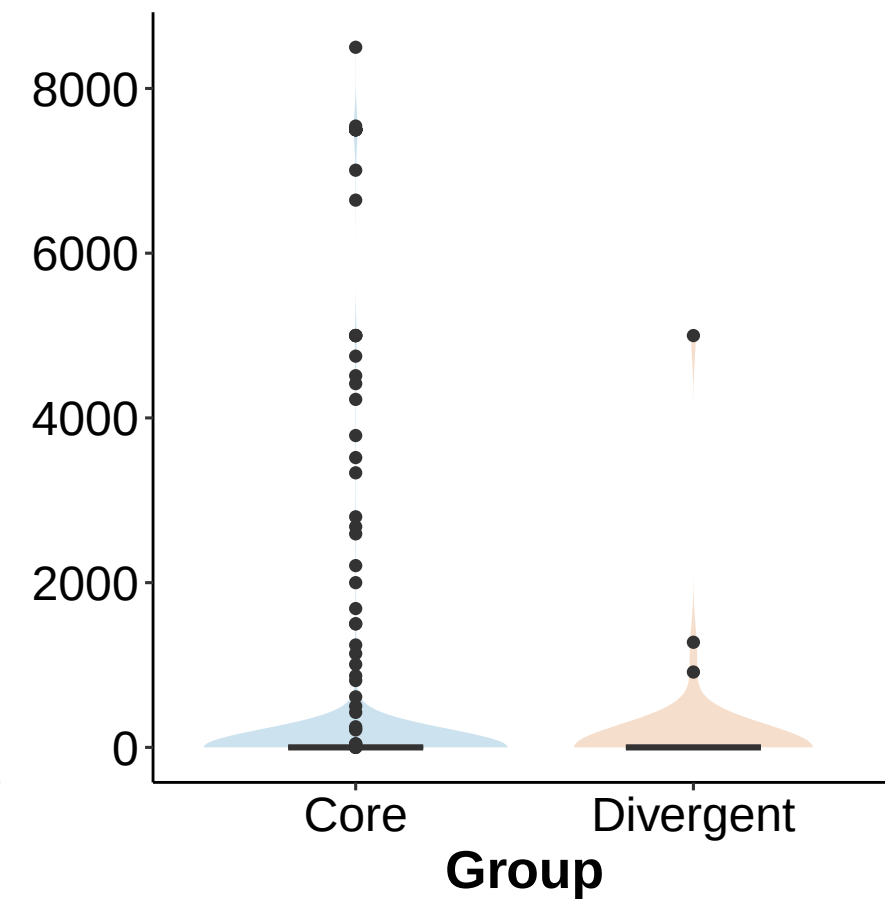
SA_Steroid_metabolism_L1w

p-adj: 0.993 | .=0 (Negligible) | .%: -97.4



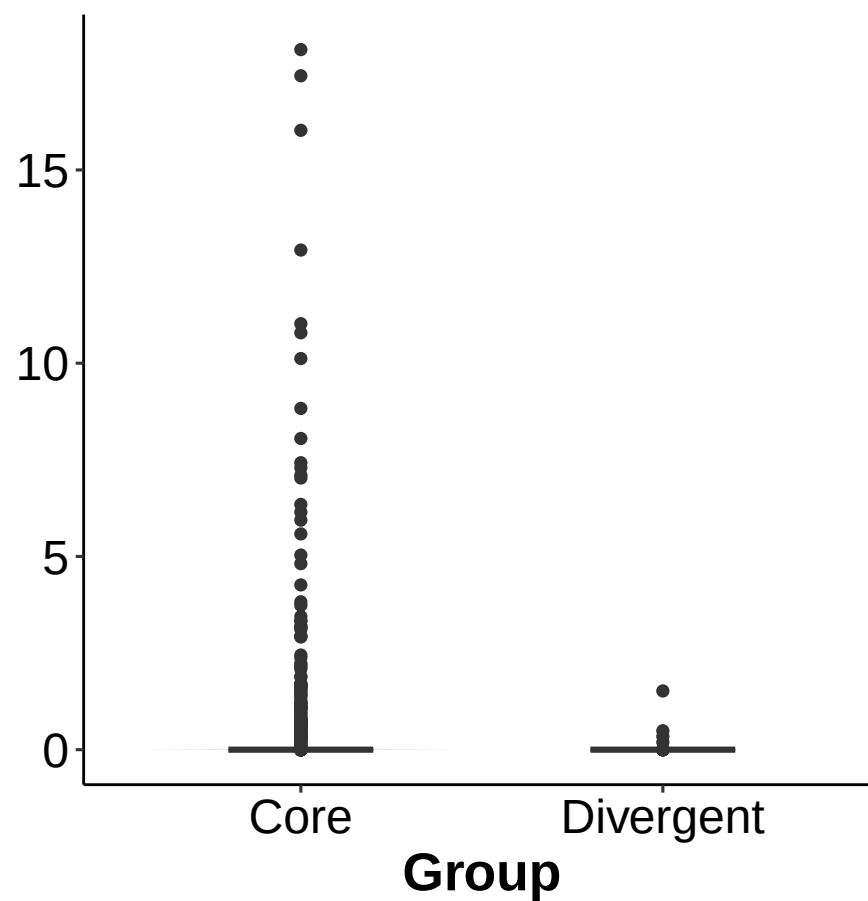
SA_Taurine_and_hypotaurine_metabolism_FBA

p-adj: 0.922 | .=0 (Negligible)



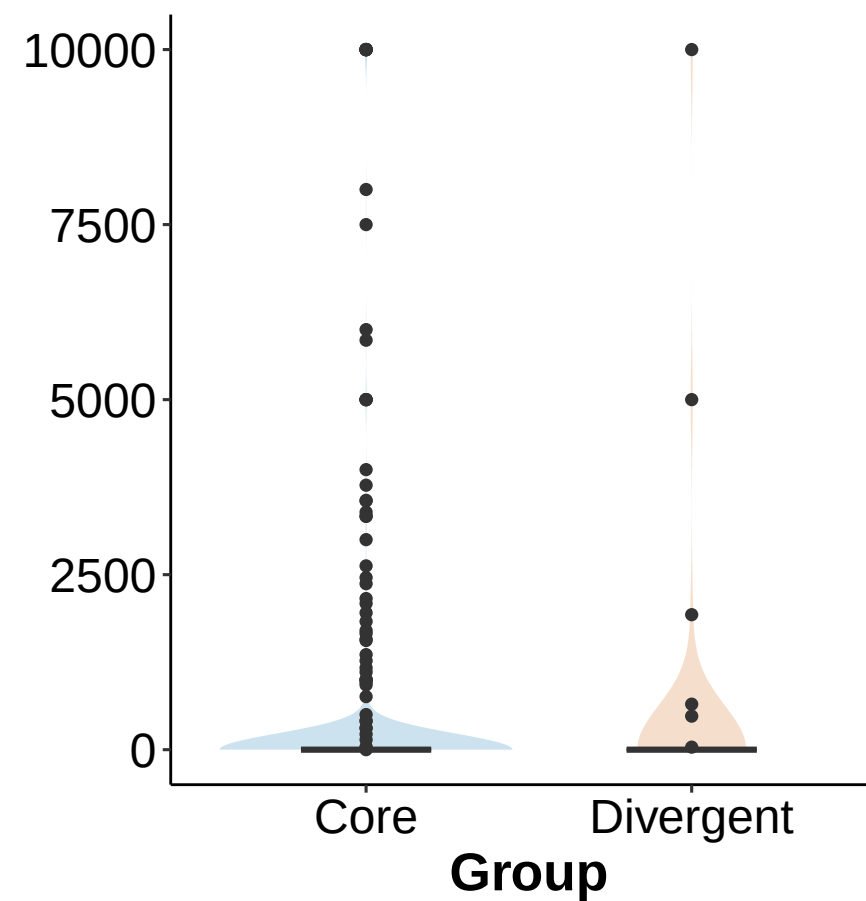
SA_Taurine_and_hypotaurine_metabolism_L1w

p-adj: 0.943 | .=-0.01 (Negligible)



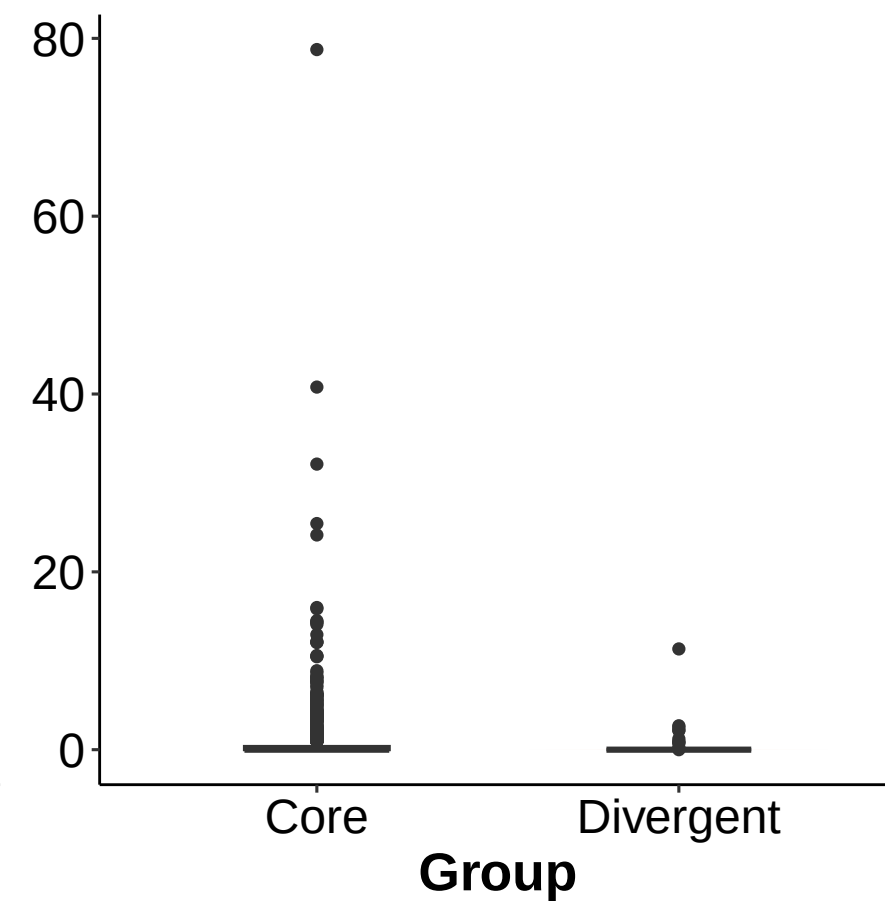
SA_Tetrahydrobiopterin_metabolism_FBA

p-adj: 0.151 | .=0.06 (Negligible)



SA_Tetrahydrobiopterin_metabolism_L1w

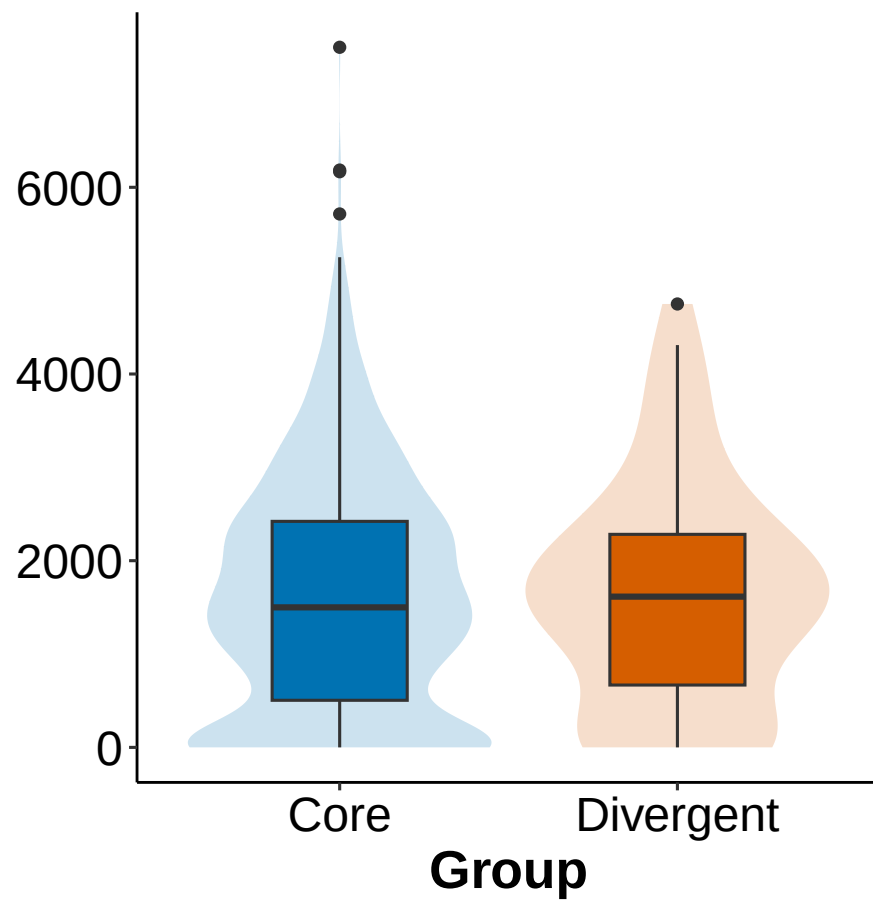
p-adj: 0.719 | .=-0.05 (Negligible) | .%: 34.6



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

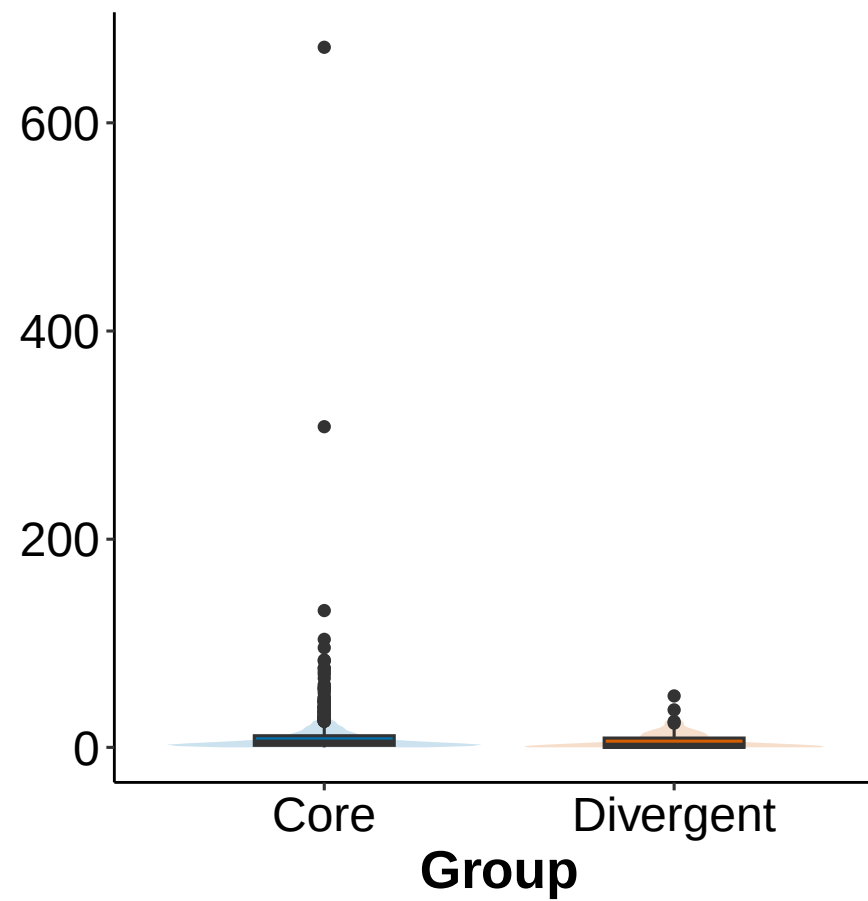
SA_Transport__endoplasmic_reticul

p-adj: 0.953 | .:=0.01 (Negligible) | .%: 7.7



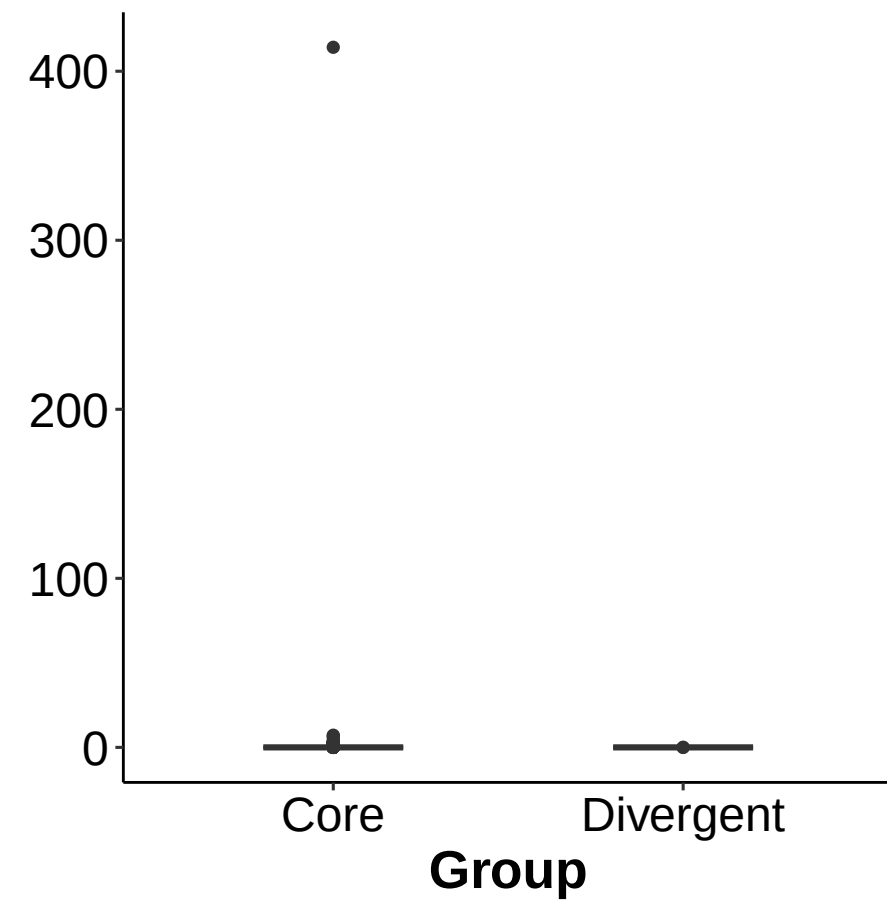
SA_Transport__endoplasmic_reticul

p-adj: 0.00143 | .:=−0.3 (Pequeño) | .%: −66.8



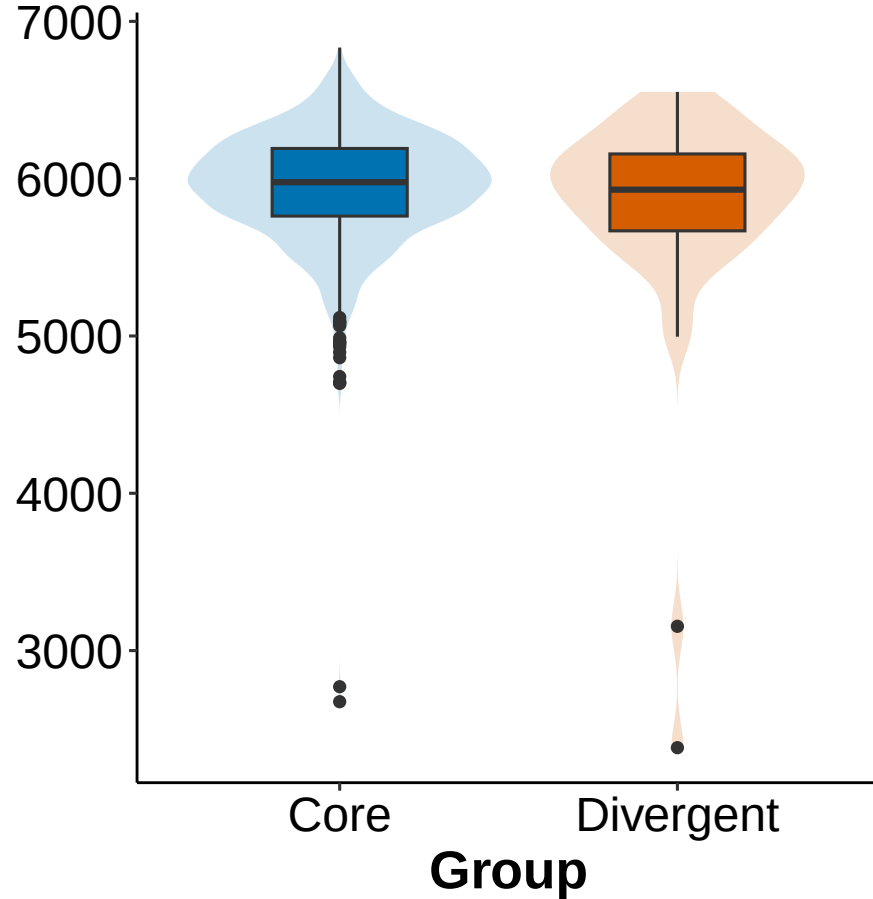
SA_Transport__endoplasmic_reticul

p-adj: 0.29 | .:=−0.06 (Negligible)



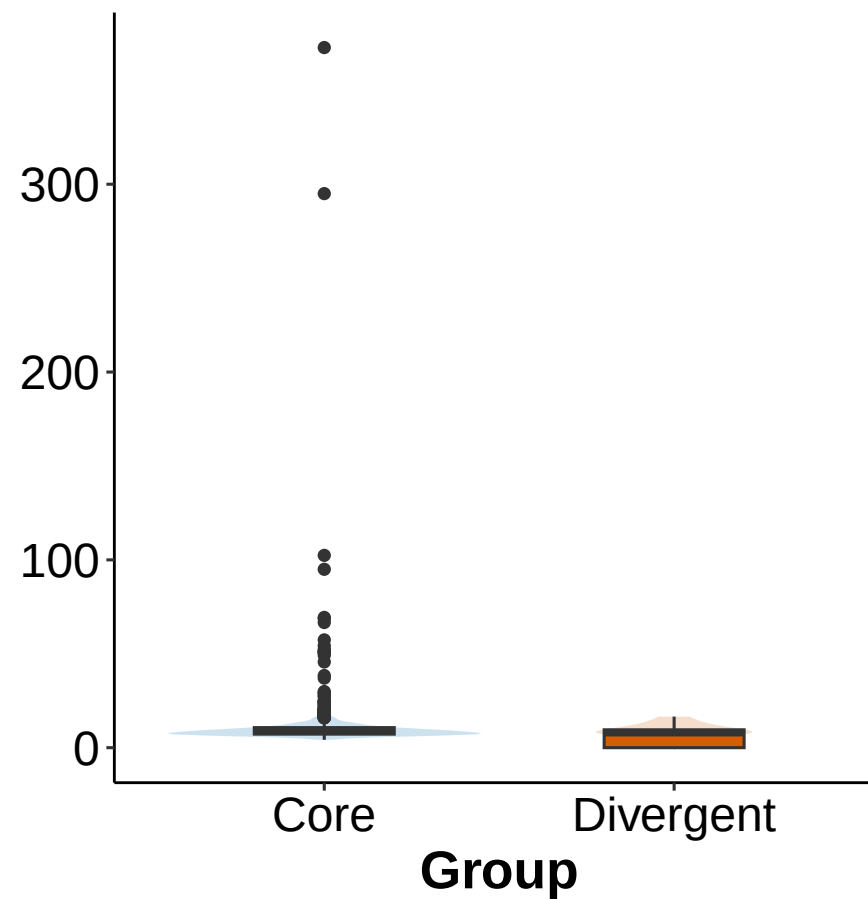
SA_Transport__extracellular_FBA

p-adj: 0.666 | .:=−0.06 (Negligible) | .%: −0.8



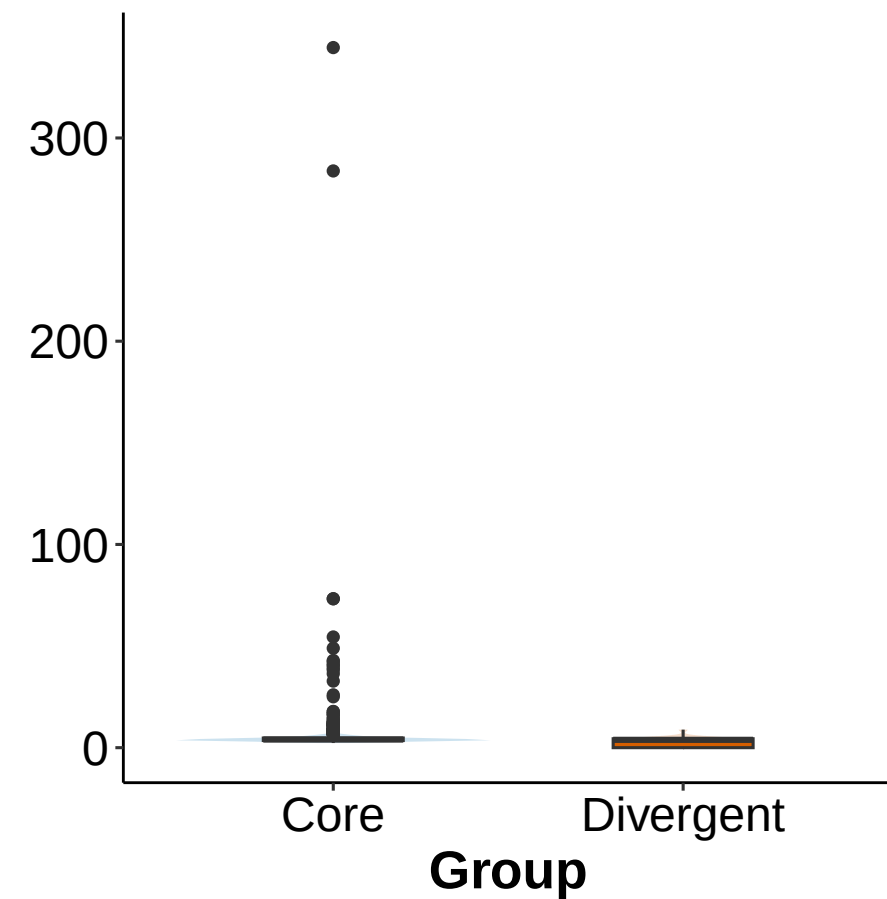
SA_Transport__extracellular_L1w

p-adj: 0.0105 | .:=−0.24 (Pequeño) | .%: −11.4



SA_Transport__extracellular_pFBA

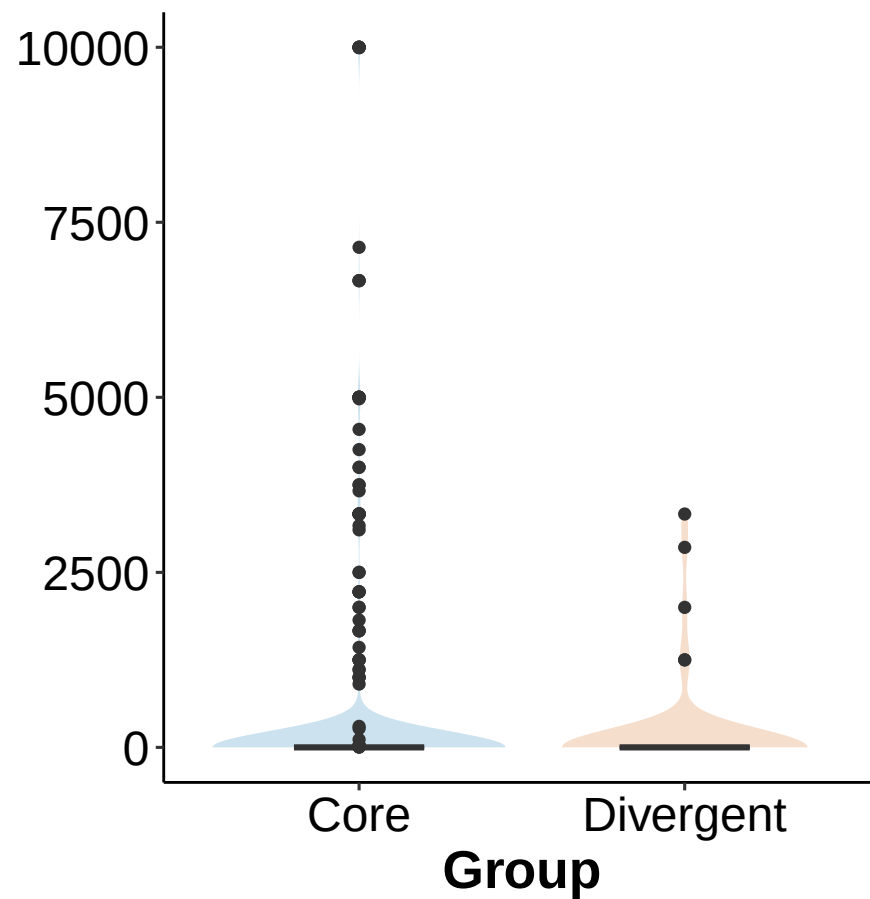
p-adj: 0.0056 | .:=−0.26 (Pequeño) | .%: −3.9



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

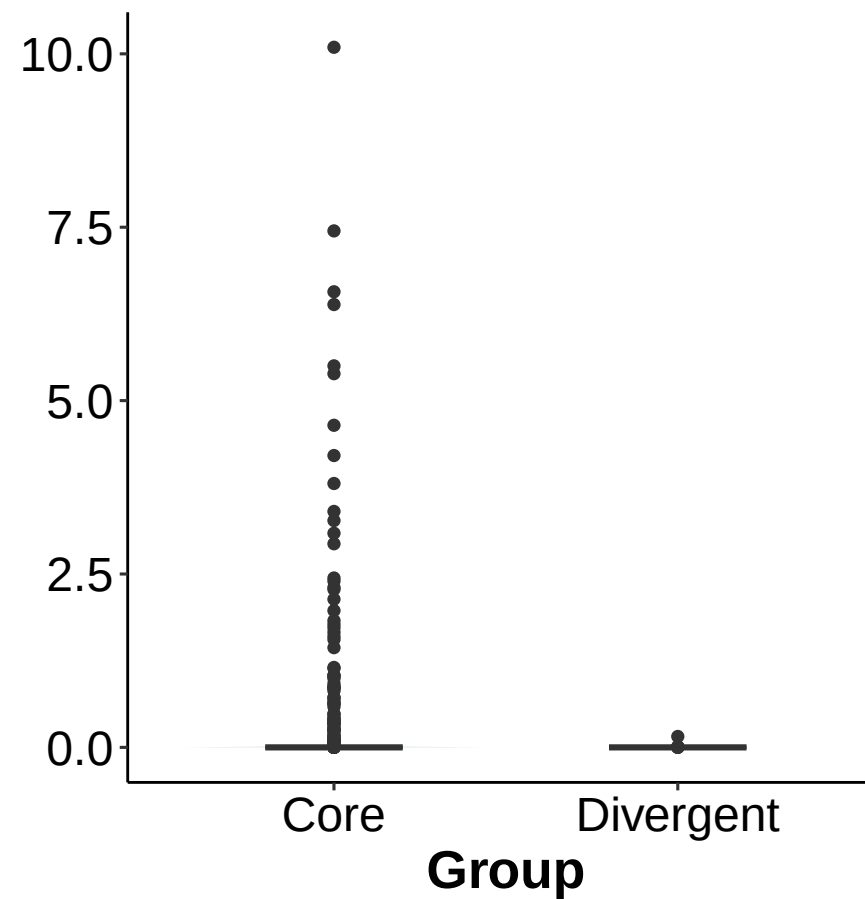
SA_Transport__golgi_apparatus_I

p-adj: 0.705 | .:=0.02 (Negligible)



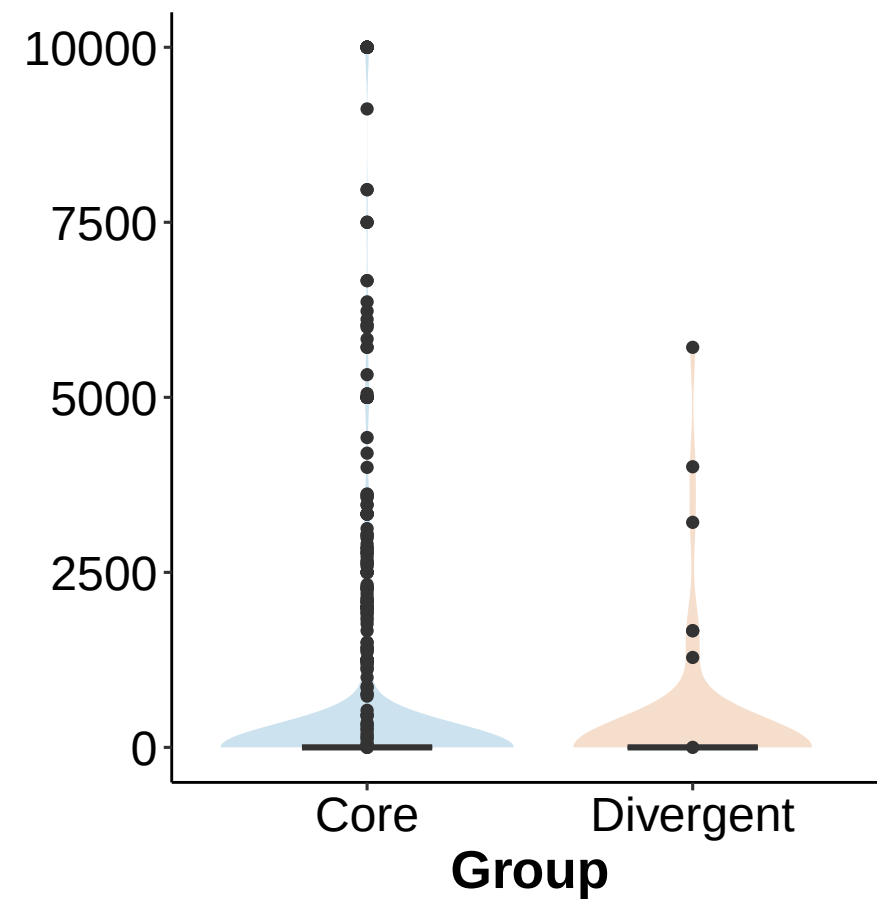
SA_Transport__golgi_apparatus_L1

p-adj: 0.838 | .:=−0.03 (Negligible) | .%: 5.4



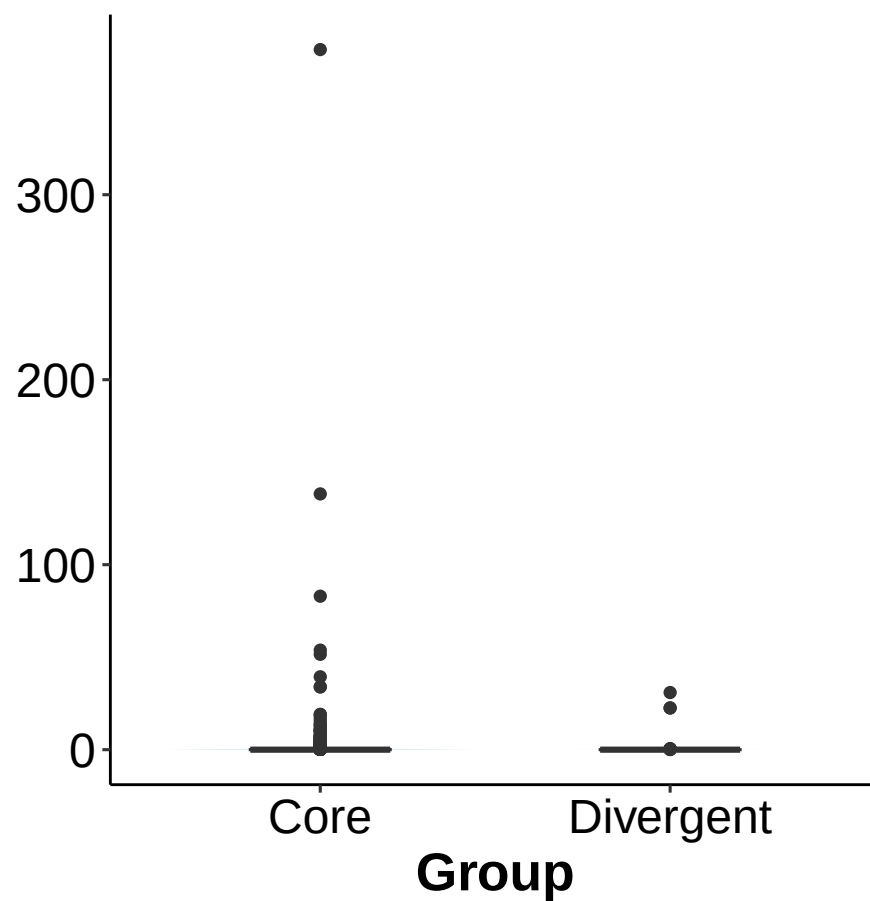
SA_Transport__lysosomal_FBA

p-adj: 0.758 | .:=−0.03 (Negligible)



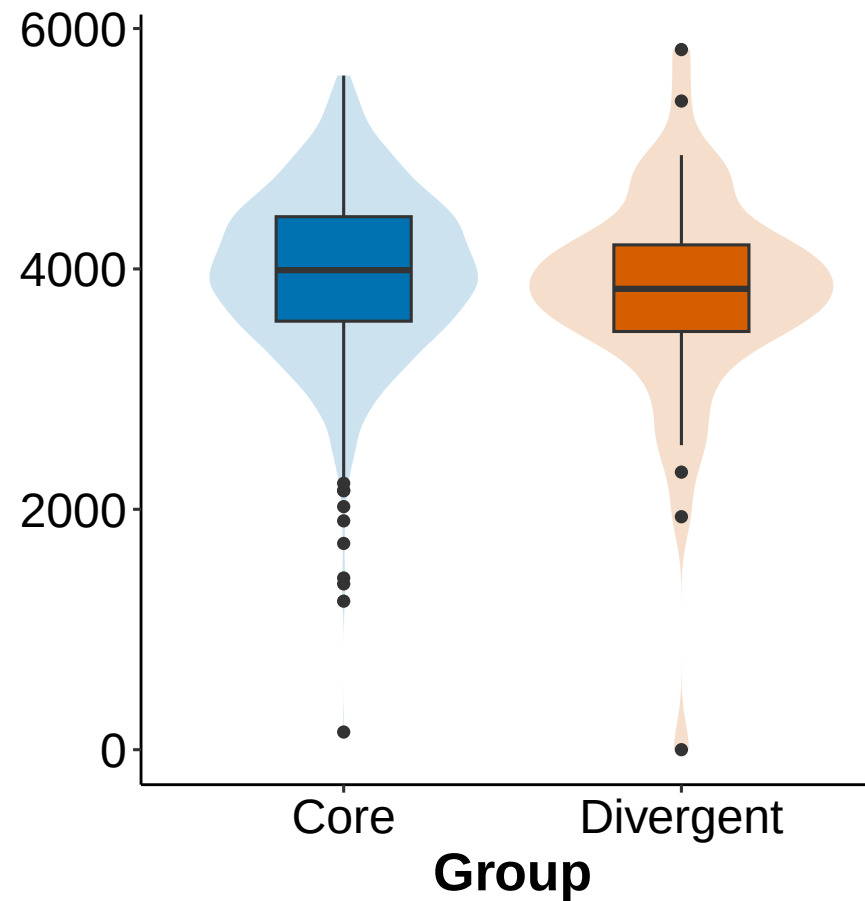
SA_Transport__lysosomal_L1w

p-adj: 0.144 | .:=−0.15 (Pequeño) | .%: −100



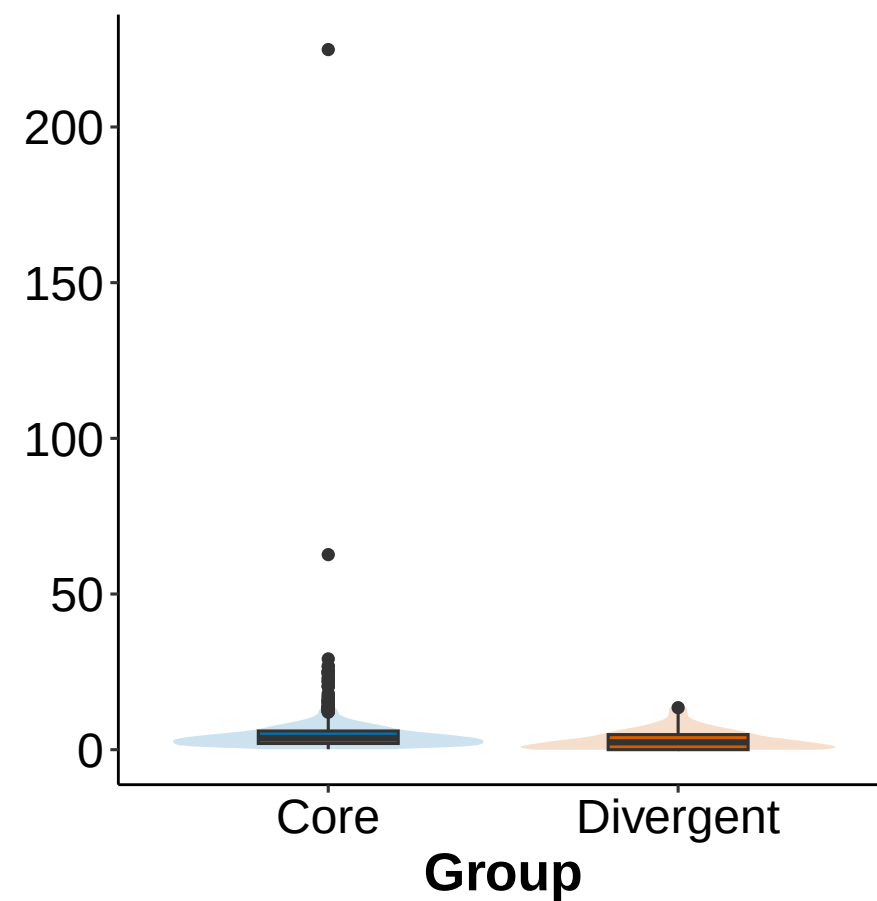
SA_Transport__mitochondrial_FBA

p-adj: 0.263 | .:=−0.12 (Negligible) | .%: −3.9



SA_Transport__mitochondrial_L1w

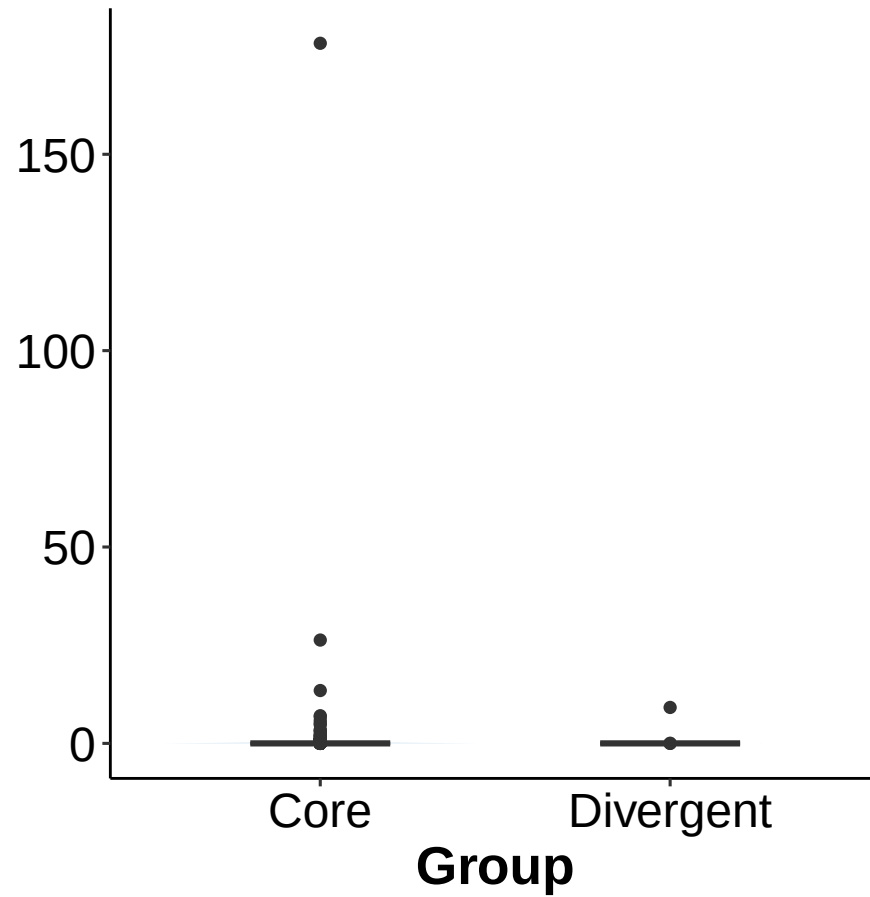
p-adj: 0.000771 | .:=−0.32 (Pequeño) | .%: −37.2



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

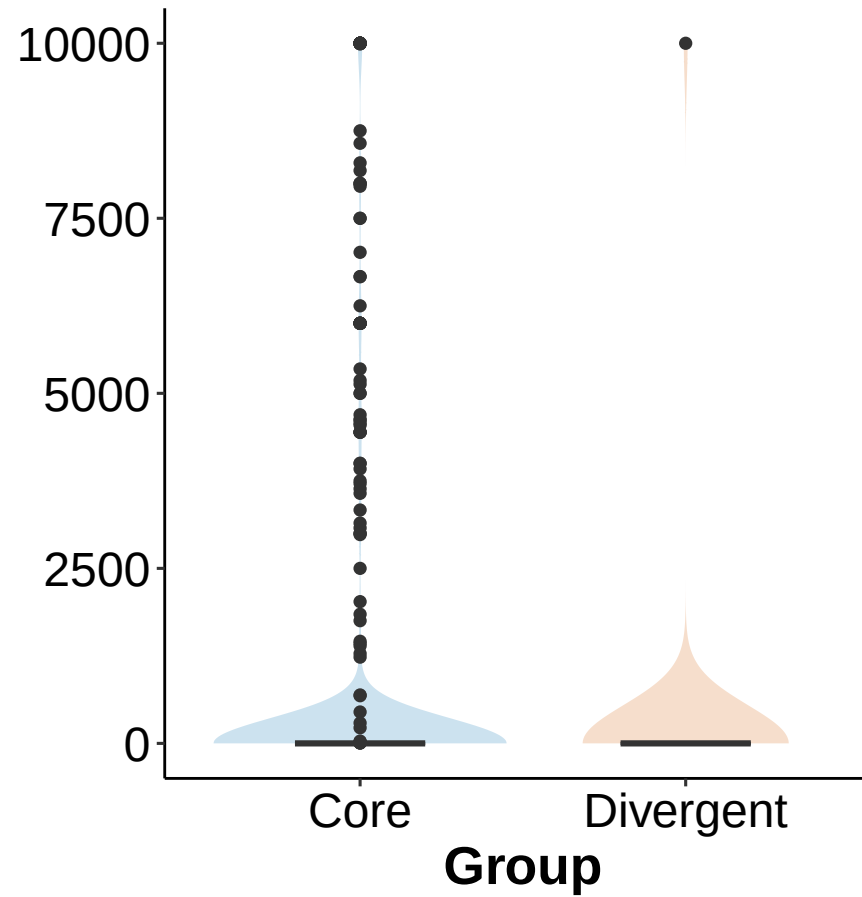
SA_Transport__mitochondrial_pFBA

p-adj: 0.00477 | .=-0.27 (Pequeño) | .%: -24.5



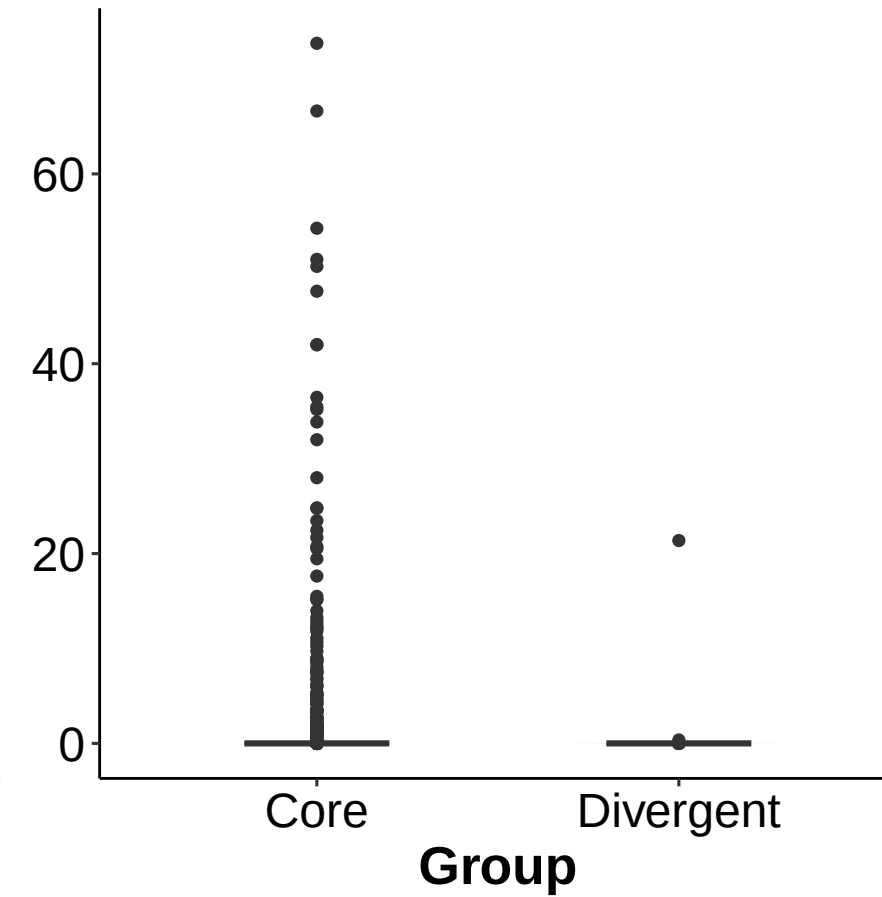
SA_Transport__nuclear_FBA

p-adj: 0.238 | .=-0.06 (Negligible)



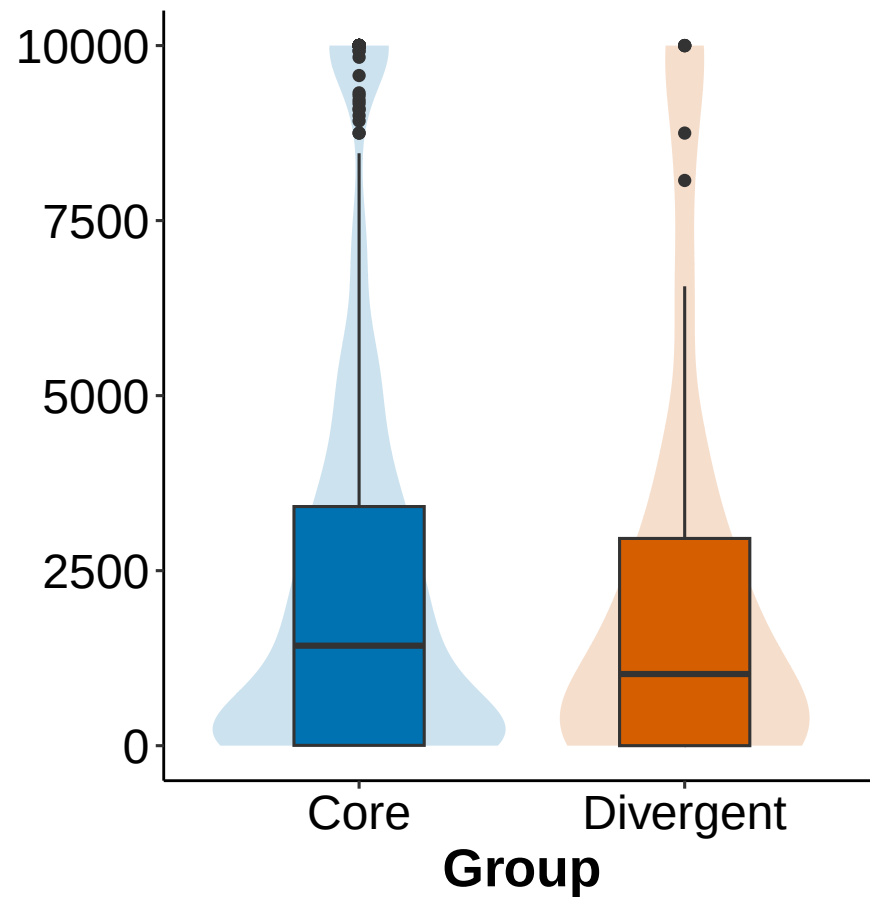
SA_Transport__nuclear_L1w

p-adj: 0.192 | .=-0.12 (Negligible)



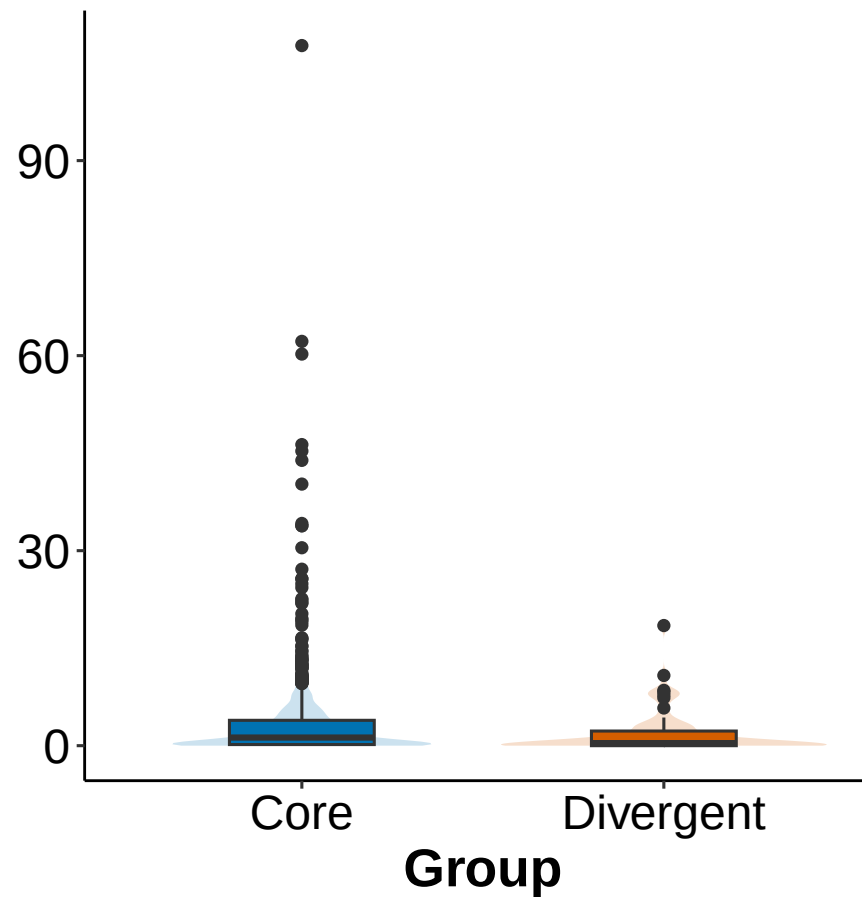
SA_Transport__peroxisomal_FBA

p-adj: 0.537 | .=-0.08 (Negligible) | .%: -28.3



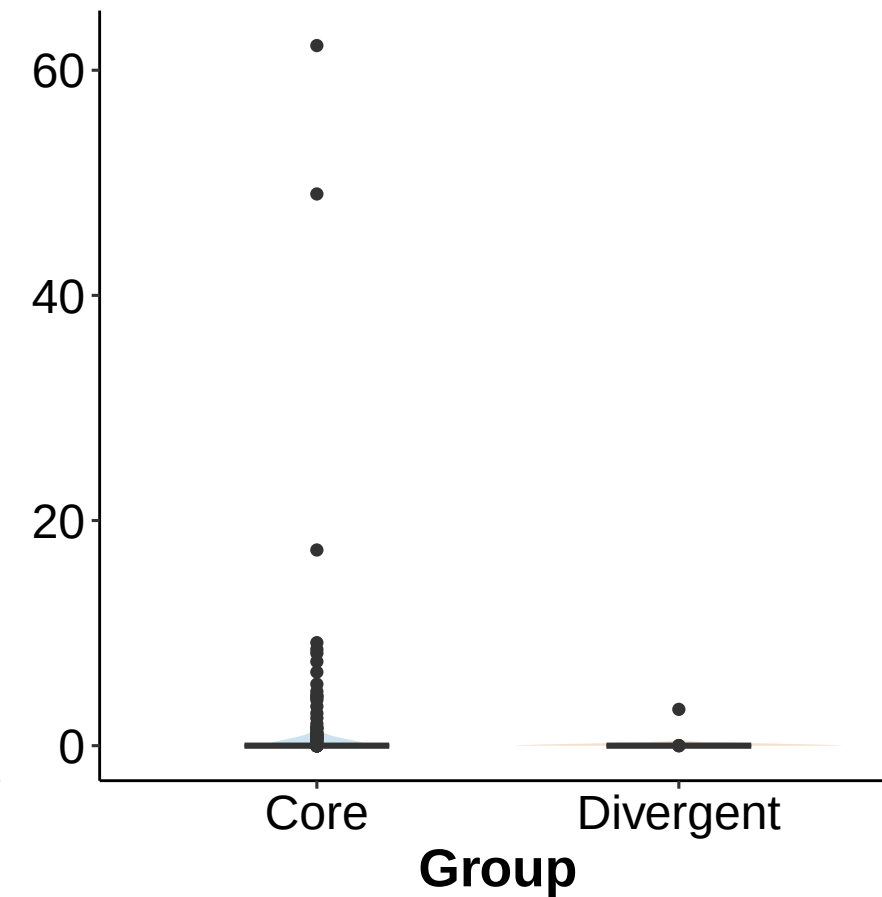
SA_Transport__peroxisomal_L1w

p-adj: 0.0136 | .=-0.24 (Pequeño) | .%: -68.6



SA_Transport__peroxisomal_pFBA

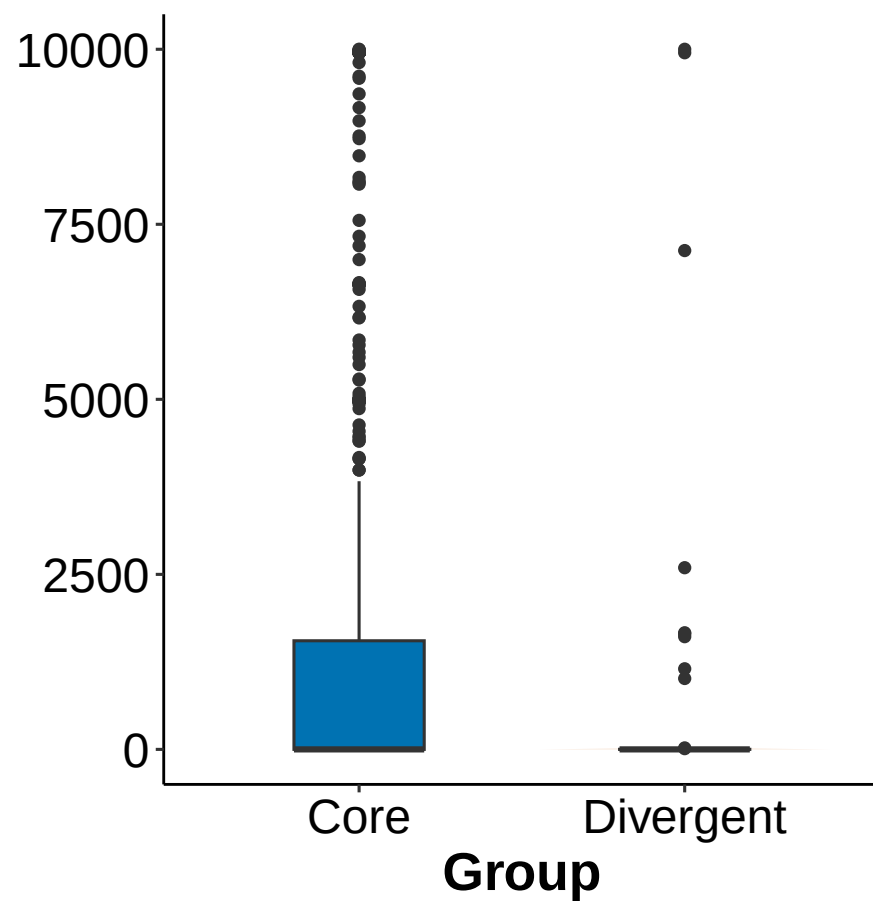
p-adj: 0.847 | .=-0.01 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

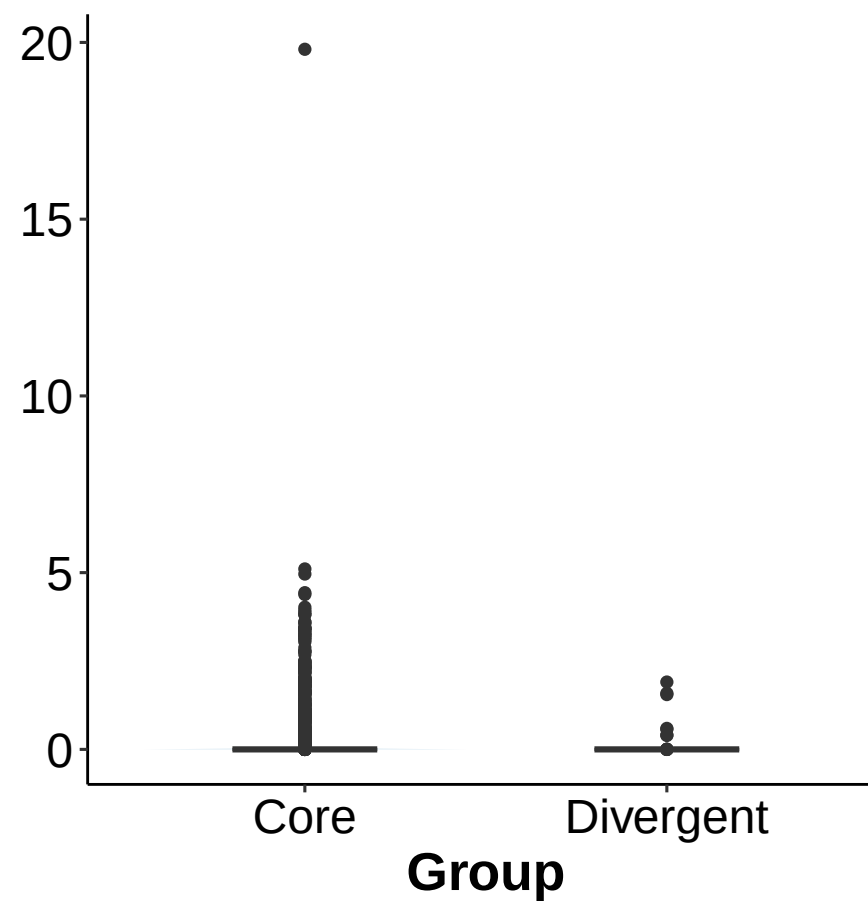
SA_Triacylglycerol_synthesis_FB/

p-adj: 0.202 | .=-0.12 (Negligible)



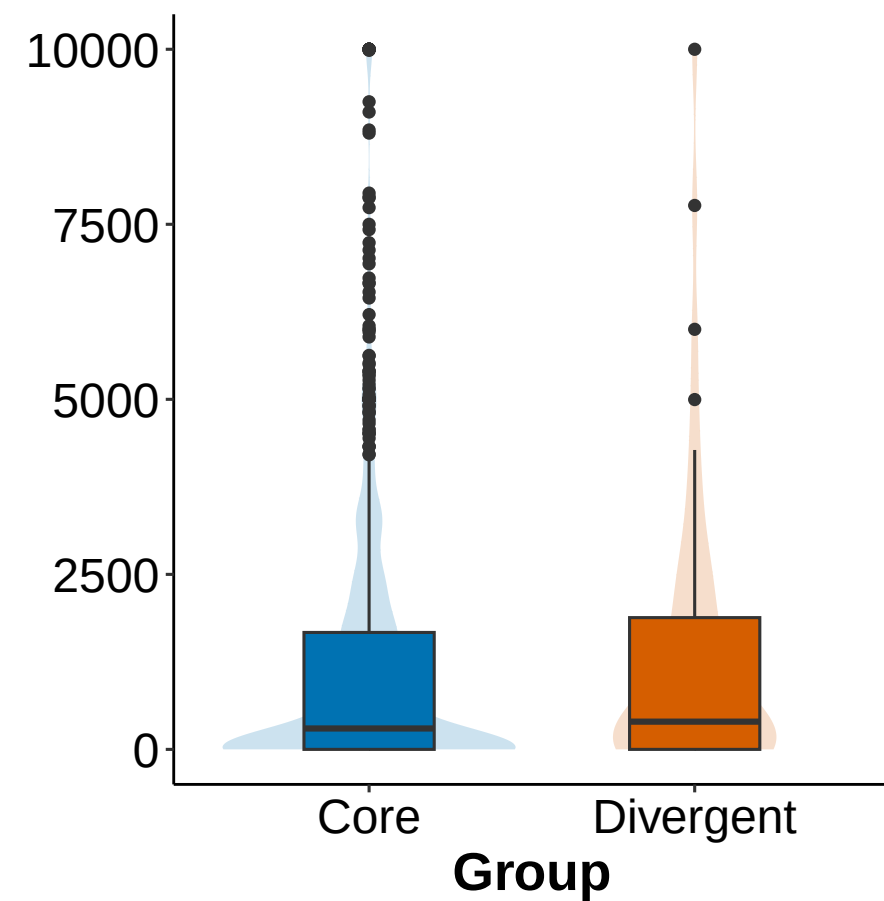
SA_Triacylglycerol_synthesis_L1w

p-adj: 0.168 | .=-0.14 (Negligible) | .%: 861.8



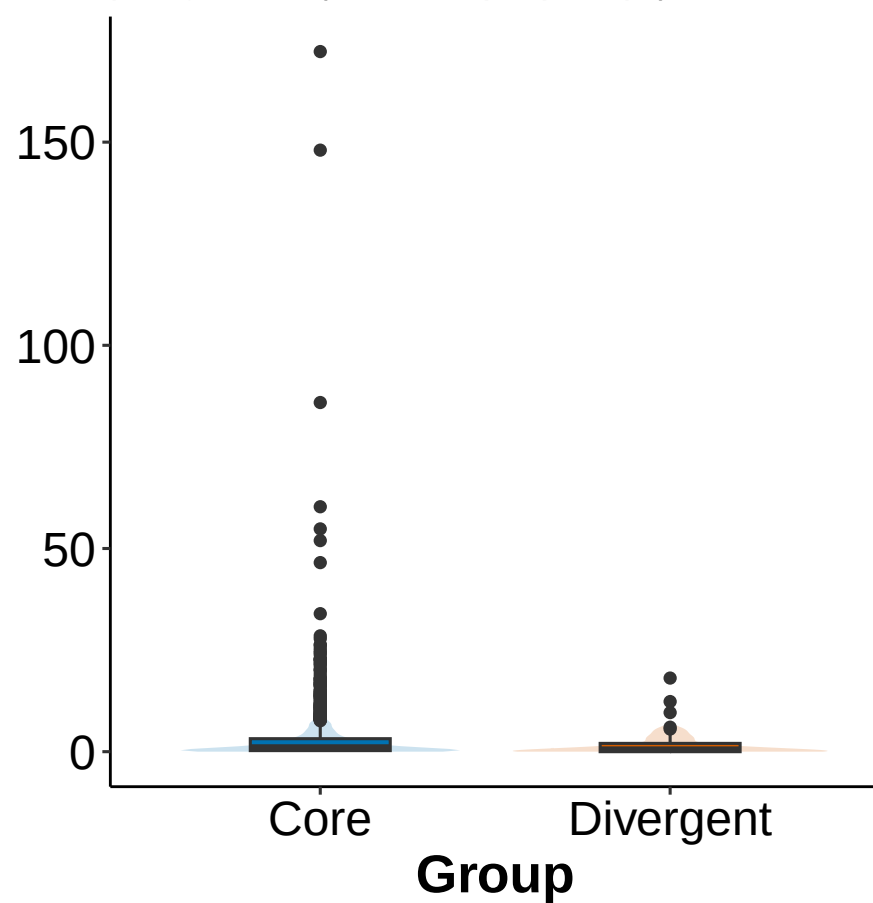
SA_Tryptophan_metabolism_FBA

p-adj: 0.705 | .=-0.05 (Negligible) | .%: 32.7



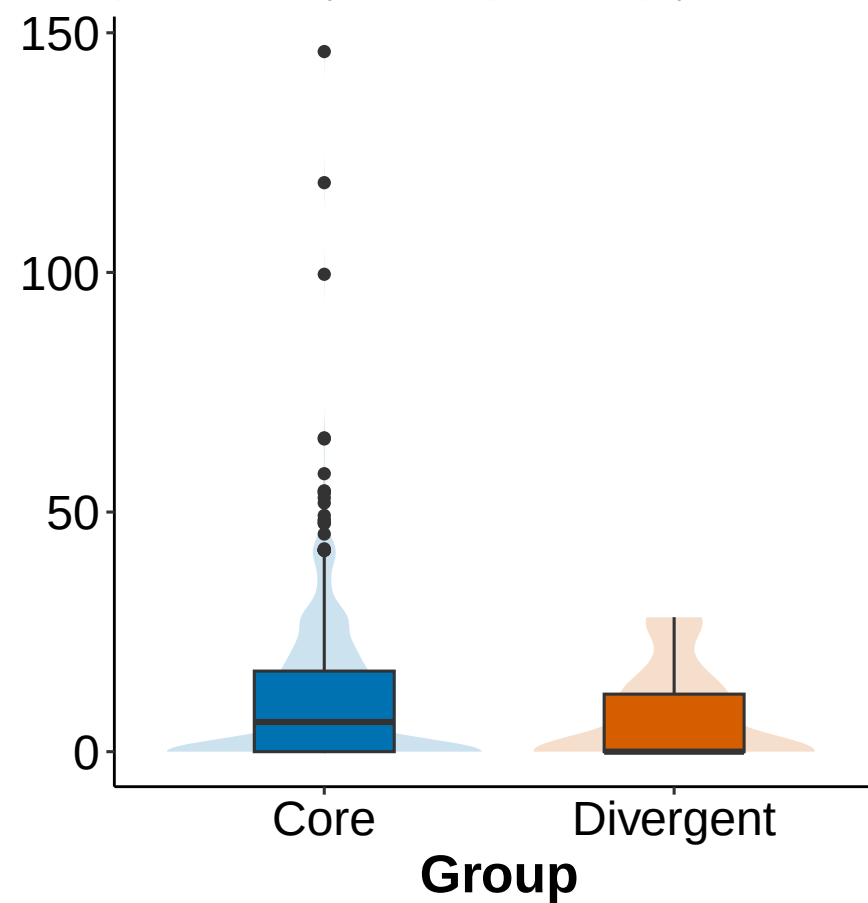
SA_Tryptophan_metabolism_L1w

p-adj: 0.028 | .=-0.21 (Pequeño) | .%: -49.1



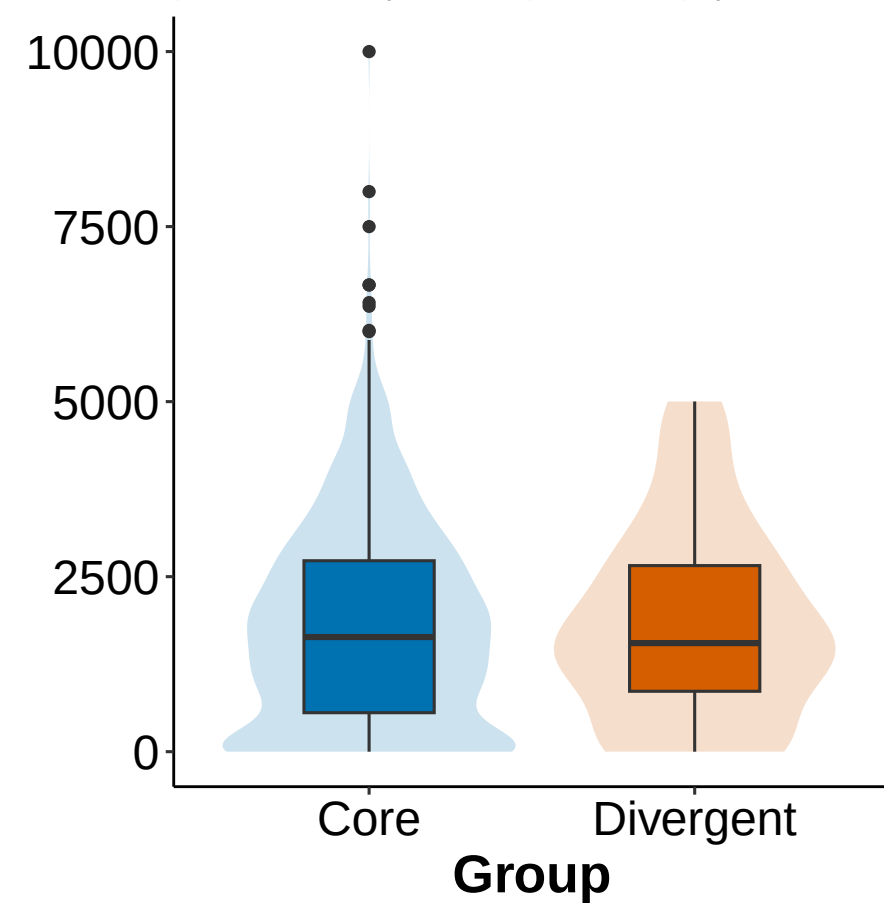
SA_Tryptophan_metabolism_pFBA

p-adj: 0.637 | .=-0.06 (Negligible) | .%: -100



SA_Tyrosine_metabolism_FBA

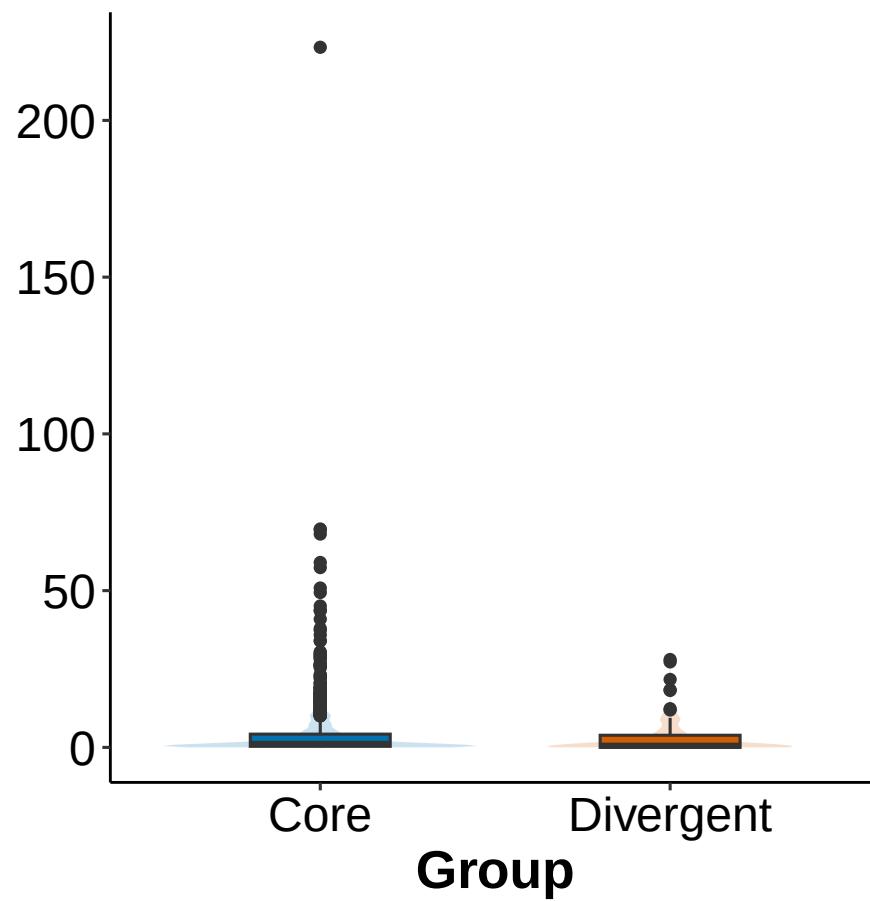
p-adj: 0.838 | .=-0.03 (Negligible) | .%: -5.3



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

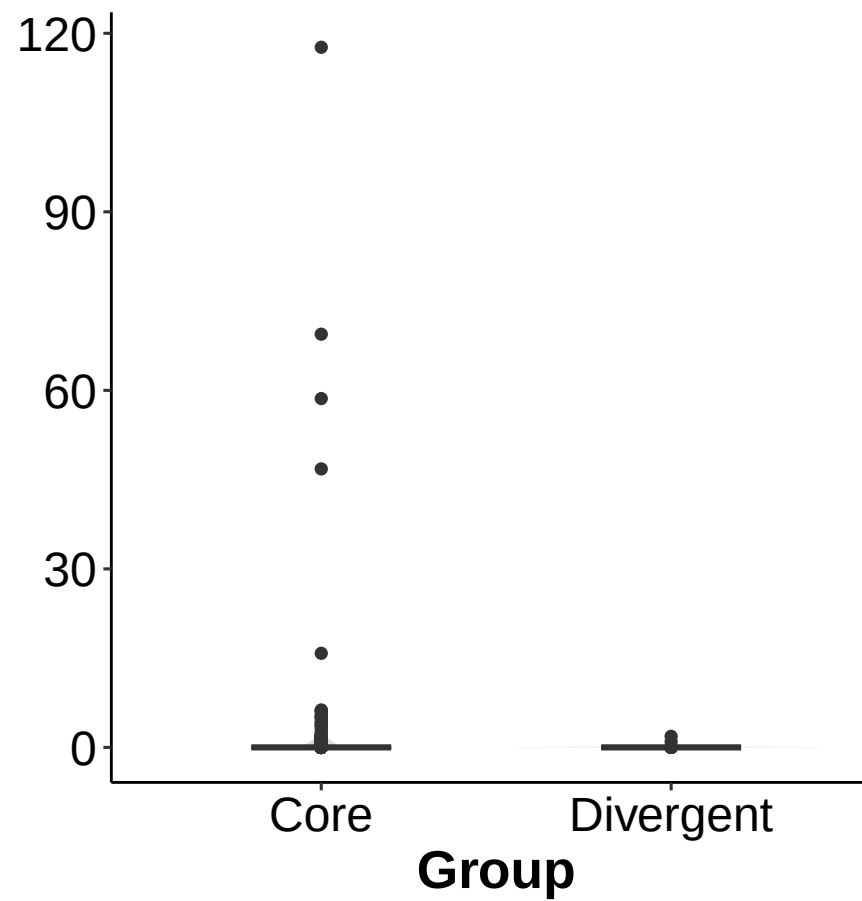
SA_Tyrosine_metabolism_L1w

p-adj: 0.0624 | .=-0.19 (Pequeño) | .%: -52.1



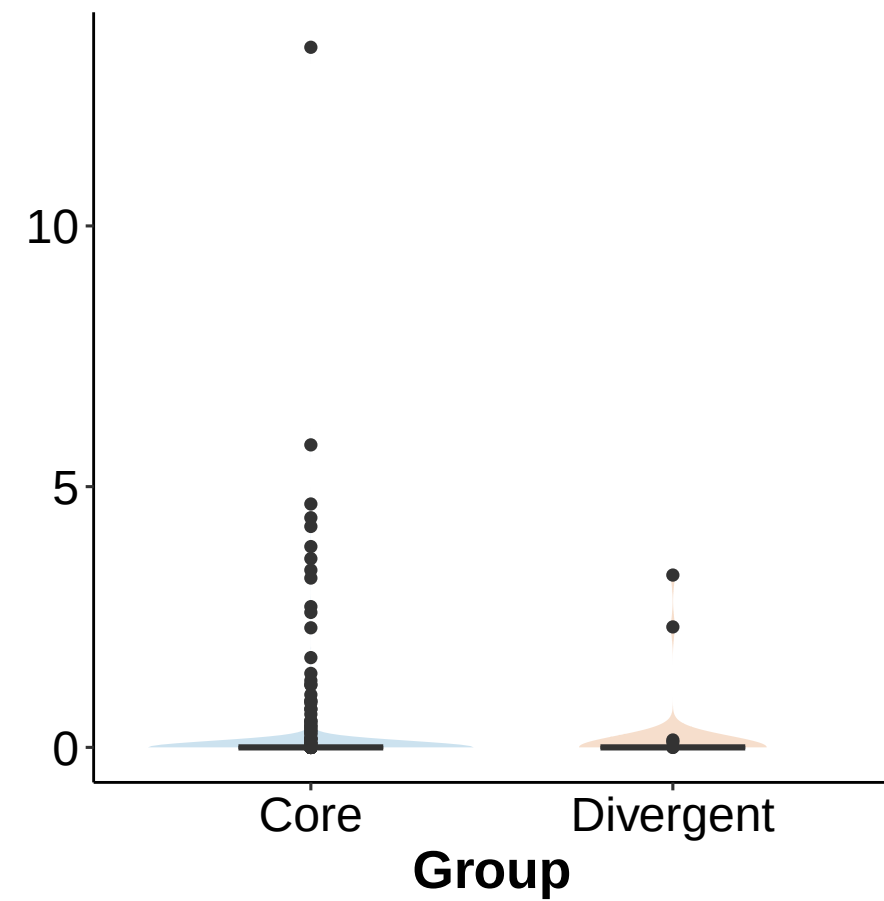
SA_Tyrosine_metabolism_pFBA

p-adj: 0.951 | .=0 (Negligible)



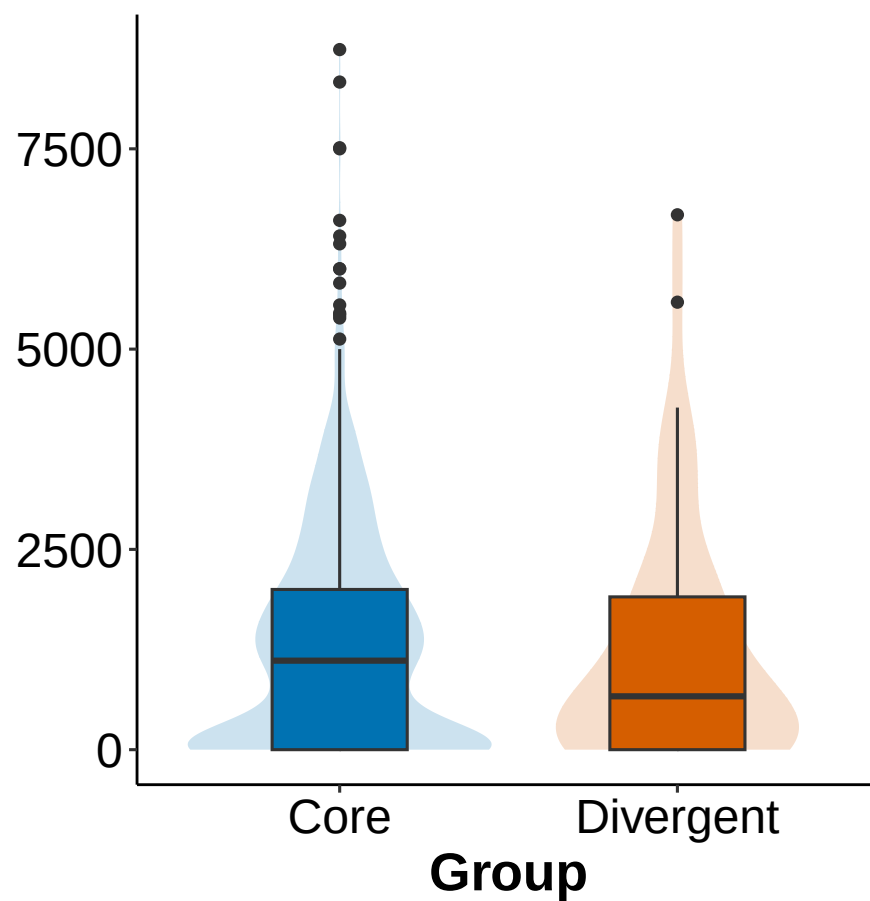
SA_Ubiquinone_synthesis_L1w

p-adj: 0.68 | .=0.03 (Negligible)



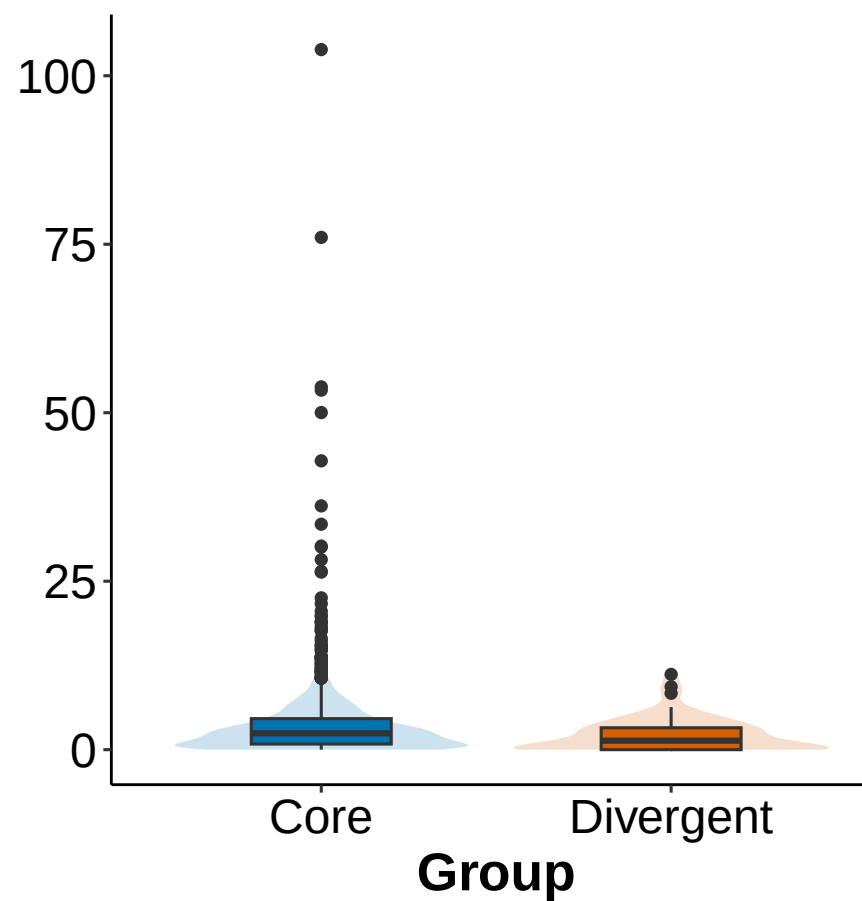
SA_Urea_cycle_FBA

p-adj: 0.542 | .=-0.08 (Negligible) | .%: -40.1



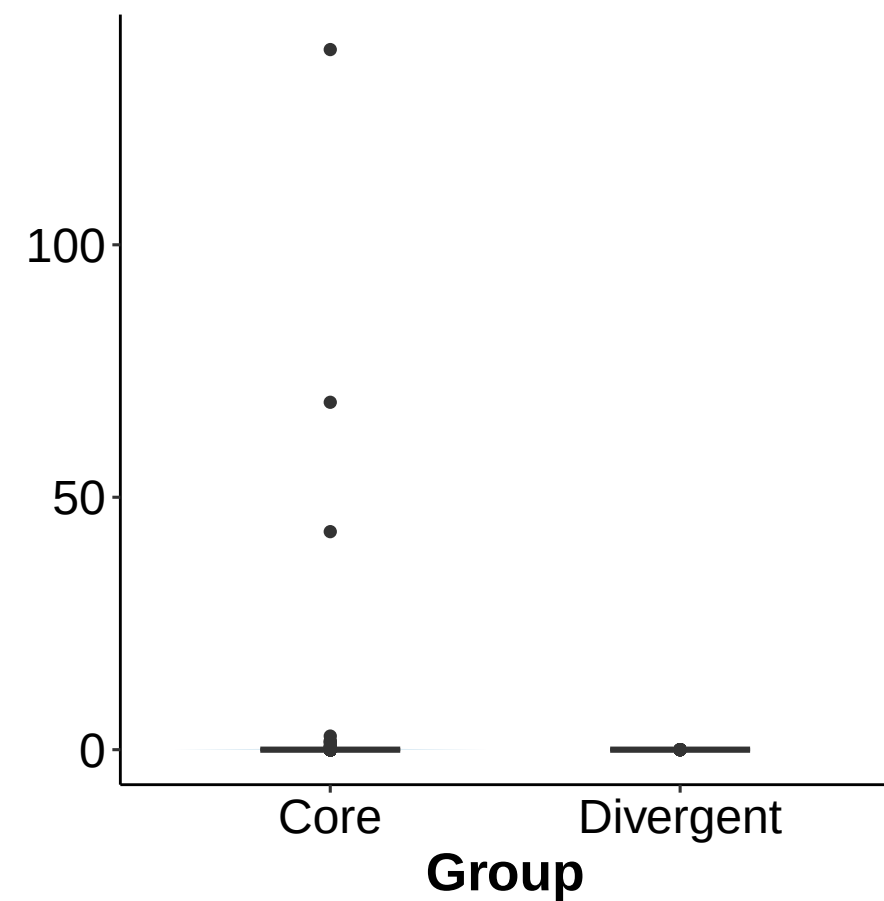
SA_Urea_cycle_L1w

p-adj: 0.00831 | .=-0.25 (Pequeño) | .%: -44.8



SA_Urea_cycle_pFBA

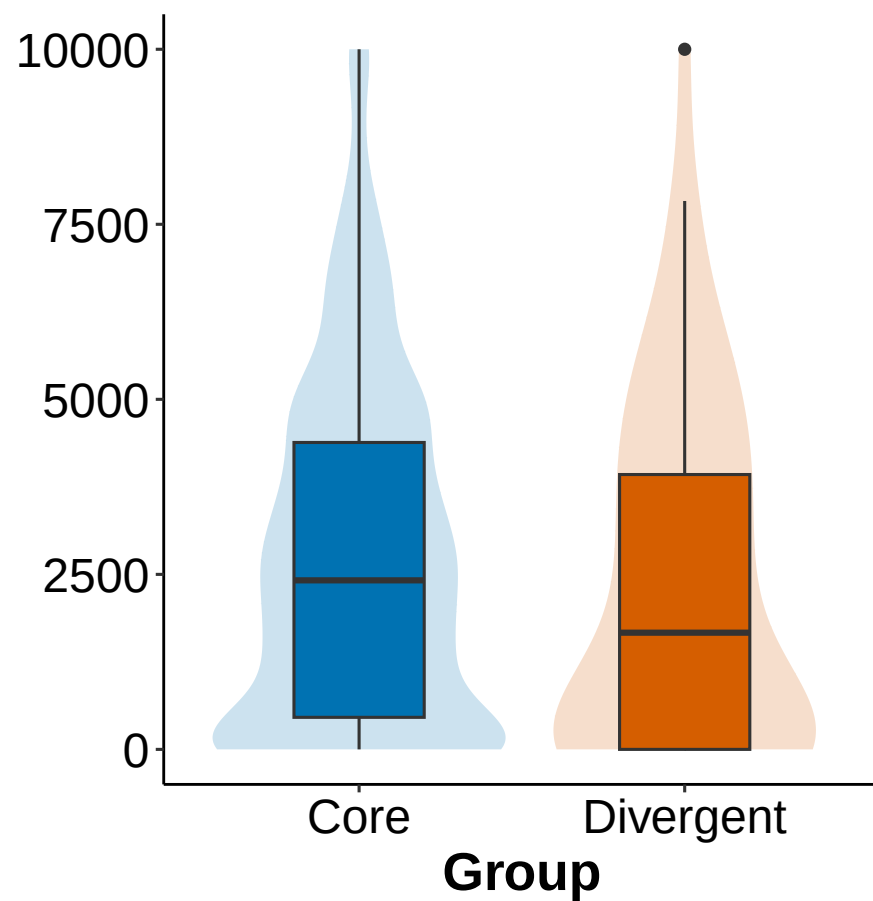
p-adj: 0.0185 | .=0.18 (Pequeño)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

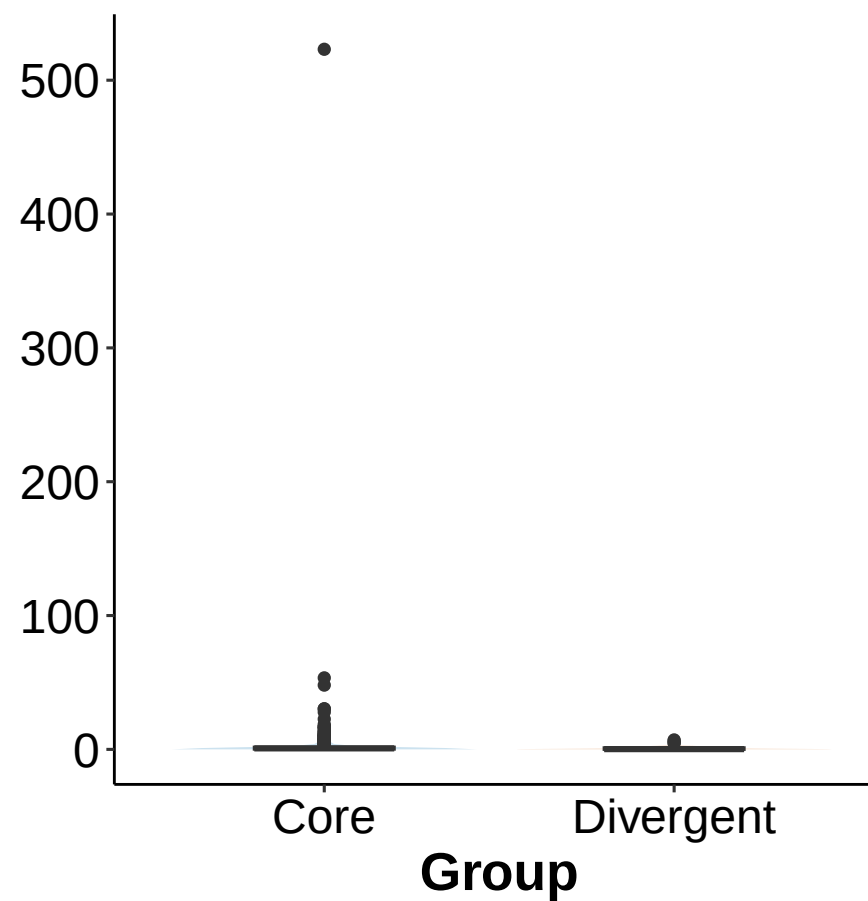
SA_Valine__leucine__and_isoleuc

p-adj: 0.238 | .=-0.13 (Negligible) | .%: -31



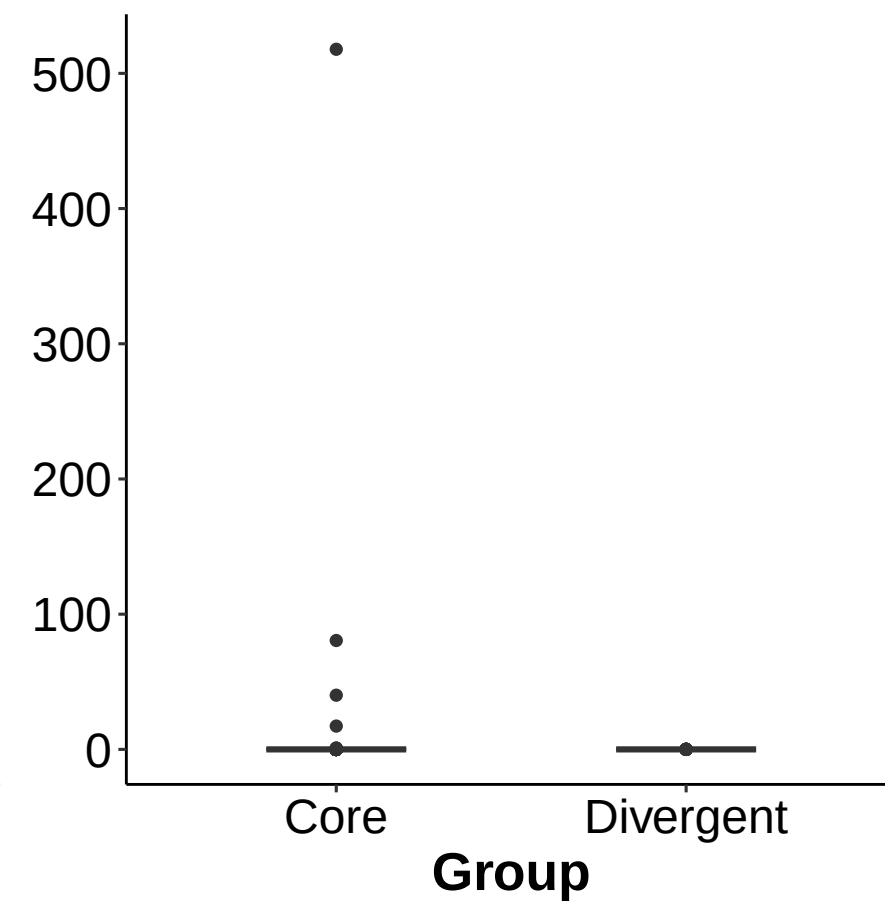
SA_Valine__leucine__and_isoleucin

p-adj: 0.00944 | .=-0.25 (Pequeño) | .%: -73



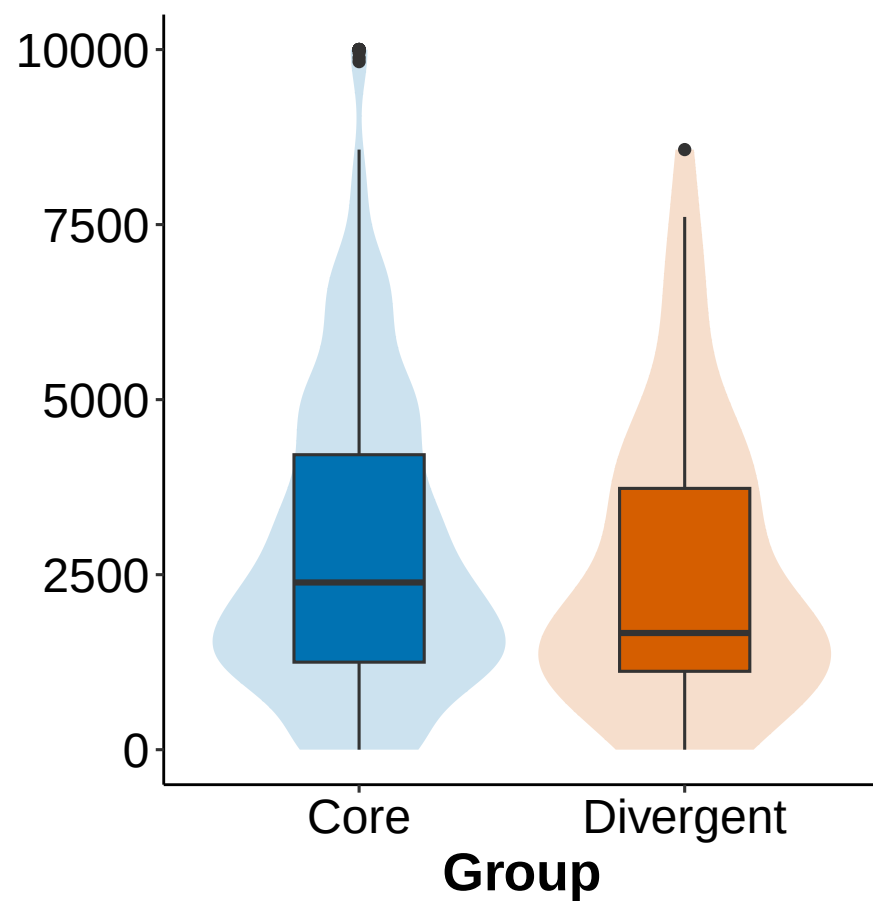
SA_Valine__leucine__and_isoleucin

p-adj: 0.537 | .-=0.05 (Negligible)



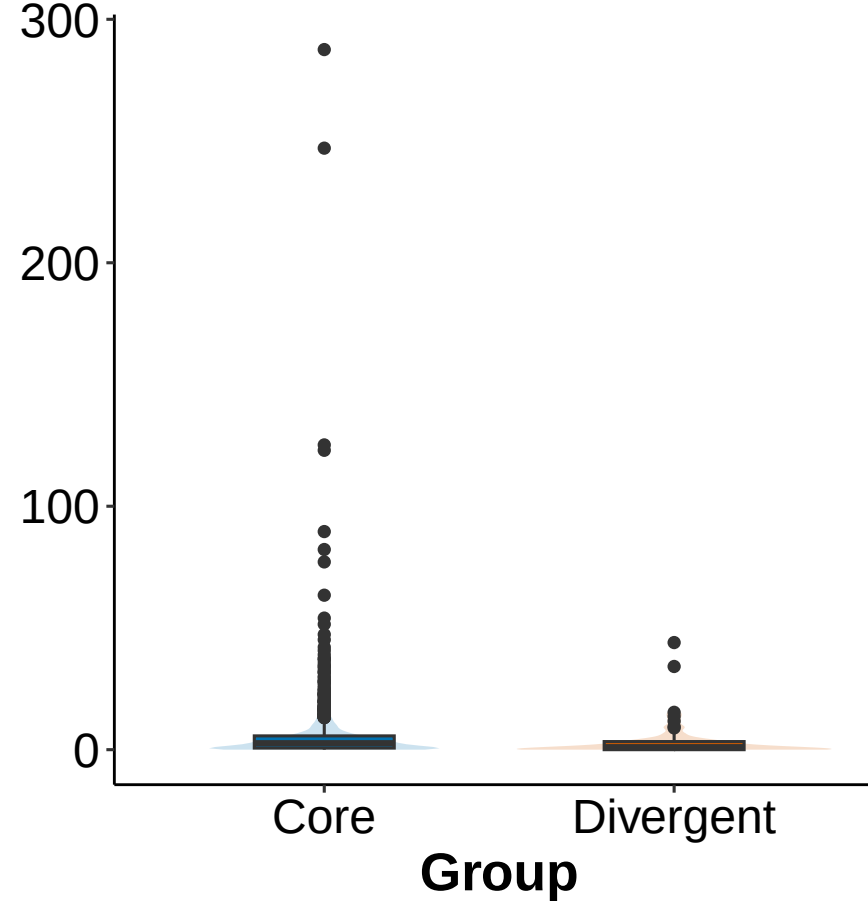
SA_Vitamin_A_metabolism_FBA

p-adj: 0.411 | .=-0.1 (Negligible) | .%: -30.2



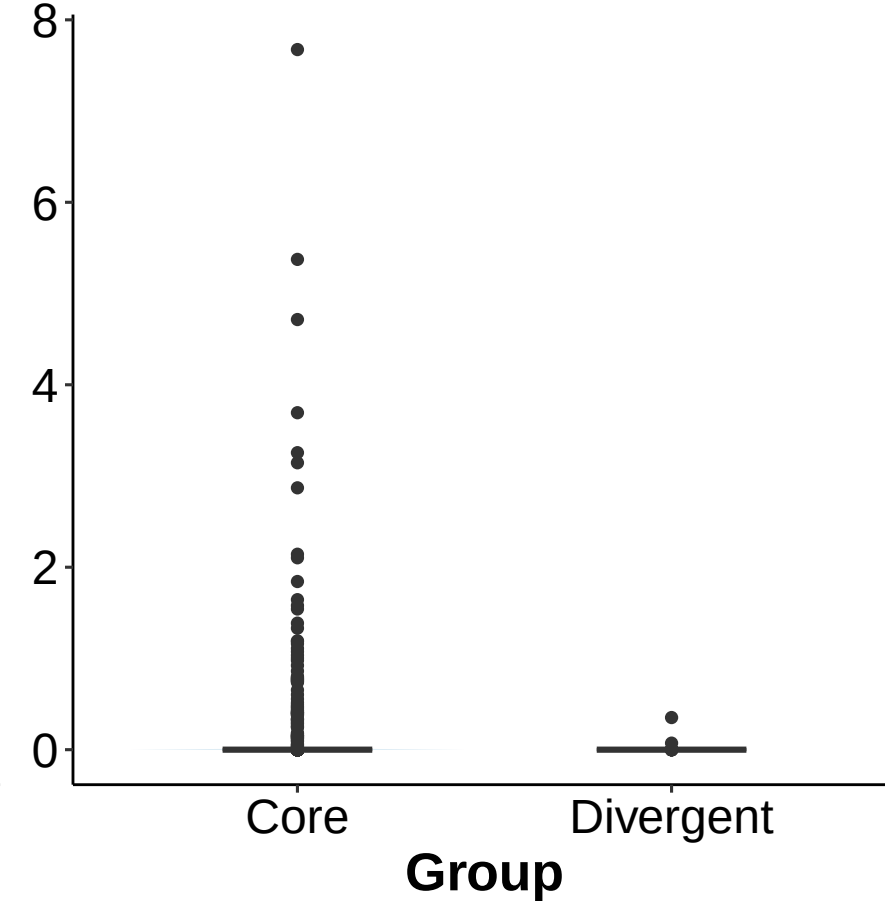
SA_Vitamin_A_metabolism_L1w

p-adj: 0.00213 | .=-0.29 (Pequeño) | .%: -61.3



SA_Vitamin_B2_metabolism_L1w

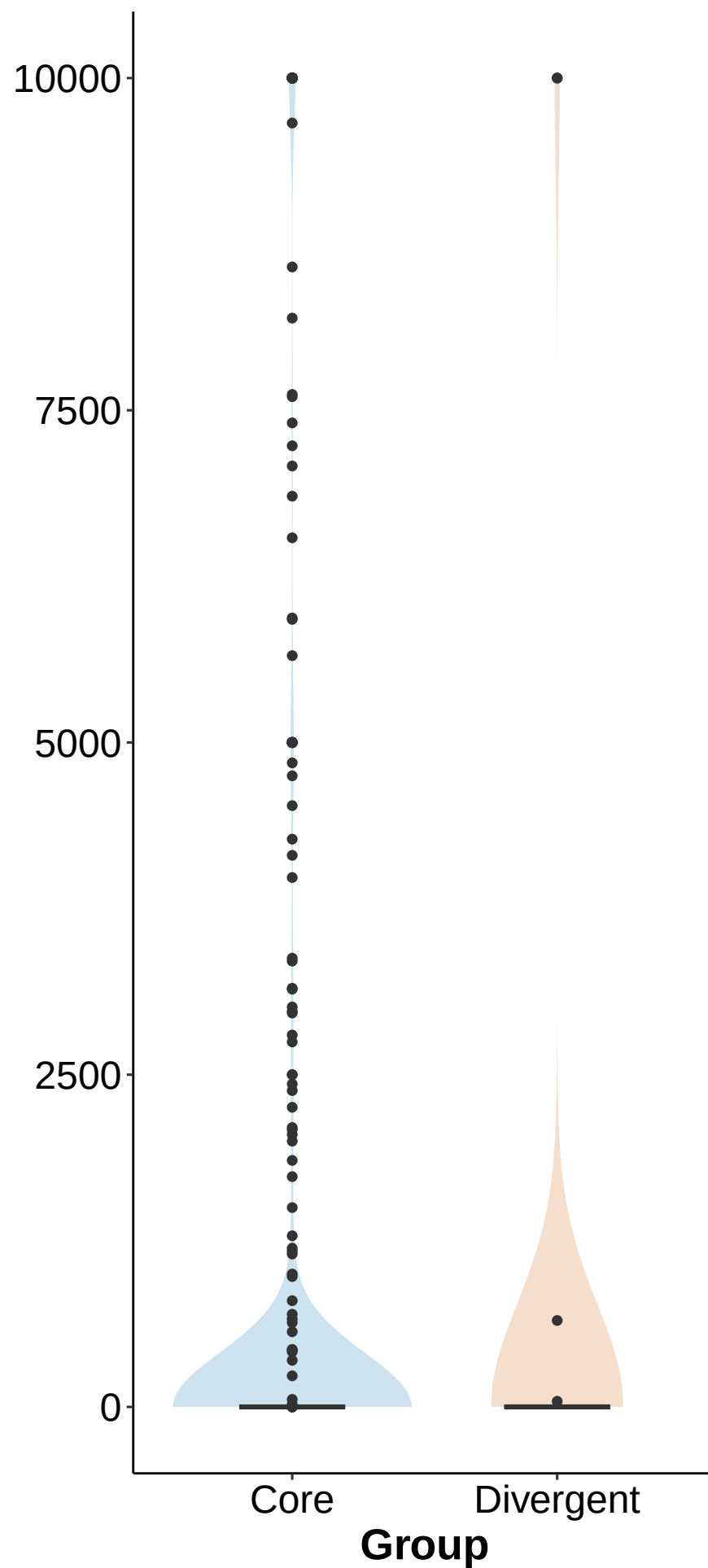
p-adj: 0.29 | .=-0.09 (Negligible)



Metabolismo – . 5. Reprogramación Sistémica (SA_*)

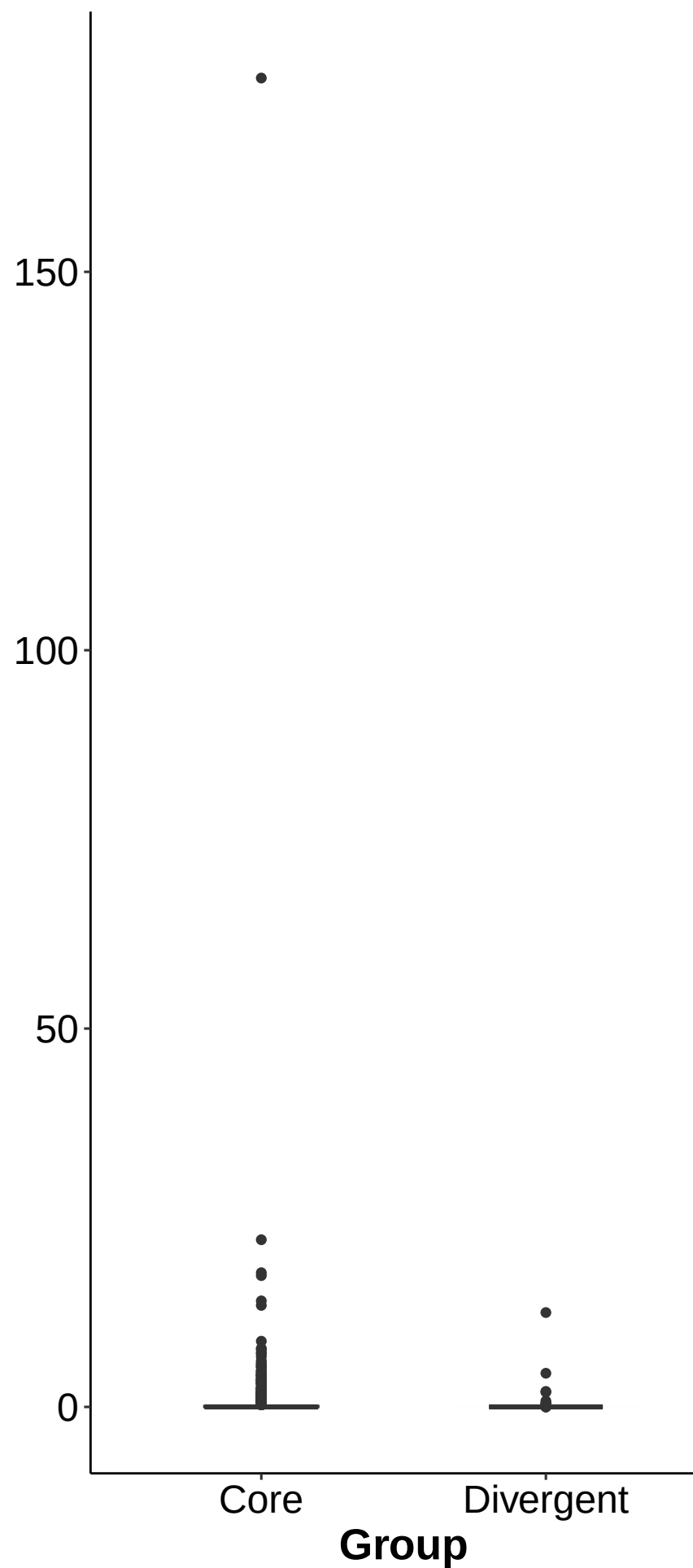
SA_Vitamin_C_metabolism_FBA

p-adj: 0.804 | .=-0.02 (Negligible)



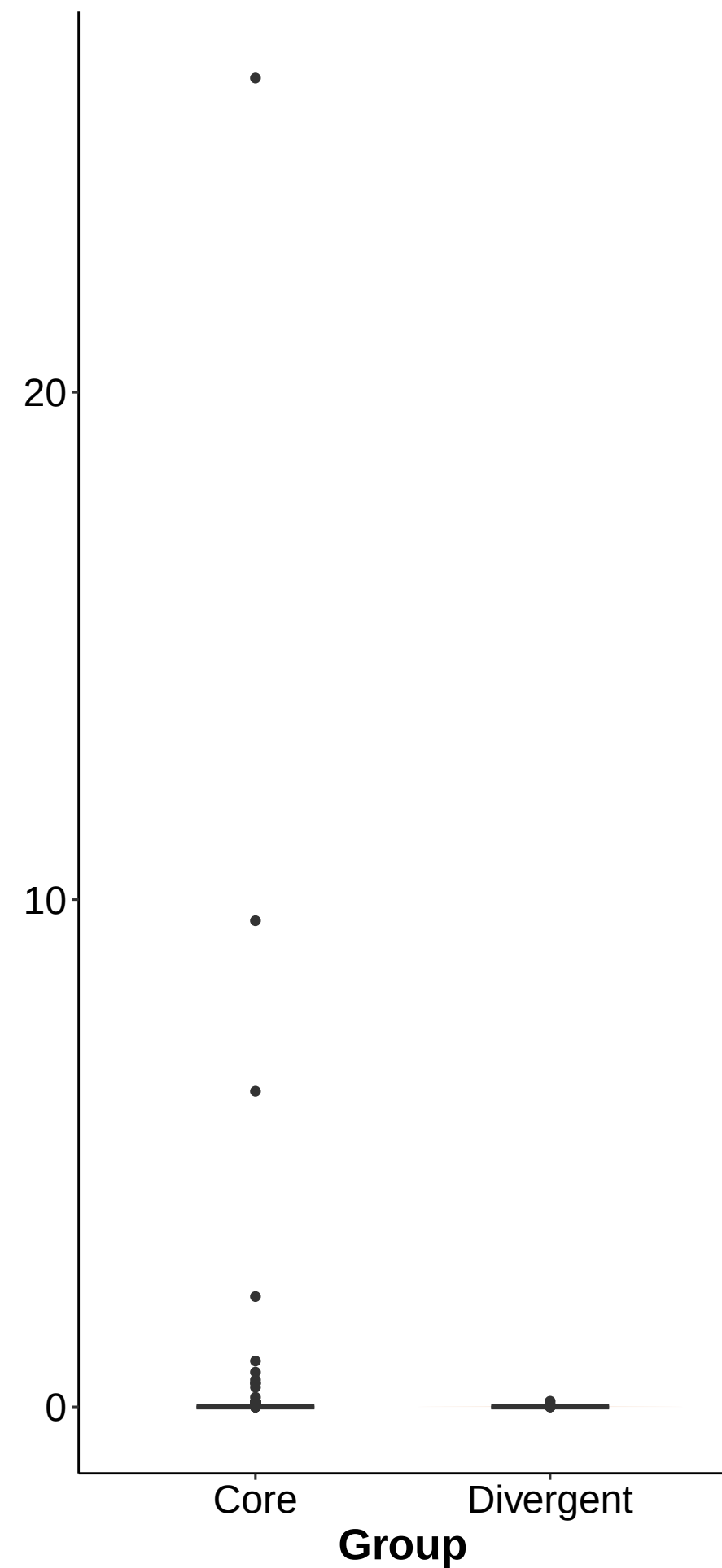
SA_Vitamin_C_metabolism_L1w

p-adj: 0.432 | .=-0.09 (Negligible) | .%: -86.9



SA_Vitamin_D_metabolism_L1w

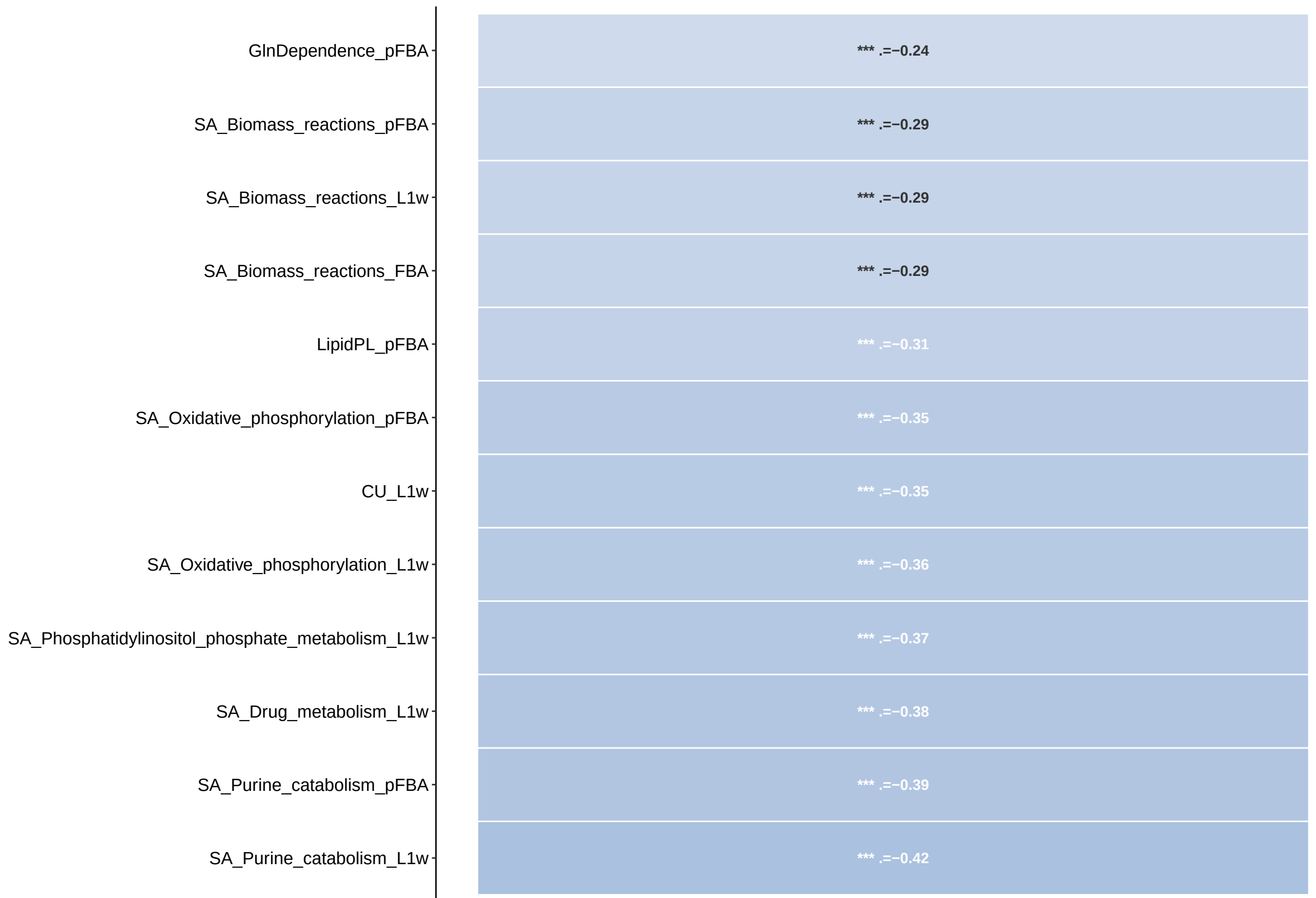
p-adj: 0.838 | .=-0.02 (Negligible)



Cliff's .: Top 12 Variables

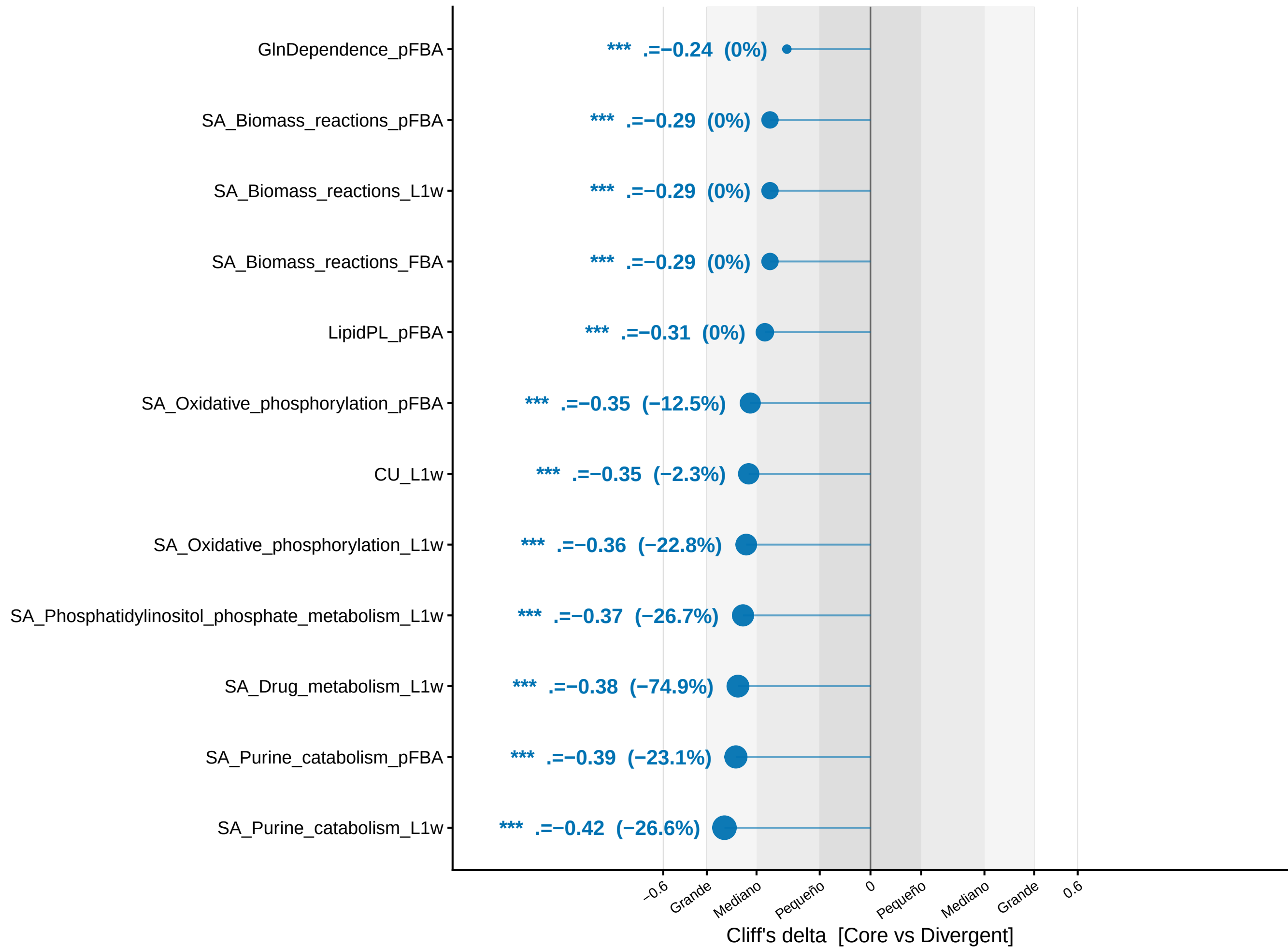
Azul = Core > Divergent | Naranja = Divergent > Core | * p_adj<0.05

Cliff's .
Div vs Core



. Top 12 Variables – Cliff's .

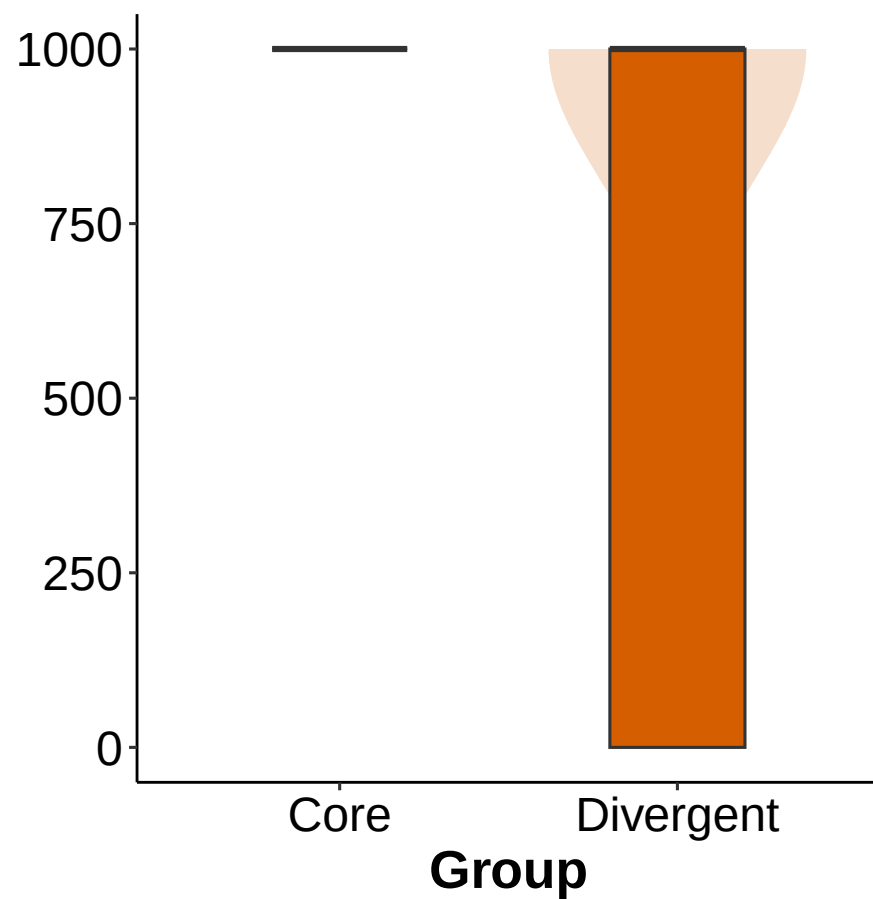
Cliff's . + .% mediana | azul = mayor en Core · naranja = mayor en Divergent



. Top 12 Variables Metabólicas (p_adj + Cliff's .)

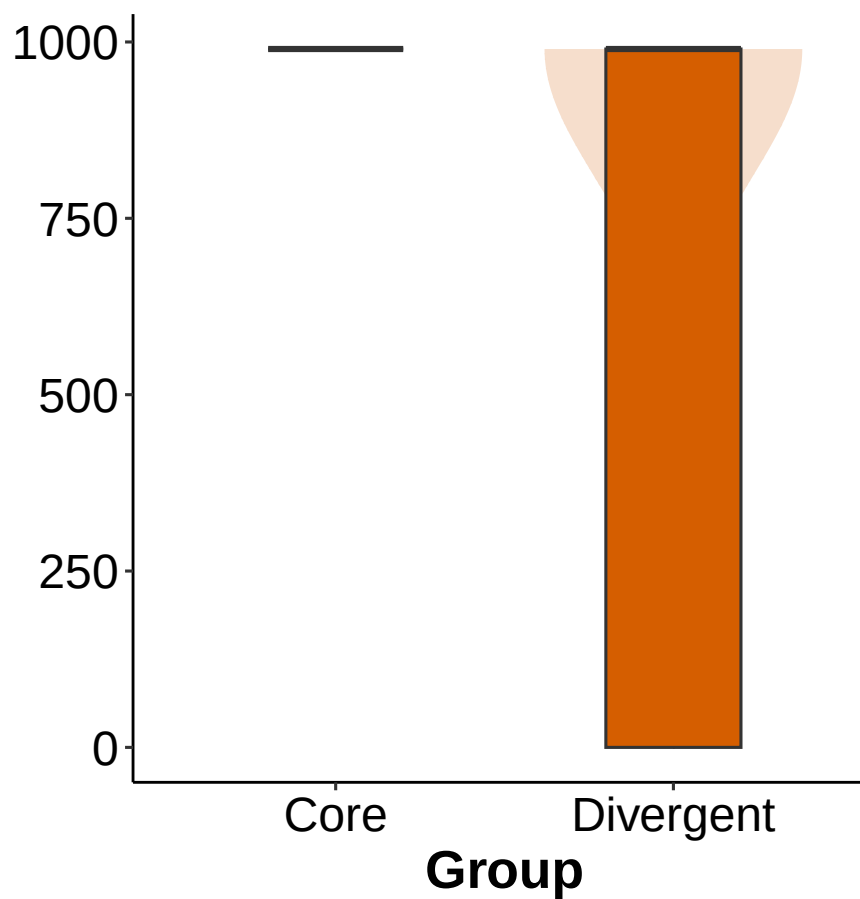
SA_Biomass_reactions_FBA

p-adj: 2.96e-71 | .=-0.29 (Pequeño) | .%: 0



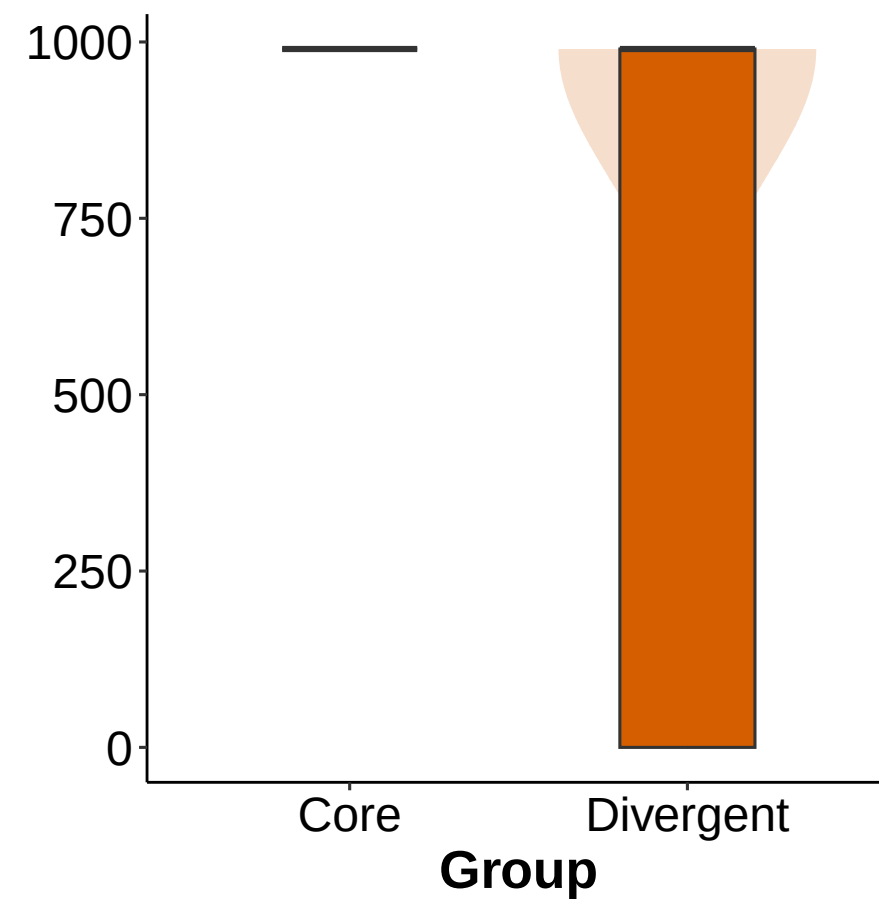
SA_Biomass_reactions_L1w

p-adj: 2.96e-71 | .=-0.29 (Pequeño) | .%: 0



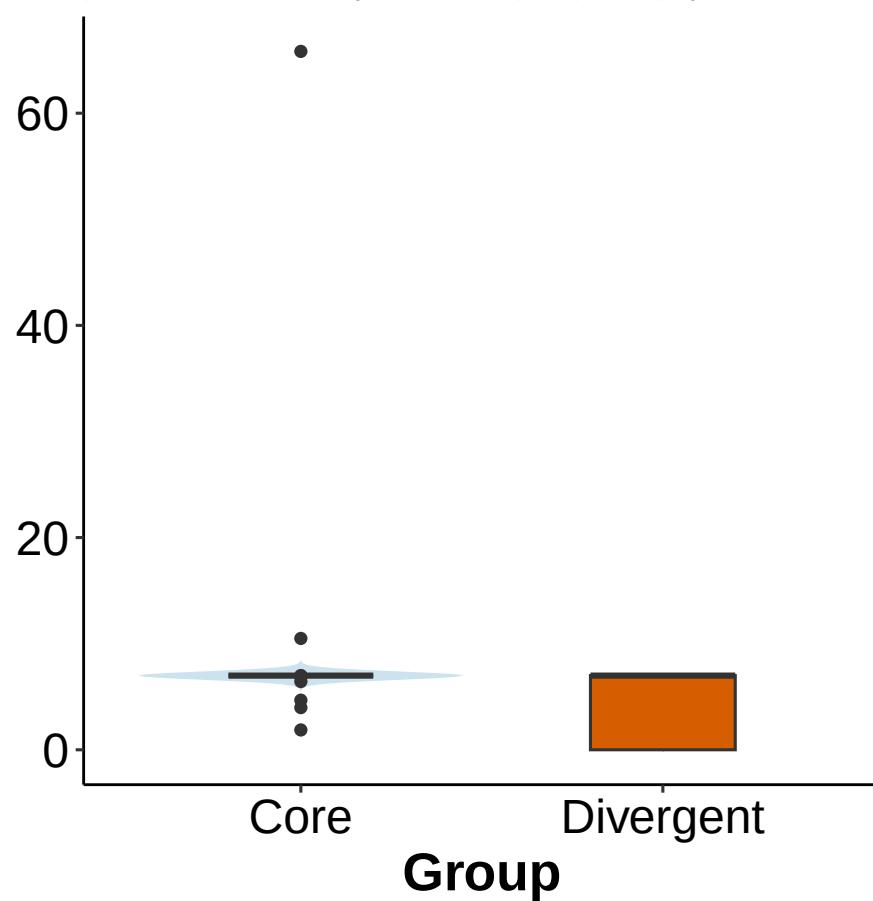
SA_Biomass_reactions_pFBA

p-adj: 2.96e-71 | .=-0.29 (Pequeño) | .%: 0



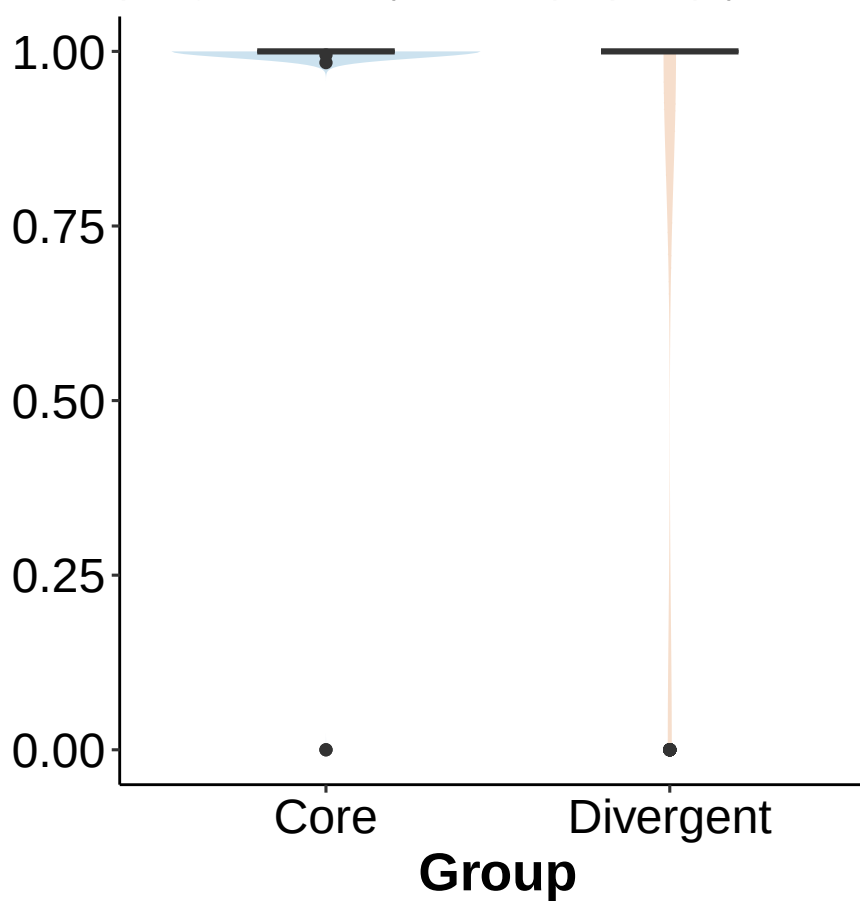
LipidPL_pFBA

p-adj: 5.79e-49 | .=-0.31 (Pequeño) | .%: 0



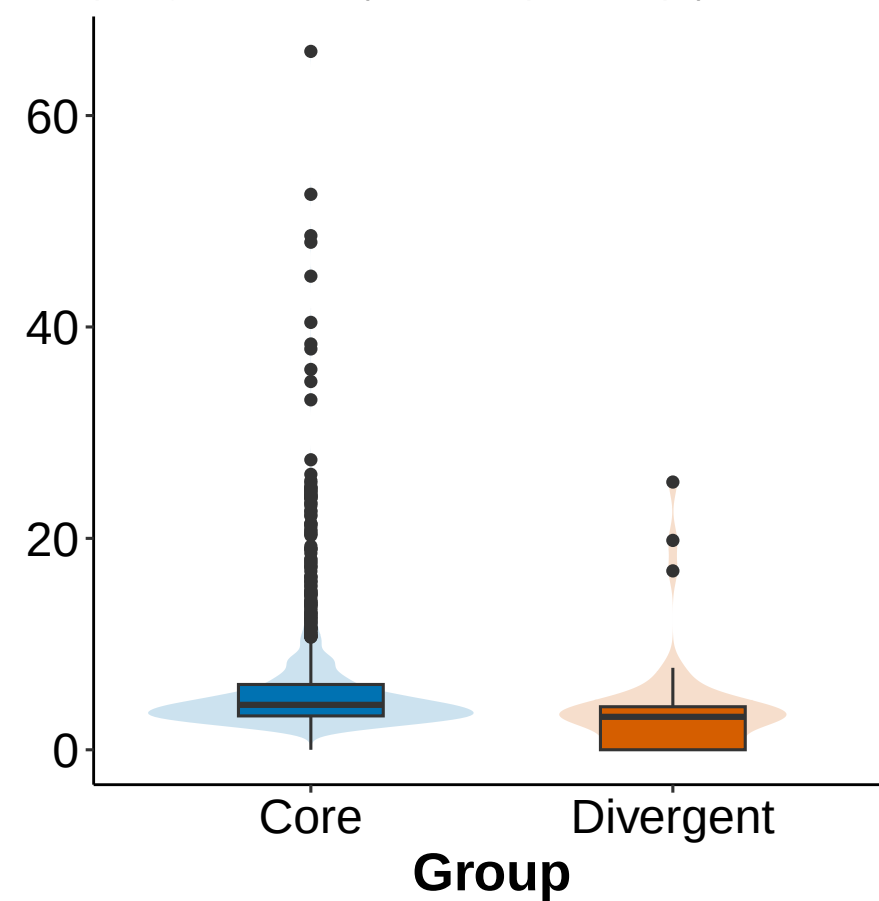
GlnDependence_pFBA

p-adj: 1.69e-44 | .=-0.24 (Pequeño) | .%: 0



SA_Purine_catabolism_L1w

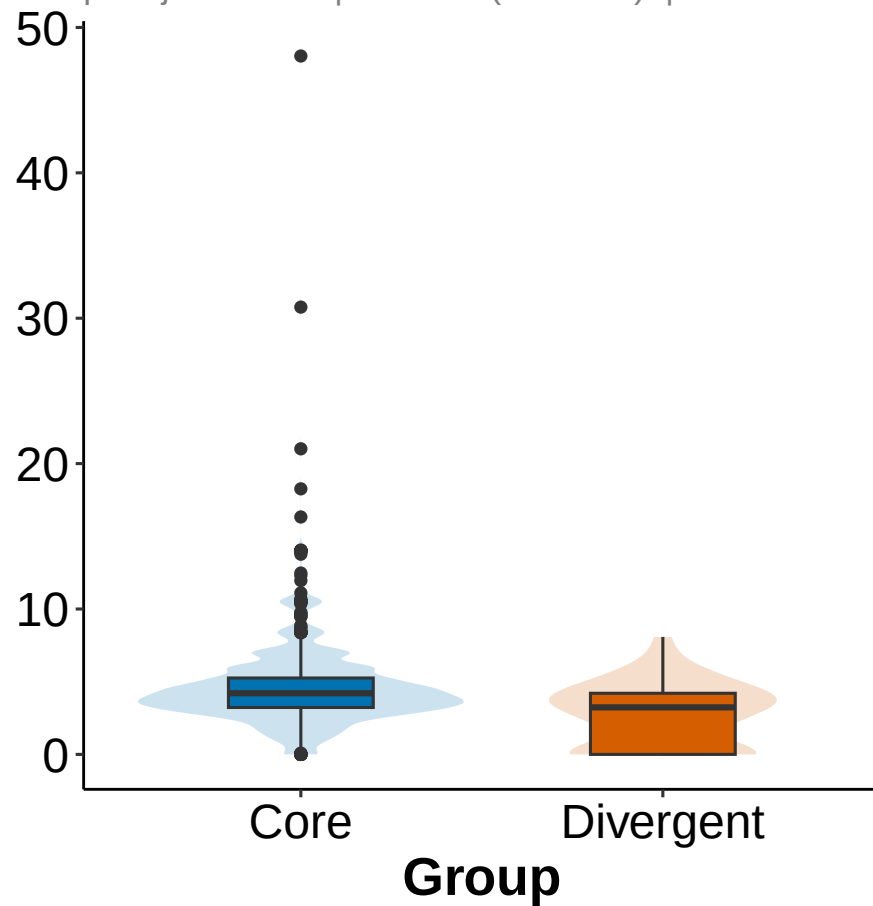
p-adj: 4.83e-06 | .=-0.42 (Mediano) | .%: -26.6



. Top 12 Variables Metabólicas (p_adj + Cliff's .)

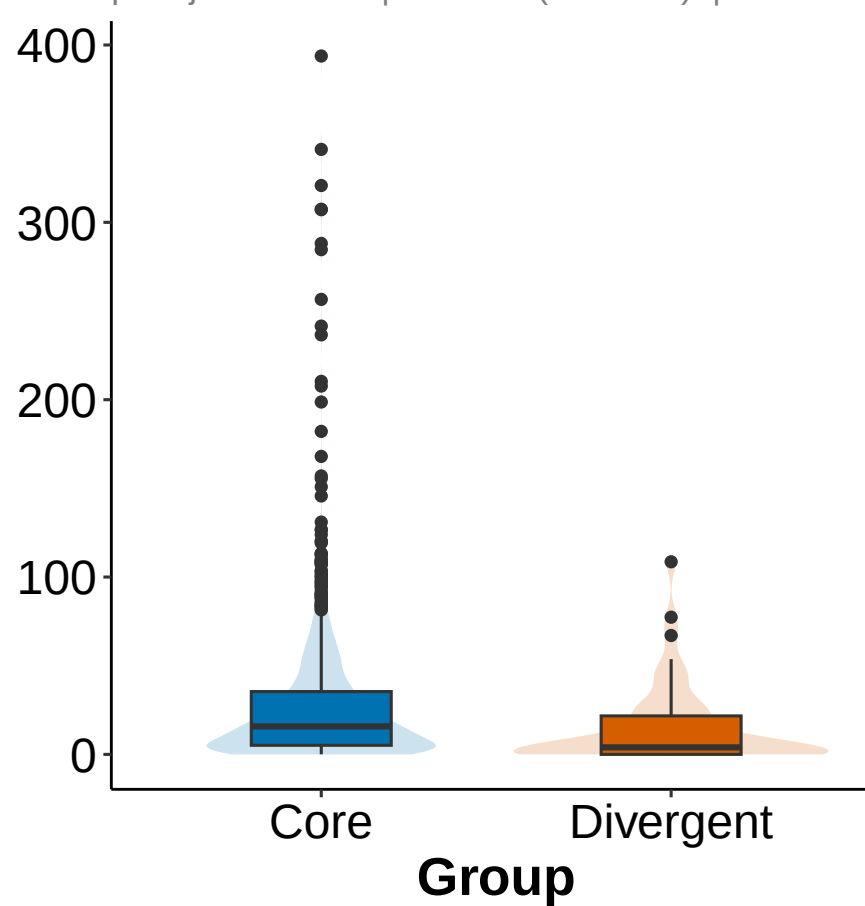
SA_Purine_catabolism_pFBA

p-adj: 3.6e-05 | .=-0.39 (Mediano) | .%: -23.1



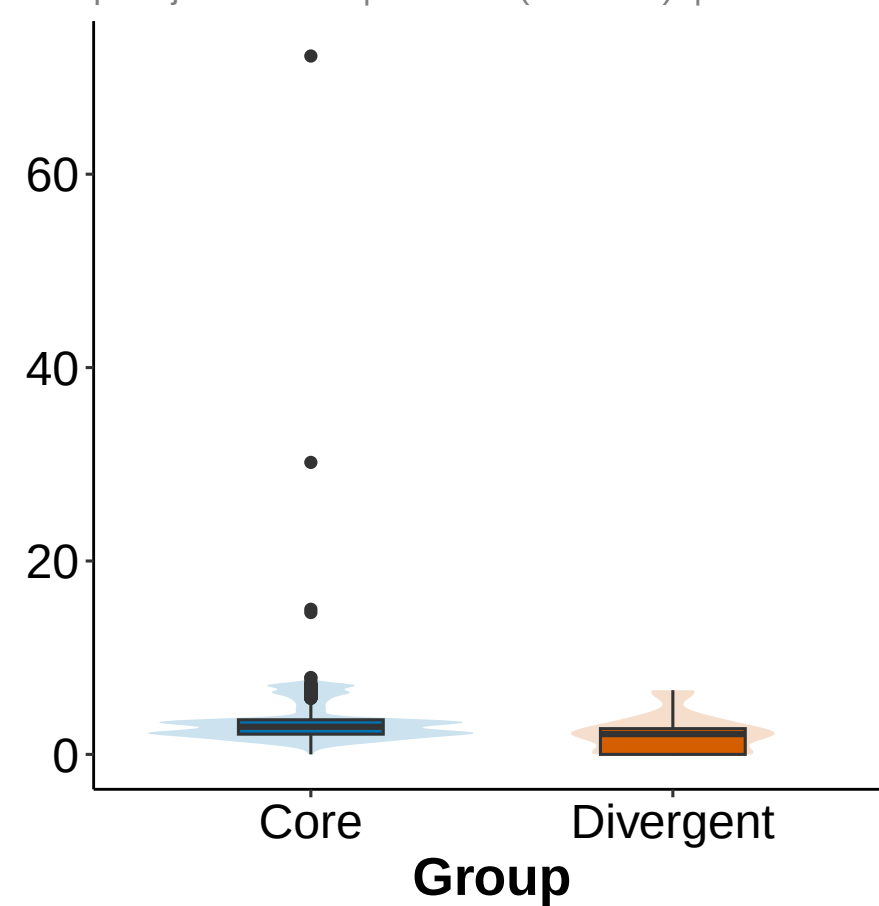
SA_Drug_metabolism_L1w

p-adj: 4.71e-05 | .=-0.38 (Mediano) | .%: -74.9



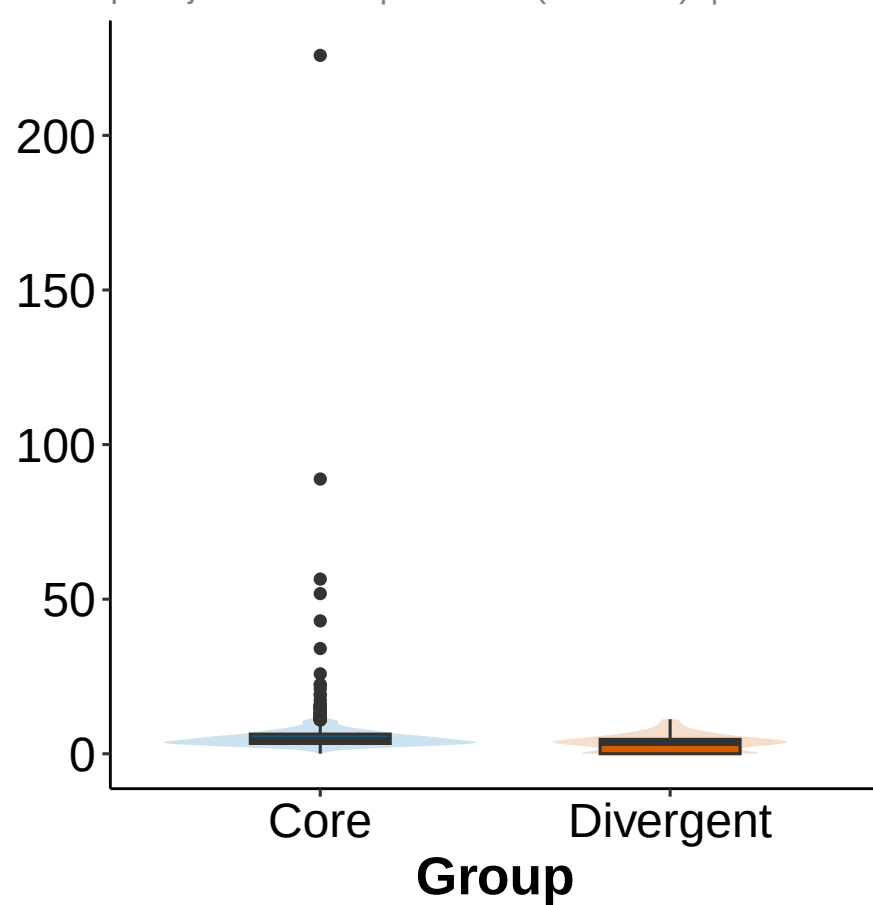
SA_Phosphatidylinositol_phosphate

p-adj: 0.000103 | .=-0.37 (Mediano) | .%: -26.7



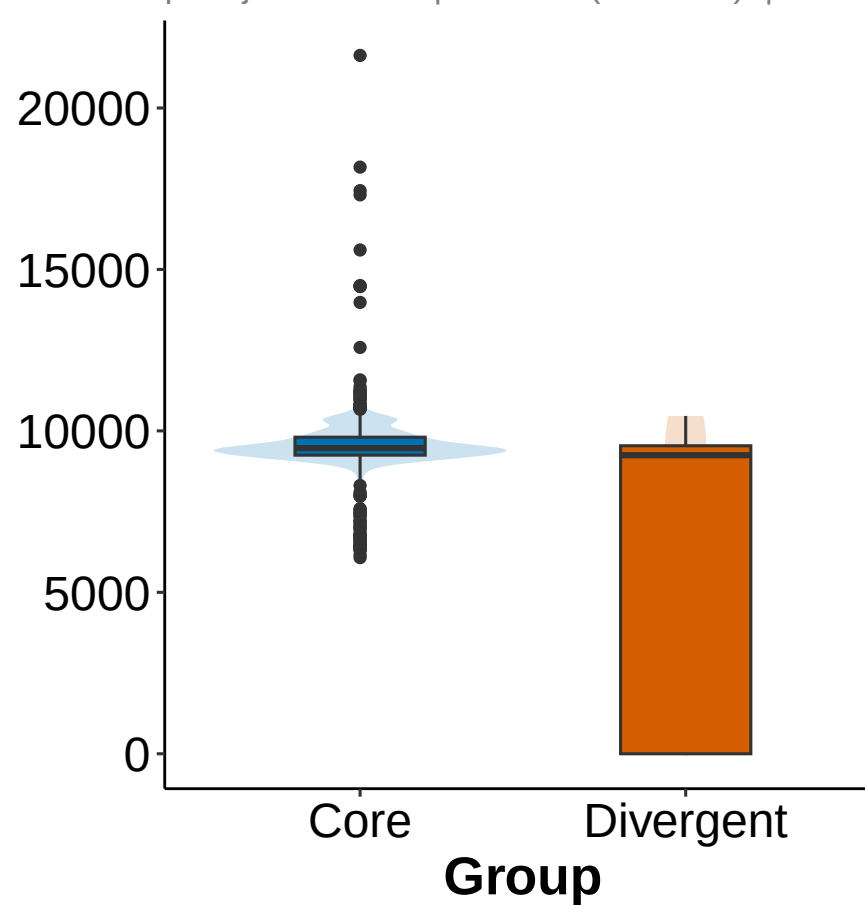
SA_Oxidative_phosphorylation_L1w

p-adj: 0.000162 | .=-0.36 (Mediano) | .%: -22.8



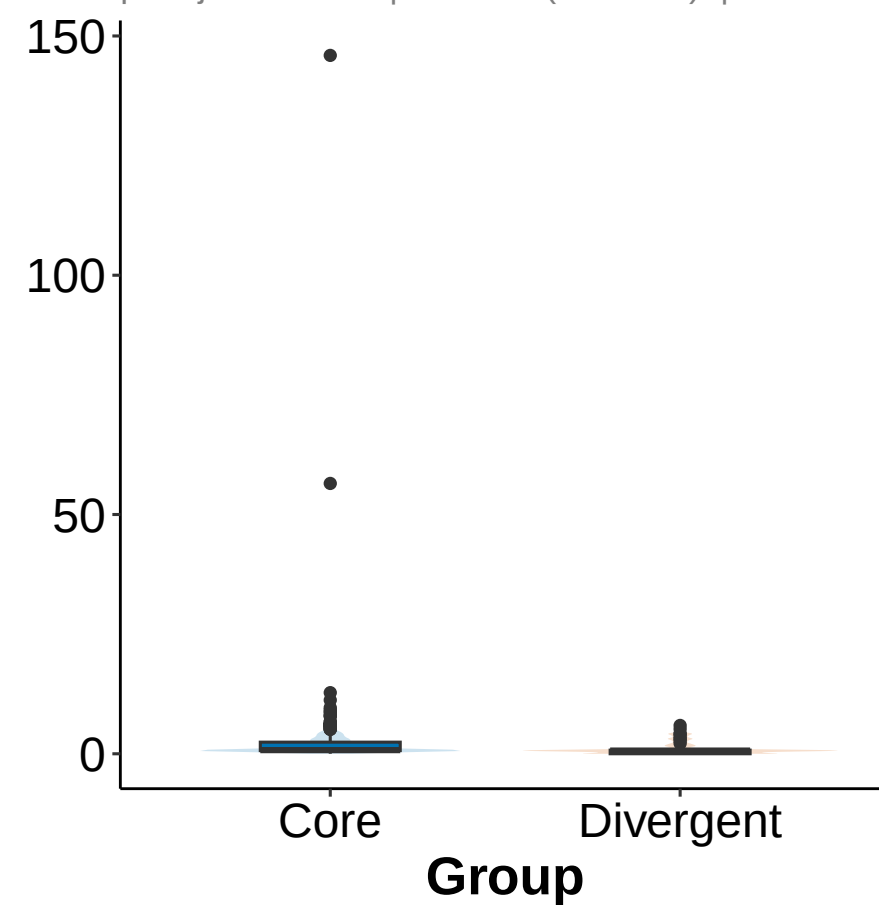
CU_L1w

p-adj: 0.000223 | .=-0.35 (Mediano) | .%: -2.



SA_Oxidative_phosphorylation_pFB

p-adj: 0.000264 | .=-0.35 (Mediano) | .%: -12.5



SECCIÓN III

Análisis Integrado: Clínica <—> Metabolismo

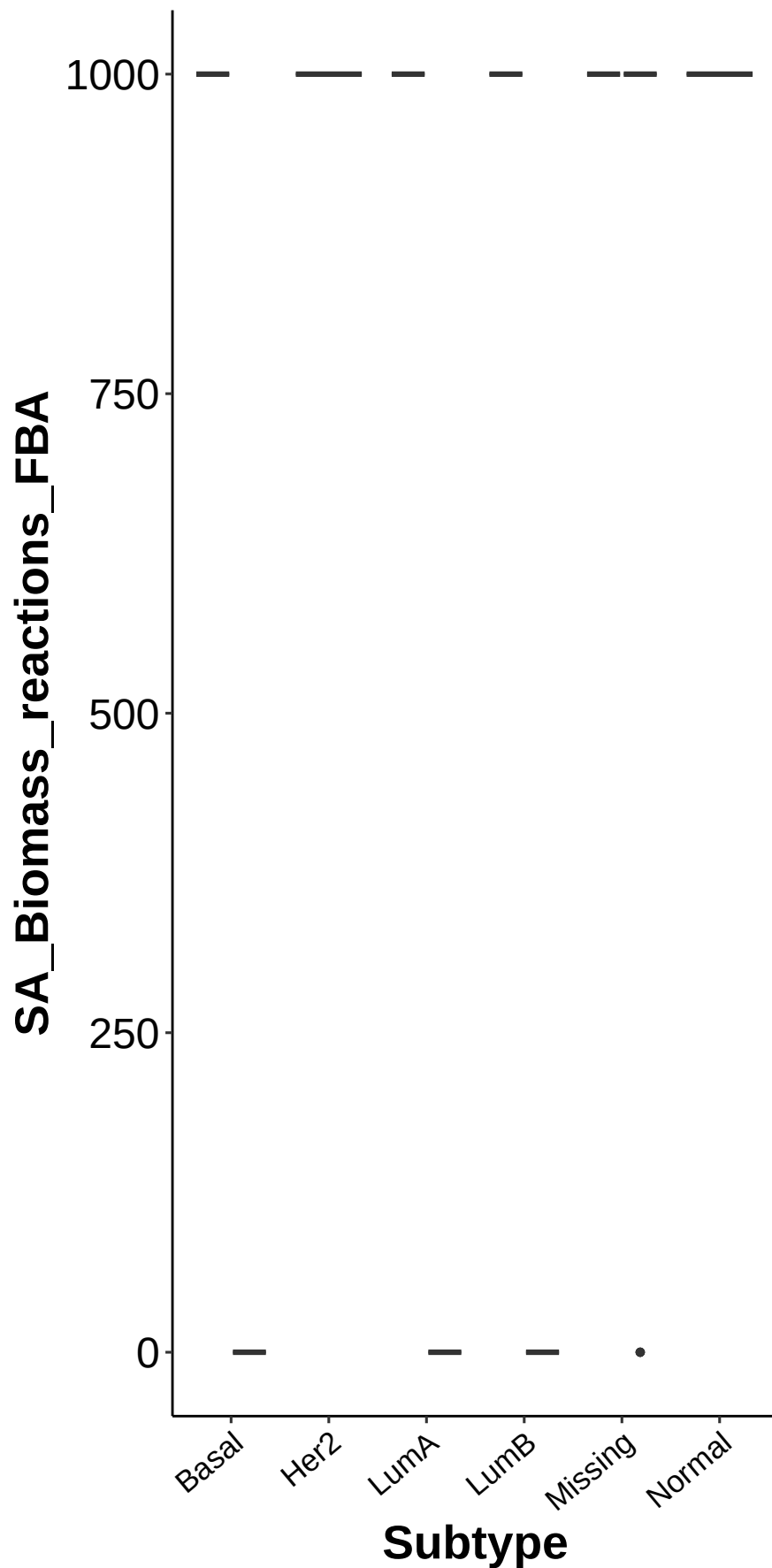
¿Core vs Divergent es reflejo del subtipo? ¿O reprogramación metabólica independiente?

Integración: Top metabólicas × Subtype

SA_Biomass_reacti

¿La diferencia Core/Divergent es

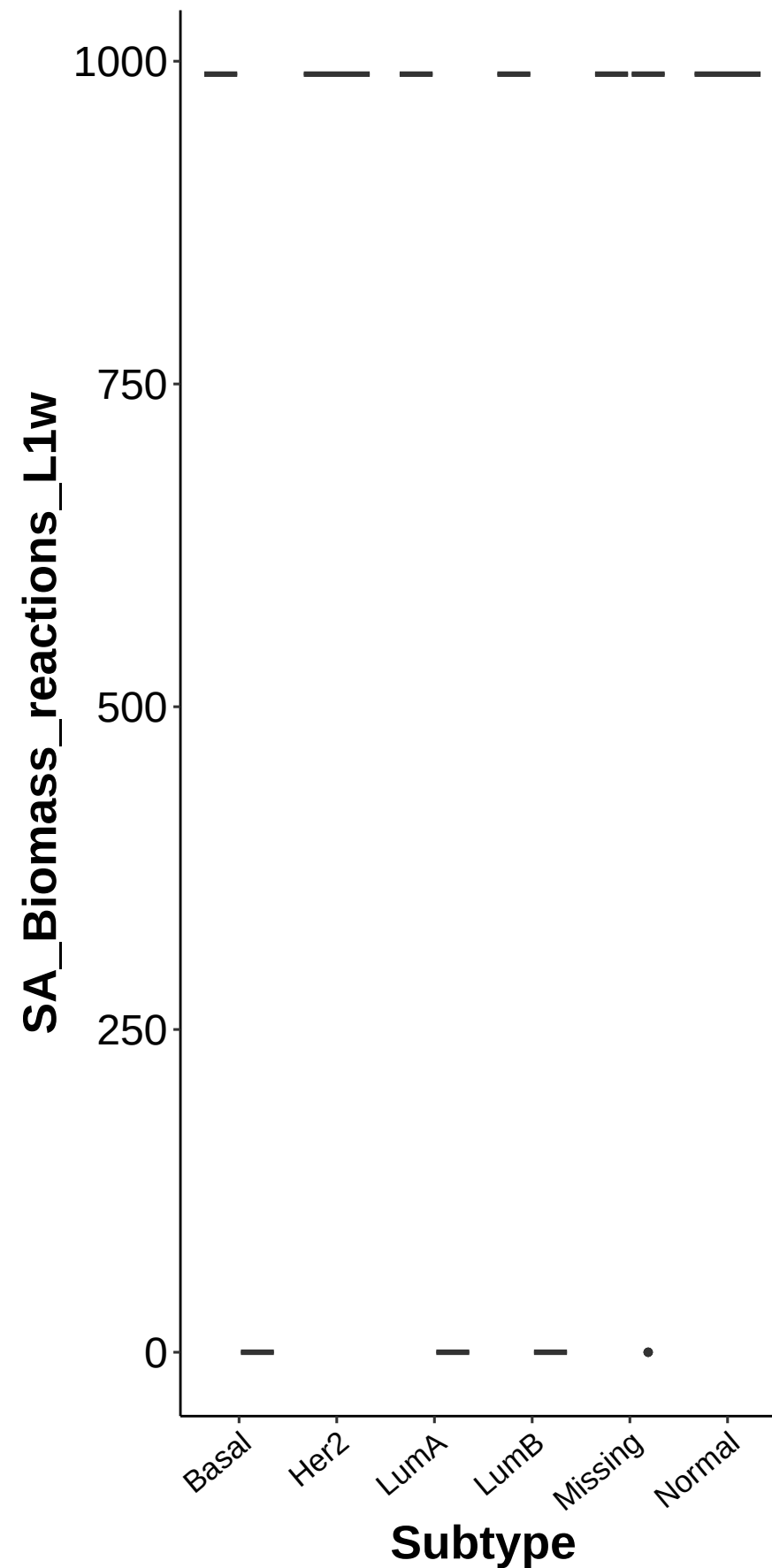
Group Core Divergent



SA_Biomass_reacti

¿La diferencia Core/Divergent es

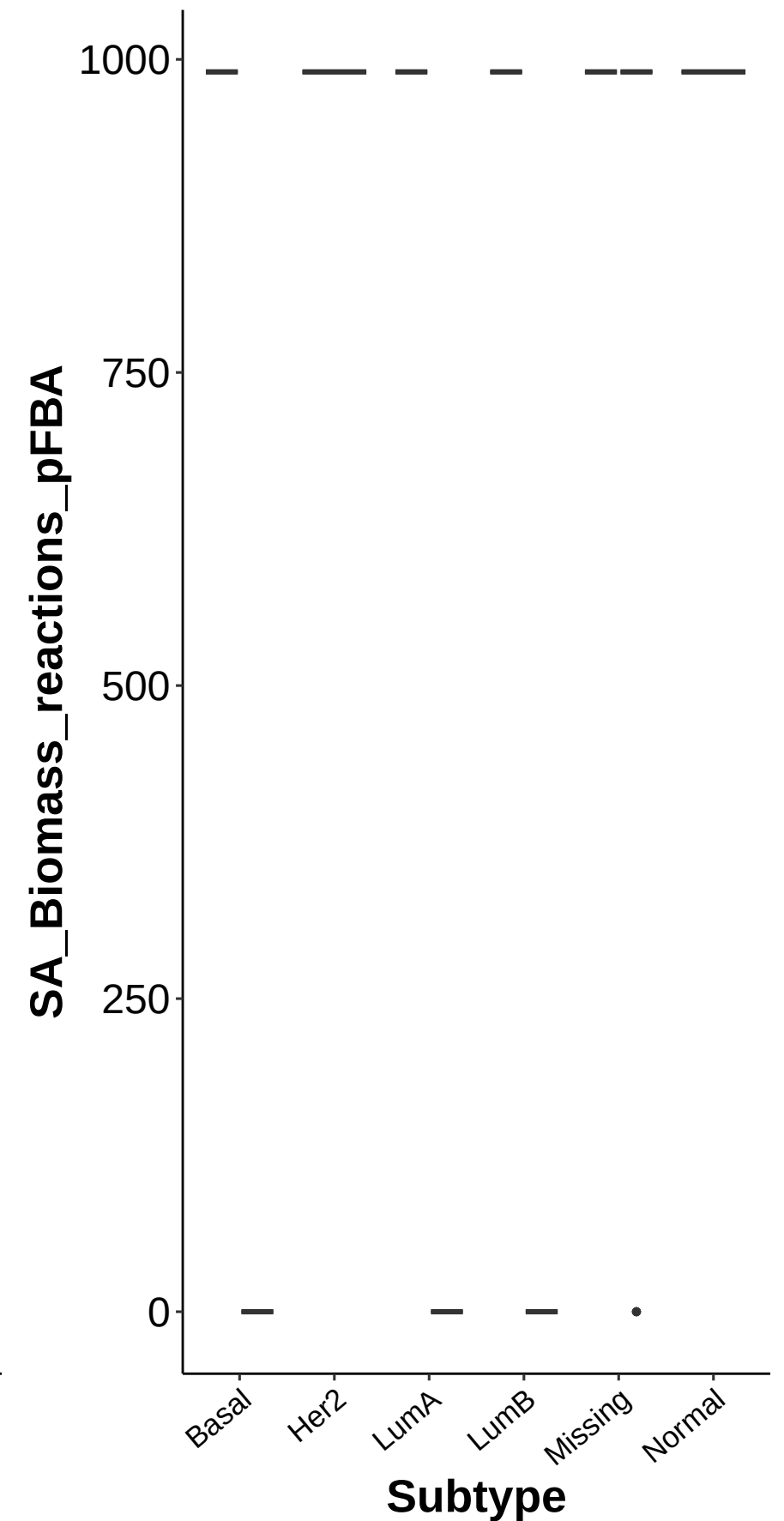
Group Core Divergent



SA_Biomass_reacti

¿La diferencia Core/Divergent es

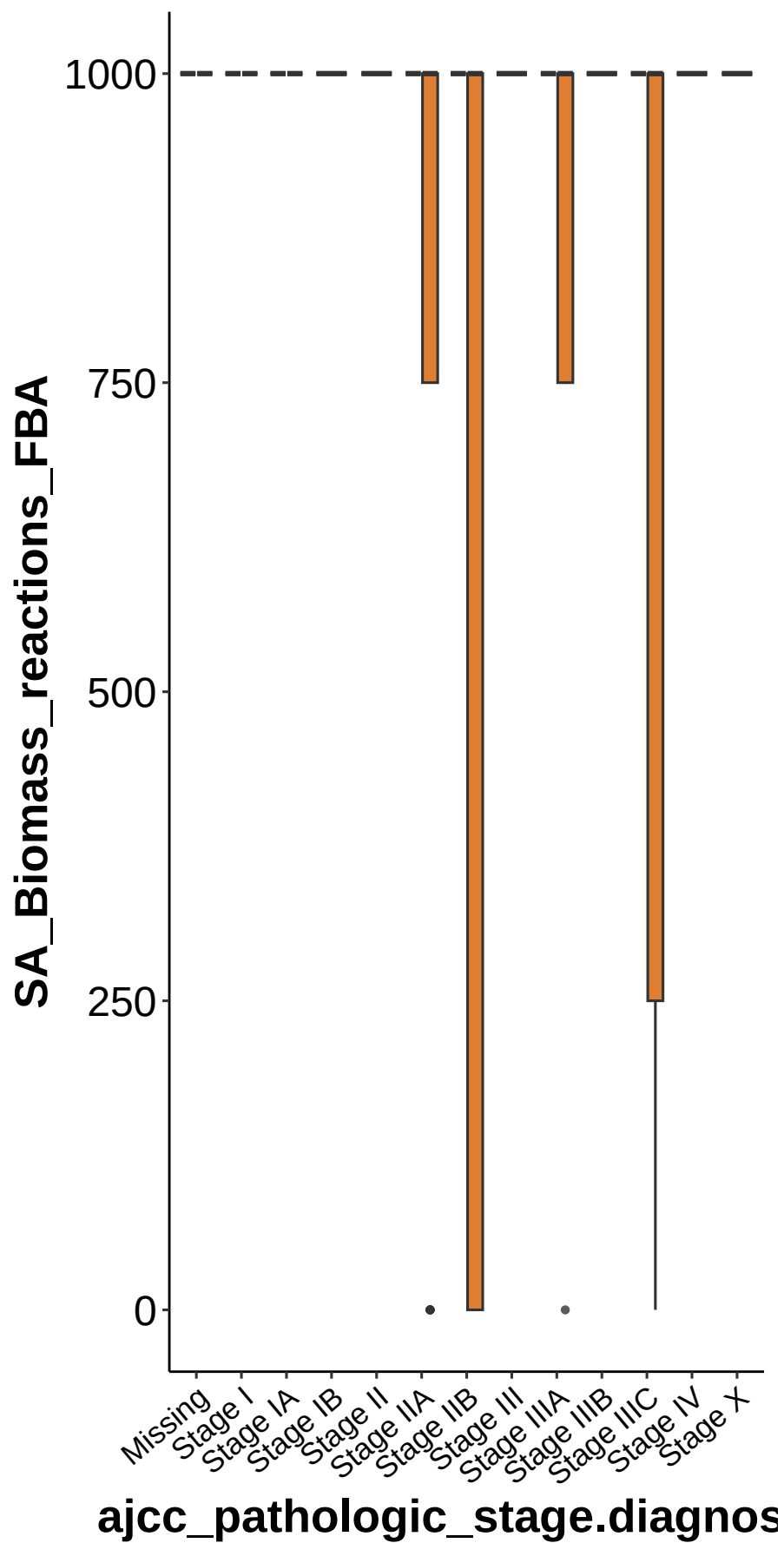
Group Core Divergent



SA_Biomass_reacti

¿La diferencia Core/Divergent es

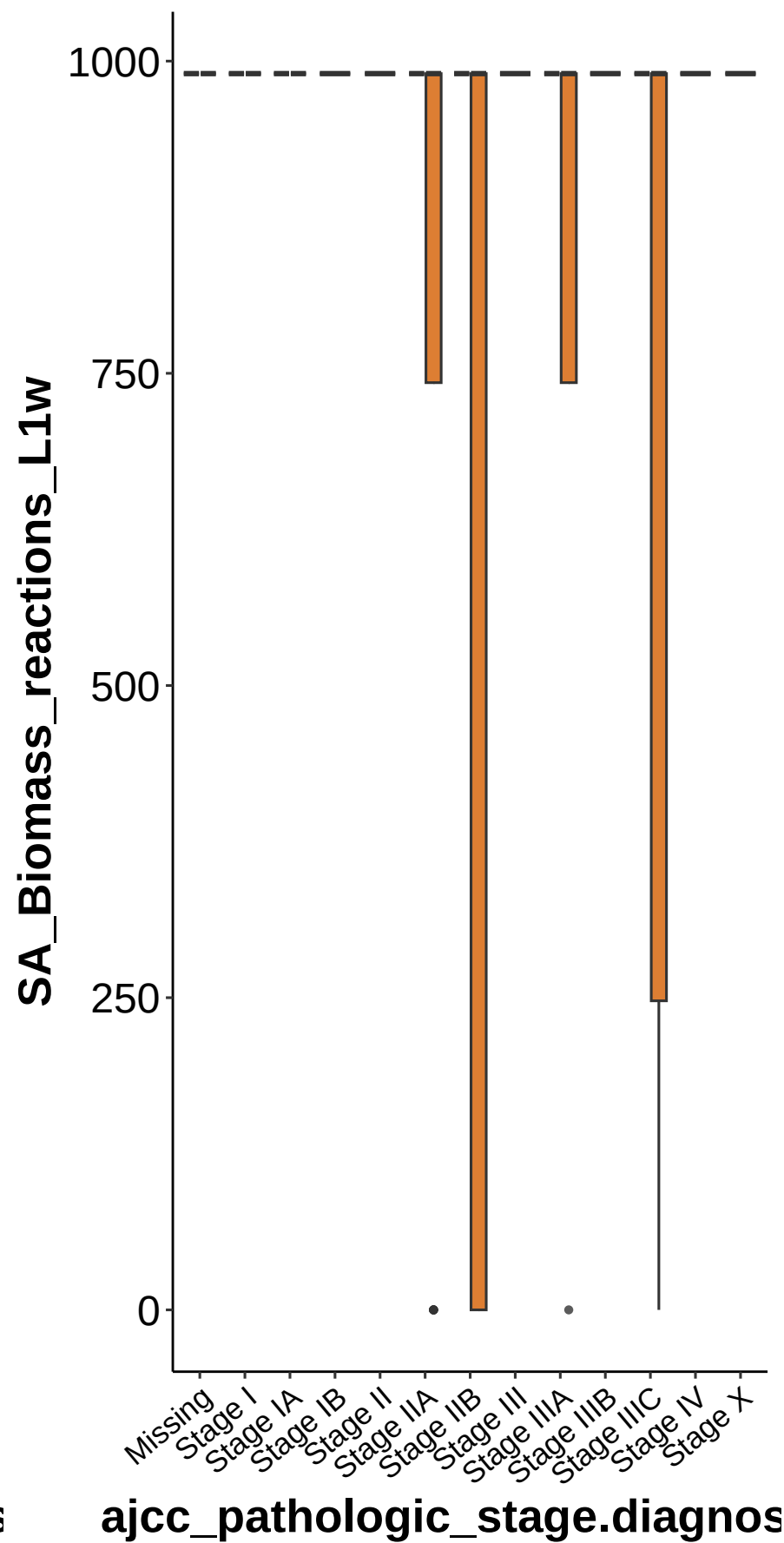
Group Core Divergent



SA_Biomass_reacti

¿La diferencia Core/Divergent es

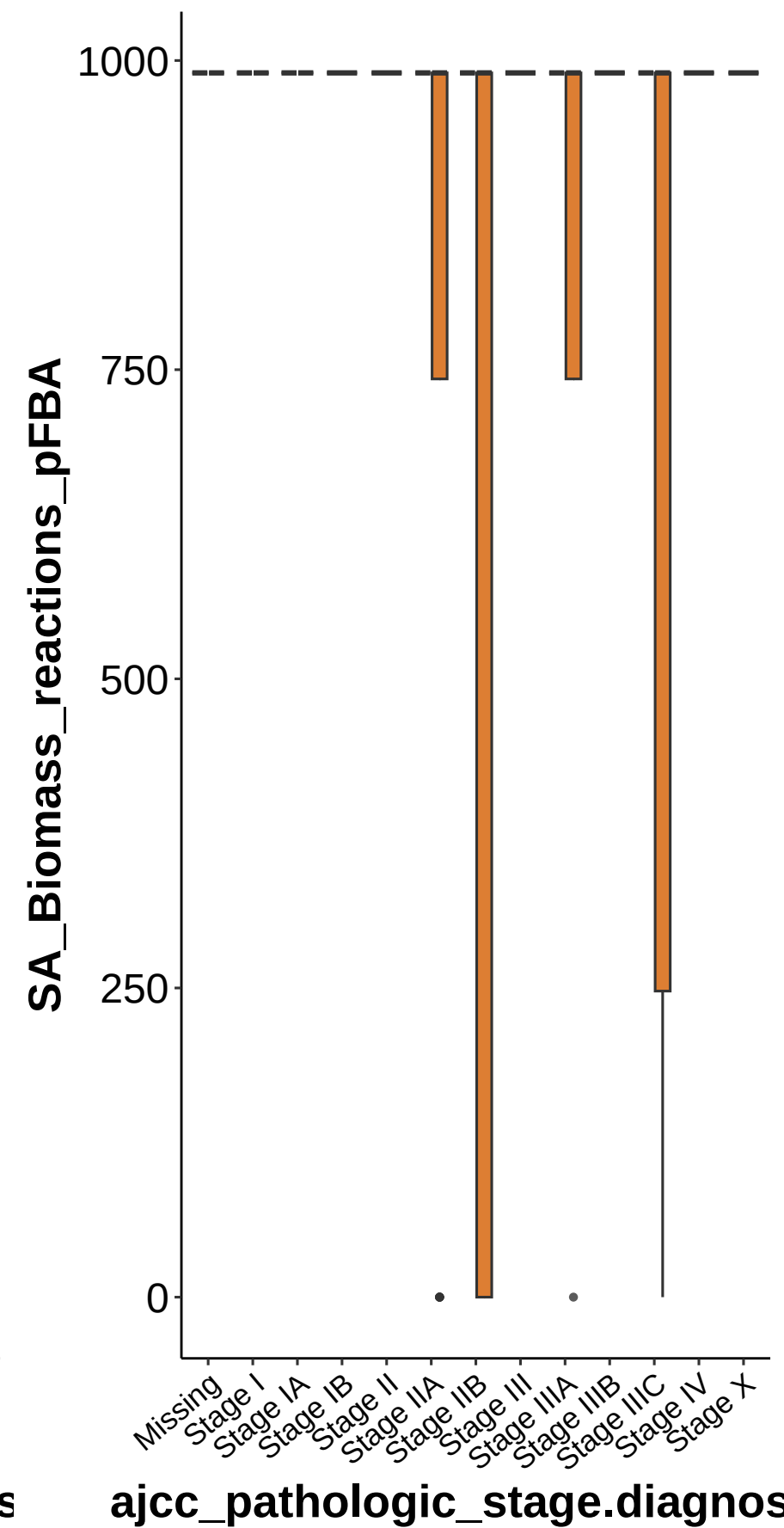
Group Core Divergent



SA_Biomass_reacti

¿La diferencia Core/Divergent es

Group Core Divergent

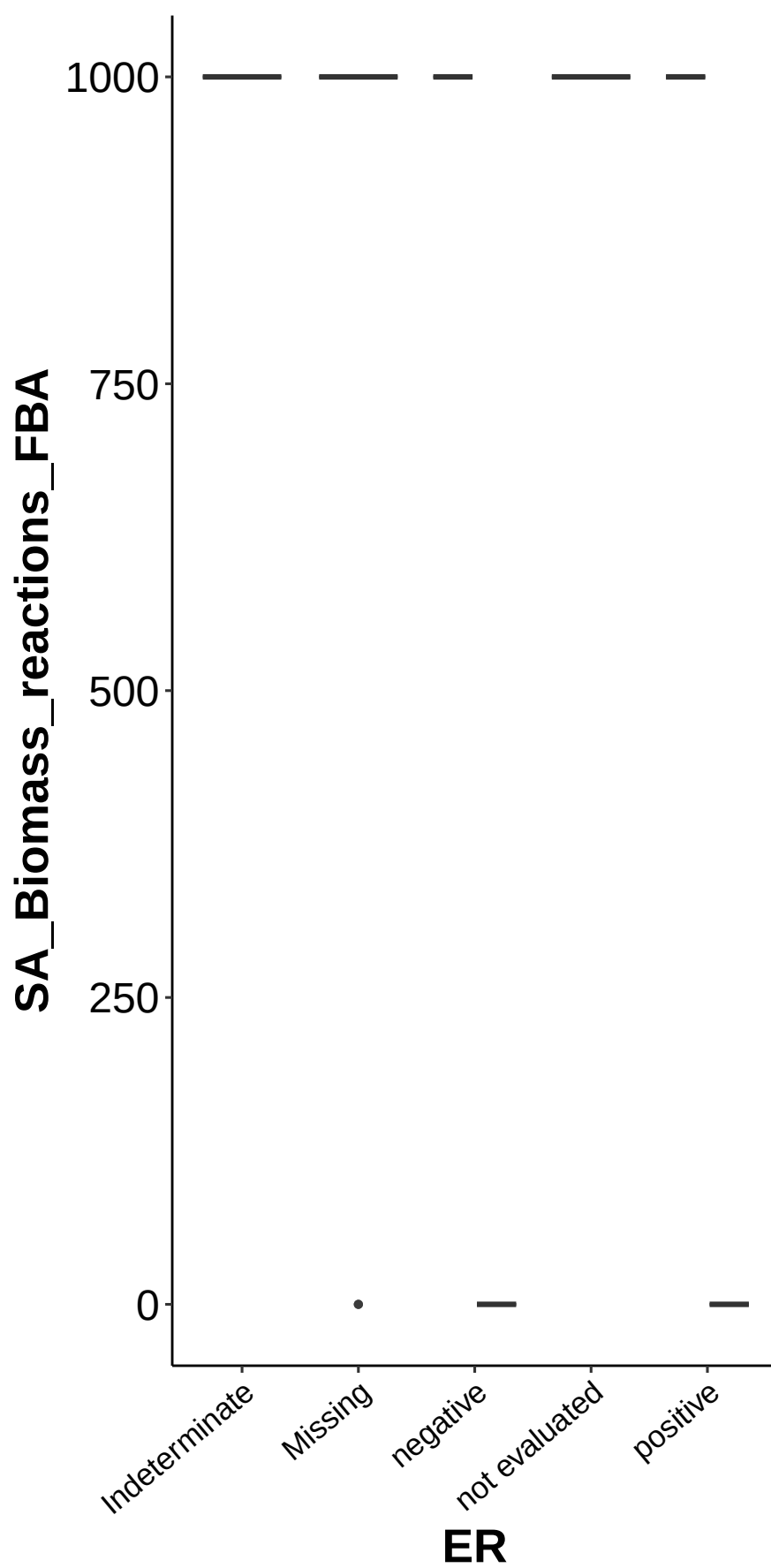


Integración: Top metabólicas × ER

SA_Biomass_reacti

¿La diferencia Core/Divergent es

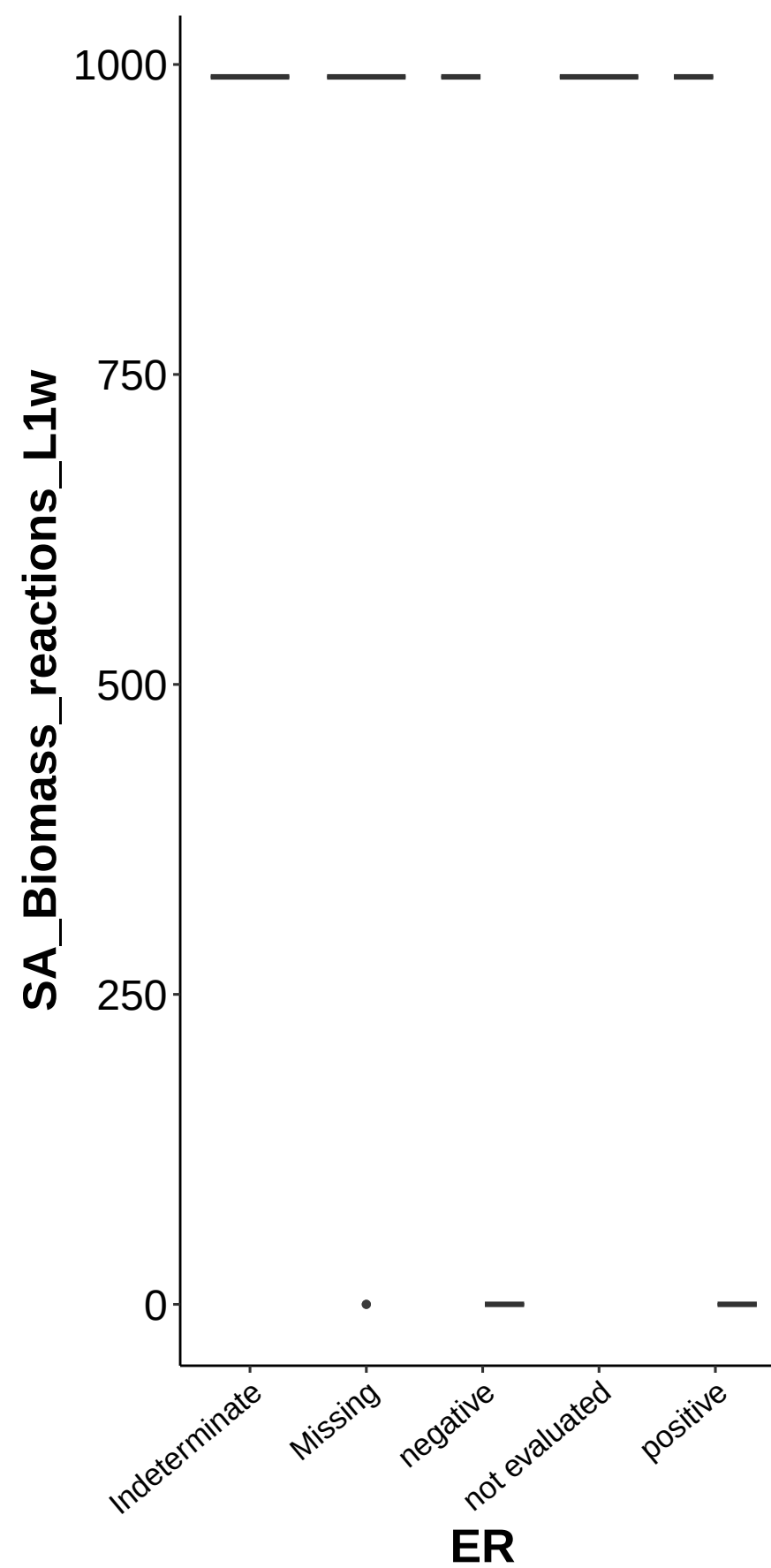
Group Core Divergent



SA_Biomass_reacti

¿La diferencia Core/Divergent es

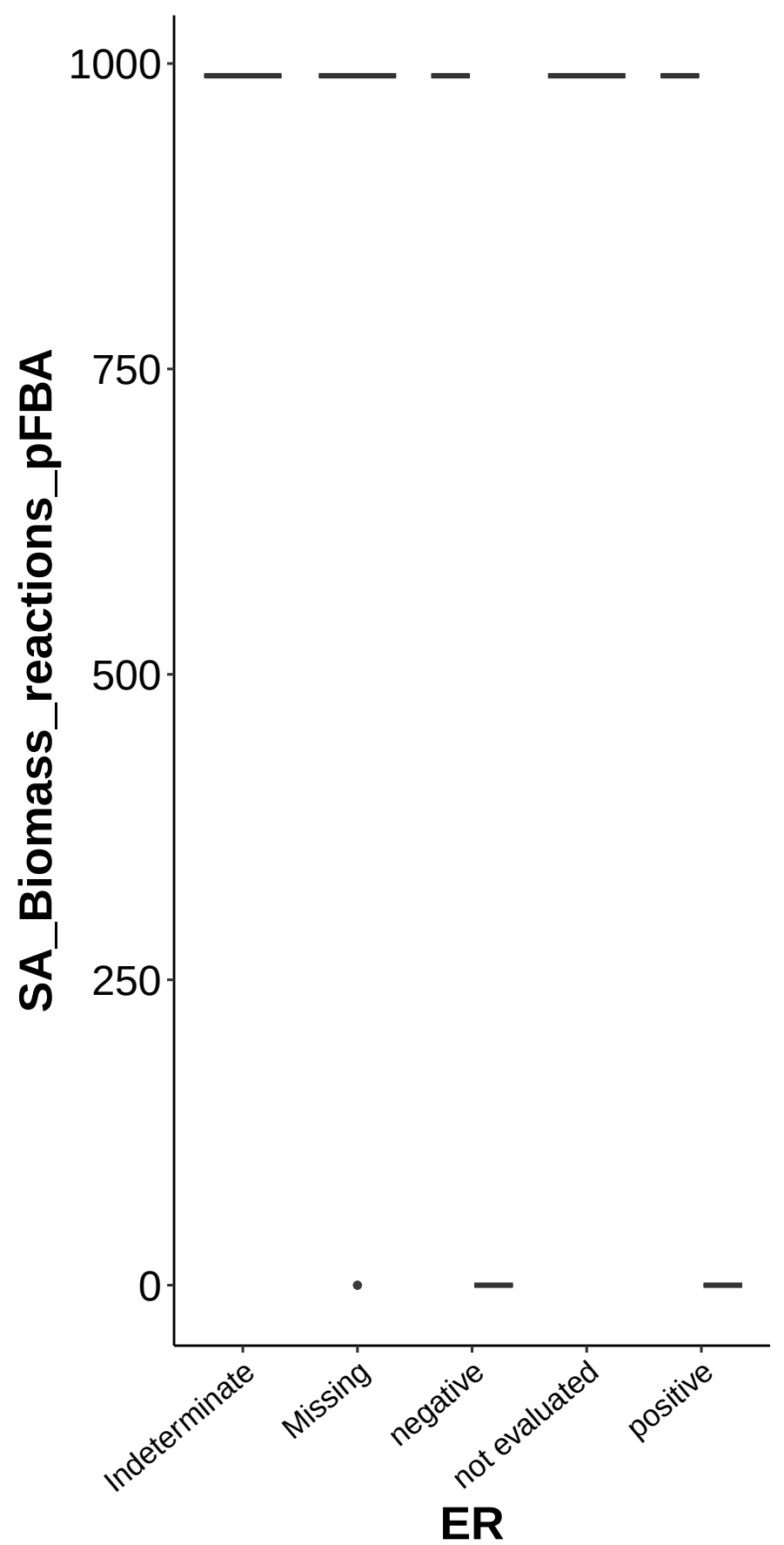
Group Core Divergent



SA_Biomass_reacti

¿La diferencia Core/Divergent es

Group Core Divergent

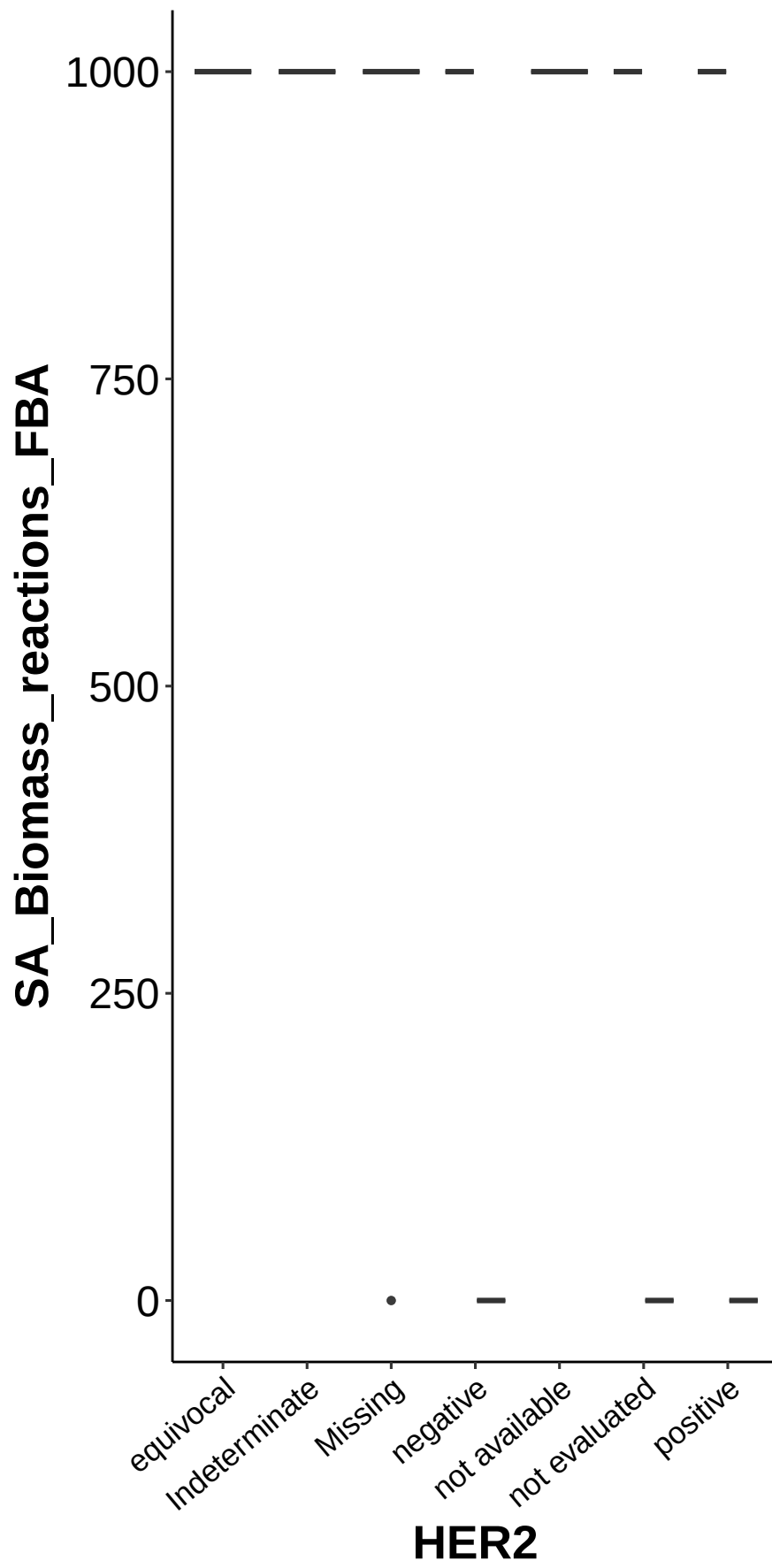


Integración: Top metabólicas × HER2

SA_Biomass_reacti

¿La diferencia Core/Divergent es

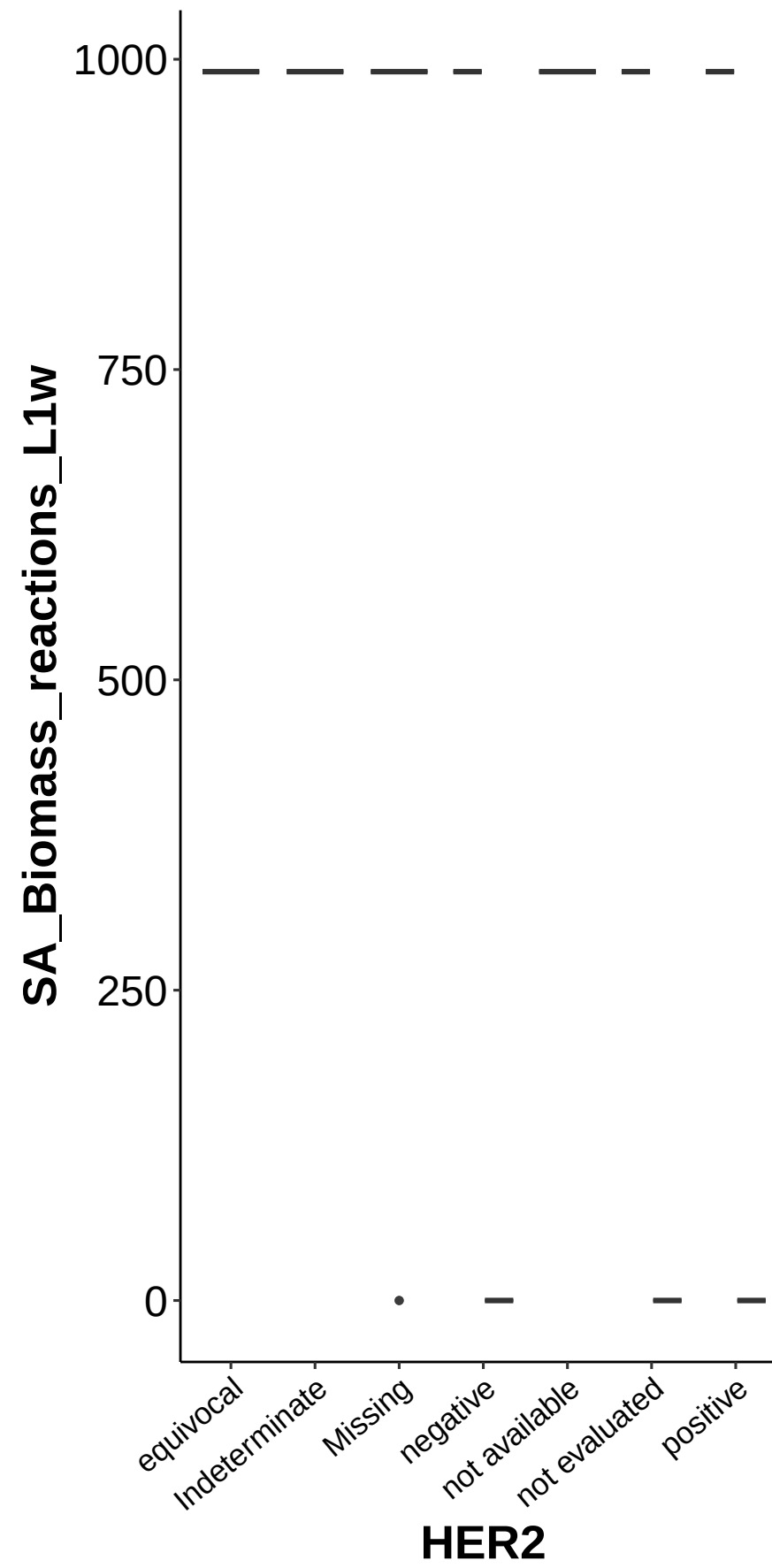
Group Core Divergent



SA_Biomass_reacti

¿La diferencia Core/Divergent es

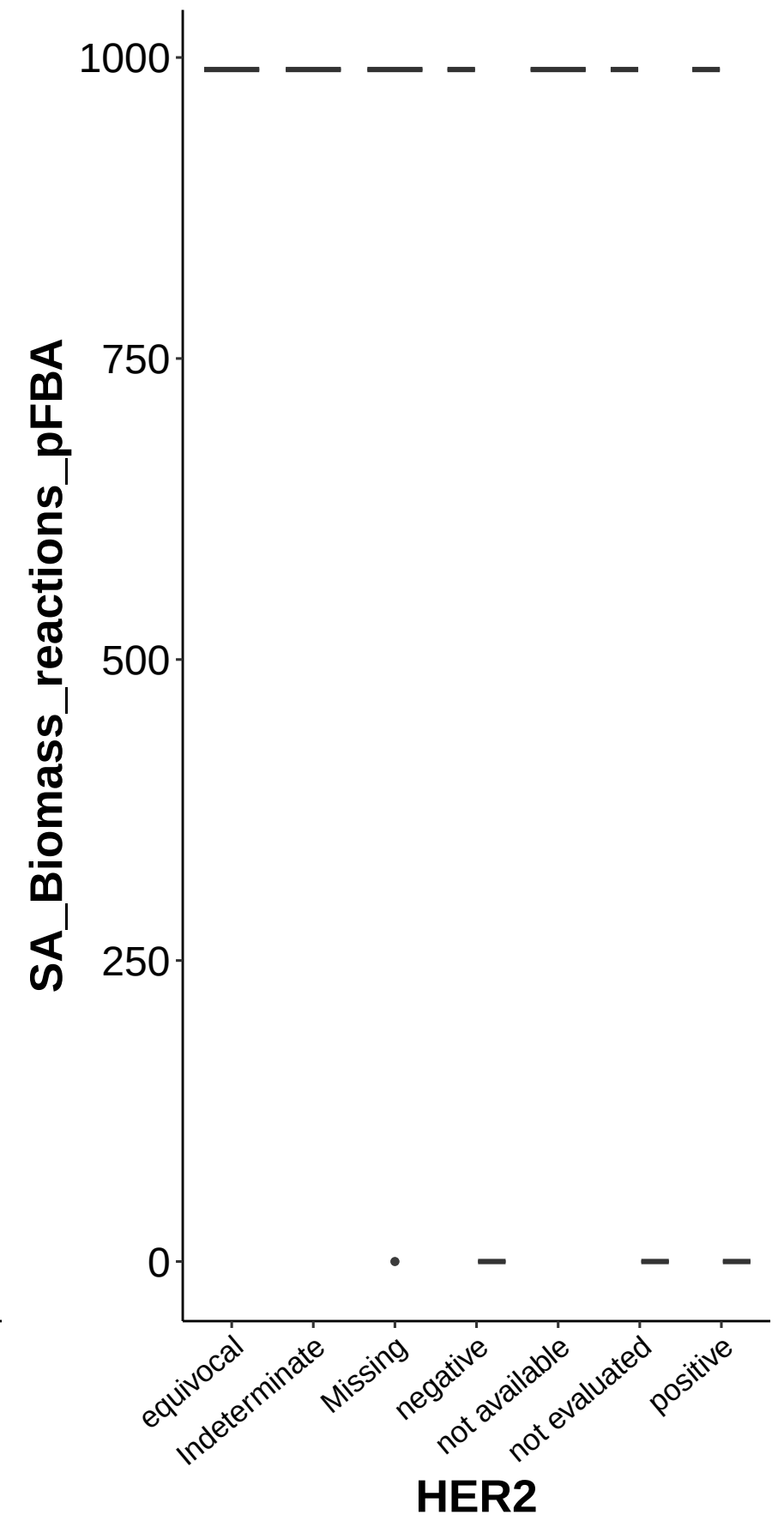
Group Core Divergent



SA_Biomass_reacti

¿La diferencia Core/Divergent es

Group Core Divergent

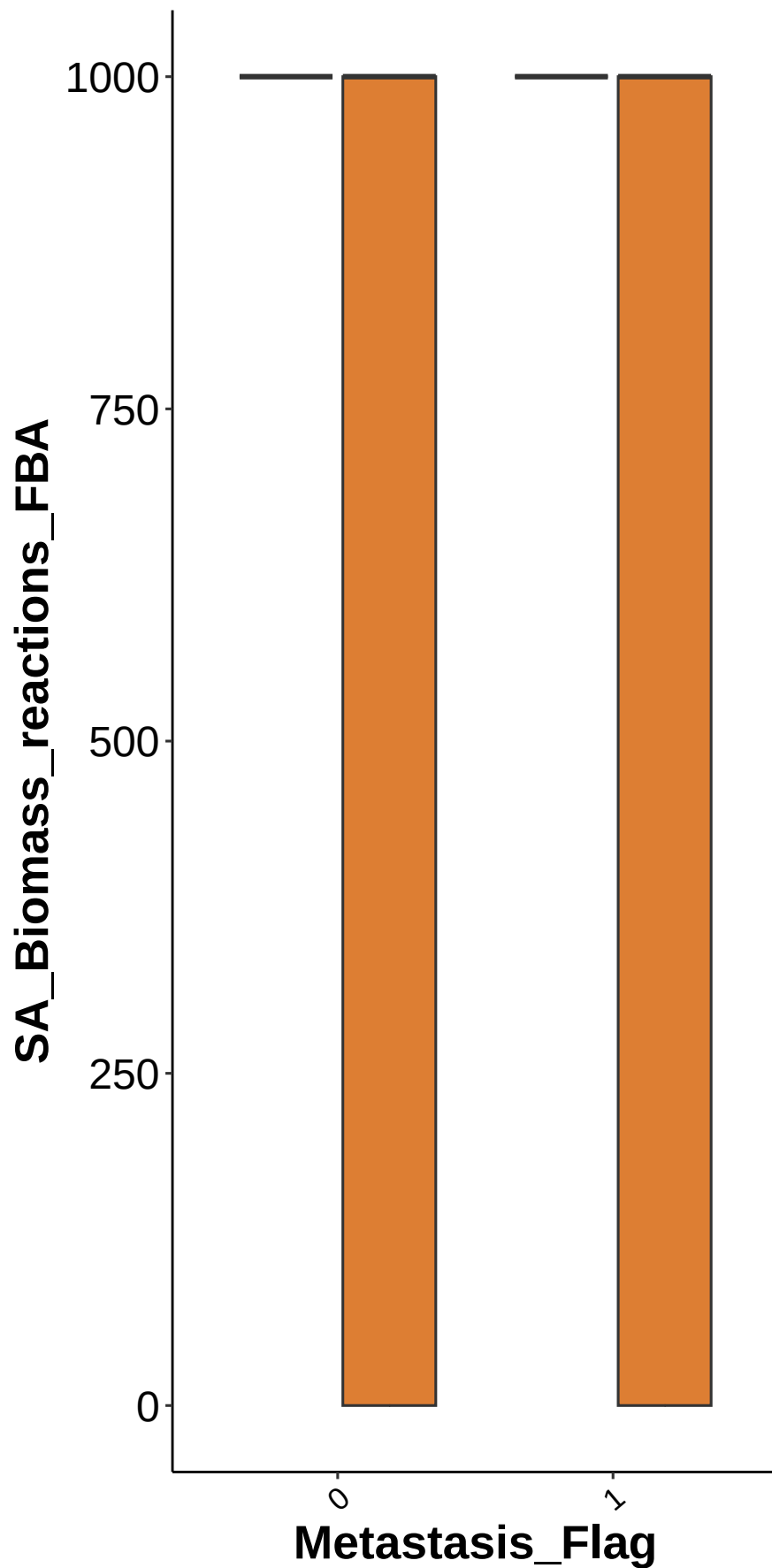


Integración: Top metabólicas × Metastasis_Flag

SA_Biomass_reacti

¿La diferencia Core/Divergent es

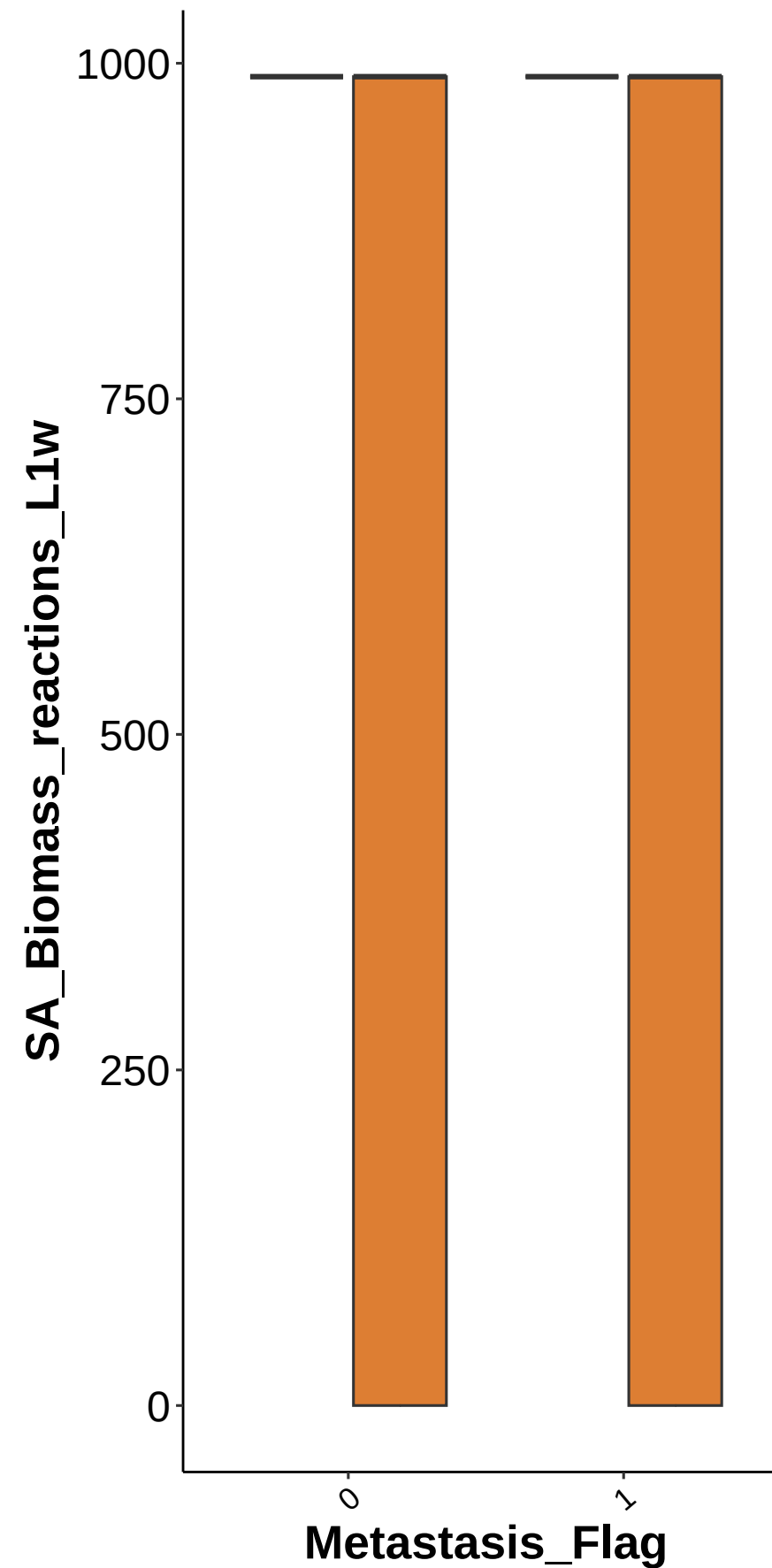
Group Core Divergent



SA_Biomass_reacti

¿La diferencia Core/Divergent es

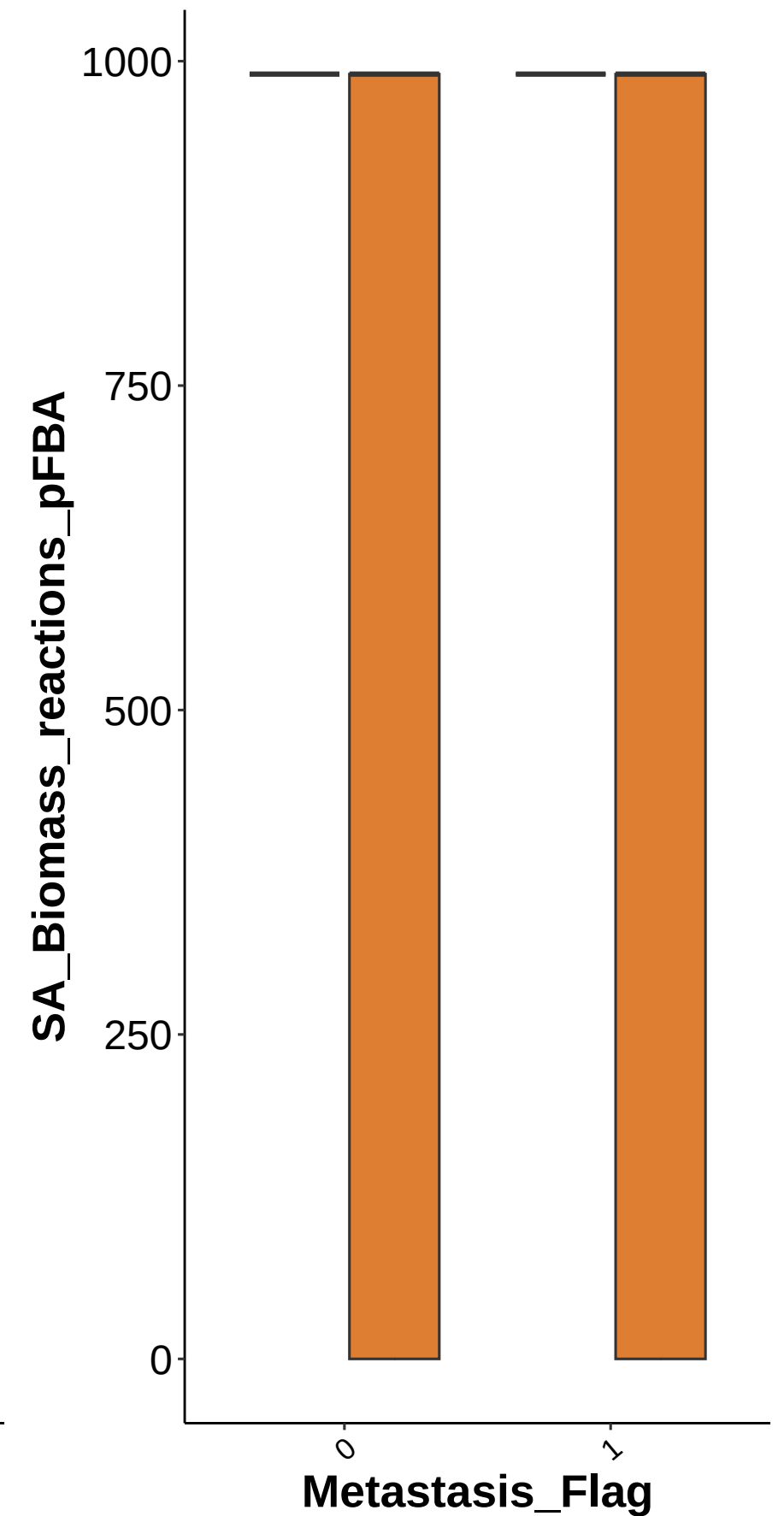
Group Core Divergent



SA_Biomass_reacti

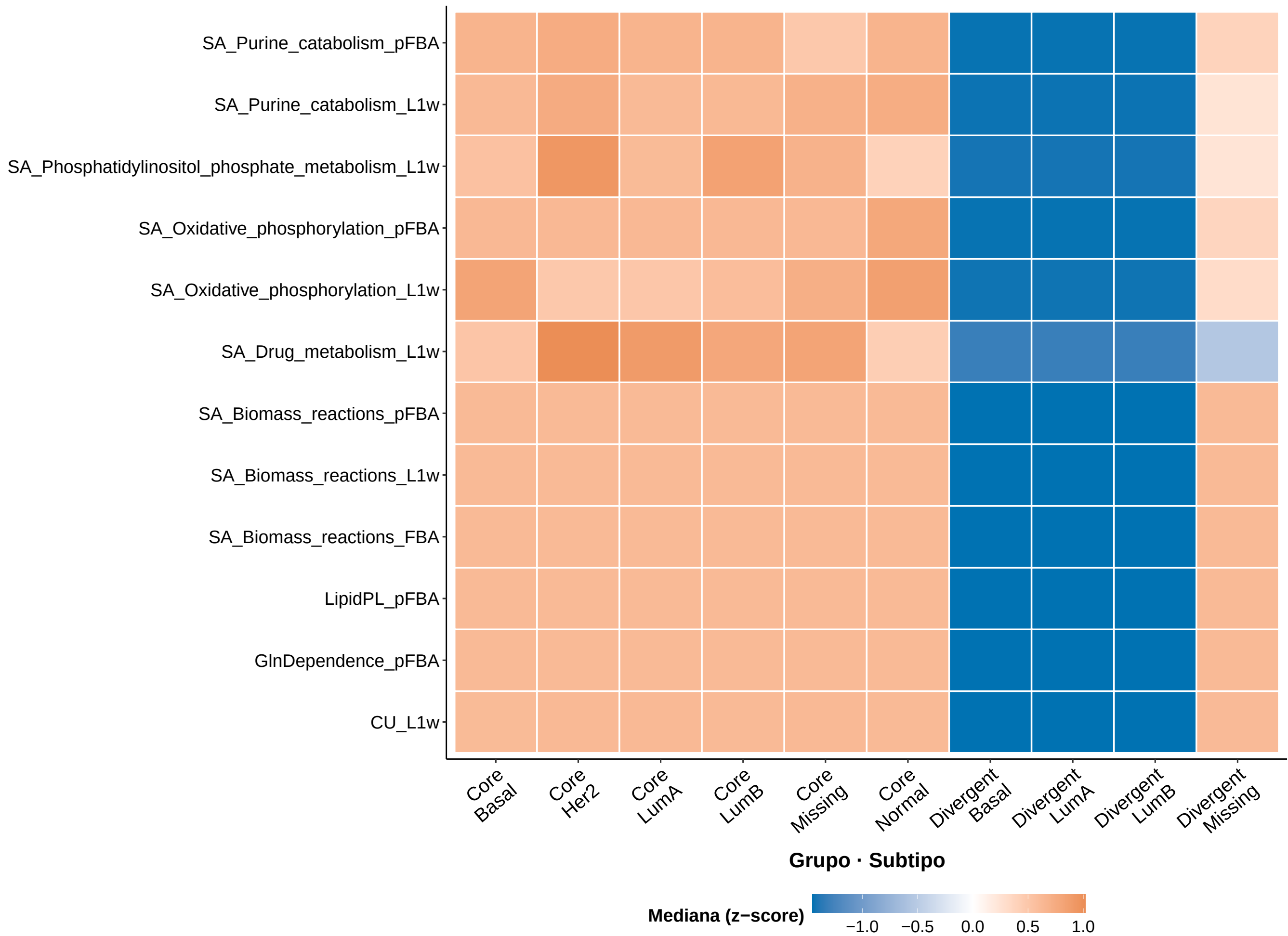
¿La diferencia Core/Divergent es

Group Core Divergent



Firma Metabólica por Subtipo Molecular

Inversión metabólica sistemática entre grupos Core y Divergent



SECCIÓN IV

Análisis de Supervivencia

¿El fenotipo metabólico Core/Divergent predice pronóstico?

Curva de Supervivencia: Core vs Divergent

